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Communication systems having pluggable modules

Abstract

A system includes a housing having a front panel, a substrate that is positioned at a distance from the front panel, and a data processor mounted on the substrate. The system includes a pluggable module having an optical module, at least one first optical connector, a first fiber optic cable optically coupled between the optical module and the first optical connector, and a fiber guide positioned between the optical module and the first optical connector and provides mechanical support for the optical module and the first optical connector. The optical module receives optical signals from the first optical connector and generates electrical signals based on the received optical signals, and the electrical signals are transmitted to the data processor. The pluggable module has a shape that enables the pluggable module to pass through an opening in the front panel to enable the optical module to be coupled to the substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. provisional application 63/212,013, filed on Jun. 17, 2021, U.S. provisional patent application 63/223,685, filed on Jul. 20, 2021, U.S. provisional patent application 63/225,779, filed on Jul. 26, 2021, U.S. provisional patent application 63/245,005, filed on Sep. 16, 2021, U.S. provisional patent application 63/245,011, filed on Sep. 16, 2021, U.S. provisional patent application 63/245,559, filed on Sep. 17, 2021, U.S. provisional patent application 63/272,025, filed on Oct. 26, 2021, U.S. provisional patent application 63/316,551, filed on Mar. 4, 2022, and U.S. provisional patent application 63/324,429, filed on Mar. 28, 2022. The entire disclosures of the above applications are hereby incorporated by reference.

TECHNICAL FIELD
(1) This document describes communication systems having pluggable modules.
BACKGROUND
(2) This section introduces aspects that can help facilitate a better understanding of the disclosure. Accordingly, the statements of this section are to be read in this light and are not to be understood as admissions about what is in the prior art or what is not in the prior art.
(3) For example, a data center can include servers installed in a rack, each server includes one or more data processors mounted on a circuit board disposed in an enclosure. Each server includes one or more optical communication modules for converting input optical signals received from optical fiber cables into input electrical signals that are provided to the one or more data processors, and converting output electrical signals from the one or more data processors to output optical signals that are output to the optical fiber cables. In some examples, pluggable optical modules (e.g., small form-factor pluggable (SFP) modules) can be used as network interface modules for connecting to fiber-optic cables.
SUMMARY OF THE INVENTION
(4) In a general aspect, a system includes: a housing that has a front panel; a first substrate that is positioned at a distance from the front panel, in which a data processor is mounted on the first substrate; and a pluggable module. The pluggable module includes an optical module (e.g., a co-packaged optical module), at least one first optical connector, a first fiber optic cable that is optically coupled between the optical module and the first optical connector, and a fiber guide that is positioned between the optical module and the first optical connector and provides mechanical support for the optical module and the first optical connector. The optical module is configured to receive optical signals from the first optical connector and generate electrical signals based on the received optical signals. The electrical signals or processed versions of the electrical signals are transmitted to the data processor. The pluggable module has a shape that enables the pluggable module to pass through an opening in the front panel to enable the optical module to be coupled to the substrate.
(5) Implementations can include one or more of the following features.
(6) The first optical connector can be configured to mate with a corresponding optical connector of an external fiber optic cable.
(7) The first optical connector can include a multi-fiber push on (MPO) connector.
(8) The fiber guide can have a length configured such that when the pluggable module is inserted through the opening in the front panel and the optical module is coupled to the first substrate or a module mounted on the first substrate, the at least one first optical connector is in a vicinity of the front panel to enable a user to attach at least one external fiber optic cable to the at least one first optical connector.
(9) The fiber guide can have a length configured such that when the pluggable module is inserted through the opening in the front panel and the optical module is coupled to the first substrate or a module mounted on the first substrate, the at

least one first optical connector has a front surface that is flush with, or protrudes from, a front surface of the front panel to enable a user to attach at least one external fiber optic cable to the at least one first optical connector.
(10) The fiber guide can have a length configured such that when the pluggable module is inserted through the opening in the front panel and the optical module is coupled to the first substrate or a module mounted on the first substrate, the at least one first optical connector has a front face that is within an inch of a front surface of the front panel.
(11) The fiber guide can have a length of at least one inch.
(12) The fiber guide can have a length of at least two inches.
(13) The fiber guide can have a length of at least four inches.
(14) The pluggable module can include at least two first optical connectors, and each first optical connector can be configured to be mated with a second optical connector of an external fiber optic cable.
(15) The pluggable module can include at least four first optical connectors, and each first optical connector can be configured to be mated with a second optical connector of an external fiber optic cable.
(16) The first fiber optic cable can include a fiber pigtail.
(17) The first substrate can have a main surface that is oriented at an angle in a range of 0 to 45 degrees relative to the front panel.
(18) The first substrate can be oriented parallel to the front panel.
(19) The first substrate can have a first side and a second side that is opposite the first side, the data processor can include electrical contacts that are electrically coupled to electrical contacts on the first side of the first substrate, the pluggable module can include electrical contacts that are electrically coupled to electrical contacts on the second side of the first substrate, and at least some of the electrical contacts on the first side of the first substrate can be electrically coupled to at least some of the electrical contacts on the second side of the first substrate.
(20) The first substrate can include at least one of a ceramic substrate, an organic high density build-up substrate, or a silicon substrate.
(21) The system can include a second substrate, the data processor can include electrical contacts that are electrically coupled to electrical contacts on the first substrate, the pluggable module can include electrical contacts that are electrically coupled to electrical contacts on the second substrate, and at least some of the electrical contacts on the first substrate can be electrically coupled to at least some of the electrical contacts on the second substrate.
(22) The first substrate can be mounted on a first side of a third substrate or circuit board, and the second substrate can be mounted on a second side of the third substrate or circuit board.
(23) Each of the first and second substrate can include at least one of a ceramic substrate, an organic high density build-up substrate, or a silicon substrate.
(24) The system can include an inlet fan mounted near the front panel and configured to increase an air flow across a surface of at least one of (i) the optical module, or (ii) a heat dissipating device thermally coupled to the optical module.
(25) The system can include two or more pluggable modules. Each pluggable module can include an optical module, at least one first optical connector, a first fiber optic cable that is optically coupled between the optical module and the first optical connector, and a fiber guide that is positioned between the optical module and the first optical connector. The fiber guides can be configured to allow air blown from the inlet fan to flow past the fiber guides and carry away heat generated by the optical module.
(26) The system can include a laser module configured to provide optical power to the optical module.
(27) The system can include a second optical connector optically coupled to the laser module. The pluggable module can include a third optical connector that is configured to mate with the second optical connector when the pluggable module is coupled to the first substrate. The first optical connector can be optically coupled to the optical module to enable the optical module to receive the optical power from the laser module.
(28) The system can include a first heat dissipating device and a second heat dissipating device, the first heat dissipating device can be thermally isolated from the second heat dissipating device, the first heat dissipating device can be thermally coupled to the optical module, and the second heat dissipating device can be thermally coupled to the laser module.
(29) The system can provide an air gap between the first heat dissipating device and the second heat dissipating device.
(30) The system can include a thermally insulating material positioned between the first heat dissipating device and the second heat dissipating device.
(31) In some examples, each of the heat dissipating device and the second heat dissipating device can be made of a material having a thermal conductivity greater than 50 W/mK.
(32) In some examples, each of the heat dissipating device and the second heat dissipating device can be made of a material having a thermal conductivity greater than 100 W/mK.
(33) In some examples, each of the heat dissipating device and the second heat dissipating device can be made of a material having a thermal conductivity greater than 200 W/mK.
(34) In some examples, the thermally insulating material can have a thermal conductivity less than 10 W/mK.
(35) In some examples, the thermally insulating material can have a thermal conductivity less than 1 W/mK.
(36) The fiber guide can include at least one of metal or a thermal conductive material.
(37) The fiber guide can include a hollow tube.
(38) The fiber guide can be rigid along a direction from the first optical connector to the optical module and can have a strength sufficient to withstand a compression force exerted on the pluggable module when the pluggable module is

inserted through the opening in the front panel and coupled to the first substrate.
(39) The fiber guide can have a spatial fan-out design such that a first portion of the fiber guide near the optical module has a smaller dimension compared to the dimension of a second portion of the fiber guide near the at least one first optical connector.
(40) The at least one first optical connector can have an overall footprint that is larger than a footprint of the optical module.
(41) The data processor can include at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, an application specific integrated circuit (ASIC), or a storage device.
(42) A photon supply can be disposed in, on, or near the fiber guide, and the photon supply can be configured to provide optical power supply light to the optical module.
(43) The photon supply can be thermally coupled to an inner surface or an outer surface of the fiber guide, and the fiber guide can be configured to assist in dissipating heat from the photon supply.
(44) The system can include guide rails configured to guide the optical module as the optical module move from a first position near the front panel to a second position near the first substrate.
(45) The optical module can include a co-packaged optical module including a photonic integrated circuit and one or more electrical integrated circuits that condition electrical signals transmitted to or from the photonic integrated circuit.
(46) The system can include a co-packaged optical module (CPO) mount attached to the first substrate, and the guide rails can be configured to provide rigid connections between the CPO mount and the front panel or a front portion of the fiber guide.
(47) The photonic integrated circuit can include at least one of (i) a photodetector to convert optical signals to electrical signals, or (ii) a modulator to convert electrical signals to optical signals.
(48) The system can include a co-packaged optical module (CPO) mount and a bolster plate, in which the co-packaged optical module is mounted on the substrate through the CPO mount, and the bolster plate is positioned to the rear of the substrate and configured to exert a force in a front direction when the guide rails are fastened to a front portion of the fiber guide or to the front panel.
(49) The optical module can have a first side and a second side, the first fiber optical cable can have a first end that has a two-dimensional arrangement of optical fiber cores, the first side of the optical module can be optically coupled to the two-dimensional arrangement of optical fiber cores, and the second side of the optical module can have a two-dimensional arrangement of electrical contacts that are configured to mate with a two-dimensional arrangement of electrical contacts on the first substrate.
(50) In some examples, the two-dimensional arrangement of electrical contacts of the optical module can include at least two rows of electrical contacts, and each row can include at least two electrical contacts.
(51) In some examples, the two-dimensional arrangement of electrical contacts of the optical module can include at least four rows of electrical contacts, and each row can include at least four electrical contacts.
(52) In some examples, the two-dimensional arrangement of electrical contacts of the optical module can include at least ten rows of electrical contacts, and each row can include at least ten electrical contacts.
(53) In another general aspect, an apparatus includes: a pluggable module including a co-packaged optical module, at least one first optical connector, a first fiber optic cable that is optically coupled between the co-packaged optical module and the first optical connector, and a fiber guide that is positioned between the co-packaged optical module and the first optical connector and provides mechanical support for the co-packaged optical module and the first optical connector. The co-packaged optical module is configured to receive optical signals from the at least one first optical connector, and generate electronic signals based on the optical signals.
(54) Implementations can include one or more of the following features. The fiber guide can include at least one of metal or a thermal conductive material.
(55) The fiber guide can include a hollow tube.
(56) The fiber guide can be rigid along a direction from the first optical connector to the co-packaged optical module and can have a strength sufficient to withstand a compression force exerted on the pluggable module when the pluggable module is inserted through an opening in a front panel of a housing and coupled to the substrate.
(57) The fiber guide can have a spatial fan-out design such that a first portion of the fiber guide near the co-packaged optical module has a smaller dimension compared to the dimension of a second portion of the fiber guide near the at least one first optical connector.
(58) The at least one first optical connector can have an overall footprint that is larger than a footprint of the co-packaged optical module.
(59) The co-packaged optical module can have a first side and a second side, the first fiber optical cable can have a first end that has a two-dimensional arrangement of optical fiber cores, the first side of the optical module can be optically coupled to the two-dimensional arrangement of optical fiber cores, and the second side of the optical module can have a two-dimensional arrangement of electrical contacts.
(60) In some examples, the two-dimensional arrangement of electrical contacts of the optical module can include at least two rows of electrical contacts, and each row can include at least two electrical contacts.
(61) In some examples, the two-dimensional arrangement of electrical contacts of the optical module can include at least

four rows of electrical contacts, and each row can include at least four electrical contacts.
(62) In some examples, the two-dimensional arrangement of electrical contacts of the optical module can include at least ten rows of electrical contacts, and each row can include at least ten electrical contacts.
(63) In another general aspect, a rackmount server includes: a housing having a front panel and a rear panel. The front panel defines an opening, and the rear panel is at a first distance from the front panel. The rackmount server includes a substrate that is positioned at a second distance from the front panel. The second distance is less than one-third of the first distance. The rackmount server includes a data processor that is mounted on the substrate. The substrate has a main surface that is oriented at an angle in a range of 0 to 45 degrees relative to the front panel. In some examples, the substrate can have electrical contacts that are configured to the electrically coupled to electrical contacts of a co-packaged optical module. In some examples, a first module is mounted on the substrate, and the first module has electrical contacts that are configured to the electrically coupled to electrical contacts of a co-packaged optical module.
(64) Implementations can include one or more of the following features. The substrate can be oriented substantially parallel to the front panel.
(65) The opening in the front panel can be configured to allow a pluggable module that includes the co-packaged optical module to be inserted through the opening to enable the co-packaged optical module to be electrically coupled to the electrical contacts on the substrate or the electrical contacts on the first module mounted on the substrate.
(66) The rackmount server can include the pluggable module.
(67) The pluggable module can include the co-packaged optical module, at least one first optical connector, a first fiber optic cable that is optically coupled between the co-packaged optical module and the first optical connector, and a fiber guide that is positioned between the co-packaged optical module and the first optical connector and provides mechanical support for the co-packaged optical module and the first optical connector.
(68) The co-packaged optical module can be configured to receive optical signals from the first optical connector, generate electrical signals based on the received optical signals, and transmit the electrical signals to the data processor.
(69) The data processor can include at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC).
(70) The substrate can have a two-dimensional arrangement of electrical contacts that are configured to be electrically coupled to a two-dimensional arrangement of electrical contacts of the co-package optical module.
(71) In some examples, the two-dimensional arrangement of electrical contacts of the substrate can include at least two rows of electrical contacts, and each row can include at least two electrical contacts.
(72) In some examples, the two-dimensional arrangement of electrical contacts of the substrate can include at least four rows of electrical contacts, and each row can include at least four electrical contacts.
(73) In some examples, the two-dimensional arrangement of electrical contacts of the substrate can include at least ten rows of electrical contacts, and each row can include at least ten electrical contacts.
(74) The substrate can have a plurality of groups of two-dimensional arrangement of electrical contacts that are configured to be electrically coupled to a corresponding plurality of groups of two-dimensional arrangement of electrical contacts of co-package optical modules.
(75) In some examples, the plurality of groups of two-dimensional arrangement of electrical contacts can include at least four groups of two-dimensional arrangement of electrical contacts, each group of two-dimensional arrangement of electrical contacts can include at least four rows of electrical contacts, and each row can include at least four electrical contacts.
(76) In some examples, the plurality of groups of two-dimensional arrangement of electrical contacts can include at least ten groups of two-dimensional arrangement of electrical contacts, each group of two-dimensional arrangement of electrical contacts can include at least ten rows of electrical contacts, and each row can include at least ten electrical contacts.
(77) In another general aspect, a system includes: a first substrate including at least one of a ceramic substrate, an organic high density build-up substrate, or a silicon substrate; a data processor mounted on a rear side of the first substrate; and a co-packaged optical module. The co-packaged optical module is removably coupled to a front side of the first substrate and configured to receive optical signals from an optical connector, generate electrical signals based on the received optical signals, and transmit the electrical signals to the data processor. The system includes a printed circuit board attached to the rear side of the first substrate, in which the printed circuit board includes an opening, and the data processor protrudes or partially protrudes through the opening, and the printed circuit board provides electrical power to the data processor through signal lines or traces in or on the first substrate.
(78) In another general aspect, an apparatus including an optical transceiver module is provided. The optical transceiver module includes a photonic integrated circuit configured to perform at least one of (i) converting optical signals to electrical signals, or (ii) converting electrical signals to optical signals; and at least one optical connector, in which the photonic integrated circuit is configured to receive optical signals from the at least one optical connector or transmit optical signals to the at least one optical connector. The optical transceiver module includes a plurality of electrical contacts, in which the photonic integrated circuit is configured to receive electrical signals from the plurality of electrical contacts or provide electrical signals to the plurality of electrical contacts. The optical transceiver module includes at least one electronic component positioned in an electrical signal path between the photonic integrated circuit and the plurality

of electrical contacts and configured to process electrical signals sent to or from the photonic integrated circuit; and at least one laser configured to provide optical power supply light to the photonic integrated circuit. The optical transceiver module includes a first thermal path and a second thermal path, in which the second thermal path is thermally isolated from the first thermal path, the first thermal path enables heat from the at least one laser to be conducted outside of the optical module, and the second thermal path enables heat from the at least one electronic component to be conducted outside of the optical module.
(79) Implementations can include one or more of the following features. The optical transceiver module can include a pluggable optical transceiver module, the plurality of electrical contacts of the pluggable optical transceiver module are configured to be removably and electrically coupled to corresponding electrical contacts of a data processing apparatus.
(80) The plurality of electrical contacts of the optical transceiver module can be configured to be fixedly and electrically coupled to corresponding electrical contacts of a data processing apparatus.
(81) The at least one electronic component can include at least one of a serializer, a deserializer, a serializer/deserializer, a digital signal processor, a driver module, or an amplifier module.
(82) The at least one laser can be positioned closer to the at least one optical connector and farther away from the plurality of electrical contacts.
(83) The optical transceiver module can have a form factor that complies with at least one of SFP (small form-factor pluggable), SFP+ (or 10 Gb SFP), SFP28, OSFP (octal SFP), OSFP-XD (OSFP extra dense), QSFP (quad small form-factor pluggable), QSFP+, QSFP28, QSFP56, or QSFP-DD (quad small form-factor pluggable double density) standard.
(84) The at least one optical connector can have a first end that has a two-dimensional arrangement of optical fiber cores, and the photonic integrated circuit can be optically coupled to the two-dimensional arrangement of optical fiber cores using a two-dimensional arrangement of optical couplers.
(85) The optical transceiver module can include a housing, the at least one electrical component and the at least one laser can be positioned inside the housing, and the housing can define an opening. The optical transceiver module can include a first heat dissipating device and a second heat dissipating device, the second heat dissipating device can be thermally isolated from the first heat dissipating device, and the second heat dissipating device can be thermally coupled to the housing. The first thermal path can extend from the at least one laser through the opening defined by the housing to the first heat dissipating device, and the second thermal path can extend from the at least one electrical component through the housing to the second heat dissipating device.
(86) The optical transceiver module can provide an air gap between the first heat dissipating device and the second heat dissipating device.
(87) The optical transceiver module can include a thermally insulating material positioned between the first heat dissipating device and the second heat dissipating device.
(88) In some examples, each of the heat dissipating device and the second heat dissipating device can be made of a material having a thermal conductivity greater than 50 W/mK.
(89) In some examples, each of the heat dissipating device and the second heat dissipating device can be made of a material having a thermal conductivity greater than 100 W/mK.
(90) In some examples, each of the heat dissipating device and the second heat dissipating device can be made of a material having a thermal conductivity greater than 200 W/mK.
(91) In some examples, the thermally insulating material can have a thermal conductivity less than 10 W/mK.
(92) In some examples, the thermally insulating material can have a thermal conductivity less than 1 W/mK.
(93) The optical transceiver module can include a fiber guide that is positioned between the photonic integrated circuit and the at least one optical connector and can provide mechanical support for the first optical connector and the photonic integrated circuit or a module that includes the photonic integrated circuit.
(94) The fiber guide can include at least one of metal or a thermal conductive material.
(95) The fiber guide can include a hollow tube.
(96) The fiber guide can be rigid along a direction from the at least one optical connector to the photonic integrated circuit or the module that includes the photonic integrated circuit and can have a strength sufficient to withstand a compression force exerted on the optical transceiver module to cause the optical transceiver module to engage a receptor of another apparatus and cause the plurality of electrical contacts to be electrically coupled to corresponding electrical contacts of the other apparatus.
(97) The fiber guide can have a spatial fan-out design such that a first portion of the fiber guide near the photonic integrated circuit has a smaller dimension compared to the dimension of a second portion of the fiber guide near the at least one optical connector.
(98) The plurality of electrical contacts can include a two-dimensional arrangement of electrical contacts.
(99) In some examples, the two-dimensional arrangement of electrical contacts of the optical module can include at least two rows of electrical contacts, and each row can include at least two electrical contacts.
(100) In some examples, the two-dimensional arrangement of electrical contacts of the optical module can include at least four rows of electrical contacts, and each row can include at least four electrical contacts.
(101) In some examples, the two-dimensional arrangement of electrical contacts of the optical module can include at least ten rows of electrical contacts, and each row can include at least ten electrical contacts.
(102) In another general aspect, a rackmount server includes a plurality of the systems and/or apparatuses described

above.
(103) In another general aspect, a data center includes a plurality of the rackmount servers described above.
(104) In another general aspect, a method includes providing a data processing server including a housing having a front panel that defines an opening; and providing a substrate positioned in the housing spaced apart from the front panel, in which a data processor is electrically coupled to a rear side of the substrate. The method includes providing a pluggable module comprising an optical module, at least one first optical connector, a first fiber optic cable that is optically coupled between the optical module and the first optical connector, and a fiber guide that is positioned between the optical module and the first optical connector and provides mechanical support for the optical module and the first optical connector. The method includes optically coupling an external fiber optic cable to the optical connector of the pluggable module; inserting the pluggable module through the opening in the front panel and electrically coupling a two-dimensional arrangement of electrical contacts of the optical module with a corresponding two-dimensional arrangement of electrical contacts on a front side of the substrate; and establishing a communication path between the data processor and the external fiber optic cable through the pluggable module.
(105) Implementations can include one or more of the following features. In some examples, the method can include transmitting data between the data processor and the external fiber optic cable through the pluggable module with a bandwidth of at least 500 Gbps.
(106) In some examples, the method can include transmitting data between the data processor and the external fiber optic cable through the pluggable module with a bandwidth of at least 1 Tbps.
(107) The front panel can define a plurality of openings, and the front side of the substrate can include a plurality of groups of two-dimensional arrangements of electrical contacts. The method can include providing a plurality of the pluggable modules; optically coupling a plurality of external fiber optic cables to the optical connectors of the pluggable modules; inserting the pluggable modules through the openings in the front panel and electrically coupling groups of two-dimensional arrangements of electrical contacts of the optical modules with corresponding groups of two-dimensional arrangements of electrical contacts on the front side of the substrate; and establishing communication paths between the data processor and the external fiber optic cables through the pluggable modules.
(108) In some examples, the plurality of pluggable modules can include at least 10 pluggable modules, and the method can include transmitting data between the data processor and the external fiber optic cables through the pluggable modules with an aggregate bandwidth of at least 5 Tbps.
(109) In some examples, the plurality of pluggable modules can include at least 30 pluggable modules, and the method can include transmitting data between the data processor and the external fiber optic cables through the pluggable modules with an aggregate bandwidth of at least 15 Tbps.
(110) In another general aspect, a method including operating any of the systems, apparatuses, rackmount servers, and/or data centers described above is provided.
(111) In another general aspect, a method including assembling and/or constructing any of the systems, apparatuses, rackmount servers, and/or data centers described above is provided.
(112) In another general aspect, an apparatus includes a pluggable optical module. The pluggable optical module includes a fiber connector, an optical module, a fiber harness, and an edge connector. The fiber connector is configured to be optically coupled to an optical fiber cable. The optical module includes a photonic integrated circuit having a first surface. A plurality of optical couplers are provided at the first surface of the photonic integrated circuit. The fiber harness is optically coupled between the fiber connector and the first surface of the photonic integrated circuit. The fiber harness includes a plurality of optical fibers and an optical fiber connector. The optical fiber connector is configured to optically couple the plurality of optical fibers to the first surface of the photonic integrated circuit. The optical fiber connector includes a two-dimensional arrangement of fiber ports. The two-dimensional arrangement of fiber ports and the optical couplers at the first surface of the photonic integrated circuit are configured to enable light signals to be transmitted between the photonic integrated circuit and the plurality of optical fibers. The edge connector includes conductive pads configured to be electrically coupled to conductive pads of a receptacle when the edge connector is mated with the receptacle. The conductive pads of the edge connector are electrically coupled to the optical module.
(113) Implementations can include one or more of the following features. In some examples, the two-dimensional arrangement of fiber ports can include at least two rows of fiber ports, and each row can include at least eight fiber ports.
(114) In some examples, the two-dimensional arrangement of fiber ports can include at least three rows of fiber ports, and each row can include at least eight fiber ports.
(115) In some examples, the two-dimensional arrangement of fiber ports can include at least four rows of fiber ports, and each row can include at least eight fiber ports.
(116) The pluggable optical module can comply with a small form factor pluggable module specification including at least one of SFP (small form-factor pluggable), SFP+, 10 Gb SFP, SFP28, OSFP (octal SFP), OSFP-XD (OSFP extra dense), QSFP (quad small form-factor pluggable), QSFP+, QSFP28, QSFP56, or QSFP-DD (quad small form-factor pluggable double density).
(117) The pluggable optical module can have a length not more than 200 mm, a width not more than 50 mm, and a height not more than 26 mm.
(118) The pluggable optical module can include a housing having an inner upper wall and an inner lower wall, the edge connector can have an upper surface extending along a first plane that is at a first distance d**1** relative to the inner upper

wall, the edge connector can have a lower surface extending along a second plane that is at a second distance d**2** relative to the inner lower wall. The fiber harness can be substantially vertically coupled to the first surface of the photonic integrated circuit such that light from the fiber harness is directed toward the first surface of the photonic integrated circuit at an angle θ**1** relative to a direction vertical to the first surface of the photonic integrated circuit, 0<θ**1**<10°. The fiber harness when extending from the first surface of the photonic integrated circuit and bending to a direction parallel to the first surface can require a clearance distance of at least d**3** so as to not damage the optical fibers in the fiber harness, and d**1**<d**3**, and d**2**<d**3**.
(119) The housing can have a first inner side wall and a second inner side wall, the substrate or circuit board can be attached to the first inner side wall, a distance from the first surface of the photonic integrated circuit to the second inner side wall can be d**4** in which d**3**<d**4**.
(120) The first surface of the photonic integrated circuit can be oriented at an angle θ**2** relative to the inner upper wall, and 45°<θ**2**<135°.
(121) In some examples, 70°<θ**2**<110°. In some examples, 80°<θ**2**<100°. In some examples, 85°<θ**2**<95°.
(122) The photonic integrated circuit can be mounted on a substrate or circuit board that is electrically coupled to the edge connector by one or more flexible cables.
(123) The photonic integrated circuit can be mounted on an upper surface of a substrate or circuit board, the edge connector can have an upper surface and a lower surface, the lower surface of the edge connector can be attached to the upper surface of the substrate or circuit board. The upper surface of the substrate or circuit board can be at a distance d**4** relative to the inner upper wall of the housing in which d**3**<d**4**.
(124) The photonic integrated circuit can be configured to perform at least one of (i) convert optical signals received from the optical fiber cable to electrical signals that are transmitted to the edge connector, or (ii) convert electrical signals that are received from the edge connector to optical signals that are transmitted to the optical fiber cable.
(125) The optical module can include a first set of at least two electrical integrated circuits that are mounted on the first surface of the photonic integrated circuit.
(126) The first set of at least two electrical integrated circuits can include two electrical integrated circuits that are positioned on opposite sides of the optical fiber connector along a plane parallel to the first surface of the photonic integrated circuit.
(127) In some examples, the first set of at least one electrical integrated circuit can include four electrical integrated circuits that surround four sides of the optical fiber connector along the plane parallel to the first surface of the photonic integrated circuit.
(128) The optical module can include a substrate or circuit board. The photonic integrated circuit is mounted on the substrate or circuit board. The optical module can include a second set of at least one electrical integrated circuit mounted on the substrate or circuit board and electrically coupled to the photonic integrated circuit through one or more signal conductors and/or traces.
(129) The photonic integrated circuit can include at least one of a photodetector or an optical modulator, and the first set of at least one integrated circuit can include at least one of a transimpedance amplifier configured to amplify a current generated by the photodetector or a driver configured to drive the optical modulator.
(130) The second set of at least one electrical integrated circuit can include a serializers/deserializers module.
(131) The pluggable optical module can include at least one laser source that is configured to provide power supply light to the photonic integrated circuit.
(132) The fiber harness can include at least one optical fiber that optically couples the at least one laser source to the photonic integrated circuit.
(133) The optical fiber connector can include at least one power supply fiber port.
(134) The apparatus can include a second circuit board and a cage mounted on the second circuit board. The pluggable optical module can be plugged into the cage, and the receptacle is located inside the cage.
(135) The apparatus can include a server computer including a first data processor. The second circuit board can be part of the server computer, the pluggable optical module can be configured to provide a communication interface that enables the first data processor to communicate with a second data processor through the optical fiber cable.
(136) The first data processor can include at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC).
(137) The apparatus can include at least one of a supercomputer, an autonomous vehicle, or a robot. The supercomputer, the autonomous vehicle, or the robot can include the server computer.
(138) The server computer can include a plurality of cages and a plurality of pluggable optical modules, the plurality of pluggable optical modules can be plugged into the plurality of cages, each pluggable optical module can be plugged into a corresponding cage.
(139) In another general aspect, an apparatus includes a system that includes a data center. The data includes: a plurality of server computers described above; and a plurality of pluggable optical modules described above. Each server computer communicates with one or more other server computers through one or more optical fiber cables and the plurality of pluggable optical modules.
(140) In another general aspect, an apparatus includes a pluggable optical module that includes: a fiber connector

configured to be optically coupled to an optical fiber cable, an optical module, a fiber harness, and an edge connector. The optical module includes a photonic integrated circuit having a first surface; and a first set of at least two electrical integrated circuits that are mounted on the first surface of the photonic integrated circuit. The fiber harness is optically coupled between the fiber connector and the first surface of the photonic integrated circuit. The edge connector includes conductive pads configured to be electrically coupled to conductive pads of a receptacle when the edge connector is mated with the receptacle. The conductive pads of the edge connector are electrically coupled to the optical module.

(141) Implementations can include one or more of the following features. The pluggable optical module can comply with a small form factor pluggable module specification including at least one of SFP (small form-factor pluggable), SFP+, 10 Gb SFP, SFP28, OSFP (octal SFP), OSFP-XD (OSFP extra dense), QSFP (quad small form-factor pluggable), QSFP+, QSFP28, QSFP56, or QSFP-DD (quad small form-factor pluggable double density).

(142) The fiber harness can include an optical connector that is coupled to the photonic integrated circuit, the first set of at least two electrical integrated circuits can include two electrical integrated circuits that are positioned on opposite sides of the optical connector along a plane parallel to the first surface of the photonic integrated circuit.

(143) In some examples, the first set of at least one electrical integrated circuit can include four electrical integrated circuits that surround four sides of the optical connector along the plane parallel to the first surface of the photonic integrated circuit.

(144) The optical module can include a substrate or circuit board. The photonic integrated circuit can be mounted on the substrate or circuit board. The optical module can include a second set of at least one electrical integrated circuit mounted on the substrate or circuit board and electrically coupled to the photonic integrated circuit through one or more signal conductors and/or traces.

(145) The photonic integrated circuit can include at least one of a photodetector or an optical modulator. The first set of at least one integrated circuit includes at least one of a transimpedance amplifier configured to amplify a current generated by the photodetector or a driver configured to drive the optical modulator.

(146) The second set of at least one electrical integrated circuit can include a serializers/deserializers module.

(147) The pluggable optical module can include a housing having an inner bottom wall, an inner upper wall, and inner side walls. The inner bottom, upper, and side walls can define a space to accommodate the optical module. The optical module can be oriented relative to the housing such that the first surface of the photonic integrated circuit is at an angle between 45° to 135° relative to the bottom surface of the housing.

(148) The optical module can be oriented relative to the housing such that the first surface of the photonic integrated circuit is at an angle between 70° to 110° relative to the bottom surface of the housing.

(149) In some examples, the optical module can be oriented relative to the housing such that the first surface of the photonic integrated circuit is at an angle between 80° to 100° relative to the bottom surface of the housing.

(150) In some examples, the optical module can be oriented relative to the housing such that the first surface of the photonic integrated circuit is at an angle between 85° to 95° relative to the bottom surface of the housing.

(151) The pluggable optical module can include a housing having an inner upper wall and an inner lower wall, the edge connector can have an upper surface extending along a first plane that is at a first distance d**1** relative to the inner upper wall, the edge connector can have a lower surface extending along a second plane that is at a second distance d**2** relative to the inner lower wall. The fiber harness can be substantially vertically coupled to the first surface of the photonic integrated circuit such that light from the fiber harness is directed toward the first surface of the photonic integrated circuit at an angle θ**1** relative to a direction vertical to the first surface of the photonic integrated circuit, 0<θ**1**<10°. The fiber harness when extending from the first surface of the photonic integrated circuit and bending to a direction parallel to the first surface can require a clearance distance of at least d**3** so as to not damage the optical fibers in the fiber harness, in which d**1**<d**3**, and d**2**<d**3**.

(152) The housing can have a first inner side wall and a second inner side wall. The substrate or circuit board can be attached to the first inner side wall. A distance from the first surface of the photonic integrated circuit to the second inner side wall can be d**4**, in which d**3**<d**4**.

(153) The photonic integrated circuit can be configured to perform at least one of (i) convert optical signals received from the optical fiber cable to electrical signals that are transmitted to the edge connector, or (ii) convert electrical signals that are received from the edge connector to optical signals that are transmitted to the optical fiber cable.

(154) In another general aspect, a method includes: transmitting signals between an optical fiber cable and a data processing apparatus through a pluggable optical module having a photonic integrated circuit. The method includes transmitting optical signals between the optical fiber cable and the photonic integrated circuit through a fiber harness and a plurality of optical couplers provided at a first surface of the photonic integrated circuit; and transmitting electrical signals between the photonic integrated circuit and the data processing apparatus through an edge connector of the pluggable optical module. The fiber harness includes a plurality of optical fibers and an optical fiber connector that optically couples the plurality of optical fibers to the plurality of optical couplers at the first surface of the photonic integrated circuit. The optical fiber connector includes a two-dimensional arrangement of fiber ports that are optically coupled to the optical couplers at the first surface of the photonic integrated circuit.

(155) Other aspects include other combinations of the features recited above and other features, expressed as methods, apparatus, systems, program products, and in other ways.

(156) By using pluggable modules each including a co-packaged optical module, one or more multi-fiber push on (MPO)

connectors, a fiber guide that mechanically connects the co-packaged optical module to the one or more multi-fiber push on connectors, and a fiber pigtail that optically connects the co-packaged optical module to the one or more multi-fiber push on connectors, operators can conveniently connect or disconnect optical links to a data processor mounted on a substrate or circuit board positioned at a distance behind the front panel of the housing of a data processing server without the need to open the front panel. This allows the operators to quickly reconfigure network connections in, e.g., a data center that has a large number of fiber optic cables that provide optical communication links between a large number of rackmount servers.
(157) By using a pluggable optical module having a two-dimensional fiber array interfacing to the photonic integrated circuit in the pluggable optical module, the bandwidth supported by the pluggable optical module can be significantly increased, as compared to a pluggable optical module having a one-dimensional fiber array interfacing to the photonic integrated circuit. By using a vertically oriented substrate or circuit board, the photonic integrated circuit and the two-dimensional fiber array coupled to the photonic integrated circuit can fit in the housing of a pluggable optical module that complies with a small form factor pluggable module specification, e.g., SFP, SFP+, 10 Gb SFP, SFP28, OSFP, OSFP-XD, QSFP, QSFP+, QSFP28, QSFP56, or QSFP-DD. Similarly, mounting the substrate or circuit board on the side wall of the housing, or placing the substrate or circuit board at a farther distance from the upper inner wall of the housing, allows a pluggable optical module to comply with a small form factor pluggable module specification while also providing sufficient space inside the housing to accommodate the photonic integrated circuit and the two-dimensional fiber array that is coupled to the photonic integrated circuit.
(158) Particular embodiments of the subject matter described in this specification can be implemented to realize one or more of the following advantages. The data processing system has a high power efficiency, a low construction cost, a low operation cost, and high flexibility in reconfiguring optical network connections.
(159) The details of one or more embodiments of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the invention will become apparent from the description, the drawings, and the claims.
(160) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. In case of conflict with patent applications, patent application publications, or patents incorporated herein by reference, the present specification, including definitions, will control.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. The dimensions of the various features can be arbitrarily expanded or reduced for clarity.
(2) FIG. **1** is a block diagram of an example optical communication system.
(3) FIG. **2** is a schematic side view of an example data processing system.
(4) FIG. **3** is a schematic side view of an example integrated optical device.
(5) FIG. **4** is a schematic side view of an example data processing system.
(6) FIG. **5** is a schematic side view of an example integrated optical device.
(7) FIGS. **6** and **7** are schematic side views of examples of data processing systems.
(8) FIG. **8** is an exploded perspective view of an integrated optical communication device.
(9) FIGS. **9** and **10** are diagrams of example layout patterns of optical and electrical terminals of integrated optical devices.
(10) FIGS. **11**, **12**, **13**, and **14** are schematic side views of examples of data processing systems.
(11) FIGS. **15** and **16** are bottom views of examples of integrated optical devices.
(12) FIG. **17** is a diagram showing various types of integrated optical communication devices that can be used in a data processing system.
(13) FIG. **18** is a diagram of an example octal serializers/deserializers block.
(14) FIG. **19** is a diagram of an example electronic communication integrated circuit.
(15) FIG. **20** is a functional block diagram of an example data processing system.
(16) FIG. **21** is a diagram of an example rackmount data processing system.
(17) FIGS. **22**, **23**, **24**, **25**, **26**A, **26**B, **26**C, **27**, **28**A, and **28**B are top view diagrams of examples of rackmount data processing systems incorporating optical interconnect modules.
(18) FIGS. **29**A and **29**B are diagrams of an example rackmount data processing system incorporating multiple optical interconnect modules.
(19) FIGS. **30** and **31** are block diagrams of example data processing systems.
(20) FIG. **32** is a schematic side view of an example data processing system.
(21) FIG. **33** is a diagram of an example electronic communication integrated circuit that includes octal serializers/deserializers blocks.
(22) FIG. **34** is a flow diagram of an example process for processing optical and electrical signals using a data processing

system.
(23) FIG. **35**A is a diagram an optical communications system.
(24) FIGS. **35**B and **35**C are diagrams of co-packaged optical interconnect modules.
(25) FIGS. **36** and **37** are diagrams of examples of optical communications systems.
(26) FIGS. **38** and **39** are diagrams of examples of serializers/deserializers blocks.
(27) FIGS. **40**A, **40**B, **41**A, **41**B, and **42** are diagrams of examples of bus processing units.
(28) FIG. **43** is an exploded view of an example of a front-mounted module of a data processing system.
(29) FIG. **44** is an exploded view of an example of the internals of an optical module.
(30) FIG. **45** is an assembled view of the internals of an optical module.
(31) FIG. **46** is an exploded view of an optical module.
(32) FIG. **47** is an assembled view of an optical module.
(33) FIG. **48** is a diagram of a portion of a grid structure and a circuit board.
(34) FIG. **49** is a diagram showing a lower mechanical part prior to insertion into the grid structure.
(35) FIG. **50** is a diagram of an example of a partially populated front-view of an assembled system.
(36) FIG. **51**A is a front view of an example of the mounting of the module.
(37) FIG. **51**B is a side view of an example of the mounting of the module.
(38) FIG. **52**A is a front view of an example of the mechanical connector structure and an optical module mounted within a grid structure.
(39) FIG. **52**B is a side view of an example of the mechanical connector structure and an optical module mounted within a grid structure.
(40) FIGS. **53** and **54** are diagrams of an example of an assembly that includes a fiber cable, an optical fiber connector, a mechanical connector module, and a grid structure.
(41) FIGS. **55**A and **55**B are perspective views of the mechanisms shown in FIGS. **53** and **54** before the optical fiber connector is inserted into the mechanical connector structure.
(42) FIG. **56** is a perspective view showing that the optical module and the mechanical connector structure are inserted into the grid structure.
(43) FIG. **57** is a perspective view showing that the optical fiber connector is mated with the mechanical connector structure.
(44) FIGS. **58**A to **58**D are diagrams of an example an optical module that includes a latch mechanism.
(45) FIG. **59** is a diagram of an alternative example of the optical module.
(46) FIGS. **60**A and **60**B are diagrams of an example implementation of the lever and the latch mechanism in the optical module with connector.
(47) FIG. **61** is a diagram of cross section of the module viewed from the front mounted in the assembly with the connector.
(48) FIGS. **62** to **65** are diagrams showing cross-sectional views of an example of a fiber cable connection design.
(49) FIG. **66** is a map of electrical contact pads.
(50) FIG. **67** is a top view of an example of a rackmount server.
(51) FIG. **68**A is a top view of an example of a rackmount server.
(52) FIG. **68**B is a diagram of an example of a front panel of the rackmount server.
(53) FIG. **68**C is a perspective view of an example of a heat sink.
(54) FIG. **69**A is a top view of an example of a rackmount server.
(55) FIG. **69**B is a diagram of an example of a front panel of the rackmount server.
(56) FIG. **70** is a top view of an example of a rackmount server.
(57) FIG. **71**A is a top view of an example of a rackmount server.
(58) FIG. **71**B is a front view of the rackmount server.
(59) FIG. **72** is a top view of an example of a rackmount server.
(60) FIG. **73**A is a top view of an example of a rackmount server.
(61) FIG. **73**B is a front view of the rackmount server.
(62) FIG. **74**A is a top view of an example of a rackmount server.
(63) FIG. **74**B is a front view of the rackmount server.
(64) FIG. **75**A is a top view of an example of a rackmount server.
(65) FIG. **75**B is a front view of the rackmount server.
(66) FIG. **75**C is a diagram of the air flow in the rackmount server.
(67) FIG. **76** is a diagram of a network rack that includes a plurality of rackmount servers.
(68) FIG. **77**A is a side view of an example of a rackmount server.
(69) FIG. **77**B is a top view of the rackmount server.
(70) FIG. **78** is a top view of an example of a rackmount server.
(71) FIG. **79** is a block diagram of an example of an optical communication system.
(72) FIG. **80**A is a diagram of an example of an optical communication system.
(73) FIG. **80**B is a diagram of an example of an optical cable assembly used in the optical communication system of FIG. **80**A.

(74) FIG. **80**C is an enlarged diagram of the optical cable assembly of FIG. **80**B.
(75) FIG. **80**D is an enlarged diagram of the upper portion of the optical cable assembly of FIG. **80**B.
(76) FIG. **80**E is an enlarged diagram of the lower portion of the optical cable assembly of FIG. **80**B.
(77) FIG. **80**F is an enlarged view of the diagram of FIG. **80**D.
(78) FIG. **80**G is an enlarged view of the diagram of FIG. **80**E.
(79) FIG. **81** is a block diagram of an example of an optical communication system.
(80) FIG. **82**A is a diagram of an example of an optical communication system.
(81) FIG. **82**B is a diagram of an example of an optical cable assembly.
(82) FIG. **82**C is an enlarged diagram of the optical cable assembly of FIG. **82**B.
(83) FIG. **82**D is an enlarged diagram of the upper portion of the optical cable assembly of FIG. **82**B.
(84) FIG. **82**E is an enlarged diagram of the lower portion of the optical cable assembly of FIG. **82**B.
(85) FIG. **82**F is an enlarged view of a portion of the diagram of FIG. **82**A.
(86) FIG. **82**G is an enlarged view of the diagram of FIG. **82**D.
(87) FIG. **82**H is an enlarged view of the diagram of FIG. **82**E.
(88) FIG. **83** is a block diagram of an example of an optical communication system.
(89) FIG. **84**A is a diagram of an example of an optical communication system.
(90) FIG. **84**B is a diagram of an example of an optical cable assembly.
(91) FIG. **84**C is an enlarged diagram of the optical cable assembly of FIG. **84**B.
(92) FIGS. **85** to **87**B are diagrams of examples of data processing systems.
(93) FIG. **88** is a diagram of an example of connector port mapping for an optical fiber interconnection cable.
(94) FIGS. **89** and **90** are diagrams of examples of fiber port mapping for optical fiber interconnection cables.
(95) FIGS. **91** and **92** are diagrams of examples of viable port mapping for optical fiber connectors of universal optical fiber interconnection cables.
(96) FIG. **93** is a diagram of an example of a port mapping for an optical fiber connector that is not appropriate for a universal optical fiber interconnection cable.
(97) FIGS. **94** and **95** are diagrams of examples of viable port mapping for optical fiber connectors of universal optical fiber interconnection cables.
(98) FIG. **96** is a top view of an example of a rackmount server.
(99) FIG. **97**A is a perspective view of the rackmount server of FIG. **96**.
(100) FIG. **97**B is a perspective view of the rackmount server of FIG. **96** with the top panel removed.
(101) FIG. **98** is a diagram of the front portion of the rackmount server of FIG. **96**.
(102) FIG. **99** includes perspective front and rear views of the front panel of the rackmount server of FIG. **96**.
(103) FIG. **100** is a top view of an example of a rackmount server.
(104) FIGS. **101**, **102**, **103**A, and **103**B are diagrams of examples of optical fiber connectors.
(105) FIGS. **104** and **105** are a top view and a front view, respectively, of an example of a rackmount device that includes a vertical printed circuit board on which co-packaged optical modules are mounted.
(106) FIG. **106** is a diagram of an example of an optical cable assembly.
(107) FIG. **107** is a front view diagram of the rackmount device with the optical cable assembly.
(108) FIG. **108** is a top view diagram of an example of a rackmount device that includes a vertical printed circuit board on which co-packaged optical modules are mounted.
(109) FIG. **109** is a front view diagram of the rackmount device with the optical cable assembly.
(110) FIGS. **110** and **111** are a top view and a front view, respectively, of an example of a rackmount device.
(111) FIG. **112** is diagram of an example of a rackmount device with example parameter values.
(112) FIGS. **113** and **114** show another example of a rackmount device with example parameter values.
(113) FIGS. **115** and **116** are a top view and a front view, respectively, of an example of a rackmount device.
(114) FIGS. **117** to **122** are diagrams of examples of systems that include co-packaged optical modules.
(115) FIG. **123** is a diagram of an example of a vertically mounted processor blade.
(116) FIG. **124** is a top view of an example of a rack system that includes several vertically mounted processor blades.
(117) FIG. **125**A is a side view of an example of a rackmount server that has a hinged front panel.
(118) FIG. **125**B is a diagram of an example of a rackmount server that has pluggable modules.
(119) FIGS. **126**A to **127** are diagrams of examples of rackmount servers that have pluggable modules.
(120) FIG. **128** is a diagram of an example of a fiber guide that includes one or more photon supplies.
(121) FIG. **129** is a diagram of an example of a rackmount server that includes guide rails/cage to assist the insertion of fiber guides.
(122) FIG. **130** is a side view of an example of a rackmount server that has a hinge-mounted front panel.
(123) FIG. **131** is a top view of an example of a rackmount server that has a hinge-mounted front panel.
(124) FIG. **132** is a diagram of an example of an optical cable.
(125) FIG. **133** shows a top view diagram and a side view diagram of a rackmount server that has a hinged front panel.
(126) FIG. **134** shows a top view, a vertical application specific integrated circuit (VASIC)-plane front view, and a front-panel front view of an example of a rackmount server.
(127) FIG. **135** shows a top view, a VASIC-plane front view, and a front-panel front view of an example of another

rackmount server.
(128) FIG. **136**A is a diagram of an example of a data processing system.
(129) FIGS. **136**B to **136**H are diagrams of portions of the data processing system of FIG. **136**A.
(130) FIG. **137** is a diagram of optical fiber connectors.
(131) FIG. **138** is a diagram of an example of a wavelength division multiplexing (WDM) data processing system.
(132) FIG. **139**A is a diagram of an example of a switch rack WDM translator.
(133) FIG. **139**B is a diagram of an example of a 4×4 wavelength/space shuffle matrix.
(134) FIG. **140**A is a diagram of an example of a wavelength division multiplexing data processing system.
(135) FIGS. **140**B to **140**H are diagrams of portions of the WDM data processing system of FIG. **140**A.
(136) FIG. **141** is a diagram of optical fiber connectors.
(137) FIG. **142** is an enlarged view of the diagram of FIG. **89**.
(138) FIG. **143** is an enlarged view of the diagram of FIG. **90**.
(139) FIG. **144** shows the diagram of FIG. **91**.
(140) FIG. **145** shows the diagram of FIG. **92**.
(141) FIG. **146** shows the diagram of FIG. **93**.
(142) FIGS. **147** to **151** are diagrams of examples of a system that can provide a large memory bank or memory pool.
(143) FIG. **152**A is a diagram of an example pluggable optical module.
(144) FIG. **152**B is a cross-sectional diagram of the pluggable optical module.
(145) FIG. **153**A is a side view of an example pluggable optical module.
(146) FIG. **153**B is a perspective view of a rear portion of the pluggable optical module.
(147) FIG. **153**C is a side view cross-sectional diagram of the pluggable optical module plugged into a cage.
(148) FIG. **154**A is a rear view of an example OSFP optical transceiver module.
(149) FIG. **154**B is a rear view of an example OSFP-XD optical transceiver module.
(150) FIG. **154**C is a diagram of an example co-packaged optical module that can fit inside the housing of the OSFP module with a vertically oriented substrate or circuit board.
(151) FIG. **154**D is a diagram of an example co-packaged optical module that can fit inside the housing of the OSFP-XD module with a vertically oriented substrate or circuit board.
(152) FIG. **154**E is a cross-sectional diagram of an example OSFP pluggable optical module.
(153) FIG. **154**F is a cross-sectional diagram of an example OSFP-XD pluggable optical module.
(154) FIG. **155**A is a side view cross-sectional diagram of an example pluggable optical module plugged into a cage.
(155) FIG. **155**B is a front view of an example co-packaged optical module.
(156) FIG. **155**C is a rear view of an example connector module.
(157) FIG. **155**D is a side view of an example co-packaged optical module, flexible RF cables, and a connector module.
(158) FIG. **155**E is a diagram of an example of the fiber port mapping for the optical fiber connector.
(159) FIG. **156**A is a side view cross-sectional diagram of an example pluggable optical module plugged into a cage.
(160) FIG. **156**B is a front view of an example co-packaged optical module.
(161) FIG. **156**C is a rear view of an example connector module.
(162) FIG. **156**D is a side view of the co-packaged optical module and the connector module.
(163) FIG. **157**A is a side view cross-sectional diagram of an example pluggable optical module plugged into a cage.
(164) FIG. **157**B is a top view cross-sectional diagram of the pluggable optical module.
(165) FIG. **157**C is a side view of the pluggable optical module.
(166) FIG. **157**D is a diagram of an example fiber port mapping for an optical fiber connector.
(167) FIG. **158** shows an example of the OSFP module pinout configuration.
(168) FIG. **159**A is a perspective view of an example 1×1 cage mounted on a circuit board.
(169) FIG. **159**B is a perspective view of an example 1×4 cage mounted on a circuit board.
(170) FIG. **159**C is a perspective view of an example OSFP module inserted into the 1×1 cage.
(171) FIG. **160** is a top view cross-sectional diagram of an example pluggable optical module.
(172) FIG. **161** is a side view cross-sectional diagram of an example pluggable optical module.
(173) FIG. **162** is a diagram of an example co-packaged optical module.
(174) FIGS. **163**A, **163**B, **164**A, and **164**B are perspective views of an example co-packaged optical module.
(175) FIG. **165** is a top view of an example placement of electrical integrated circuits on a photonic integrated circuit.
(176) FIGS. **166**A to **166**D are diagrams of examples of a photonic integrated circuit and a fiber connection.
(177) FIG. **167** is a diagram of an example of a pluggable module.
(178) FIG. **168**A is a diagram of another example of a pluggable module.
(179) FIG. **168**B is a diagram of an example of the top panel of the housing of the pluggable module of FIG. **168**A.
(180) FIG. **169** is a cross-sectional diagram of the pluggable module of FIG. **168**A.
(181) FIG. **170**A is a diagram of an example of results of a simulation of thermal distribution of a portion of the pluggable module of FIG. **168**A.
(182) FIG. **170**B is a diagram of a segment of the laser heat sink of the pluggable module of FIG. **168**A.

DETAILED DESCRIPTION

(183) This document describes a novel system for high bandwidth data processing, including novel input/output interface

modules for coupling bundles of optical fibers to data processing integrated circuits (e.g., network switches, central processing units, graphics processor units, tensor processing units, digital signal processors, and/or other application specific integrated circuits (ASICs)) that process the data transmitted through the optical fibers. In some implementations, the data processing integrated circuit is mounted on a circuit board (or substrate or a combination of circuit board(s) and substrate(s)) positioned near the input/output interface module through a relatively short electrical signal path on the circuit board (or substrate or a combination of circuit board(s) and substrate(s)). The input/output interface module includes a first connector that allows a user to conveniently connect or disconnect the input/output interface module to or from the circuit board (or substrate or a combination of circuit board(s) and substrate(s)). The input/output interface module can also include a second connector that allows the user to conveniently connect or disconnect the bundle of optical fibers to or from the input/output interface module. In some implementations, a rack mount system having a front panel is provided in which the circuit board (which supports the input/output interface modules and the data processing integrated circuits) (or substrate or a combination of circuit board(s) and substrate(s)) is vertically mounted in an orientation substantially parallel to, and positioned near, the front panel. In some examples, the circuit board (or substrate or a combination of circuit board(s) and substrate(s)) functions as the front panel or part of the front panel. The second connectors of the input/output interface modules face the front side of the rack mount system to allow the user to conveniently connect or disconnect bundles of optical fibers to or from the system.
(184) In some implementations, a feature of the high bandwidth data processing system is that, by vertically mounting the circuit board that supports the input/output interface modules and the data processing integrated circuits to be near the front panel, or configuring the circuit board as the front panel or part of the front panel, the optical signals can be routed from the optical fibers through the input/output interface modules to the data processing integrated circuits through relatively short electrical signal paths. This allows the signals transmitted to the data processing integrated circuits to have a high bit rate (e.g., over 50 Gbps) while maintaining low crosstalk, distortion, and noise, hence reducing power consumption and footprint of the data processing system.
(185) In some implementations, a feature of the high bandwidth data processing system is that the cost of maintenance and repair can be lower compared to traditional systems. For example, the input/output interface modules and the fiber optic cables are configured to be detachable, a defective input/output interface module can be replaced without taking apart the data processing system and without having to re-route any optical fiber. Another feature of the high bandwidth data processing system is that, because the user can easily connect or disconnect the bundles of the optical fibers to or from the input/output interface modules through the front panel of the rack mount system, the configurations for routing of high bit rate signals through the optical fibers to the various data processing integrated circuits is flexible and can easily be modified. For example, connecting a bundle of hundreds of strands of optical fibers to the optical connector of the rack mount system can be almost as simple as plugging a universal serial bus (USB) cable into a USB port. A further feature of the high bandwidth data processing system is that the input/output interface module can be made using relatively standard, low cost, and energy efficient components so that the initial hardware costs and subsequent operational costs of the input/output interface modules can be relatively low, compared to conventional systems.
(186) In some implementations, optical interconnects can co-package and/or co-integrate optical transponders with electronic processing chips. It is useful to have transponder solutions that consume relatively low power and that are sufficiently robust against significant temperature variations as may be found within an electronic processing chip package. In some implementations, high speed and/or high bandwidth data processing systems can include massively spatially parallel optical interconnect solutions that multiplex information onto relatively few wavelengths and use a relatively large number of parallel spatial paths for chip-to-chip interconnection. For example, the relatively large number of parallel spatial paths can be arranged in two-dimensional arrays using connector structures such as those disclosed in U.S. patent application Ser. No. 16/816,171, filed on Mar. 11, 2020, published as US 2021/0286140, and incorporated herein by reference in its entirety.
(187) FIG. **1** shows a block diagram of a communication system **100** that incorporates one or more novel features described in this document. In some implementations, the system **100** includes nodes **101_1** to **101_6** (collectively referenced as **101**), which in some embodiments can each include one or more of: optical communication devices, electronic and/or optical switching devices, electronic and/or optical routing devices, network control devices, traffic control devices, synchronization devices, computing devices, and data storage devices. The nodes **101_1** to **101_6** can be suitably interconnected by optical fiber links **102_1** to **102_12** (collectively referenced as **102**) establishing communication paths between the communication devices within the nodes. The optical fiber links **102** can include the fiber-optic cables described in U.S. Pat. No. 11,194,109, issued on Dec. 7, 2021, titled "Optical Fiber Cable and Raceway Therefor," and incorporated herein by reference in its entirety. The system **100** can also include one or more optical power supply modules **103** producing one or more light outputs, each light output comprising one or more continuous-wave (CW) optical fields and/or one or more trains of optical pulses for use in one or more of the optical communication devices of the nodes **101_1** to **101_6**. For illustration purposes, only one such optical power supply module **103** is shown in FIG. **1**. A person of ordinary skill in the art will understand that some embodiments can have more than one optical power supply module **103** appropriately distributed over the system **100** and that such multiple power supply modules can be synchronized, e.g., using some of the techniques disclosed in U.S. Pat. No. 11,153,670, issued on Oct. 19, 2021, titled "Communication System Employing Optical Frame Templates," incorporated herein by reference in its entirety.
(188) Some end-to-end communication paths can pass through an optical power supply module **103** (e.g., see the

communication path between the nodes **101_2** and **101_6**). For example, the communication path between the nodes **101_2** and **101_6** can be jointly established by the optical fiber links **102_7** and **102_8**, whereby light from the optical power supply module **103** is multiplexed onto the optical fiber links **102_7** and **102_8**.
(189) Some end-to-end communication paths can pass through one or more optical multiplexing units **104** (e.g., see the communication path between the nodes **101_2** and **101_6**). For example, the communication path between the nodes **101_2** and **101_6** can be jointly established by the optical fiber links **102_10** and **102_11**. Multiplexing unit **104** is also connected, through the link **102_9**, to receive light from the optical power supply module **103** and, as such, can be operated to multiplex said received light onto the optical fiber links **102_10** and **102_11**.
(190) Some end-to-end communication paths can pass through one or more optical switching units **105** (e.g., see the communication path between the nodes **101_1** and **101_4**). For example, the communication path between the nodes **101_1** and **101_4** can be jointly established by the optical fiber links **102_3** and **102_12**, whereby light from the optical fiber links **102_3** and **102_4** is either statically or dynamically directed to the optical fiber link **102_12**.
(191) As used herein, the term "network element" refers to any element that generates, modulates, processes, or receives light within the system **100** for the purpose of communication. Example network elements include the node **101**, the optical power supply module **103**, the optical multiplexing unit **104**, and the optical switching unit **105**.
(192) Some light distribution paths can pass through one or more network elements. For example, optical power supply module **103** can supply light to the node **101_4** through the optical fiber links **102_7**, **102_4**, and **102_12**, letting the light pass through the network elements **101_2** and **105**.
(193) Various elements of the communication system **100** can benefit from the use of optical interconnects, which can use photonic integrated circuits comprising optoelectronic devices, co-packaged and/or co-integrated with electronic chips comprising integrated circuits.
(194) As used herein, the term "photonic integrated circuit" (or PIC) should be construed to cover planar lightwave circuits (PLCs), integrated optoelectronic devices, wafer-scale products on substrates, individual photonic chips and dies, and hybrid devices. A substrate can be made of, e.g., one or more ceramic materials, or organic "high density build-up" (HDBU). A substrate can be, e.g., a silicon substrate. Example material systems that can be used for manufacturing various photonic integrated circuits can include but are not limited to III-V semiconductor materials, silicon photonics, silica-on-silicon products, silica-glass-based planar lightwave circuits, polymer integration platforms, lithium niobate and derivatives, nonlinear optical materials, etc. Both packaged devices (e.g., wired-up and/or encapsulated chips) and unpackaged devices (e.g., dies) can be referred to as planar lightwave circuits.
(195) Photonic integrated circuits are used for various applications in telecommunications, instrumentation, and signal-processing fields. In some implementations, a photonic integrated circuit uses optical waveguides to implement and/or interconnect various circuit components, such as for example, optical switches, couplers, routers, splitters, multiplexers/demultiplexers, filters, modulators, phase shifters, lasers, amplifiers, wavelength converters, optical-to-electrical (O/E) and electrical-to-optical (E/O) signal converters, etc. For example, a waveguide in a photonic integrated circuit can be an on-chip solid light conductor that guides light due to an index-of-refraction contrast between the waveguide's core and cladding. A photonic integrated circuit can include a planar substrate onto which optoelectronic devices are grown by an additive manufacturing process and/or into which optoelectronic devices are etched by a subtractive manufacturing processes, e.g., using a multi-step sequence of photolithographic and chemical processing steps.
(196) In some implementations, an "optoelectronic device" can operate on both light and electrical currents (or voltages) and can include one or more of: (i) an electrically driven light source, such as a laser diode; (ii) an optical amplifier; (iii) an optical-to-electrical converter, such as a photodiode; and (iv) an optoelectronic component that can control the propagation and/or certain properties (e.g., amplitude, phase, polarization) of light, such as an optical modulator or a switch. The corresponding optoelectronic circuit can additionally include one or more optical elements and/or one or more electronic components that enable the use of the circuit's optoelectronic devices in a manner consistent with the circuit's intended function. Some optoelectronic devices can be implemented using one or more photonic integrated circuits.
(197) As used herein, the term "integrated circuit" (IC) should be construed to encompass both a non-packaged die and a packaged die. In a typical integrated circuit-fabrication process, dies (chips) are produced in relatively large batches using wafers of silicon or other suitable material(s). Electrical and optical circuits can be gradually created on a wafer using a multi-step sequence of photolithographic and chemical processing steps. Each wafer is then cut ("diced") into many pieces (chips, dies), each containing a respective copy of the circuit that is being fabricated. Each individual die can be appropriately packaged prior to being incorporated into a larger circuit or be left non-packaged.
(198) The term "hybrid circuit" can refer to a multi-component circuit constructed of multiple monolithic integrated circuits, and possibly some discrete circuit components, all attached to each other to be mountable on and electrically connectable to a common base, carrier, or substrate. A representative hybrid circuit can include (i) one or more packaged or non-packaged dies, with some or all of the dies including optical, optoelectronic, and/or semiconductor devices, and (ii) one or more optional discrete components, such as connectors, resistors, capacitors, and inductors. Electrical connections between the integrated circuits, dies, and discrete components can be formed, e.g., using patterned conducting (such as metal) layers, ball-grid arrays, solder bumps, wire bonds, etc. Electrical connections can also be removable, e.g., by using land-grid arrays and/or compression interposers. The individual integrated circuits can include any combination of one or more respective substrates, one or more redistribution layers (RDLs), one or more interposers, one or more laminate plates, etc.

(199) In some embodiments, individual chips can be stacked. As used herein, the term “stack” refers to an orderly arrangement of packaged or non-packaged dies in which the main planes of the stacked dies are substantially parallel to each other. A stack can typically be mounted on a carrier in an orientation in which the main planes of the stacked dies are parallel to each other and/or to the main plane of the carrier.
(200) A “main plane” of an object, such as a die, a photonic integrated circuit, a substrate, or an integrated circuit, is a plane parallel to a substantially planar surface thereof that has the largest sizes, e.g., length and width, among all exterior surfaces of the object. This substantially planar surface can be referred to as a main surface. The exterior surfaces of the object that have one relatively large size, e.g., length, and one relatively small size, e.g., height, are typically referred to as the edges of the object.
(201) FIG. **2** is a schematic cross-sectional diagram of a data processing system **200** that includes an integrated optical communication device **210** (also referred to as an optical interconnect module), a fiber-optic connector assembly **220**, a package substrate **230**, and an electronic processor integrated circuit **240**. The data processing system **200** can be used to implement, e.g., one or more of devices **101_1** to **101_6** of FIG. **1**. FIG. **3** shows an enlarged cross-sectional diagram of the integrated optical communication device **210**.
(202) Referring to FIGS. **2** and **3**, the integrated optical communication device **210** includes a substrate **211** having a first main surface **211_1** and a second main surface **211_2**. The main surfaces **211_1** and **211_2**, respectively, include arrays of electrical contacts **212_1** and **212_2**. In some embodiments, the minimum spacing d.sub.1 between any two contacts within the array of contacts **212_1** is larger than the minimum spacing d.sub.2 between any two contacts within the array of contacts **212_2**. In some embodiments the minimum spacing between any two contacts within the array of contacts **212_2** is between 40 and 200 micrometers. In some embodiments, the minimum spacing between any two contacts within the array of contacts **212_1** is between 200 micrometers and 1 millimeter. At least some of the contacts **212_1** are electrically connected through the substrate **211** with at least some of the contacts **212_2**. In some embodiments, the contacts **212_1** can be permanently attached to a corresponding array of electrical contacts **232_1** on the package substrate **230**. In some embodiments, the contacts **212_1** can include mechanisms to allow the device **210** to be removably connected to the package substrate **230**, as indicated by a double arrow **233**. For example, the system can include mechanical mechanisms (e.g., one or more snap-on or screw-on mechanisms) to hold the various modules in place. In some embodiments, the contacts **212_1**, **212_2**, and/or **232_1** can include one or more of solder balls, metal pillars, and/or metal pads, etc. In some embodiments, the contacts **212_1**, and/or **232_1** can include one or more of spring-loaded elements, compression interposers, and/or land-grid arrays.
(203) In some embodiments, the integrated optical communication device **210** can be connected to the electronic processor integrated circuit **240** using traces **231** embedded in one or more layers of the package substrate **230**. In some embodiments, the processor integrated circuit **240** can include monolithically embedded therein an array of serializers/deserializers (SerDes) **247** electrically coupled to the traces **231**. In some embodiments, the processor integrated circuit **240** can include electronic switching circuitry, electronic routing circuitry, network control circuitry, traffic control circuitry, computing circuitry, synchronization circuitry, time stamping circuitry, and data storage circuitry. In some implementations, the processor integrated circuit **240** can be a network switch, a central processing unit, a graphics processor unit, a tensor processing unit, a digital signal processor, or an application specific integrated circuit (ASIC).
(204) Because the electronic processor integrated circuit **240** and the integrated communication device **210** are both mounted on the package substrate **230**, the electrical connectors or traces **231** can be made shorter, as compared to mounting the electronic processor integrated circuit **240** and the integrated communication device **210** on separate circuit boards. Shorter electrical connectors or traces **231** can transmit signals that have a higher data rate with lower noise, lower distortion, and/or lower crosstalk.
(205) In some implementations, the electrical connectors or traces can be configured as differential pairs of transmission lines, e.g., in a ground-signal-ground-signal-ground configuration. In some examples, the speed of such signal links can be 10 Gbps or more; 56 Gbps or more; 112 Gbps or more; or 224 Gbps or more.
(206) In some implementations, the integrated optical communication device **210** further includes a first optical connector part **213** having a first surface **213_1** and a second surface **213_2**. The connector part **213** is configured to receive a second optical connector part **223** of the fiber-optic connector assembly **220**, optically coupled to the connector part **213** through the surfaces **213_1** and **223_2**. In some embodiments the connector part **213** can be removably attached to the connector part **223**, as indicated by a double-arrow **234**, e.g., through a hole **235** in the package substrate **230**. In some embodiments the connector part **213** can be permanently attached to the connector part **223**. In some embodiments, the connector parts **213** and **223** can be implemented as a single connector element combining the functions of both the connector parts **213** and **223**.
(207) In some implementations, the optical connector part **223** is attached to an array of optical fibers **226**. In some embodiments, the array of optical fibers **226** can include one or more of: single-mode optical fiber, multi-mode optical fiber, multi-core optical fiber, polarization-maintaining optical fiber, dispersion-compensating optical fiber, hollow-core optical fiber, or photonic crystal fiber. In some embodiments, the array of optical fibers **226** can be a linear (1D) array. In some other embodiments, the array of optical fibers **226** can be a two-dimensional (2D) array. For example, the array of optical fibers **226** can include 2 or more optical fibers, 4 or more optical fibers, 10 or more optical fibers, 100 or more optical fibers, 500 or more optical fibers, or 1000 or more optical fibers. Each optical fiber can include, e.g., 2 or more

cores, or 10 or more cores, in which each core provides a distinct light path. Each light path can include a multiplex of, e.g., 2 or more, 4 or more, 8 or more, or 16 or more serial optical signals, e.g., by use of wavelength division multiplexing channels, polarization-multiplexed channels, coherent quadrature-multiplexed channels. The connector parts **213** and **223** are configured to establish light paths through the first main surface **211_1** of the substrate **211**. For example, the array of optical fibers **226** can include n1 optical fibers, each optical fiber can include n2 cores, and the connector parts **213** and **223** can establish n1×n2 light paths through the first main surface **211_1** of the substrate **211**. Each light path can include a multiplex of n3 serial optical signals, resulting in a total of n1×n2×n3 serial optical signals passing through the connector parts **213** and **223**. In some embodiments, the connector parts **213** and **223** can be implemented, e.g., as disclosed in U.S. patent application Ser. No. 16/816,171.

(208) In some implementations, the integrated optical communication device **210** further includes a photonic integrated circuit **214** having a first main surface **214_1** and a second main surface **214_2**. The photonic integrated circuit **214** is optically coupled to the connector part **213** through its first main surface **214_1**, e.g., as disclosed in U.S. patent application Ser. No. 16/816,171. For example, the connector part **213** can be configured to optically couple light to the photonic integrated circuit **214** using optical coupling interfaces, e.g., vertical grating couplers or turning mirrors. In the example above, a total of n1×n2×n3 serial optical signals can be coupled through the connector parts **213** and **223** to the photonic integrated circuit **214**. Each serial optical signal is converted to a serial electrical signal by the photonic integrated circuit **214**, and each serial electrical signal is transmitted from the photonic integrated circuit **214** to a deserializer unit, or a serializer/deserializer unit, described below.

(209) In some embodiments, the connector part **213** can be mechanically connected (e.g., glued) to the photonic integrated circuit **214**. The photonic integrated circuit **214** can contain active and/or passive optical and/or opto-electronic components including optical modulators, optical detectors, optical phase shifters, optical power splitters, optical wavelength splitters, optical polarization splitters, optical filters, optical waveguides, or lasers. In some embodiments, the photonic integrated circuit **214** can further include monolithically integrated active or passive electronic elements such as resistors, capacitors, inductors, heaters, or transistors.

(210) In some implementations, the integrated optical communication device **210** further includes an electronic communication integrated circuit **215** configured to facilitate communication between the array of optical fibers **226** and the electronic processor integrated circuit **240**. A first main surface **215_1** of the electronic communication integrated circuit **215** is electrically coupled to the second main surface **214_2** of the photonic integrated circuit **214**, e.g., through solder bumps, copper pillars, etc. The first main surface **215_1** of the electronic communication integrated circuit **215** is further electrically connected to the second main surface **211_2** of the substrate **211** through the array of electrical contacts **212_2**. In some embodiments, the electronic communication integrated circuit **215** can include electrical pre-amplifiers and/or electrical driver amplifiers electrically coupled, respectively, to photodetectors and modulators within the photonic integrated circuit **214** (see also FIG. **14**). In some embodiments, the electronic communication integrated circuit **215** can include a first array of serializers/deserializers (SerDes) **216** (also referred to as a serializers/deserializers module) whose serial inputs/outputs are electrically connected to the photodetectors and the modulators of the photonic integrated circuit **214** and a second array of serializers/deserializers **217**, whose serial inputs/outputs are electrically coupled to the contacts **212_1** through the substrate **211**. Parallel inputs of the array of serializers/deserializers **216** can be connected to parallel outputs of the array of serializers/deserializers **217** and vice versa through a bus processing unit **218**, which can be, e.g., a parallel bus of electrical lanes, a cross-connect device, or a re-mapping device (gearbox). For example, the bus processing unit **218** can be configured to enable switching of the signals, allowing the routing of signals to be re-mapped. For example, N×50 Gbps electrical lanes can be remapped into N/2×100 Gbps electrical lanes, N being a positive even integer. An example of a bus processing unit **218** is shown in FIG. **40**A.

(211) For example, the electronic communication integrated circuit **215** includes a first serializers/deserializers module that includes multiple serializer units and multiple deserializer units, and a second serializers/deserializers module that includes multiple serializer units and multiple deserializer units. The first serializers/deserializers module includes the first array of serializers/deserializers **216**. The second serializers/deserializers module includes the second array of serializers/deserializers **217**.

(212) In some implementations, the first and second serializers/deserializers modules have hardwired functional units so that which units function as serializers and which units function as deserializers are fixed. In some implementations, the functional units can be configurable. For example, the first serializers/deserializers module is capable of operating as serializer units upon receipt of a first control signal, and operating as deserializer units upon receipt of a second control signal. Likewise, the second serializers/deserializers module is capable of operating as serializer units upon receipt of a first control signal, and operating as deserializer units upon receipt of a second control signal.

(213) Signals can be transmitted between the optical fibers **226** and the electronic processor integrated circuit **240**. For example, signals can be transmitted from the optical fibers **226** to the photonic integrated circuit **214**, to the first array of serializers/deserializers **216**, to the second array of serializers/deserializers **217**, and to the electronic processor integrated circuit **240**. Similarly, signals can be transmitted from the electronic processor integrated circuit **240** to the second array of serializers/deserializers **217**, to the first array of serializers/deserializers **216**, to the photonic integrated circuit **214**, and to the optical fibers **226**.

(214) In some implementations, the electronic communication integrated circuit **215** is implemented as a first integrated circuit and a second integrated circuit that are electrically coupled each other. For example, the first integrated circuit

includes the array of serializers/deserializers **216**, and the second integrated circuit includes the array of serializers/deserializers **217**.
(215) In some implementations, the integrated optical communication device **210** is configured to receive optical signals from the array of optical fibers **226**, generate electrical signals based on the optical signals, and transmit the electrical signals to the electronic processor integrated circuit **240** for processing. In some examples, the signals can also flow from the electronic processor integrated circuit **240** to the integrated optical communication device **210**. For example, the electronic processor integrated circuit **240** can transmit electronic signals to the integrated optical communication device **210**, which generates optical signals based on the received electronic signals, and transmits the optical signals to the array of optical fibers **226**.
(216) In some implementations, the photodetectors of the photonic integrated circuit **214** convert the optical signals transmitted in the optical fibers **226** to electrical signals. In some examples, the photonic integrated circuit **214** can include transimpedance amplifiers for amplifying the currents generated by the photodetectors, and drivers for driving output circuits (e.g., driving optical modulators). In some examples, the transimpedance amplifiers and drivers are integrated with the electronic communication integrated circuit **215**. For example, the optical signal in each optical fiber **226** can be converted to one or more serial electrical signals. For example, one optical fiber can carry multiple signals by use of wavelength division multiplexing. The optical signals (and the serial electrical signals) can have a high data rate, such as 50 Gbps, 100 Gbps, or more. The first serializers/deserializers module **216** converts the serial electrical signals to sets of parallel electrical signals. For example, each serial electrical signal can be converted to a set of N parallel electrical signals, in which N can be, e.g., 2, 4, 8, 16, or more. The first serializers/deserializers module **216** conditions the serial electrical signals upon conversion into sets of parallel electrical signals, in which the signal conditioning can include, e.g., one or more of clock and data recovery, and signal equalization. The first serializers/deserializers module **216** sends the sets of parallel electrical signals to the second serializers/deserializers module **217** through the bus processing unit **218**. The second serializers/deserializers module **217** converts the sets of parallel electrical signals to high speed serial electrical signals that are output to the electrical contacts **212_2** and **212_1**.
(217) The serializers/deserializers module (e.g., **216**, **217**) can perform functions such as fixed or adaptive signal pre-distortion on the serialized signal. Also, the parallel-to-serial mapping can use a serialization factor M different from N, e.g., 50 Gbps at the input to the first serializers/deserializers module **216** can become 50×1 Gbps on a parallel bus, and two such parallel buses from two serializers/deserializers modules **216** having a total of 100×1 Gbps can then be mapped to a single 100 Gbps serial signal by the serializers/deserializers module **217**. An example of the bus processing unit **218** for performing such mapping is shown in FIG. **40**B. Also, the high-speed modulation on the serial side can be different, e.g., the serializers/deserializers module **216** can use 50 Gbps Non-Return-to-Zero (NRZ) modulation whereas the serializers/deserializers module **217** can use 100 Gbps Pulse-Amplitude Modulation 4-Level (PAM4) modulation. In some implementations, coding (line coding or error-correction coding) can be performed at the bus processing unit **218**. The first and second serializers/deserializers modules **216** and **217** can be commercially available high quality, low power serializers/deserializers that can be purchased in bulk at a low cost.
(218) In some implementations, the package substrate **230** can include connectors on the bottom side that connects the package substrate **230** to another circuit board, such as a motherboard. The connection can use, e.g., fixed (e.g., by use of solder connection) or removable (e.g., by use of one or more snap-on or screw-on mechanisms). In some examples, another substrate can be provided between the electronic processor integrated circuit **240** and the package substrate **230**.
(219) Referring to FIG. **4**, in some implementations, a data processing system **250** includes an integrated optical communication device **252** (also referred to as an optical interconnect module), a fiber-optic connector assembly **220**, a package substrate **230**, and an electronic processor integrated circuit **240**. The data processing system **250** can be used, e.g., to implement one or more of devices **101_1** to **101_6** of FIG. **1**. The integrated optical communication device **252** is configured to receive optical signals, generate electrical signals based on the optical signals, and transmit the electrical signals to the electronic processor integrated circuit **240** for processing. In some examples, the signals can also flow from the electronic processor integrated circuit **240** to the integrated optical communication device **252**. For example, the electronic processor integrated circuit **240** can transmit electronic signals to the integrated optical communication device **252**, which generates optical signals based on the received electronic signals, and transmits the optical signals to the array of optical fibers **226**.
(220) The system **250** is similar to the data processing system **200** of FIG. **2** except that in the system **250**, in the direction of the cross section of the figure, a portion **254** of the top surface of the photonic integrated circuit **214** is not covered by the first serializers/deserializers module **216** and the second serializers/deserializers module **217**. For example, the portion **254** can be used to couple to other electronic components, optical components, or electro-optical components, either from the bottom (as shown in FIG. **4**) or from the top (as shown in FIG. **6**). In some examples, the first serializers/deserializers module **216** can have a high temperature during operation. The portion **254** is not covered by the first serializers/deserializers module **216** and can be less thermally coupled to the first serializers/deserializers module **216**. In some examples, the photonic integrated circuit **214** can include modulators that modulate the phases of optical signals by modifying the temperature of waveguides and thereby modifying the refractive indices of the waveguides. In such devices, using the design shown in the example of FIG. **4** can allow the modulators to operate in a more thermally stable environment.
(221) FIG. **5** shows an enlarged cross-sectional diagram of the integrated optical communication device **252**. In some

implementations, the substrate **211** includes a first slab **256** and a second slab **258**. The first slab **256** provides electrical connectors to fan out the electrical contacts, and the second slab **258** provides a removable connection to the package substrate **230**. The first slab **256** includes a first set of contacts arranged on the top surface and a second set of contacts arranged on the bottom surface, in which the first set of contacts has a fine pitch and the second set of contacts has a coarse pitch. The minimum distance between contacts in the second set of contacts is greater than the minimum distance between contacts in the first set of contacts. The second slab **258** can include, e.g., spring-loaded contacts **259**.

(222) Referring to FIG. **6**, in some implementations, a data processing system **260** includes an integrated optical communication device **262** (also referred to as an optical interconnect module), a fiber-optic connector assembly **270**, a package substrate **230**, and an electronic processor integrated circuit **240**. The data processing system **260** can be used, e.g., to implement one or more of devices **101_1** to **101_6** of FIG. **1**. The integrated optical communication device **262** includes a photonic integrated circuit **264**. The photonic integrated circuit **264** can include components that perform functions similar to those of the photonic integrated circuit **214** of FIGS. **2-5**. The integrated optical communication device **262** further includes a first optical connector part **266** that is configured to receive a second optical connector part **268** of the fiber-optic connector assembly **270**. For example, snap-on or screw-on mechanisms can be used to hold the first and second optical connector parts **266** and **268** together.

(223) The connector parts **266** and **268** can be similar to the connector parts **213** and **223**, respectively, of FIG. **4**. In some examples, the optical connector part **268** is attached to an array of optical fibers **272**, which can be similar to the fibers **226** of FIG. **4**.

(224) The photonic integrated circuit **264** has a top main surface and bottom main surface. The terms “top” and “bottom” refer to the orientations shown in the figure. It is understood that the devices described in this document can be positioned in any orientation, so for example the “top surface” of a device can be oriented facing downwards or sideways, and the “bottom surface” of the device can be oriented facing upwards or sideways. A difference between the photonic integrated circuit **264** and the photonic integrated circuit **214** (FIG. **4**) is that the photonic integrated circuit **264** is optically coupled to the connector part **268** through the top main surface, whereas the photonic integrated circuit **214** is optically coupled to the connector part **213** through the bottom main surface. For example, the connector part **266** can be configured to optically couple light to the photonic integrated circuit **214** using optical coupling interfaces, e.g., vertical grating couplers or turning mirrors, similar to the way that the connector part **213** optically couples light to the photonic integrated circuit **214**.

(225) The integrated optical communication devices **252** (FIG. **4**) and **262** (FIG. **6**) provide flexibility in the design of the data processing systems, allowing the fiber-optic connector assembly **220** or **270** to be positioned on either side of the package substrate **230**.

(226) Referring to FIG. **7**, in some implementations, a data processing system **280** includes an integrated optical communication device **282** (also referred to as an optical interconnect module), a fiber-optic connector assembly **270**, a package substrate **230**, and an electronic processor integrated circuit **240**. The data processing system **280** can be used, e.g., to implement one or more of devices **101_1** to **101_6** of FIG. **1**.

(227) The integrated optical communication device **282** includes a photonic integrated circuit **284**, a circuit board **286**, a first serializers/deserializers module **216**, a second serializers/deserializers module **217**, and a control circuit **287**. The photonic integrated circuit **284** can include components that perform functions similar to those of the photonic integrated circuit **214** (FIGS. **2-5**) and **264** (FIG. **6**). The control circuit **287** controls the operation of the photonic integrated circuit **284**. For example, the control circuit **287** can control one or more photodetector and/or modulator bias voltages, heater voltages, etc., either statically or adaptively based on one or more sensor voltages that the control circuit **287** can receive from the photonic integrated circuit **284**. The integrated optical communication device **282** further includes a first optical connector part **288** that is configured to receive a second optical connector part **268** of the fiber-optic connector assembly **270**. The optical connector part **268** is attached to an array of optical fibers **272**.

(228) The circuit board **286** has a top main surface **290** and a bottom main surface **292**. The photonic integrated circuit **284** has a top main surface **294** and bottom main surface **296**. The first and second serializers/deserializers modules **216**, **217** are mounted on the top main surface **290** of the circuit board **286**. The top main surface **294** of the photonic integrated circuit **284** has electrical terminals that are electrically coupled to corresponding electrical terminals on the bottom main surface **292** of the circuit board **286**. In this example, the photonic integrated circuit **284** is mounted on a side of the circuit board **286** that is opposite to the side of the circuit board **286** on which the first and second serializers/deserializers modules **216**, **217** are mounted. The photonic integrated circuit **284** is electrically coupled to the first serializers/deserializers **216** by electrical connectors **300** that pass through the circuit board **286** in the thickness direction. In some embodiments, the electrical connectors **300** can be implemented as vias.

(229) The connector part **288** has dimensions that are configured such that the fiber-optic connector assembly **270** can be coupled to the connector part **288** without bumping into other components of the integrated optical communication device **282**. The connector part **288** can be configured to optically couple light to the photonic integrated circuit **284** using optical coupling interfaces, e.g., vertical grating couplers or turning mirrors, similar to the way that the connector part **213** or **266** optically couples light to the photonic integrated circuit **214** or **264**, respectively.

(230) When the integrated optical communication device **282** is coupled to the package substrate **230**, the photonic integrated circuit **284** and the control circuit **287** are positioned between the circuit board **286** and the package substrate **230**. The integrated optical communication device **282** includes an array of contacts **298** arranged on the bottom main

surface **292** of the circuit board **286**. The array of contacts **298** is configured such that after the circuit board **286** is coupled to the package substrate **230**, the array of contacts **298** maintains a thickness d**3** between the circuit board **286** and the package substrate **230**, in which the thickness d**3** is slightly larger than the thicknesses of the photonic integrated circuit **284** and the control circuit **287**.
(231) FIG. **8** is an exploded perspective view of the integrated optical communication device **282** of FIG. **7**. The photonic integrated circuit **284** includes an array of optical coupling components **310**, e.g., vertical grating couplers or turning mirrors, as disclosed in U.S. patent application Ser. No. 16/816,171, that are configured to optically couple light from the optical connector part **288** to the photonic integrated circuit **214**. The optical coupling components **310** are densely packed and have a fine pitch so that optical signals from many optical fibers can be coupled to the photonic integrated circuit **284**. For example, the minimum distance between adjacent optical coupling components **310** can be as small as, e.g., 5 μm, 10 μm, 50 μm, or 100 μm.
(232) An array of electrical terminals **312** arranged on the top main surface **294** of the photonic integrated circuit **284** are electrically coupled to an array of electrical terminals **314** arranged on the bottom main surface **292** of the circuit board **286**. The array of electrical terminals **312** and the array of electrical terminals **314** have a fine pitch, in which the minimum distance between two adjacent electrical terminals can be as small as, e.g., 10 μm, 40 μm, or 100 μm. An array of electrical terminals **316** arranged on the bottom main surface of the first serializers/deserializers **216** are electrically coupled to an array of electrical terminals **318** arranged on the top main surface **290** of the circuit board **286**. An array of electrical terminals **320** arranged on the bottom main surface of the second serializers/deserializers module **217** are electrically coupled an array of electrical terminals **322** arranged on the top main surface **290** of the circuit board **286**.
(233) For example, the arrays of electrical terminals **312**, **314**, **316**, **318**, **320**, and **322** have a fine pitch (or fine pitches). For simplicity of description, in the example of FIG. **8**, for each of the arrays of electrical terminals **312**, **314**, **316**, **318**, **320**, and **322**, the minimum distance between adjacent terminals is d**2**, which can be in the range of, e.g., 10 μm to 200 μm. In some examples, the minimum distance between adjacent terminals for different arrays of electrical terminals can be different. For example, the minimum distance between adjacent terminals for the arrays of electrical terminals **314** (which are arranged on the bottom surface of the circuit board **286**) can be different from the minimum distance between adjacent terminals for the arrays of electrical terminals **318** arranged on the top surface of the circuit board **286**. The minimum distance between adjacent terminals for the arrays of electrical terminals **316** of the first serializers/deserializers **216** can be different from the minimum distance between adjacent terminals for the arrays of electrical terminals **320** of the second serializers/deserializers module **217**.
(234) An array of electrical terminals **324** arranged on the bottom main surface of the circuit board **286** are electrically coupled to the array of contacts **298**. The array of electrical terminals **324** can have a coarse pitch. For example, the minimum distance between adjacent electrical terminals is d**1**, which can be in the range of, e.g., 200 μm to 1 mm. The array of contacts **298** can be configured as a module that maintains a distance that is slightly larger than the thicknesses of the photonic integrated circuit **284** and the control circuit **287** (which is not shown in FIG. **8**) between the integrated optical communication device **282** and the package substrate **230** after the integrated optical communication device **282** is coupled to the package substrate **230**. The array of contacts **298** can include, e.g., a substrate that has embedded spring loaded connectors.
(235) FIG. **9** is a diagram of an example layout design for optical and electrical terminals of the integrated optical communication device **282** of FIGS. **7** and **8**. FIG. **9** shows the layout of the optical and electrical terminals when viewed from the top or bottom side of the device **282**. In this example, the photonic integrated circuit **284** has a width of about 5 mm and a length of about 2.2 mm to 18 mm. For the example in which the length of the photonic integrated circuit **284** is about 2.2 mm, the optical signals provided to the photonic integrated circuit **284** can have a total bandwidth of about 1.6 Tbps. For the example in which the length of the photonic integrated circuit is about 18 mm, the optical signals provided to the photonic integrated circuit can have a total bandwidth of about 12.8 Tbps. The width of the integrated optical communication device **282** can be about 8 mm.
(236) An array **330** of optical coupling components **310** is provided to allow optical signals to be provided to the photonic integrated circuit **284** in parallel. The first serializers/deserializers **216** include an array **332** of electrical terminals **316** arranged on the bottom surface of the first serializers/deserializers **216**. The second serializers/deserializers module **217** include an array **334** of electrical terminals **320** arranged on the bottom surface of the second serializers/deserializers module **217**. The arrays **332** and **334** of electrical terminals **316**, **320** have a fine pitch, and the minimum distance between adjacent terminals can be in the range of, e.g., 40 μm to 200 μm. An array **336** of electrical terminals **324** is arranged on the bottom main surface of the circuit board **286**. The array **336** of electrical terminals **324** has a coarse pitch, and the minimum distance between adjacent terminals can be in the range of, e.g., 200 μm to 1 mm. For example, the array **336** of electrical terminals **324** can be part of a compression interposer that has a pitch of about 400 μm between terminals.
(237) FIG. **10** is a diagram of an example layout design for optical and electrical terminals of the integrated optical communication device **210** of FIG. **2**. FIG. **10** shows the layout of the optical and electrical terminals when viewed from the top or bottom side of the device **210**. In this embodiment, the photonic integrated circuit **214** is implemented as a single chip. In some embodiments, the photonic integrated circuit **214** can be tiled across multiple chips. Likewise, the electronic communication integrated circuit **215** is implemented as a single chip in this embodiment. In some embodiments, the electronic communication integrated circuit **215** can be tiled cross multiple chips. In this embodiment, the electronic communication integrated circuit **215** is implemented using 16 serializers/deserializers blocks **216_1** to

216_16 that are electrically connected to the photonic integrated circuit **214** and 16 serializers/deserializers blocks **217_1** to **217_16**, which are electrically connected to an array of contacts **212_1** by electrical connectors that pass through the substrate **211** in the thickness direction. The 16 serializers/deserializers blocks **216_1** to **216_16** are electrically coupled to the 16 serializers/deserializers blocks **217_1** to **217_16** by bus processing units **218_1** to **218_16**, respectively. In this embodiment, each serializers/deserializers block (**216** or **217**) is implemented using 8 serial differential transmitters (TX) and 8 serial differential receivers (RX). In order to transfer the electrical signals from the serializers/deserializers blocks **217** to ASIC **240**, a total of 8×16×2=256 electrical differential signal contacts **212_1** in addition to 8×17×2=272 ground (GND) contacts **212_1** can be used. Other contact arrangements that beneficially reduce crosstalk, e.g., placing a ground contact between every pair of TX and RX contacts, can also be used as will be appreciated by a person skilled in the art. The transmitter contacts are collectively referenced as **340**, the receiver contacts are collectively referenced as **342**, and the ground contacts are collectively referenced as **344**.

(238) The electrical contacts of the serializers/deserializers blocks **216_1** to **216_12** and **217_1** to **217_12** have a fine pitch, and the minimum distance between adjacent terminals can be in the range of, e.g., 40 μm to 200 μm. The electrical contacts **212_1** have a coarse pitch, and the minimum distance between adjacent terminals can be in the range of, e.g., 200 μm to 1 mm.

(239) FIG. **11** is a schematic side view of an example data processing system **350**, which includes an integrated optical communication device **374**, a package substrate **230**, and a host application specific integrated circuit **240**. The integrated optical communication device **374** and the host application specific integrated circuit **240** are mounted on the top side of the package substrate **230**. The integrated optical communication device **374** includes a first optical connector **356** that allows optical signals transmitted in optical fibers to be coupled to the integrated optical communication device **374**, in which a portion of the optical fibers connected to the first optical connector **356** are positioned at a region facing the bottom side of the package substrate **230**.

(240) The integrated optical communication device **374** includes a photonic integrated circuit **352**, a combination of drivers and transimpedance amplifiers (D/T) **354**, a first serializers/deserializers module **216**, a second serializers/deserializers module **217**, the first optical connector **356**, a control module **358**, and a substrate **360**. The host application specific integrated circuit **240** includes an embedded third serializers/deserializers module **247**.

(241) In this example, the photonic integrated circuit **352**, the drivers and transimpedance amplifiers **354**, the first serializers/deserializers module **216**, and the second serializers/deserializers module **217** are mounted on the top side of the substrate **360**. In some embodiments, the drivers and transimpedance amplifiers **354**, the first serializers/deserializers module **216**, and the second serializers/deserializers module **217** can be monolithically integrated into a single electrical chip. The first optical connector **356** is optically coupled to the bottom side of the photonic integrated circuit **352**. The control module **358** is electrically coupled to electrical terminals arranged on the bottom side of the substrate **360**, whereas the photonic integrated circuit **352** is connected to electrical terminals arranged on the top side of the substrate **360**. The control module **358** is electrically coupled to the photonic integrated circuit **352** through electrical connectors **362** that pass through the substrate **360** in the thickness direction. In some embodiments, the substrate **360** can be removably connected to the package substrate **230**, e.g., using a compression interposer or a land grid array.

(242) The photonic integrated circuit **352** is electrically coupled to the drivers and transimpedance amplifiers **354** through electrical connectors **364** on or in the substrate **360**. The drivers and transimpedance amplifiers **354** are electrically coupled to the first serializers/deserializers module **216** by electrical connectors **366** on or in the substrate **360**. The second serializers/deserializers module **216** has electrical terminals **370** on the bottom side that are electrically coupled to electrical terminals **366** arranged on the bottom side of the substrate **360** through electrical connectors **368** that pass through the substrate **360** in the thickness direction. The electrical terminals **370** have a fine pitch, whereas the electrical terminals **366** have a coarse pitch. The electrical terminals **366** are electrically coupled to the third serializers/deserializers module **247** through electrical connectors or traces **372** on or in the package substrate **230**.

(243) In some implementations, optical signals are converted by the photonic integrated circuit **352** to electrical signals, which are conditioned by the first serializers/deserializers module **216** (or the second serializers/deserializers module **217**), and processed by the host application specific integrated circuit **240**. The host application specific integrated circuit **240** generates electrical signals that are converted by the photonic integrated circuit **352** into optical signals.

(244) FIG. **12** is a schematic side view of an example data processing system **380**, which includes an integrated optical communication device **382**, a package substrate **230**, and a host application specific integrated circuit **240**. The integrated optical communication device **382** is similar to the integrated optical communication device **374** (FIG. **11**), except that the transimpedance amplifiers and drivers are implemented in a separate chip **384** from the chip housing the serializers/deserializers modules **216** and **217**.

(245) FIG. **13** is a schematic side view of an example data processing system **390** that includes an integrated optical communication device **402**, a package substrate **230**, and a host application specific integrated circuit (not shown in the figure). The integrated optical communication device **402** includes photonic integrated circuit **392**, a first serializers/deserializers module **394**, a second serializers/deserializers module **396**, a third serializers/deserializers module **398**, and a fourth serializers/deserializers module **400** that are mounted on a substrate **410**. The photonic integrated circuit **392** can include transimpedance amplifiers and drivers, or such amplifiers and/or drivers can be included in the serializers/deserializers modules **394** and **398**. The first serializers/deserializers module **394** and the second serializers/deserializers module **396** are positioned on the right side of the photonic integrated circuit **392**. The third

serializers/deserializers module **398** and the fourth serializers/deserializers module **400** are positioned on the left side of the photonic integrated circuit **392**. Here, the term "left" and "right" refer to the relative positions shown in the figure. It is understood that the system **390** can be positioned in any orientation so that the first serializers/deserializers module **394** and the second serializers/deserializers module **396** are not necessarily at the right side of the photonic integrated circuit **392**, and the third serializers/deserializers module **398** and the fourth serializers/deserializers module **400** are not necessarily at the left side of the photonic integrated circuit **392**.

(246) The photonic integrated circuit **392** receives optical signals from a first optical connector **404**, generates serial electrical signals based on the optical signals, sends the serial electrical signals to the first and second serializers/deserializers modules **394** and **398**. The first and second serializers/deserializers modules **394** and **398** generate parallel electrical signals based on the received serial electrical signals, and send the parallel electrical signals to the third and fourth serializers/deserializers modules **396** and **400**, respectively. The third and fourth serializers/deserializers modules **396** and **400** generate serial electrical signals based on the received parallel electrical signals, and send the serial electrical signals to electrical terminals **406** and **408**, respectively, arranged on the bottom side of the substrate **410**.

(247) The first optical connector **404** is optically coupled to the bottom side of the photonic integrated circuit **392**. In some embodiments, the optical connector **404** can also be placed on the top of the photonic integrated circuit **392** and couple light to the top side of the photonic integrated circuit **392** (not shown in the figure). The first optical connector **404** is optically coupled to a second optical connector, which in turn is optically coupled to a plurality of optical fibers. In the configuration shown in FIG. **13**, the first optical connector **404**, the second optical connector, and/or the optical fibers pass through an opening **412** in the package substrate **230**. The electrical terminals **406** are arranged on the right side of the first optical connector **404**, and the electrical terminals **408** are arranged on the left side of the first optical connector **404**. The electrical terminals **406** and **408** are configured such that the substrate **410** can be removably coupled to the package substrate **230**.

(248) FIG. **14** is a schematic side view of an example data processing system **420** that includes an integrated optical communication device **428**, a package substrate **230**, and a host application specific integrated circuit (not shown in the figure). The integrated optical communication device **428** includes a photonic integrated circuit **422** (which does not include a transimpedance amplifier and driver), a first serializers/deserializers module **394**, a second serializers/deserializers module **396**, a third serializers/deserializers module **398**, and a fourth serializers/deserializers module **400** that are mounted on a substrate **410**. The integrated optical communication device **428** includes a first set of transimpedance amplifiers and driver circuits **424** positioned at the right of the photonic integrated circuit **422**, and a second set of transimpedance amplifiers and driver circuits **426** positioned at the left of the photonic integrated circuit **422**. The first set of transimpedance amplifiers and driver circuits **424** is positioned between the photonic integrated circuit **422** and a first serializers/deserializers module **394**. The second set of transimpedance amplifiers and driver circuits **424** is positioned between the photonic integrated circuit **422** and a third serializers/deserializers module **398**.

(249) In some implementations, the integrated optical communication device **402** (or **408**) can be modified such that the first optical connector **404** couples optical signals to the top side of the photonic integrated circuit **392** (or **422**).

(250) FIG. **32** is a schematic side view of an example data processing system **510** that includes an integrated optical communication device **512**, a package substrate **230**, and a host application specific integrated circuit (not shown in the figure). The integrated optical communication device **512** includes a substrate **514** that includes a first slab **516** and a second slab **518**. The first slab **516** provides electrical connectors to fan out the electrical contacts. The first slab **516** includes a first set of contacts arranged on the top surface and a second set of contacts arranged on the bottom surface, in which the first set of contacts has a fine pitch and the second set of contacts has a coarse pitch. The second slab **518** provides a removable connection to the package substrate **230**. A photonic integrated circuit **524** is mounted on the bottom side of the first slab **516**. A first optical connector **520** passes through an opening in the substrate **514** and couples optical signals to the top side of the photonic integrated circuit **524**.

(251) A first serializers/deserializers module **394**, a second serializers/deserializers module **396**, a third serializers/deserializers module **398**, and a fourth serializers/deserializers module **400** are mounted on the top side of the first slab **516**. The photonic integrated circuit **524** is electrically coupled to the first and third serializers/deserializers modules **394** and **398** by electrical connectors **522** that pass through the substrate **514** in the thickness direction. For example, the electrical connectors **522** can be implemented as vias. In some examples, drivers and transimpedance amplifiers can be integrated in the photonic integrated circuit **524**, or integrated in the serializers/deserializers modules **394** and **398**. In some examples, the drivers and transimpedance amplifiers can be implemented in a separate chip (not shown in the figure) positioned between the photonic integrated circuit **524** and the serializers/deserializers modules **394** and **398**, similar to the example in FIG. **14**. A control chip (not shown in the figure) can be provided to control the operation of the photonic integrated circuit **512**.

(252) FIG. **15** is a bottom view of an example of the integrated optical communication device **428** of FIG. **14**. The photonic integrated circuit **422** includes modulator and photodetector blocks on both sides of a center line **432** in the longitudinal direction. The photonic integrated circuit **422** includes a fiber coupling region **430** arranged either at the bottom side of the photonic integrated circuit **392** or at the top side of the photonic integrated circuit (see FIG. **32**), in which the fiber coupling region **430** includes multiple optical coupling elements **310**, e.g., receiver optical coupling elements (RX), transmitter optical coupling elements (TX), and remote optical power supply (e.g., **103** in FIG. **1**) optical coupling elements (PS).

(253) Complementary metal oxide semiconductor (CMOS) transimpedance amplifier and driver blocks **424** are arranged on the right side of the photonic integrated circuit **424**, and CMOS transimpedance amplifier and driver blocks **426** are arranged on the left side of the photonic integrated circuit **424**. A first serializers/deserializers module **394** and a second serializers/deserializers module **396** are arranged on the right side of the CMOS transimpedance amplifier and driver blocks **424**. A third serializers/deserializers module **398** and a fourth serializers/deserializers module **400** are arranged on the left side of the CMOS transimpedance amplifier and driver blocks **426**.
(254) In this example, each of the first, second, third, and fourth serializers/deserializers module **394**, **396**, **398**, **400** includes 8 serial differential transmitter blocks and 8 serial differential receiver blocks. The integrated optical communication device **428** has a width of about 3.5 mm and a length of slightly more than about 3.6 mm.
(255) FIG. **16** is a bottom view of an example of the integrated optical communication device **428** of FIG. **14**, in which the electrical terminals **406** and **408** are also shown. As shown in the figure, the electrical terminals **406** and **408** have a coarse pitch, the minimum distance between terminals in the array of electrical terminals **406** or **408** is much larger than the minimum distance between terminals in the array of electrical terminals of the first, second, third, and fourth serializers/deserializers modules **394**, **396**, **398**, and **400**. For example, the array of electrical terminals **406** and **408** can be part of a compression interposer that has a pitch of about 400 μm between terminals.
(256) In some implementations, the electrical terminals (e.g., **406** and **408**) can be arranged in a configuration as shown in FIG. **66**. FIG. **66** shows a pad map **1020** that shows the locations of various contact pads as viewed from the bottom of the package. The contact pads occupy an area that is about 9.8 mm×9.8 mm, in which 400 μm pitch pads are used.
(257) The middle rectangle **1022** is a cutout that connects the photonic integrated circuit to the optics that leave from the top of the module. The bigger rectangle **1024** represents the photonic integrated circuit. The two gray rectangles **1026***a*, **1026***b* represent circuitry in a serializers/deserializers chip **1028***a*. The two gray rectangles **1026***c*, **1026***d* represent circuitry in another serializers/deserializers chip **1028***b*. The serializers/deserializers chips are positioned on the top of the package, and the photonic integrated circuit is positioned on the bottom of the package. The overlap between the photonic integrated circuit and the serializers/deserializers chips **1028***a*, **1028***b* is designed so that vias (not shown in the figure) can directly connect these integrated circuits through the package. In some implementations, the serializers/deserializers chips **1028***a*, **1028***b* and/or other electronic integrated circuits can be placed around three or four sides of the optical connector (represented by the rectangle **1022**).
(258) In the examples of the data processing systems shown in FIGS. **2-8**, **11-14**, and **32**, the integrated optical communication device (e.g., **210**, **252**, **262**, **282**, **374**, **382**, **402**, **428**, **512**, which includes the photonic integrated circuit and the serializers/deserializers modules) is mounted on the package substrate **230** on the same side (top side in the examples shown in the figures) as the electronic processor integrated circuit (or host application specific integrated circuit) **240**. The data processing systems can also be modified such that the integrated optical communication device is mounted on the package substrate **230** on the opposite side as the electronic processor integrated circuit (or host application specific integrated circuit) **240**. For example, the electronic processor integrated circuit **240** can be mounted on the top side of the package substrate **230** and one or more integrated optical communication devices of the form disclosed in FIGS. **2-8**, **11-14**, and **32** can be mounted on the bottom side of the package substrate **230**.
(259) FIG. **17** is a diagram showing four types of integrated optical communication devices that can be used in a data processing system **440**. In these examples, the integrated optical communication device does not include serializers/deserializers modules. At least some of the signal conditioning is performed by the serializers/deserializers module(s) in the digital application specific integrated circuit. The integrated optical communication device is mounted on the side of the printed circuit board that is opposite to the side on which the digital application specific integrated circuit is mounted, allowing the connectors to be short.
(260) In a first example, the data processing system includes a digital application specific integrated circuit **444** mounted on the top side of a substrate **442**, and an integrated optical communication device **448** mounted on the bottom side of the first circuit board. In some implementations, the integrated optical communication device **448** includes a photonic integrated circuit **450** and a set of transimpedance amplifiers and drivers **452** that are mounted on the bottom side of a substrate **454** (e.g., a second circuit board). The top side of the photonic integrated circuit **450** is electrically coupled to the bottom side of the substrate **454**. A first optical connector part **456** is optically coupled to the bottom side of the photonic integrated circuit **450**. The first optical connector part **456** is configured to be optically coupled to a second optical connector part **458** that is optically coupled to a plurality of optical fibers (not shown in the figure). An array of electrical terminals **460** is arranged on the top side of the substrate **454** and configured to enable the integrated optical communication device **448** to be removably coupled to the substrate **442**.
(261) The optical signals from the optical fibers are processed by the photonic integrated circuit **450**, which generates serial electrical signals based on the optical signals. The serial electrical signals are amplified by the set of transimpedance amplifiers and drivers **452**, which drives the output signals that are transmitted to a serializers/deserializers module **446** embedded in the digital application specific integrated circuit **444**.
(262) In a second example, an integrated optical communication device **462** can be mounted on the bottom side of the substrate **442** to provide an optical/electrical communications interface between the optical fibers and the digital application specific integrated circuit **444**. The integrated optical communication device **462** includes a photonic integrated circuit **464** that is mounted on the bottom side of a substrate **454** (e.g., a second circuit board). The top side of the photonic integrated circuit **464** is electrically coupled to the bottom side of the substrate **454**. A first optical connector

part **456** is optically coupled to the bottom side of the photonic integrated circuit **450**. An array of electrical terminals **460** is arranged on the top side of the substrate **454** and configured to enable the integrated optical communication device **462** to be removably coupled to the substrate **442**. The integrated optical communication device **462** is similar to the integrated optical communication device **448**, except that either the photonic integrated circuit **464** or the serializers/deserializers module **446** includes the set of transimpedance amplifiers and driver circuitry. In some examples, the serializers/deserializers module **446** is configured to directly accept electrical signals emerging from photonic integrated circuit **464**, e.g., by having a high enough receiver input impedance that converts the photocurrent generated within the photonic integrated circuit **464** to a voltage swing suitable for further electrical processing. For example, the serializers/deserializers module **446** is configured to have a low transmitter output impedance, and provide an output voltage swing that allows direct driving of optical modulators embedded within the photonic integrated circuit **464**.
(263) In a third example, an integrated optical communication device **466** can be mounted on the bottom side of the substrate **442** to provide an optical/electrical communications interface between the optical fibers and the digital application specific integrated circuit **444**. The integrated optical communication device **466** includes a photonic integrated circuit **468** that is mounted on the top side of a substrate **470** (e.g., a second circuit board). The bottom side of the photonic integrated circuit **468** is electrically coupled to the top side of the substrate **470**. A first optical connector part **456** is optically coupled to the bottom side of the photonic integrated circuit **468**. An array of electrical terminals **460** is arranged on the top side of the substrate **470** and configured to enable the integrated optical communication device **466** to be removably coupled to the substrate **442**. In some examples, either the photonic integrated circuit **468** or the serializers/deserializers module **446** includes the set of transimpedance amplifiers and driver circuitry. In some examples, the serializers/deserializers module **446** is configured to directly accept electrical signals emerging from the photonic integrated circuit **464**.
(264) In a fourth example, an integrated optical communication device **472** can be mounted on the bottom side of the substrate **442** to provide an optical/electrical communications interface between the optical fibers and the digital application specific integrated circuit **444**. The integrated optical communication device **472** includes a photonic integrated circuit **474** and a set of transimpedance amplifiers and drivers **476** that are mounted on the top side of a substrate **470** (e.g., a second circuit board). The bottom side of the photonic integrated circuit **474** is electrically coupled to the top side of the substrate **470**. A first optical connector part **456** is optically coupled to the bottom side of the photonic integrated circuit **468**. An array of electrical terminals **460** is arranged on the top side of the substrate **470** and configured to enable the integrated optical communication device **466** to be removably coupled to the substrate **442**. The integrated optical communication device **472** is similar to the integrated optical communication device **466**, except that neither the photonic integrated circuit **464** nor the serializers/deserializers module **446** include a set of transimpedance amplifiers and driver circuitry, and the set of transimpedance amplifiers and drivers **476** is implemented as a separate integrated circuit.
(265) FIG. **18** is a diagram of an example octal serializers/deserializers block **480** that includes 8 serial differential transmitters (TX) **482** and 8 serial differential receivers (RX) **484**. Each serial differential receiver **484** receives a serial differential signal, generates parallel signals based on the serial differential signal, and provides the parallel signals on the parallel bus **488**. Each serial differential transmitter **482** receives parallel signals from the parallel bus **488**, generates a serial differential signal based on the parallel signals, and provides the serial differential signal on an output electrical terminal **490**. The serializers/deserializers block **480** outputs and/or receives parallel signals through a parallel bus interface **492**.
(266) In the examples described above, such as those shown in FIGS. **2-14**, the integrated optical communication device (e.g., **210**, **252**, **262**, **282**, **374**, **382**, **402**, **428**) includes a first serializers/deserializers module (e.g., **216**, **394**, **398**) and a second serializers/deserializers module (e.g., **217**, **396**, **400**). The first serializers/deserializers module serially interfaces with the photonic integrated circuit, and the second serializers/deserializers module serially interfaces with the electronic processor integrated circuit or host application specific integrated circuit (e.g., **240**). In some implementations, the electronic communication integrated circuit **215** includes an array of serializers/deserializers that can be logically partitioned into a first sub-array of serializers/deserializers and a second sub-array of serializers/deserializers. The first sub-array of serializers/deserializers corresponds to the serializers/deserializers module (e.g., **216**, **394**, **398**), and the second sub-array of serializers/deserializers corresponds to the second serializers/deserializers module (e.g., **217**, **396**, **400**).
(267) FIG. **38** is a diagram of an example octal serializers/deserializers block **480** coupled to a bus processing unit **218**. The octal serializers/deserializers block **480** includes 8 serial differential transmitters (TX**1** to TX**8**) **482** and 8 serial differential receivers (RX**1** to RX**4**) **484**. In some implementations, the transmitters and receivers are partitioned such that the transmitters TX**1**, TX**2**, TX**3**, TX**4** and receivers RX**1**, RX**2**, RX**3**, RX**4** form a first serializers/deserializers module **840**, and the transmitters TX**5**, TX**6**, TX**7**, TX**8** and receivers RX**5**, RX**6**, RX**7**, RX**8** form a second serializers/deserializers module **842**. Serial electrical signals received at the receivers RX**1**, RX**2**, RX**3**, RX**4** are converted to parallel electrical signals and routed by the bus processing unit **218** to the transmitters TX**5**, TX**6**, TX**7**, TX**8**, which convert the parallel electrical signals to serial electrical signals. For example, the photonic integrated circuit can send serial electrical signals to the receivers RX**1**, RX**2**, RX**3**, RX**4**, and the transmitters TX**5**, TX**6**, TX**7**, TX**8** can transmit serial electrical signals to the electronic processor integrated circuit or host application specific integrated circuit.
(268) For example, the bus processing unit **218** can re-map the lanes of signals and perform coding on the signals, such that the bit rate and/or modulation format of the serial signals output from the transmitters TX**5**, TX**6**, TX**7**, TX**8** can be

different from the bit rate and/or modulation format of the serial signals received at the receivers RX**1**, RX**2**, RX**3**, RX**4**. For example, 4 lanes of T Gbps NRZ serial signals received at the receivers RX**1**, RX**2**, RX**3**, RX**4** can be re-encoded and routed to transmitters TX**5**, TX**6** to output 2 lanes of 2×T Gbps PAM4 serial signals.

(269) Similarly, serial electrical signals received at the receivers RX**5**, RX**6**, RX**7**, RX**8** are converted to parallel electrical signals and routed by the bus processing unit **218** to the transmitters TX**1**, TX**2**, TX**3**, TX**4**, which convert the parallel electrical signals to serial electrical signals. For example, the electronic processor integrated circuit or host application specific integrated circuit can send serial electrical signals to the receivers RX**5**, RX**6**, RX**7**, RX**8**, and the transmitters TX**1**, TX**2**, TX**3**, TX**4** can transmit serial electrical signals to the photonic integrated circuit.

(270) For example, the bus processing unit **218** can re-map the lanes of signals and perform coding on the signals, such that the bit rate and/or modulation format of the serial signals output from the transmitters TX**1**, TX**2**, TX**3**, TX**4** can be different from the bit rate and/or modulation format of the serial signals received at the receivers RX**5**, RX**6**, RX**7**, RX**8**. For example, 2 lanes of 2×T Gbps PAM4 serial signals received at receivers RX**5**, RX**6** can be re-encoded and routed to the transmitters TX**5**, TX**6**, TX**7**, TX**8** to output 4 lanes of T Gbps NRZ serial signals.

(271) FIG. **39** is a diagram of another example octal serializers/deserializers block **480** coupled to a bus processing unit **218**, in which the transmitters and receivers are partitioned such that the transmitters TX**1**, TX**2**, TX**5**, TX**6** and receivers RX**1**, RX**2**, RX**5**, RX**6** form a first serializers/deserializers module **850**, and the transmitters TX**3**, TX**4**, TX**7**, TX**8** and receivers RX**3**, RX**4**, RX**7**, RX**8** form a second serializers/deserializers module **852**. Serial electrical signals received at the receivers RX**1**, RX**2**, RX**5**, RX**6** are converted to parallel electrical signals and routed by the bus processing unit **218** to the transmitters TX**3**, TX**4**, TX**7**, TX**8**, which convert the parallel electrical signals to serial electrical signals. For example, the photonic integrated circuit can send serial electrical signals to the receivers RX**1**, RX**2**, RX**5**, RX**6**, and the transmitters TX**3**, TX**4**, TX**7**, TX**8** can transmit serial electrical signals to the electronic processor integrated circuit or host application specific integrated circuit.

(272) Similarly, serial electrical signals received at the receivers RX**3**, RX**4**, RX**7**, RX**8** are converted to parallel electrical signals and routed by the bus processing unit **218** to the transmitters TX**1**, TX**2**, TX**5**, TX**6**, which convert the parallel electrical signals to serial electrical signals. For example, the electronic processor integrated circuit or host application specific integrated circuit can send serial electrical signals to the receivers RX**3**, RX**4**, RX**7**, RX**8**, and the transmitters TX**1**, TX**2**, TX**5**, TX**6** can transmit serial electrical signals to the photonic integrated circuit.

(273) In some implementations, the bus processing unit **218** can re-map the lanes of signals and perform coding on the signals, such that the bit rate and/or modulation format of the serial signals output from the transmitters TX**3**, TX**4**, TX**7**, TX**8** can be different from the bit rate and/or modulation format of the serial signals received at the receivers RX**1**, RX**2**, RX**5**, RX**6**. Similarly, the bus processing unit **218** can re-map the lanes of signals and perform coding on the signals such that the bit rate and/or modulation format of the serial signals output from the transmitters TX**1**, TX**2**, TX**5**, TX**6** can be different from the bit rate and/or modulation format of the serial signals received at the receivers RX**4**, RX**4**, RX**7**, RX**8**.

(274) FIGS. **38** and **39** show two examples of how the receivers and transmitters can be partitioned to form the first serializers/deserializers module and the second serializers/deserializers module. The partitioning can be arbitrarily determined based on application, and is not limited to the examples shown in FIGS. **38** and **39**. The partitioning can be programmable and dynamically changed by the system.

(275) FIG. **19** is a diagram of an example electronic communication integrated circuit **480** that includes a first octal serializers/deserializers block **482** electrically coupled to a second octal serializers/deserializers block **484**. For example, the electronic communication integrated circuit **480** can be used as the electronic communication integrated circuit **215** of FIGS. **2** and **3**. The first octal serializers/deserializers block **482** can be used as the first serializers/deserializers module **216**, and the second octal serializers/deserializers block **484** can be used as the second serializers/deserializers module **217**. For example, the first octal serializers/deserializers block **482** can receive 8 serial differential signals, e.g., through electrical terminals arranged at the bottom side of the block, and generate 8 sets of parallel signals based on the 8 serial differential signals, in which each set of parallel signals is generated based on the corresponding serial differential signal. The first octal serializers/deserializers block **482** can condition serial electrical signals upon conversion into the 8 sets of parallel signals, such as performing clock and data recovery, and/or signal equalization. The first octal serializers/deserializers block **482** transmits the 8 sets of parallel signals to the second octal serializers/deserializers block **484** through a parallel bus **485** and a parallel bus **486**. The second octal serializers/deserializers block **484** can generate 8 serial differential signals based on the 8 sets of parallel signals, in which each serial differential signal is generated based on the corresponding set of parallel signals. The second octal serializers/deserializers block **484** can output the 8 serial differential signals through, e.g., electrical terminals arranged at the bottom side of the block.

(276) Multiple serializers/deserializers blocks can be electrically coupled to multiple serializers/deserializers blocks through a bus processing unit that can be, e.g., a parallel bus of electrical lanes, a static or a dynamically reconfigurable cross-connect device, or a re-mapping device (gearbox). FIG. **33** is a diagram of an example electronic communication integrated circuit **530** that includes a first octal serializers/deserializers block **532** and a second octal serializers/deserializers block **534** electrically coupled to a third octal serializers/deserializers block **536** through a bus processing unit **538**. In this example, the bus processing unit **538** is configured to enable switching of the signals, allowing the routing of signals to be re-mapped, in which 8×50 Gbps serial electrical signals using NRZ modulation that are serially interfaced to the first and second octal serializers/deserializers blocks **532** and **534** are re-routed or combined into 8×100 Gbps serial electrical signals using PAM4 modulation that are serially interfaced to the third octal serializers/deserializers

block **536**. An example of the bus processing unit **538** is shown in FIG. **41**A. In some examples, the bus processing unit **538** enables N lanes of T Gbps serial electrical signals to be remapped into N/M lanes of M×T Gbps serial electrical signals, N and M being positive integers, T being a real value, in which the N serially interfacing electrical signals can be modulated using a first modulation format and the M serially interfacing electrical signals can be modulated using a second modulation format.

(277) In some other examples, the bus processing unit **538** can allow for redundancy to increase reliability. For example, the first and the second serializers/deserializers blocks **532** and **534** can be jointly configured to serially interface to a total of N lanes of T×N/(N−k) Gbps electrical signals, while the third serializers/deserializers block **536** can be configured to serially interface to N lanes of T Gbps electrical signals. The bus processing unit **538** can then be configured to remap the data from only N−k out of the N lanes serially interfacing to the first and the second serializers/deserializers blocks **532** and **534** (carrying an aggregate bit rate of (N−k)×T×N/(N−k)=T×N) to the third serializers/deserializers block **536**. This way, the bus processing unit **538** allows for k out of N serially interfacing electrical links to the first and the second serializers/deserializers blocks **532** and **534** to fail while still maintaining an aggregate of T×N Gbps of data serially interfacing to the third serializers/deserializers block **536**. The number k is a positive integer. In some embodiments, k can be approximately 1% of N. In some other embodiments, k can be approximately 10% of N. In some embodiment, the selection of which N−k of the N serially interfacing electrical links to the first and the second serializers/deserializers blocks **532** and **534** to remap to the third serializers/deserializers block **536** using bus processing unit **538** can be dynamically selected, e.g., based on signal integrity and signal performance information extracted from the serially interfacing signals by the serializers/deserializers blocks **532** and **534**. An example of the bus processing unit **538** is shown in FIG. **41**B, in which N=16, k=2, T=50 Gbps.

(278) In some examples, using the redundancy technique discussed above, the bus processing unit **538** enables N lanes of T×N/(N−k) Gbps serial electrical signals to be remapped into N/M lanes of M×T Gbps serial electrical signals. The bus processing unit **538** enables k out of N serially interfacing electrical links to fail while still maintaining an aggregate of T×N Gbps of data serially interfacing to the third serializers/deserializers block **536**.

(279) FIG. **20** is a functional block diagram of an example data processing system **200**, which can be used to implement, e.g., one or more of devices **101_1** to **101_6** of FIG. **1**. Without implied limitation, the data processing system **200** is shown as part of the node **101_1** for illustration purposes. The data processing system **200** can be part of any other network element of the system **100**. The data processing system **200** includes an integrated communication device **210**, a fiber-optic connector assembly **220**, a package substrate **230**, and an electronic processor integrated circuit **240**.

(280) The connector assembly **220** includes a connector **223** and a fiber array **226**. The connector **223** can include multiple individual fiber-optic connectors **423**_*i* (i∈{R**1** . . . RM; S**1** . . . SK; T**1** . . . TN} with K, M, and N being positive integers). In some embodiments, some or all of the individual connectors **423**_*i* can form a single physical entity. In some embodiments some or all of the individual connectors **423**_*i* can be separate physical entities. When operating as part of the network element **101_1** of the system **100**, (i) the connectors **423**_S**1** through **423**_SK can be connected to optical power supply **103**, e.g., through link **102_6**, to receive supply light; (ii) the connectors **423**_R**1** through **423**_RM can be connected to the transmitters of the node **101_2**, e.g., through the link **102_1**, to receive from the node **101_2** optical communication signals; and (iii) the connectors **423**_T**1** through **423**_TN can be connected to the receivers of the node **101_2**, e.g., through the link **102_1**, to transmit to the node **101_2** optical communication signals.

(281) In some implementations, the communication device **210** includes an electronic communication integrated circuit **215**, a photonic integrated circuit **214**, a connector part **213**, and a substrate **211**. The connector part **213** can include multiple individual optical connectors **413**_*i* to photonic integrated circuit **214** (i∈{R**1** . . . RM; S**1** . . . SK; T**1** . . . TN} with K, M, and N being positive integers). In some embodiments, some or all of the individual connectors **413**_*i* can form a single physical entity. In some embodiments some or all of the individual connectors **413**_*i* can be separate physical entities. The optical connectors **413**_*i* are configured to optically couple light to the photonic integrated circuit **214** using optical coupling interfaces **414**, e.g., vertical grating couplers, turning mirrors, etc., as disclosed in U.S. patent application Ser. No. 16/816,171.

(282) In operation, light entering the photonic integrated circuit **214** from the link **102_6** through coupling interfaces **414**_S**1** through **414**_SK can be split using an optical splitter **415**. The optical splitter **415** can be an optical power splitter, an optical polarization splitter, an optical wavelength demultiplexer, or any combination or cascade thereof, e.g., as disclosed in U.S. Pat. No. 11,153,670 and in U.S. patent application Ser. No. 16/888,890, filed on Jun. 1, 2020, published as US 2021/0376950, which is incorporated herein by reference in its entirety. In some embodiments, one or more splitting functions of the splitter **415** can be integrated into the optical coupling interfaces **414** and/or into optical connectors **413**. For example, in some embodiments, a polarization-diversity vertical grating coupler can be configured to simultaneously act as a polarization splitter **415** and as a part of optical coupling interface **414**. In some other embodiments, an optical connector that includes a polarization-diversity arrangement can simultaneously act as an optical connector **413** and as a polarization splitter **415**.

(283) In some embodiments, light at one or more outputs of the splitter **415** can be detected using a receiver **416**, e.g., to extract synchronization information as disclosed in U.S. Pat. No. 11,153,670. In various embodiments, the receiver **416** can include one or more p-i-n photodiodes, one or more avalanche photodiodes, one or more self-coherent receivers, or one or more analog (heterodyne/homodyne) or digital (intradyne) coherent receivers. In some embodiments, one or more opto-electronic modulators **417** can be used to modulate onto light at one or more outputs of the splitter **415** data for

communication to other network elements.
(284) Modulated light at the output of the modulators **417** can be multiplexed in polarization or wavelength using a multiplexer **418** before leaving the photonic integrated circuit **214** through optical coupling interfaces **414**_T**1** through **414**_TN. In some embodiments, the multiplexer **418** is not provided, i.e., the output of each modulator **417** can be directly coupled to a corresponding optical coupling interface **414**.
(285) On the receiver side, light entering the photonic integrated circuit **214** through a coupling interfaces **414**_R**1** through **414**_RM from, e.g., the link **101_2**, can first be demultiplexed in polarization and/or in wavelength using an optical demultiplexer **419**. The outputs of the demultiplexer **419** are then individually detected using receivers **421**. In some embodiments, the demultiplexer **419** is not provided, i.e., the output of each coupling interface **414**_R**1** through **414**_RM can be directly coupled to a corresponding receiver **421**. In various embodiments, the receiver **421** can include one or more p-i-n photodiodes, one or more avalanche photodiodes, one or more self-coherent receivers, or one or more analog (heterodyne/homodyne) or digital (intradyne) coherent receivers.
(286) The photonic integrated circuit **214** is electrically coupled to the integrated circuit **215**. In some implementations, the photonic integrated circuit **214** provides a plurality of serial electrical signals to the first serializers/deserializers module **216**, which generates sets of parallel electrical signals based on the serial electrical signals, in which each set of parallel electrical signal is generated based on a corresponding serial electrical signal. The first serializers/deserializers module **216** conditions the serial electrical signals, demultiplexes them into the sets of parallel electrical signals and sends the sets of parallel electrical signals to the second serializers/deserializers module **217** through a bus processing unit **218**. In some implementations, the bus processing unit **218** enables switching of signals and performs line coding and/or error-correcting coding functions. An example of the bus processing unit **218** is shown in FIG. **42**.
(287) The second serializers/deserializers module **217** generates a plurality of serial electrical signals based on the sets of parallel electrical signals, in which each serial electrical signal is generated based on a corresponding set of parallel electrical signal. The second serializers/deserializers module **217** sends the serial electrical signals through electrical connectors that pass through the substrate **211** in the thickness direction to an array of electrical terminals **500** that are arranged on the bottom surface of the substrate **211**. For example, the array of electrical terminals **500** configured to enable the integrated communication device **210** to be easily coupled to, or removed from, the package substrate **230**.
(288) In some implementations, the electronic processor integrated circuit **240** includes a data processor **502** and an embedded third serializers/deserializers module **504**. The third serializers/deserializers module **504** receives the serial electrical signals from the second serializers/deserializers module **217**, and generates sets of parallel electrical signals based on the serial electrical signals, in which each set of parallel electrical signal is generated based on a corresponding serial electrical signal. The data processor **502** processes the sets of parallel signals generated by the third serializers/deserializers module **504**.
(289) In some implementations, the data processor **502** generates sets of parallel electrical signals, and the third serializers/deserializers module **504** generates serial electrical signals based on the sets of parallel electrical signals, in which each serial electrical signal is generated based on a corresponding set of parallel electrical signal. The serial electrical signals are sent to the second serializers/deserializers module **217**, which generates sets of parallel electrical signals based on the serial electrical signals, in which each set of parallel electrical signal is generated based on a corresponding serial electrical signal. The second serializers/deserializers module **217** sends the sets of parallel electrical signals to the first serializers/deserializers module **216** through the bus processing unit **218**. The first serializers/deserializers module **216** generates serial electrical signals based on the sets of parallel electrical signals, in which each serial electrical signal is generated based on a corresponding set of parallel electrical signals. The first serializers/deserializers module **216** sends the serial electrical signals to the photonic integrated circuit **214**. The opto-electronic modulators **417** modulate optical signals based on the serial electrical signals, and the modulated optical signals are output from the photonic integrated circuit **214** through optical coupling interfaces **414**_T**1** through **414**_TN.
(290) In some embodiments, supply light from the optical power supply **103** includes an optical pulse train, and synchronization information extracted by the receiver **416** can be used by the serializers/deserializers module **216** to align the electrical output signals of the serializers/deserializers module **216** with respective copies of the optical pulse trains at the outputs of the splitter **415** at the modulators **417**. For example, the optical pulse train can be used as an optical power supply at the optical modulator. In some such implementations, the first serializers/deserializers module **216** can include interpolators or other electrical phase adjustment elements.
(291) Referring to FIG. **21**, in some implementations, a data processing system **540** includes an enclosure or housing **542** that has a front panel **544**, a bottom panel **546**, side panels **548** and **550**, a rear panel **552**, and a top panel (not shown in the figure). The system **540** includes a printed circuit board **558** that extends substantially parallel to the bottom panel **546**. A data processing chip **554** is mounted on the printed circuit board **558**, in which the chip **554** can be, e.g., a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC).
(292) At the front panel **544** are pluggable input/output interfaces **556** that allow the data processing chip **554** to communicate with other systems and devices. For example, the input/output interfaces **556** can receive optical signals from outside of the system **540** and convert the optical signals to electrical signals for processing by the data processing chip **554**. The input/output interfaces **556** can receive electrical signals from the data processing chip **554** and convert the

electrical signals to optical signals that are transmitted to other systems or devices. For example, the input/output interfaces **556** can include one or more of small form-factor pluggable (SFP), SFP+, SFP28, QSFP, QSFP28, or QSFP56 transceivers. The electrical signals from the transceiver outputs are routed to the data processing chip **554** through electrical connectors on or in the printed circuit board **558**.

(293) In the examples shown in FIGS. **21** to **29**B, **69**A, **70**, **71**A, **72**, **72**A, **74**A, **75**A, **75**C, **76**, **77**A, **77**B, **78**, **96** to **98**, **100**, **110**, **112**, **113**, **115**, **117** to **122**, **125**A to **127**, and **129** to **131**, various embodiments can have various form factors, e.g., in some embodiments the top panel and the bottom panel **546** can have the largest area, in other embodiments the side panels **548** and **550** can have the largest area, and in yet other embodiments the front panel **544** and the rear panel **552** can have the largest area. In various embodiments, the printed circuit board **558** can be substantially parallel to the two side panels, e.g., the data processing system **540** as shown in FIG. **21** can stand on one of its side panels during normal operation (such that the side panel **550** is positioned at the bottom, and the bottom panel **546** is positioned at the side). In various embodiments, the data processing system **540** can comprise two or more printed circuit boards some of which can be substantially parallel to the bottom panel and some of which can be substantially parallel to the side panels. For example, some computer systems for machine learning/artificial intelligence applications have vertical circuit boards that are plugged into the systems. As used herein, the distinction between “front” and “back” is made based on where the majority of input/output interfaces **556** are located, irrespective of what a user may consider the front or back of data processing system **540**.

(294) FIG. **22** is a diagram of a top view of an example data processing system **560** that includes a housing **562** having side panels **564** and **566**, and a rear panel **568**. The system **560** includes a vertically mounted printed circuit board **570** that can also function as the front panel. The surface of the printed circuit board **570** is substantially perpendicular to the bottom panel of the housing **562**. The term “substantially perpendicular” is meant to take into account of manufacturing and assembly tolerances, so that if a first surface is substantially perpendicular to a second surface, the first surface is at an angle in a range from 85° to 95° relative to the second surface. On the printed circuit board **570** are mounted a data processing chip **572** and an integrated communication device **574**. In some examples, the data processing chip **572** and the integrated communication device **574** are mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached (e.g., electrically coupled) to the printed circuit board **570**. The data processing chip **572** can be, e.g., a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC). A heat sink **576** is provided on the data processing chip **572**.

(295) In some implementations, the integrated communication device **574** includes a photonic integrated circuit **586** and an electronic communication integrated circuit **588** mounted on a substrate **594**. The electronic communication integrated circuit **588** includes a first serializers/deserializers module **590** and a second serializers/deserializers module **592**. The printed circuit board **570** can be similar to the package substrate **230** (FIGS. **2**, **4**, **11-14**), the data processing chip **572** can be similar to the electronic processor integrated circuit or application specific integrated circuit **240**, and the integrated communication device **574** can be similar to the integrated communication device **210**, **252**, **374**, **382**, **402**, **428**. In some embodiments, the integrated communication device **574** is soldered to the printed circuit board **570**. In some other embodiments, the integrated communication device **574** is removably connected to the printed circuit board **570**, e.g., via a land grid array or a compression interposer. Related holding fixtures including snap-on or screw-on mechanisms are not shown in the figure.

(296) In some examples, the integrated communication device **574** includes a photonic integrated circuit without serializers/deserializers modules, and drivers/transimpedance amplifiers (TIA) are provided separately. In some examples, the integrated communication device **574** includes a photonic integrated circuit and drivers/transimpedance amplifiers but without serializers/deserializers modules.

(297) The integrated communication device **574** includes a first optical connector **578** that is configured to receive a second optical connector **580** that is coupled to a bundle of optical fibers **582**. The integrated communication device **574** is electrically coupled to the data processing chip **572** through electrical connectors or traces **584** on or in the printed circuit board **570**. Because the data processing chip **572** and the integrated communication device **574** are both mounted on the printed circuit board **570**, the electrical connectors or traces **584** can be made shorter, compared to the electrical connectors that electrically couple the transceivers **556** to the data processing chip **554** of FIG. **21**. Using shorter electrical connectors or traces **584** allows the signals to have a higher data rate with lower noise, lower distortion, and/or lower crosstalk. Mounting the printed circuit board **570** perpendicular to the bottom panel of the housing allows for more easily accessible connections to the integrated communication device **574** that may be removed and re-connected without, e.g., removing the housing from a rack.

(298) In some examples, the bundle of optical fibers **582** can be firmly attached to the photonic integrated circuit **586** without the use of the first and second optical connectors **578**, **580**.

(299) The printed circuit board **570** can be secured to the side panels **564** and **566**, and the bottom and top panels of the housing using, e.g., brackets, screws, clips, and/or other types of fastening mechanisms. The surface of the printed circuit board **570** can be oriented perpendicular to bottom panel of the housing, or at an angle (e.g., between −60° to 60°) relative to the vertical direction (the vertical direction being perpendicular to the bottom panel). The printed circuit board **570** can have multiple layers, in which the outermost layer (i.e., the layer facing the user) has an exterior surface that is configured to be aesthetically pleasing.

(300) The first optical connector **578**, the second optical connector **580**, and the bundle of optical fibers **582** can be similar to those shown in FIGS. **2**, **4**, and **11-16**. As described above, the bundle of fibers **582** can include 10 or more optical fibers, 100 or more optical fibers, 500 or more optical fibers, or 1000 or more optical fibers. The optical signals provided to the photonic integrated circuit **586** can have a high total bandwidth, e.g., about 1.6 Tbps, or about 12.8 Tbps, or more.
(301) Although FIG. **22** shows one integrated communication device **574**, there can be additional integrated communication devices **574** that are electrically coupled to the data processing chip **572**. The data processing system **560** can include a second printed circuit board (not shown in the figure) oriented parallel to the bottom panel of the housing **562**. The second printed circuit board can support other optical and/or electronic devices, such as storage devices, memory chips, controllers, power supply modules, fans, and other cooling devices.
(302) In some examples of the data processing system **540** (FIG. **21**), the transceiver **556** can include circuitry (e.g., integrated circuits) that perform some type of processing of the signals and/or the data contained in the signals. The signals output from the transceiver **556** need to be routed to the data processing chip **554** through longer signal paths that place a limit on the data rate. In some data processing systems, the data processing chip **554** outputs processed data that are routed to one of the transceivers and transmitted to another system or device. Again, the signals output from the data processing chip **554** need to be routed to the transceiver **556** through longer signal paths that place a limit on the data rate. By comparison, in the data processing system **560** (FIG. **22**), the electrical signals that are transmitted between the integrated communication devices **574** and the data processing chip **572** pass through shorter signal paths and thus support a higher data rate.
(303) FIG. **23** is a diagram of a top view of an example data processing system **600** that includes a housing **602** having side panels **604** and **606**, and a rear panel **608**. The system **600** includes a vertically mounted printed circuit board **610** that functions as the front panel. The surface of the printed circuit board **610** is substantially perpendicular to the bottom panel of the housing **602**. A data processing chip **572** is mounted on an interior side of the printed circuit board **610**, and an integrated communication device **612** is mounted on an exterior side of the printed circuit board **610**. In some examples, the data processing chip **572** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the printed circuit board **610**. In some embodiments, the integrated communication device **612** is soldered to the printed circuit board **610**. In some other embodiments, the integrated communication device **612** is removably connected to the printed circuit board **610**, e.g., via a land grid array or a compression interposer. Related holding fixtures including snap-on or screw-on mechanisms are not shown in the figure. A heat sink **576** is provided on the data processing chip **572**.
(304) In some implementations, the integrated communication device **612** includes a photonic integrated circuit **614** and an electronic communication integrated circuit **588** mounted on a substrate **618**. The electronic communication integrated circuit **588** includes a first serializers/deserializers module **590** and a second serializers/deserializers module **592**. The integrated communication device **612** includes a first optical connector **578** that is configured to receive a second optical connector **580** that is coupled to a bundle of optical fibers **582**. The integrated communication device **612** is electrically coupled to the data processing chip **572** through electrical connectors or traces **616** that pass through the printed circuit board **610** in the thickness direction. Because the data processing chip **572** and the integrated communication device **612** are both mounted on the printed circuit board **610**, the electrical connectors or traces **616** can be made shorter, thereby allowing the signals to have a higher data rate with lower noise, lower distortion, and/or lower crosstalk. Mounting the integrated communication device **612** on the outside of the printed circuit board **610** perpendicular to the bottom panel of the housing and accessible from outside the housing allows for more easily accessible connections to the integrated communication device **612** that may be removed and re-connected without, e.g., removing the housing from a rack.
(305) In some examples, the integrated communication device **612** includes a photonic integrated circuit without serializers/deserializers modules, and drivers and transimpedance amplifiers (TIA) are provided separately. In some examples, the integrated communication device **612** includes a photonic integrated circuit and drivers/transimpedance amplifiers but without serializers/deserializers modules. In some examples, the bundle of optical fibers **582** can be firmly attached to the photonic integrated circuit **614** without the use of the first and second optical connectors **578**, **580**.
(306) In some examples, the data processing chip **572** is mounted on the rear side of the substrate, and the integrated communication device **612** are removably attached to the front side of the substrate, in which the substrate provides high speed connections between the data processing chip **572** and the integrated communication device **612**. For example, the substrate can be attached to a front side of a printed circuit board, in which the printed circuit board includes an opening that allows the data processing chip **572** to be mounted on the rear side of the substrate. The printed circuit board can provide from a motherboard electrical power to the substrate (and hence to the data processing chip **572** and the integrated communication device **612**, and allow the data processing chip **572** and the integrated communication device **612** to connect to the motherboard using low-speed electrical links.
(307) The printed circuit board **610** can be secured to the side panels **604** and **606**, and the bottom and top panels of the housing using, e.g., brackets, screws, clips, and/or other types of fastening mechanisms. The surface of the printed circuit board **610** can be oriented perpendicular to bottom panel of the housing, or at an angle (e.g., between −60° to 60°) relative to the vertical direction (the vertical direction being perpendicular to the bottom panel). The printed circuit board **610** can have multiple layers, in which the portion of the outermost layer (i.e., the layer facing the user) not covered by the integrated communication device **612** has an exterior surface that is configured to be aesthetically pleasing.
(308) FIGS. **24-27** below illustrate four general designs in which the data processing chips are positioned near the input/output communication interfaces. FIG. **24** is a top view of an example data processing system **630** in which a data

processing chip **640** is mounted near an optical/electrical communication interface **644** to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **640** and the optical/electrical communication interface **644**. In this example, the data processing chip **640** and the optical/electrical communication interface **644** are mounted on a circuit board **642** that functions as the front panel of an enclosure **632** of the system **630**, thus allowing optical fibers to be easily coupled to the optical/electrical communication interface **644**. In some examples, the data processing chip **640** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **642**.

(309) The enclosure **632** has side panels **634** and **636**, a rear panel **638**, a top panel, and a bottom panel. In some examples, the circuit board **642** is perpendicular to the bottom panel. In some examples, the circuit board **642** is oriented at an angle in a range −60° to 60° relative to a vertical direction of the bottom panel. The side of the circuit board **642** facing the user is configured to be aesthetically pleasing.

(310) The optical/electrical communication interface **644** is electrically coupled to the data processing chip **640** by electrical connectors or traces **646** on or in the circuit board **642**. The circuit board **642** can be a printed circuit board that has one or more layers. The electrical connectors or traces **646** can be signal lines printed on the one or more layers of the printed circuit board **642** and provide high bandwidth data paths (e.g., one or more Gigabits per second per data path) between the data processing chip **640** and the optical/electrical communication interface **644**.

(311) In a first example, the data processing chip **640** receives electrical signals from the optical/electrical communication interface **644** and does not send electrical signals to the optical/electrical communication interface **644**. In a second example, the data processing chip **640** receives electrical signals from, and sends electrical signals to, the optical/electrical communication interface **644**. In the first example, the optical/electrical communication interface **644** receives optical signals from optical fibers, generates electrical signals based on the optical signals, and sends the electrical signals to the data processing chip **640**. In the second example, the optical/electrical communication interface **644** also receives electrical signals from the data processing chip, generates optical signals based on the electrical signals, and sends the optical signals to the optical fibers.

(312) An optical connector **648** is provided to couple optical signals from the optical fibers to the optical/electrical communication interface **644**. In this example, the optical connector **648** passes through an opening in the circuit board **642**. In some examples, the optical connector **648** is securely fixed to the optical/electrical communication interface **644**. In some examples, the optical connector **648** is configured to be removably coupled to the optical/electrical communication interface **644**, e.g., by using a pluggable and releasable mechanism, which can include one or more snap-on or screw-on mechanisms. In some other examples, an array of 10 or more fibers is securely or fixedly attached to the optical connector **648**.

(313) The optical/electrical communication interface **644** can be similar to, e.g., the integrated communication device **210** (FIG. **2**), **252** (FIG. **4**), **374** (FIG. **11**), **382** (FIG. **12**), **402** (FIG. **13**), and **428** (FIG. **14**). In some examples, the optical/electrical communication interface **644** can be similar to the integrated optical communication device **448**, **462**, **466**, **472** (FIG. **17**), except that the optical/electrical communication interface **644** is mounted on the same side of the circuit board **642** as the data processing chip **640**. The optical connector **648** can be similar to, e.g., the first optical connector part **213** (FIGS. **2**, **4**), the first optical connector **356** (FIGS. **11**, **12**), the first optical connector **404** (FIGS. **13**, **14**), and the first optical connector part **456** (FIG. **17**). In some examples, a portion of the optical connector **648** can be part of the optical/electrical communication interface **644**. In some examples, the optical connector **648** can also include the second optical connector part **223** (FIGS. **2**, **4**), **458** (FIG. **17**) that is optically coupled to the optical fibers. FIG. **24** shows that the optical connector **648** passes through the circuit board **642**. In some examples, the optical connector **648** can be short so that the optical fibers pass through, or partly through, the circuit board **642**. In some examples, the optical connector is not attached vertically to a photonic integrated circuit that is part of the optical/electrical communication interface **644** but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc. Any such solution is conceptually included in the vertical optical coupling attachment schematically visualized in FIGS. **24-27**.

(314) FIG. **25** is a top view of an example data processing system **650** in which a data processing chip **670** is mounted near an optical/electrical communication interface **652** to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **670** and the optical/electrical communication interface **652**. In this example, the data processing chip **670** and the optical/electrical communication interface **652** are mounted on a circuit board **654** that is positioned near a front panel **656** of an enclosure **658** of the system **630**, thus allowing optical fibers to be easily coupled to the optical/electrical communication interface **652**. In some examples, the data processing chip **670** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **654**.

(315) The enclosure **658** has side panels **660** and **662**, a rear panel **664**, a top panel, and a bottom panel. In some examples, the circuit board **654** and the front panel **656** are perpendicular to the bottom panel. In some examples, the circuit board **654** and the front panel **656** are oriented at an angle in a range −60° to 60° relative to a vertical direction of the bottom panel. In some examples, the circuit board **654** is substantially parallel to the front panel **656**, e.g., the angle between the surface of the circuit board **654** and the surface of the front panel **656** can be in a range of −5° to 5°. In some

examples, the circuit board **654** is at an angle relative to the front panel **656**, in which the angle is in a range of −45° to 45°.

(316) The optical/electrical communication interface **652** is electrically coupled to the data processing chip **670** by electrical connectors or traces **666** on or in the circuit board **654**, similar to those of the system **630**. The signal path between the data processing chip **670** and the optical/electrical communication interface **652** can be unidirectional or bidirectional, similar to that of the system **630**.

(317) An optical connector **668** is provided to couple optical signals from the optical fibers to the optical/electrical communication interface **652**. In this example, the optical connector **668** passes through an opening in the front panel **656** and an opening in the circuit board **654**. The optical connector **668** can be securely fixed, or releasably connected, to the optical/electrical communication interface **652**, similar to that of the system **630**.

(318) The optical/electrical communication interface **652** can be similar to, e.g., the integrated communication device **210** (FIG. **2**), **252** (FIG. **4**), **374** (FIG. **11**), **382** (FIG. **12**), **402** (FIG. **13**), and **428** (FIG. **14**). In some examples, the optical/electrical communication interface **652** can be similar to the integrated optical communication device **448**, **462**, **466**, **472** (FIG. **17**), except that the optical/electrical communication interface **652** is mounted on the same side of the circuit board **654** as the data processing chip **640**. The optical connector **668** can be similar to, e.g., the first optical connector part **213** (FIGS. **2**, **4**), the first optical connector **356** (FIGS. **11**, **12**), the first optical connector **404** (FIGS. **13**, **14**), and the first optical connector part **456** (FIG. **17**). In some examples, the optical connector is not attached vertically to a photonic integrated circuit that is part of the optical/electrical communication interface **652** but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc. In some examples, a portion of the optical connector **668** can be part of the optical/electrical communication interface **652**. In some examples, the optical connector **668** can also include the second optical connector part **223** (FIGS. **2**, **4**), **458** (FIG. **17**) that is optically coupled to the optical fibers. FIG. **25** shows that the optical connector **668** passes through the front panel **656** and the circuit board **654**. In some examples, the optical connector **668** can be short so that the optical fibers pass through, or partly through, the front panel **656**. The optical fibers can also pass through, or partly through, the circuit board **654**.

(319) In the examples of FIGS. **24** and **25**, only one optical/electrical communication interface (**544**, **652**) is shown in the figures. It is understood that the systems **630**, **650** can include multiple optical/electrical communication interfaces that are mounted on the same circuit board as the data processing chip to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip and each of the optical/electrical communication interfaces.

(320) FIG. **26**A is a top view of an example data processing system **680** in which a data processing chip **682** is mounted near optical/electrical communication interfaces **684**A, **684**B, **684**C (collectively referenced as **684**) to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **682** and each of the optical/electrical communication interfaces **684**. The data processing chip **682** is mounted on a first side of a circuit board **686** that functions as a front panel of an enclosure **688** of the system **680**. In some examples, the data processing chip **682** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **686**. The optical/electrical communication interfaces **684** are mounted on a second side of the circuit board **686**, in which the second side faces the exterior of the enclosure **688**. In this example, the optical/electrical communication interfaces **684** are mounted on an exterior side of the enclosure **688**, allowing optical fibers to be easily coupled to the optical/electrical communication interfaces **684**.

(321) The enclosure **688** has side panels **690** and **692**, a rear panel **694**, a top panel, and a bottom panel. In some examples, the circuit board **686** is perpendicular to the bottom panel. In some examples, the circuit board **686** is oriented at an angle in a range −60° to 60° (or −30° to 30°, or −10° to 10°, or −1° to 1°) relative to a vertical direction of the bottom panel.

(322) Each of the optical/electrical communication interfaces **684** is electrically coupled to the data processing chip **682** by electrical connectors or traces **696** that pass through the circuit board **686** in the thickness direction. For example, the electrical connectors or traces **696** can be configured as vias of the circuit board **686**. The signal paths between the data processing chip **682** and each of the optical/electrical communication interfaces **684** can be unidirectional or bidirectional, similar to those of the systems **630** and **650**.

(323) For example, the system **680** can be configured such that signals are transmitted unidirectionally between the data processing chip **682** and one of the optical/electrical communication interfaces **684**, and bidirectionally between the data processing chip **682** and another one of the optical/electrical communication interfaces **684**. For example, the system **680** can be configured such that signals are transmitted unidirectionally from the optical/electrical communication interface **684**A to the data processing chip **682**, and unidirectionally from the data processing chip to the optical/electrical communication interface **684**B and/or optical/electrical communication interface **684**C.

(324) Optical connectors **698**A, **698**B, **698**C (collectively referenced as **698**) are provided to couple optical signals from the optical fibers to the optical/electrical communication interfaces **684**A, **684**B, **684**C, respectively. The optical connectors **698** can be securely fixed, or releasably connected, to the optical/electrical communication interfaces **684**, similar to those of the systems **630** and **650**.

(325) The optical/electrical communication interface **684** can be similar to, e.g., the integrated communication device **210** (FIG. **2**), **252** (FIG. **4**), **374** (FIG. **11**), **382** (FIG. **12**), **402** (FIG. **13**), **428** (FIG. **14**), and **512** (FIG. **32**), except that the optical/electrical communication interface **684** is mounted on the side of the circuit board **686** opposite to the side of the data processing chip **682**. In some examples, the optical/electrical communication interface **684** can be similar to the integrated optical communication device **448**, **462**, **466**, **472** (FIG. **17**). The optical connector **698** can be similar to, e.g., the first optical connector part **213** (FIGS. **2**, **4**), the first optical connector **356** (FIGS. **11**, **12**), the first optical connector **404** (FIGS. **13**, **14**), the first optical connector part **456** (FIG. **17**), and the first optical connector part **520** (FIG. **32**). In some examples, the optical connector is not attached vertically to a photonic integrated circuit that is part of the optical/electrical communication interface **684** but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc. In some examples, a portion of the optical connector **668** can be part of the optical/electrical communication interface **652**. In some examples, the optical connector **668** can also include the second optical connector part **223** (FIGS. **2**, **4**), **458** (FIG. **17**) that is optically coupled to the optical fibers.

(326) In some examples, the optical/electrical communication interfaces **684** are securely fixed (e.g., by soldering) to the circuit board **686**. In some examples, the optical/electrical communication interfaces **684** are removably connected to the circuit board **686**, e.g., by use of mechanical mechanisms such as one or more snap-on or screw-on mechanisms. An advantage of the system **680** is that in case of a malfunction at one of the optical/electrical communication interfaces **684**, the faulty optical/electrical communication interface **684** can be replaced without opening the enclosure **688**.

(327) FIG. **26**B is a top view of an example data processing system **690***b* in which a data processing chip **691***b* is mounted near optical/electrical communication interfaces **692***a*, **692***b*, **692***c* (collectively referenced as **692**) to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **691***b* and each of the optical/electrical communication interfaces **692**. The data processing chip **691***b* is mounted on a first side of a circuit board **693***b* that functions as a front panel of an enclosure **694***b* of the system **690***b*. In this example, the optical/electrical communication interface **692***a* is mounted on the first side of the circuit board **693***b* and the optical/electrical communication interfaces **692***b* and **692***c* are mounted on a second side of the circuit board **693***b*, in which the second side faces the exterior of the enclosure **694***b*. In this example, the optical/electrical communication interfaces **692***b* and **692***c* are mounted on an exterior side of the enclosure **694***b*, allowing connection to optical fiber from the front of the enclosure **694***b* while the optical/electrical communication interface **692***a* is located internal to the enclosure **694***b*, for example, to allow connection to optical fiber at the rear of the enclosure **694***b*. In some examples, two or more of the optical/electrical communication interfaces **692** can be located internal to the enclosure **694***b* and connect to optical fibers at the rear of the enclosure **694***b*.

(328) The enclosure **694***b* has side panels **695***b* and **696***b*, a rear panel **697***b*, a top panel, and a bottom panel. In some examples, the circuit board **693***b* is perpendicular to the bottom panel. In some examples, the circuit board **693***b* is oriented at an angle in a range −60° to 60° (or −30° to 30°, or −10° to 10°, or −1° to 1°) relative to a vertical direction of the bottom panel.

(329) Each of the optical/electrical communication interfaces **692** is electrically coupled to the data processing chip **691***b* by electrical connectors or traces **698***b* that pass through the circuit board **693***b* in the thickness direction. For example, the electrical connectors or traces **698***b* can be configured as vias of the circuit board **693***b*. In this example, the electrical connectors or traces **698***b* extend to both sides of the circuit board **693***b* (e.g., for connecting to optical/electrical communication interfaces **692** located internal to and external of the enclosure **694***b*). The signal paths between the data processing chip **691***b* and each of the optical/electrical communication interfaces **692** can be unidirectional or bidirectional, similar to those of the systems **630**, **650** and **680**.

(330) For example, the system **690***b* can be configured such that signals are transmitted unidirectionally between the data processing chip **691***b* and one of the optical/electrical communication interfaces **692**, and bidirectionally between the data processing chip **691***b* and another one of the optical/electrical communication interfaces **692**. For example, the system **690***b* can be configured such that signals are transmitted unidirectionally from the optical/electrical communication interface **692***a* to the data processing chip **691***b*, and unidirectionally from the data processing chip **691***b* to the optical/electrical communication interface **692***b* and/or optical/electrical communication interface **692***c*.

(331) Optical connectors **699***a*, **699***b*, **699***c* (collectively referenced as **699**) are provided to couple optical signals from the optical fibers to the optical/electrical communication interfaces **692***a*, **692***b*, **692***c*, respectively. The optical connectors **699** can be securely fixed, or releasably connected, to the optical/electrical communication interfaces **692**, similar to those of the systems **630**, **650**, and **680**. In this example, optical connector **699***b* and optical connector **699***c* can connect to optical fibers at the front of the enclosure **694***b* and the optical connector **699***a* can connect to optical fibers at the rear of the enclosure **694***b*. In the illustrated example, the optical connector **699***a* connects to an optical fiber at the rear of the enclosure **694***b* by being connected to a fiber **1000***b* that connects to a rear panel interface **1001***b* (e.g., a backplane, etc.) that is mounted to the rear panel **697***b*. In some examples, the optical connectors **699** can be securely or fixedly attached to communication interfaces **692**. In some examples, the optical connectors **699** can be securely or fixedly attached to an array of optical fibers.

(332) The optical/electrical communication interface **692** can be similar to, e.g., the integrated communication device **210**

(FIG. **2**), **252** (FIG. **4**), **374** (FIG. **11**), **382** (FIG. **12**), **402** (FIG. **13**), **428** (FIG. **14**), and **512** (FIG. **32**), except that the optical/electrical communication interfaces **692***b* and **692***c* are mounted on the side of the circuit board **693***b* opposite to the side of the data processing chip **691***b*. In some examples, the optical/electrical communication interface **692** can be similar to the integrated optical communication device **448**, **462**, **466**, **472** (FIG. **17**). The optical connector **699** can be similar to, e.g., the first optical connector part **213** (FIGS. **2**, **4**), the first optical connector **356** (FIGS. **11**, **12**), the first optical connector **404** (FIGS. **13**, **14**), the first optical connector part **456** (FIG. **17**), and the first optical connector part **520** (FIG. **32**). In some examples, the optical connector is not attached vertically to a photonic integrated circuit that is part of the optical/electrical communication interface **692** but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc. In some examples, a portion of the optical connector **699** can be part of the optical/electrical communication interface **692**. In some examples, the optical connector **699** can also include the second optical connector part **223** (FIGS. **2**, **4**), **458** (FIG. **17**) that is optically coupled to the optical fibers.

(333) In some examples, the optical/electrical communication interfaces **692** are securely fixed (e.g., by soldering) to the circuit board **693***b*. In some examples, the optical/electrical communication interfaces **692** are removably connected to the circuit board **693***b*, e.g., by use of mechanical mechanisms such as one or more snap-on or screw-on mechanisms. An advantage of the system **690***b* is that in case of a malfunction at one of the optical/electrical communication interfaces **692**, the faulty optical/electrical communication interface **692** can be replaced without opening the enclosure **694***b*.

(334) FIG. **26**C is a top view of an example data processing system **690***c* in which a data processing chip **691***c* is mounted near optical/electrical communication interfaces **692***d*, **692***e*, **692***f* (collectively referenced as **692**) to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **691***c* and each of the optical/electrical communication interfaces **692**. The data processing chip **691***c* is mounted on a first side of a circuit board **693***c* that functions as a front panel of an enclosure **694***c* of the system **690***c*. In this example, the optical/electrical communication interface **692***d* is mounted on the first side of the circuit board **693***c* and the optical/electrical communication interfaces **692***e* and **692***f* are mounted on a second side of the circuit board **693***c*, in which the second side faces the exterior of the enclosure **694***c*. In this example, the optical/electrical communication interfaces **692***e* and **692***f* are mounted on an exterior side of the enclosure **694***c*, allowing connection to optical fibers from the front of the enclosure **694***c* while the optical/electrical communication interface **692***d* is located internal to the enclosure **694***c*, for example, to allow connection to optical fiber at the rear of the enclosure **694***c*. In some examples, two or more of the optical/electrical communication interfaces **692** can be located internal to the enclosure **694***c* and connect to optical fibers at the rear of the enclosure **694***c*.

(335) The enclosure **694***c* has side panels **695***c* and **696***c*, a rear panel **697***c*, a top panel, and a bottom panel. In some examples, the circuit board **693***c* is perpendicular to the bottom panel. In some examples, the circuit board **693***c* is oriented at an angle in a range −60° to 60° (or −30° to 30°, or −10° to 10°, or −1° to 1°) relative to a vertical direction of the bottom panel.

(336) Each of the optical/electrical communication interfaces **692** is electrically coupled to the data processing chip **691***c* by electrical connectors or traces **698***c* that pass through the circuit board **693***c* in the thickness direction. For example, the electrical connectors or traces **698***c* can be configured as vias of the circuit board **693***c*. In this example, the electrical connectors or traces **698***c* extend to both sides of the circuit board **693***b* (e.g., for connecting to optical/electrical communication interfaces **692** located internal to and external of the enclosure **694***b*. The signal paths between the data processing chip **691***c* and each of the optical/electrical communication interfaces **692** can be unidirectional or bidirectional, similar to those of the systems **630**, **650** and **680**.

(337) For example, the system **690***c* can be configured such that signals are transmitted unidirectionally between the data processing chip **691***c* and one of the optical/electrical communication interfaces **692**, and bidirectionally between the data processing chip **691***c* and another one of the optical/electrical communication interfaces **692**. For example, the system **690***c* can be configured such that signals are transmitted unidirectionally from the optical/electrical communication interface **692***d* to the data processing chip **691***c*, and unidirectionally from the data processing chip **691***c* to the optical/electrical communication interface **692***e* and/or optical/electrical communication interface **692***f*.

(338) Optical connectors **699***d*, **699***e*, **699***f* (collectively referenced as **699**) are provided to couple optical signals from the optical fibers to the optical/electrical communication interfaces **692***d*, **692***e*, **692***f*, respectively. The optical connectors **699** can be securely fixed, or releasably connected, to the optical/electrical communication interfaces **692**, similar to those of the systems **630**, **650**, and **680**. In the illustrated example, the optical/electrical communication interfaces **692***d* and optical connector **699***d* are oriented differently compared to the optical/electrical communication interfaces **692***a* and optical connector **699***a* of FIG. **26**B. Here the orientation change is a counter clockwise rotation of 90 degrees. Other types of orientation changes (e.g., rotations, pitches, tipping, etc.) may be implemented. Position changes (e.g., translations) and other types of location changes may also be employed. In this example, optical connector **699***e* and optical connector **699***f* can connect to optical fibers at the front of the enclosure **694***c* and the optical connector **699***d* can connect to optical fibers the rear of the enclosure **694***c*. In the illustrated example, the optical connector **699***d* connects to an optical fiber at the rear of the enclosure **694***c* by being connected to a fiber **1000***c* that connects to a rear panel interface **1001***c* (e.g., a backplane, etc.) that is mounted to the rear panel **697***c*.

(339) The optical/electrical communication interface **692** can be similar to, e.g., the integrated communication device **210** (FIG. **2**), **252** (FIG. **4**), **374** (FIG. **11**), **382** (FIG. **12**), **402** (FIG. **13**), **428** (FIG. **14**), and **512** (FIG. **32**), except that the optical/electrical communication interface **692***e* and **692***f* are mounted on the side of the circuit board **693***c* opposite to the side of the data processing chip **691***c*. In some examples, the optical/electrical communication interface **692** can be similar to the integrated optical communication device **448**, **462**, **466**, **472** (FIG. **17**). The optical connector **699** can be similar to, e.g., the first optical connector part **213** (FIGS. **2**, **4**), the first optical connector **356** (FIGS. **11**, **12**), the first optical connector **404** (FIGS. **13**, **14**), the first optical connector part **456** (FIG. **17**), and the first optical connector part **520** (FIG. **32**). In some examples, the optical connector is not attached vertically to a photonic integrated circuit that is part of the optical/electrical communication interface **692** but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc. In some examples, a portion of the optical connector **699** can be part of the optical/electrical communication interface **692**. In some examples, the optical connector **699** can also include the second optical connector part **223** (FIGS. **2**, **4**), **458** (FIG. **17**) that is optically coupled to the optical fibers.

(340) In some examples, the optical/electrical communication interfaces **692** are securely fixed (e.g., by soldering) to the circuit board **693***c*. In some examples, the optical/electrical communication interfaces **692** are removably connected to the circuit board **693***c*, e.g., by use of mechanical mechanisms such as one or more snap-on or screw-on mechanisms. An advantage of the system **690***c* is that in case of a malfunction at one of the optical/electrical communication interfaces **692**, the faulty optical/electrical communication interface **692** can be replaced without opening the enclosure **694***c*.

(341) FIG. **27** is a top view of an example data processing system **700** in which a data processing chip **702** is mounted near optical/electrical communication interfaces **704***a*, **704***b*, **704***c* (collectively referenced as **704**) to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **702** and each of the optical/electrical communication interfaces **704**. The data processing chip **702** is mounted on a first side of a circuit board **706** that is positioned near a front panel of an enclosure **710** of the system **700**, similar to the configuration of the system **650** (FIG. **25**). In some examples, the data processing chip **702** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **706**. The optical/electrical communication interfaces **704** are mounted on a second side of the circuit board **708**. In this example, the optical/electrical communication interfaces **704** pass through openings in the front panel **708**, allowing optical fibers to be easily coupled to the optical/electrical communication interfaces **704**.

(342) The enclosure **710** has side panels **712** and **714**, a rear panel **716**, a top panel, and a bottom panel. In some examples, the circuit board **706** and the front panel **708** are oriented at an angle in a range −60° to 60° relative to a vertical direction of the bottom panel. In some examples, the circuit board **706** is substantially parallel to the front panel **708**, e.g., the angle between the surface of the circuit board **706** and the surface of the front panel **708** can be in a range of −5° to 5°. In some examples, the circuit board **706** is at an angle relative to the front panel **708**, in which the angle is in a range of −45° to 45°.

(343) For example, the angle can refer to a rotation around an axis that is parallel to the larger dimension of the front panel (e.g., the width dimension in a typical 1 U, 2 U, or 4 U rackmount device), or a rotation around an axis that is parallel to the shorter dimension of the front panel (e.g., the height dimension in the 1 U, 2 U, or 4 U rackmount device). The angle can also refer to a rotation around an axis along any other direction. For example, the circuit board **706** is positioned relative to the front panel such that components such as the interconnection modules, including optical modules or photonic integrated circuits, mounted on or attached to the circuit board **706** can be accessed through the front side, either through one or more openings in the front panel, or by opening the front panel to expose the components, without the need to separate the top or side panels from the bottom panel. Such orientation of the circuit board (or a substrate on which a data processing module is mounted) relative to the front panel also applies to the examples shown in FIGS. **21** to **26**, **28**B to **29**B, **69**A, **70**, **71**A, **72**, **73**A, **74**A, **75**A, **75**C, **76**, **77**A, **77**B, **78**, **96** to **98**, **100**, **110**, **112**, **113**, **115**, **117** to **122**, **125**A to **127**, and **129** to **131**.

(344) Each of the optical/electrical communication interfaces **704** is electrically coupled to the data processing chip **702** by electrical connectors or traces **718** that pass through the circuit board **706** in the thickness direction, similar to those of the system **680** (FIG. **26**). The signal paths between the data processing chip **702** and each of the optical/electrical communication interfaces **704** can be unidirectional or bidirectional, similar to those of the system **630** (FIG. **24**), **650** (FIG. **25**), and **680** (FIG. **26**).

(345) Optical connectors **716***a*, **716***b*, **716***c* (collectively referenced as **716**) are provided to couple optical signals from the optical fibers to the optical/electrical communication interfaces **704***a*, **704***b*, **704***c*, respectively. The optical connectors **716** can be securely fixed, or releasably connected, to the optical/electrical communication interfaces **704**, similar to those of the systems **630**, **650**, and **680**.

(346) The optical/electrical communication interface **704** can be similar to, e.g., the integrated communication device **210** (FIG. **2**), **252** (FIG. **4**), **374** (FIG. **11**), **382** (FIG. **12**), **402** (FIG. **13**), **428** (FIG. **14**), and **512** (FIG. **32**), except that the optical/electrical communication interface **704** is mounted on the side of the circuit board **706** opposite to the side of the data processing chip **702.** In some examples, the optical/electrical communication interface **704** can be similar to the integrated optical communication device **448**, **462**, **466**, **472** (FIG. **17**). The optical connector **716** can be similar to, e.g.,

the first optical connector part **213** (FIGS. **2**, **4**), the first optical connector **356** (FIGS. **11**, **12**), the first optical connector **404** (FIGS. **13**, **14**), the first optical connector part **456** (FIG. **17**), and the first optical connector part **520** (FIG. **32**). In some examples, the optical connector is not attached vertically to a photonic integrated circuit that is part of the optical/electrical communication interface **704** but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc. In some examples, a portion of the optical connector **716** can be part of the optical/electrical communication interface **704**. In some examples, the optical connector **716** can also include the second optical connector part **223** (FIGS. **2**, **4**), **458** (FIG. **17**) that is optically coupled to the optical fibers.

(347) In some examples, the optical/electrical communication interfaces **704** are securely fixed (e.g., by soldering) to the circuit board **706**. In some examples, the optical/electrical communication interfaces **704** are removably connected to the circuit board **706**, e.g., by use of mechanical mechanisms such as one or more snap-on or screw-on mechanisms. An advantage of the system **700** is that in case of a malfunction at one of the optical/electrical communication interfaces **704**, the faulty optical/electrical communication interface **704** can unplugged or decoupled from the circuit board **706** and replaced without opening the enclosure **710**.

(348) In some implementations, the optical/electrical communication interfaces **704** do not protrude through openings in the front panel **708**. For example, each optical/electrical communication interface **704** can be at a distance behind the front panel **708**, and a fiber patchcord or pigtail can connect the optical/electrical communication interface **704** to an optical connector on the front panel **708**, similar to the examples shown in FIGS. **77**A, **77**B, **78**, **125**A, **125**B, **129**, and **130**. In some examples, the front panel **708** is configured to be removable or to be able to open to allow servicing of communication interface **704**, similar to the examples shown in FIGS. **77**A, **125**A, and **130**.

(349) FIG. **28**A is a top view of an example data processing system **720** in which a data processing chip **722** is mounted near an optical/electrical communication interface **724** to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **720** and the optical/electrical communication interface **724**. The data processing chip **722** is mounted on a first side of a circuit board **730** that functions as a front panel of an enclosure **732** of the system **720**. In some examples, the data processing chip **722** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **730**. The optical/electrical communication interface **724** is mounted on a second side of the circuit board **730**, in which the second side faces the exterior of the enclosure **732**. In this example, the optical/electrical communication interface **724** is mounted on an exterior side of the enclosure **732**, allowing optical fibers **734** to be easily coupled to the optical/electrical communication interface **724**.

(350) The enclosure **732** has side panels **736** and **738**, a rear panel **740**, a top panel, and a bottom panel. In some examples, the circuit board **730** is perpendicular to the bottom panel. In some examples, the circuit board **730** is oriented at an angle in a range −60° to 60° relative to a vertical direction of the bottom panel.

(351) The optical/electrical communication interface **724** includes a photonic integrated circuit **726** mounted on a substrate **728** that is electrically coupled to the circuit board **730**. The optical/electrical communication interface **724** is electrically coupled to the data processing chip **722** by electrical connectors or traces **742** that pass through the circuit board **730** in the thickness direction. For example, the electrical connectors or traces **742** can be configured as vias of the circuit board **730**. The signal paths between the data processing chip **722** and the optical/electrical communication interface **724** can be unidirectional or bidirectional, similar to those of the systems **630**, **650**, **680**, and **700**.

(352) An optical connector **744** is provided to couple optical signals from the optical fibers **734** to the optical/electrical communication interface **724**. The optical connector **744** can be securely fixed, or removably connected, to the optical/electrical communication interface **744**, similar to those of the systems **630**, **650**, **680**, and **700**.

(353) In some implementations, the optical/electrical communication interface **724** can be similar to, e.g., the integrated communication device **448**, **462**, **466**, and **472** of FIG. **17**. The optical signals from the optical fibers are processed by the photonic integrated circuit **726**, which generates serial electrical signals based on the optical signals. For example, the serial electrical signals are amplified by a set of transimpedance amplifiers and drivers (which can be part of the photonic integrated circuit **726** or a serializers/deserializers module in the data processing chip **722**), which drives the output signals that are transmitted to the serializers/deserializers module embedded in the data processing chip **722**.

(354) The optical connector **744** includes a first optical connector **746** and a second optical connector **748**, in which the second optical connector **748** is optically coupled to the optical fibers **734**. The first optical connector **746** can be similar to, e.g., the first optical connector part **213** (FIGS. **2**, **4**), the first optical connector **356** (FIGS. **11**, **12**), the first optical connector **404** (FIGS. **13**, **14**), the first optical connector part **456** (FIG. **17**), and the first optical connector part **520** (FIG. **32**). The second optical connector **748** can be similar to the second optical connector part **223** (FIGS. **2**, **4**) and **458** (FIG. **17**). In some examples, the optical connectors **746** and **748** can form a single piece such that the optical/electrical communication interface **724** is securely or fixedly attached to a fiber bundle. In some examples, the optical connector is not attached vertically to the photonic integrated circuit **726** but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc.

(355) In some examples, the optical/electrical communication interface **724** is securely fixed (e.g., by soldering) to the circuit board **730**. In some examples, the optical/electrical communication interface **724** is removably connected to the circuit board **730**, e.g., by use of mechanical mechanisms such as one or more snap-on or screw-on mechanisms. An advantage of the system **720** is that in case of a malfunction of the optical/electrical communication interface **724**, the faulty optical/electrical communication interface **724** can be replaced without opening the enclosure **732**.

(356) FIG. **28**B is a top view of an example data processing system **2800** that is similar to the system **720** of FIG. **28**A, except that the circuit board **730** that is recessed from a front panel **2802** of an enclosure **732** of the system **2800**. The photonic integrated circuit **726** is optically coupled through a fiber patchcord or pigtail **2804** to a first optical connector **2806** attached to the inner side of the front panel **2802**. The first optical connector **2806** is optically coupled to a second optical connector **2808** attached to the outer side of the front panel **2802**. The second optical connector **2808** is optically coupled to the exterior optical fibers **734**.

(357) The technique of using a fiber patchcord or pigtail to optically couple the photonic integrated circuit to the optical connector attached to the inner side of the front panel can also be applied to the data processing system **700** of FIG. **27**. For example, the modified system can have a recessed substrate or circuit board, multiple co-packaged optical modules (e.g., **704**) mounted on the opposite side of the data processing chip **702** relative to the substrate or circuit board, and fiber jumpers (e.g., **2804**) optically coupling the co-packaged optical modules to the front panel.

(358) In the examples of FIGS. **28**A and **28**B, the data processing chip **722** can be mounted on a substrate that is electrically coupled to the circuit board **730**.

(359) In each of the examples in FIGS. **24**, **25**, **26**, **27**, and **28**, the optical/electrical communication interface **644**, **652**, **684**, **704**, and **724** can be electrically coupled to the circuit board **642**, **654**, **686**, **706**, and **730**, respectively, using electrical contacts that include one or more of spring-loaded elements, compression interposers, and/or land-grid arrays.

(360) FIG. **29**A is a diagram of an example data processing system **750** that includes a vertically mounted circuit board **752** that enables high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between data processing chips **758** and optical/electrical communication interfaces **760**. The data processing chips **758** and the optical/electrical communication interfaces **760** are mounted on the circuit board **752**, in which each data processing chip **758** is electrically coupled to a corresponding optical/electrical communication interface **760**. The data processing chips **758** are electrically coupled to one another by electrical connectors (e.g., electrical signal lines on one or more layers of the circuit board **752**).

(361) The data processing chips **758** can be similar to, e.g., the electronic processor integrated circuit, data processing chip, or host application specific integrated circuit **240** (FIGS. **2**, **4**, **6**, **7**, **11**, **12**), digital application specific integrated circuit **444** (FIG. **17**), data processor **502** (FIG. **20**), data processing chip **572** (FIGS. **22**, **23**), **640** (FIG. **24**), **670** (FIG. **25**), **682** (FIG. **26**A), **702** (FIG. **27**), and **722** (FIG. **28**). Each of the data processing chips **758** can be, e.g., a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC).

(362) Although the figure shows that the optical/electrical communication interfaces **760** are mounted on the side of the circuit board **752** facing the front panel **754**, the optical/electrical communication interfaces **760** can also be mounted on the side of the circuit board **752** facing the interior of the enclosure **756**. The optical/electrical communication interfaces **760** can be similar to, e.g., the integrated communication devices **210** (FIGS. **2**, **3**, **10**), **252** (FIGS. **4**, **5**), **262** (FIG. **6**), the integrated optical communication devices **282** (FIGS. **7-9**), **374** (FIG. **11**), **382** (FIG. **12**), **390** (FIG. **13**), **428** (FIG. **14**), **402** (FIGS. **15**, **16**), **448**, **462**, **466**, **472** (FIG. **17**), the integrated communication devices **574** (FIG. **22**), **612** (FIG. **23**), and the optical/electrical communication interfaces **644** (FIG. **24**), **652** (FIG. **25**), **684** (FIG. **26**), **704** (FIG. **27**).

(363) The circuit board **752** is positioned near a front panel **754** of an enclosure **756**, and optical signals are coupled to the optical/electrical communication interfaces **760** through optical paths that pass through openings in the front panel **754**. This allows users to conveniently removably connect optical fiber cables **762** to the input/output interfaces **760**. The position and orientation of the circuit board **752** relative to the enclosure **756** can be similar to, e.g., those of the circuit board **654** (FIG. **25**) and **706** (FIG. **27**).

(364) In some implementations, the data processing system **750** can include multiple types of optical/electrical communication interfaces **760**. For example, some of the optical/electrical communication interfaces **760** can be mounted on the same side of the circuit board **752** as the corresponding data processing chip **758**, and some of the optical/electrical communication interfaces **760** can be mounted on the opposite side of the circuit board **752** as the corresponding data processing chip **758**. Some of the optical/electrical communication interfaces **760** can include first and second serializers/deserializers modules, and the corresponding data processing chips **758** can include third serializers/deserializers modules, similar to the examples in FIGS. **2-8**, **11-14**, **20**, **22**, and **23**. Some of the optical/electrical communication interfaces **760** can include no serializers/deserializers module, and the corresponding data processing chips **758** can include serializers/deserializers modules, similar to the example of FIG. **17**. Some of the optical/electrical communication interfaces **760** can include sets of transimpedance amplifiers and drivers, either embedded in the photonic integrated circuits or in separate chips external to the photonic integrated circuits. Some of the optical/electrical communication interfaces **760** do not include transimpedance amplifiers and drivers, in which sets of transimpedance amplifiers and drivers are included in the corresponding data processing chips **758**. The data processing system **750** can also include electrical communication interfaces that interface to electrical cables, such as high speed

PCIe cables, Ethernet cables, or Thunderbolt™ cables. The electrical communication interfaces can include modules that perform various functions, such as translation of communication protocols and/or conditioning of signals.
(365) Other types of connections may be present and associated with circuit board **752** and other boards included in the enclosure **756**. For example, two or more circuit boards (e.g., vertically mounted circuit boards) can be connected which may or may not include the circuit board **752**. For instances in which circuit board **752** is connected to at least one other circuit board (e.g., vertically mounted in the enclosure **756**), one or more connection techniques can be employed. For example, an optical/electrical communication interface (e.g., similar to optical/electrical communication interfaces **760**) can be used to connect data processing chips **758** to other circuit boards. Interfaces for such connections can be located on the same side of the circuit board **752** that the processing chips **758** are mounted. In some implementations, interfaces can be located on another portion of the circuit board (e.g., a side that is opposite from the side that the processing chips **758** are mounted). Connections can utilize other portions of the circuit board **752** and/or one or more other circuit boards present in the enclosure **756**. For example an interface can be located on an edge of one or more of the boards (e.g., an upper edge of a vertically mounted circuit board) and the interface can connect with one or more other interfaces (e.g., the optical/electrical communication interfaces **760**, another edge mounted interface, etc.). Through such connections, two or more circuit boards can connect, receive and send signals, etc.
(366) In the example shown in FIG. **29**A, the circuit board **752** is placed near the front panel **754**. In some examples, the circuit board **752** can also function as the front panel, similar to the examples in FIGS. **22-24**, **26**, and **28**.
(367) FIG. **29**B is a diagram of an example data processing system **2000** that illustrates some of the configurations described with respect to FIGS. **26**A to **26**C and FIG. **29**A along with other capabilities. The system **2000** includes a vertically mounted printed circuit board **2002** (or, e.g., a substrate) upon which is mounted a data processing chip **2004** (e.g., an ASIC), and a heat sink **2006** is thermally coupled to the data processing chip **2004**. Optical/electrical communication interfaces are mounted on both sides of the printed circuit board **2002**. In particular, optical/electrical communication interface **2008** is mounted on the same side of the printed circuit board **2002** as the data processing chip **2004**. In this example, optical/electrical communication interfaces **2010**, **2012**, and **2014** are mounted on an opposite side of the printed circuit board **2002**. To send and receive signals (e.g., with other optical/electrical communication interfaces), each of the optical/electrical communication interfaces **2010**, **2012**, and **2014** connects to optical fibers **2016**, **2018**, **2020**, respectively. Electrical connection sockets/connectors can also be mounted to one or more sides of the printed circuit board **2002** for sending and receiving electrical signals, for example. In this example, two electrical connection sockets/connectors **2022** and **2024** are mounted to the side of the printed circuit board **2002** that the data processing chip **2004** is mounted and two electrical connection sockets/connectors **2026** and **2028** are mounted to the opposite side of the printed circuit board **2002**. In this example, electrical connection sockets/connector **2028** is connected (or includes) a timing module **2030** that provides various functionality (e.g., regenerate data, retime data, maintain signal integrity, etc.). To send and receive electrical signals, each of the electrical connection sockets/connectors **2022-2028** are connected to electrical connection cables **2032**, **2034**, **2036**, **2038**, respectively. One or more types of connection cables can be implemented, for example, fly-over cables can be employed for connecting to one or more of the electrical connection sockets/connectors **2022-2028**.
(368) In this example, the system **2000** includes vertically mounted line cards **2040**, **2042**, **2044**. In this particular example, line card **2040** includes an electrical connection sockets/connector **2046** that is connected to electrical cable **2036**, and line card **2042** includes an electrical connection sockets/connector **2048** that is connected to electrical cable **2032**. Line card **2044** includes an electrical connection sockets/connector **2050**. Each of the line cards **2040**, **2042**, **2044** include pluggable optical modules **2052**, **2054**, **2056** that can implement various interface techniques (e.g., QSFP, QSFP-DD, XFP, SFP, CFP).
(369) In this particular example, the printed circuit board **2002** is approximate to a forward panel **2058** of the system **2000**; however, the printed circuit board **2002** can be positioned in other locations within the system **2000**. Multiple printed circuit boards can also be included in the system **2000**. For example, a second printed circuit board **2060** (e.g., a backplane) is included in the system **2000** and is located approximate to a back panel **2062**. By locating the printed circuit board **2060** towards the rear, signals (e.g., data signals) can be sent to and received from other systems (e.g., another switch box) located, for example, in the same switch rack or other location as the system **2000**. In this example, a data processing chip **2064** is mounted to the printed circuit board **2060** that can perform various operations (e.g., data processing, prepare data for transmission, etc.). Similar to the printed circuit board **2002** located forward in the system **2000**, the printed circuit board **2060** includes an optical/electrical communication interface **2066** that communicates with the optical/electrical communication interface **2008** (located on the same side on printed circuit board **2002** as data processing chip **2004**) using optical fibers **2068**. The printed circuit board **2060** includes electrical connection sockets/connectors **2070** that uses the electrical connection cable **2034** to send electrical signals to and receive electrical signals from the electrical connection sockets/connectors **2024**. The printed circuit board **2060** can also communicate with other components of the system **2000**, for example, one or more of the line cards. As illustrated in the figure, electrical connection sockets/connectors **2072** located on the printed circuit board **2060** uses the electrical connection cable **2074** to send electrical signals to and/or receive electrical signals from the electrical connection sockets/connector **2050** of the line card **2044**. Similar to the printed circuit board **2002**, other portions of the system **2000** can include timing modules. For example, the line cards **2040**, **2042**, and **2044** can include timing modules (respectively identified with symbol "*", "**", and "***"). Similarly, the second circuit board **2060** can include timing modules such as timing modules **2076** and **2078**

for regenerating data, re-timing data, maintaining signal integrity, etc.

(370) A feature of some of the systems described in this document is that the main data processing module(s) of a system, such as switch chip(s) in a switch server, and the communication interface modules that support the main data processing module(s), are configured to allow convenient access by users. In the examples shown in FIGS. **21** to **29**B, **69**A, **70**, **71**A, **72**, **72**A, **74**A, **75**A, **75**C, **76**, **77**A, **77**B, **78**, **96** to **98**, **100**, **110**, **112**, **113**, **115**, **117** to **122**, **125**A to **127**, **129**, **136** to **149**, **159**, and **160**, the main data processing module and the communication interface modules are positioned near the front panel, the rear panel, or both, and allow easy access by the user through the front/rear panel. However, it is also possible to position the main data processing module and the communication interface modules near one or more side panels, the top panel, the bottom panel, or two or more of the above, depending on how the system is placed in the environment. In a system that includes multiple racks of rackmount devices (see e.g., FIGS. **76** and **86**), the communication interfaces (e.g., co-packaged optical modules) in each rackmount device can be conveniently accessed without the need to remove the rackmount device from the rack and opening up the housing in order to expose the inner components.

(371) In some implementations, for a single rack of rackmount servers where there is open space at the front, rear, left, and right side of the rack, in each rackmount server, it is possible to place a first main data processing module and the communication interface modules supporting the first main data processing module near the front panel, place a second main data processing module and the communication interface modules supporting the second main data processing module near the left panel, place a third main data processing module and the communication interface modules supporting the third main data processing module near the right panel, and place a fourth main data processing module and the communication interface modules supporting the fourth main data processing module near the rear panel. The thermal solutions, including the placement of fans and heat dissipating devices, and the configuration of airflows around the main data processing modules and the communication interface modules, are adjusted accordingly.

(372) For example, if a data processing server is mounted to the ceiling of a room or a vehicle, the main data processing module and the communication interface modules can be positioned near the bottom panel for easy access. For example, if a data processing server is mounted beneath the floor panel of a room or a vehicle, the main data processing module and the communication interface modules can be positioned near the top panel for easy access. The housing of the data processing system does not have to be in a box shape. For example, the housing can have curved walls, be shaped like a globe, or have an arbitrary three-dimensional shape.

(373) FIG. **30** is a diagram of an example high bandwidth data processing system **800** that can be similar to, e.g., systems **200** (FIGS. **2**, **20**), **250** (FIG. **4**), **260** (FIG. **6**), **280** (FIG. **7**), **350** (FIG. **11**), **380** (FIG. **12**), **390** (FIG. **13**), **420** (FIG. **14**), **560** (FIG. **22**), **600** (FIG. **23**), **630** (FIG. **24**), and **650** (FIG. **25**) described above. A first optical signal **770** is transmitted from an optical fiber to a photonic integrated circuit **772**, which generates a first serial electrical signal **774** based on the first optical signal. The first serial electrical signal **774** is provided to a first serializers/deserializers module **776**, which converts the first serial electrical signal **774** to a third set of parallel signals **778**. The first serializers/deserializers module **776** conditions the serial electrical signal upon conversion into the parallel electrical signals, in which the signal conditioning can include, e.g., one or more of clock and data recovery, and signal equalization. The third set of parallel signals **778** is provided to a second serializers/deserializers module **780**, which generates a fifth serial electrical signal **782** based on the third set of parallel signals **778**. The fifth serial electrical signal **782** is provided to a third serializers/deserializers module **784**, which generates a seventh set of parallel signals **786** that is provided to a data processor **788**.

(374) In some implementations, the photonic integrated circuit **772**, the first serializers/deserializers module **776**, and the second serializers/deserializers module **780** can be mounted on a substrate of an integrated communication device, an optical/electrical communication interface, or an input/output interface module. The first serializers/deserializers module **776** and the second serializers/deserializers module **780** can be implemented in a single chip. In some implementations, the third serializers/deserializers module **784** can be embedded in the data processor **788**, or the third serializers/deserializers module **784** can be separate from the data processor **788**.

(375) The data processor **788** generates an eighth set of parallel signals **790** that is sent to the third serializers/deserializers module **784**, which generates a sixth serial electrical signal **792** based on the eighth set of parallel signals **790**. The sixth serial electrical signal **792** is provided to the second serializers/deserializers module **780**, which generates a fourth set of parallel signals **794** based on the sixth serial electrical signal **792**. The second serializers/deserializers module **780** can condition the serial electrical signal **792** upon conversion into the fourth set of parallel electrical signals **794**. The fourth set of parallel signals **794** is provided to the first serializers/deserializers module **780**, which generates a second serial electrical signal **796** based on the fourth set of parallel signals **794** that is sent to the photonic integrated circuit **772**. The photonic integrated circuit **772** generates a second optical signal **798** based on the second serial electrical signal **796**, and sends the second optical signal **798** to an optical fiber. The first and second optical signals **770**, **798** can travel on the same optical fiber or on different optical fibers.

(376) A feature of the system **800** is that the electrical signal paths traveled by the first, fifth, sixth, and second serial electrical signals **774**, **782**, **792**, **796** are short (e.g., less than 5 inches), to allow the first, fifth, sixth, and second serial electrical signals **782**, **792** to have a high data rate (e.g., up to 50 Gbps).

(377) FIG. **31** is a diagram of an example high bandwidth data processing system **810** that can be similar to, e.g., systems **680** (FIG. **26**), **700** (FIG. **27**), and **750** (FIG. **29**) described above. The system **810** includes a data processor **812** that receives and sends signals from and to multiple photonic integrated circuits. The system **810** includes a second photonic

integrated circuit **814**, a fourth serializers/deserializers module **816**, a fifth serializers/deserializers module **818**, and a sixth serializers/deserializers module **820**. The operations of the second photonic integrated circuit **814**, a fourth serializers/deserializers module **816**, a fifth serializers/deserializers module **818**, and a sixth serializers/deserializers module **820** can be similar to those of the first photonic integrated circuit **772**, the first serializers/deserializers module **776**, the second serializers/deserializers module **780**, and the third serializers/deserializers module **784**. The third serializers/deserializers module **784** and the sixth serializers/deserializers module **820** can be embedded in the data processor **812**, or be implemented in separate chips.
(378) In some examples, the data processor **812** processes first data carried in the first optical signal received at the first photonic integrated circuit **772**, and generates second data that is carried in the fourth optical signal output from the second photonic integrated circuit **814**.
(379) The examples in FIGS. **30** and **31** include three serializers/deserializers modules between the photonic integrated circuit and the data processor, it is understood that the same principles can be applied to systems that has only one serializers/deserializers module between the photonic integrated circuit and the data processor.
(380) In some implementations, signals are transmitted unidirectionally from the photonic integrated circuit **772** to the data processor **788** (FIG. **30**). In that case, the first serializers/deserializers module **776** can be replaced with a serial-to-parallel converter, the second serializers/deserializers module **780** can be replaced with a parallel-to-serial converter, and the third serializers/deserializers module **784** can be replaced with a serial-to-parallel converter. In some implementations, signals are transmitted unidirectionally from the data processor **812** (FIG. **31**) to the second photonic integrated circuit **814**. In that case, the sixth serializers/deserializers module **820** can be replaced with a parallel-to-serial converter, the fifth serializers/deserializers module **818** can be replaced with a serial-to-parallel converter, and the fourth serializers/deserializers module **816** can be replaced with a parallel-to-serial converter.
(381) It should be appreciated by those of ordinary skill in the art that the various embodiments described herein in the context of coupling light from one or more optical fibers, e.g., **226** (FIGS. **2** and **4**) or **272** (FIGS. **6** and **7**) to the photonic integrated circuit, e.g., **214** (FIGS. **2** and **4**), **264** (FIG. **6**), or **296** (FIG. **7**) will be equally operable to couple light from the photonic integrated circuit to one or more optical fibers. This reversibility of the coupling direction is a general feature of at least some embodiments described herein, including some of those using polarization diversity.
(382) The example optical systems disclosed herein should only be viewed as some of many possible embodiments that can be used to perform polarization demultiplexing and independent array pattern scaling, array geometry re-arrangement, spot size scaling, and angle-of-incidence adaptation using diffractive, refractive, reflective, and polarization-dependent optical elements, 3D waveguides and 3D printed optical components. Other implementations achieving the same set of functionalities are also covered by the spirit of this disclosure.
(383) For example, the optical fibers can be coupled to the edges of the photonic integrated circuits, e.g., using fiber edge couplers. The signal conditioning (e.g., clock and data recovery, signal equalization, or coding) can be performed on the serial signals, the parallel signals, or both. The signal conditioning can also be performed during the transition from serial to parallel signals.
(384) In some implementations, the data processing systems described above can be used in, e.g., data center switching systems, supercomputers, internet protocol (IP) routers, Ethernet switching systems, graphics processing work stations, and systems that apply artificial intelligence algorithms.
(385) In the examples described above in which the figures show a first serializers/deserializers module (e.g., **216**) placed adjacent to a second serializers/deserializers module (e.g., **217**), it is understood that a bus processing unit **218** can be positioned between the first and second serializers/deserializers modules and perform, e.g., switching, re-routing, and/or coding functions described above.
(386) In some implementations, the data processing systems described above includes multiple data generators that generate large amounts of data that are sent through optical fibers to the data processors for processing. For example, an autonomous driving vehicle (e.g., car, truck, train, boat, ship, submarine, helicopter, drone, airplane, space rover, or space ship) or a robot (e.g., an industrial robot, a helper robot, a medical surgery robot, a merchandise delivery robot, a teaching robot, a cleaning robot, a cooking robot, a construction robot, an entertainment robot) can include multiple high resolution cameras and other sensors (e.g., LIDARs (Light Detection and Ranging), radars) that generate video and other data that have a high data rate. The cameras and/or sensors can send the video data and/or sensor data to one or more data processing modules through optical fibers. The one or more data processing modules can apply artificial intelligence technology (e.g., using one or more neural networks) to recognize individual objects, collections of objects, scenes, individual sounds, collections of sounds, and/or situations in the environment of the vehicle and quickly determine appropriate actions for controlling the vehicle or robot.
(387) FIG. **34** is a flow diagram of an example process for processing high bandwidth data. A process **830** includes receiving **832** a plurality of channels of first optical signals from a plurality of optical fibers. The process **830** includes generating **834** a plurality of first serial electrical signals based on the received optical signals, in which each first serial electrical signal is generated based on one of the channels of first optical signals. The process **830** includes generating **836** a plurality of sets of first parallel electrical signals based on the plurality of first serial electrical signals, and conditioning the electrical signals, in which each set of first parallel electrical signals is generated based on a corresponding first serial electrical signal. The process **830** includes generating **838** a plurality of second serial electrical signals based on the plurality of sets of first parallel electrical signals, in which each second serial electrical signal is generated based on a

corresponding set of first parallel electrical signals.

(388) In some implementations, a data center includes multiple systems, in which each system incorporates the techniques disclosed in FIGS. **22** to **29** and the corresponding description. Each system includes a vertically mounted printed circuit board, e.g., **570** (FIG. **22**), **610** (FIG. **23**), **642** (FIG. **24**), **654** (FIG. **25**), **686** (FIG. **26**), **706** (FIG. **27**), **730** (FIG. **28**), **752** (FIG. **29**) that functions as the front panel of the housing or is substantially parallel to the front panel. At least one data processing chip and at least one integrated communication device or optical/electrical communication interface are mounted on the printed circuit board. The integrated communication device or optical/electrical communication interface can incorporate techniques disclosed in FIGS. **2-22** and **30-34** and the corresponding description. Each integrated communication device or optical/electrical communication interface includes a photonic integrated circuit that receives optical signals and generates electrical signals based on the optical signals. The optical signals are provided to the photonic integrated circuit through one or more optical paths (or spatial paths) that are provided by, e.g., cores of the fiber-optic cables, which can incorporate techniques described in U.S. patent application Ser. No. 16/822,103. A large number of parallel optical paths (or spatial paths) can be arranged in two-dimensional arrays using connector structures, which can incorporate techniques described in U.S. patent application Ser. No. 16/816,171.

(389) FIG. **35**A shows an optical communications system **1250** providing high-speed communications between a first chip **1252** and a second chip **1254** using co-packaged optical (CPO) interconnect modules **1258** similar to those shown in, e.g., FIGS. **2-5** and **17**. Each of the first and second chips **1252**, **1254** can be a high-capacity chip, e.g., a high bandwidth Ethernet switch chip. The first and second chips **1252**, **1254** communicate with each other through an optical fiber interconnection cable **1734** that includes a plurality of optical fibers. In some implementations, the optical fiber interconnection cable **1734** can include optical fiber cores that transmit data and control signals between the first and second chips **802**, **804**. As described in more detail below, optical fibers **1730** and **1732**, which in some examples can be partly bundled together with the interconnection cable **1734**, include one or more optical fiber cores that transmit optical power supply light from an optical power supply or photon supply to photonic integrated circuits that provide optoelectronic interfaces for the first and second chips **1252**, **1254**. The optical fiber interconnection cable **1734** can include single-core fibers or multi-core fibers. Similarly, the optical fibers **1730** and **1732** can include single-core fibers or multi-core fibers. Each single-core fiber includes a cladding and a core, typically made from glasses of different refractive indices such that the refractive index of the cladding is lower than the refractive index of the core to establish a dielectric optical waveguide. Each multi-core optical fiber includes a cladding and multiple cores, typically made from glasses of different refractive indices such that the refractive index of the cladding is lower than the refractive index of the core. More complex refractive index profiles, such as index trenches, multi-index profiles, or gradually changing refractive index profiles can also be used. More complex geometric structures such as non-circular cores or claddings, photonic crystal structures, photonic bandgap structures, or nested antiresonant nodeless hollow core structures can also be used.

(390) The example of FIG. **35**A illustrates a switch-to-switch use case. An external optical power supply or photon supply **1256** provides optical power supply signals, which can be, e.g., continuous-wave light, one or more trains of periodic optical pulses, or one or more trains of non-periodic optical pulses. The power supply light is provided from the photon supply **1256** to the co-packaged optical interconnect modules **1258** through optical fibers **1730** and **1732**, respectively. For example, the optical power supply **1256** can provide continuous wave light, or both pulsed light for data modulation and synchronization, as described in U.S. Pat. No. 11,153,670. This allows the first chip **1252** to be synchronized with the second chip **1254**.

(391) For example, the photon supply **1256** can correspond to the optical power supply **103** of FIG. **1**. The pulsed light from the photon supply **1256** can be provided to the link **102_6** of the data processing system **200** of FIG. **20**. In some implementations, the photon supply **1256** can provide a sequence of optical frame templates, in which each of the optical frame templates includes a respective frame header and a respective frame body, and the frame body includes a respective optical pulse train. The modulators **417** can load data into the respective frame bodies to convert the sequence of optical frame templates into a corresponding sequence of loaded optical frames that are output through optical fiber link **102_1**.

(392) The implementation shown in FIG. **35**A uses a packaging solution corresponding to FIG. **35**B, whereby in contrast to FIG. **17** substrates **454** and **460** are not used and the photonic integrated circuit **464** is directly attached to the serializers/deserializers module **446**. FIG. **35**C shows an implementation similar to FIG. **5**, in which the photonic integrated circuit **464** is directly attached to the serializers/deserializers **216**.

(393) FIG. **36** shows an example of an optical communications system **1260** providing high-speed communications between a high-capacity chip **1262** (e.g., an Ethernet switch chip) and multiple lower-capacity chips **1264***a*, **1264***b*, **1264***c*, e.g., multiple network interface chips, attached to computer servers using co-packaged optical interconnect modules **1258** similar to those shown in FIG. **35**A. The high-capacity chip **1262** communicates with the lower-capacity chips **1264***a*, **1264***b*, **1264***c* through a high-capacity optical fiber interconnection cable **1740** that later branches out into several lower-capacity optical fiber interconnection cables **1742***a*, **1742***b*, **1742***c* that are connected to the lower-capacity chips **1264***a*, **1264***b*, **1264***c*, respectively. This example illustrates a switch-to-servers use case.

(394) An external optical power supply or photon supply **1266** provides optical power supply signals, which can be continuous-wave light, one or more trains of periodic optical pulses, or one or more trains of non-periodic optical pulses. The power supply light is provided from the photon supply **1266** to the optical interconnect modules **1258** through optical fibers **1744**, **1746***a*, **1746***b*, **1746***c*, respectively. For example, the optical power supply **1266** can provide both pulsed light for data modulation and synchronization, as described in U.S. Pat. No. 11,153,670. This allows the high-capacity chip

1262 to be synchronized with the lower-capacity chips **1264***a*, **1264***b*, and **1264***c*.

(395) FIG. **37** shows an optical communications system **1270** providing high-speed communications between a high-capacity chip **1262** (e.g., an Ethernet switch chip) and multiple lower-capacity chips **1264***a*, **1264***b*, e.g., multiple network interface chips, attached to computer servers using a mix of co-packaged optical interconnect modules **1258** similar to those shown in FIG. **35** as well as conventional pluggable optical interconnect modules **1272**.

(396) An external optical power supply or photon supply **1274** provides optical power supply signals, which can be continuous-wave light, one or more trains of periodic optical pulses, or one or more trains of non-periodic optical pulses. For example, the optical power supply **1274** can provide both pulsed light for data modulation and synchronization, as described in U.S. Pat. No. 11,153,670. This allows the high-capacity chip **1262** to be synchronized with the lower-capacity chips **1264***a* and **1264***b*.

(397) Some aspects of the systems **1250**, **1260**, and **1270** are described in more detail in connection with FIGS. **79** to **84**B.

(398) FIG. **43** shows an exploded view of an example of a front-mounted module **860** of a data processing system that includes a vertically mounted printed circuit board **862**, (or substrate made of, e.g., organic or ceramic material, or a silicon substrate), a host application specific integrated circuit **864** mounted on the back-side of the circuit board **862**, and a heat sink **866**. In some examples, the host application specific integrated circuit **864** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **862**. The front-mounted module **860** can be, e.g., the front panel of the housing of the data processing system, similar to the configuration shown in FIGS. **26**A, **28**A or positioned near the front panel of the housing, similar to the configuration shown in FIGS. **27**, **28**B. Three optical modules with connectors, e.g., **868***a*, **868***b*, **868***c*, collectively referenced as **868**, are shown in the figure. Additional optical modules with connectors can be used. The data processing system can be similar to, e.g., the data processing system **680** (FIG. **26**A) or **700** (FIG. **27**). The printed circuit board **862** can be similar to, e.g., the printed circuit board **686** (FIG. **26**A) or **706** (FIG. **27**). The application specific integrated circuit **864** can be similar to, e.g., the application specific integrated circuit **682** (FIG. **26**A) or **702** (FIG. **27**). The heat sink **866** can be similar to, e.g., the heat sink **576** (FIG. **23**). The optical modules with connector **868** each include an optical module **880** (see FIGS. **44**, **45**) and a mechanical connector structure **900** (see FIGS. **46**, **47**). The optical module **880** can be similar to, e.g., the optical/electrical communication interfaces **682** (FIG. **26**A) or **704** (FIG. **27**), or the integrated optical communication device **512** of FIG. **32**.

(399) The optical module with connector **868** can be inserted into a first grid structure **870**, which can function as both (i) a heat spreader/heat sink and (ii) a mechanical holding fixture for the optical modules with connectors **868**. The first grid structure **870** includes an array of receptors, and each receptor can receive an optical module with connector **868**. When assembled, the first grid structure **870** is connected to the printed circuit board **862**. The first grid structure **870** can be firmly held in place relative to the printed circuit board **862** by sandwiching the printed circuit board **862** in between the first grid structure **870** and a second structure **872** (e.g., a second grid structure) located on the opposite side of the printed circuit board **862** and connected to the first grid structure **870** through the printed circuit board **862**, e.g., by use of screws. Thermal vias between the first grid structure **870** and the second structure **872** can conduct heat from the front-side of the printed circuit board **862** to the heat sink **866** on the back-side of the printed circuit board **862**. Additional heat sinks can also be mounted directly onto the first grid structure **870** to provide cooling in the front.

(400) The printed circuit board **862** includes electrical contacts **876** configured to electrically connect to the removable optical module with connectors **868** after the removable optical module with connectors **868** are inserted into the first grid structure **870**. The first grid structure **870** can include an opening **874** at the location in which the host application specific integrated circuit **864** is mounted on the other side of the printed circuit board **862** to allow for components such as voltage regulators, filters, and/or decoupling capacitors to be mounted on the printed circuit board **862** in immediate lateral vicinity to the host application specific integrated circuit **864**.

(401) In some examples, the host application specific integrated circuit **864** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **862**, similar to the examples shown in FIGS. **136** to **159**. The substrate can be similar to the substrate **13602** of FIGS. **136** to **159**, the second grid structure **872** can be similar to the rear lattice structure **13626**, the circuit board **862** can be similar to the printed circuit board **13604**, the host application specific integrated circuit **864** can be similar to the data processing chip **12382**, and the heat sink **866** can be similar to the heat dissipating device **13610**. The first grid structure **870** can have an overall shape similar to the front lattice structure **13606** of FIGS. **136** to **159**, except that the first grid structure **870** includes mechanisms for coupling to the removable optical module with connectors **868**.

(402) FIGS. **44** and **45** show an exploded view and an assembled view, respectively, of an example optical module **880**, which can be similar to the integrated optical communication device **512** of FIG. **32**. The optical module **880** includes an optical connector part **882** (which can be similar to the first optical connector **520** of FIG. **32**) that can either directly or through an (e.g., geometrically wider) upper connector part **884** receive light from fibers embedded in a second optical connector part (not shown in FIGS. **44**, **45**), which can be similar to, e.g., the optical connector part **268** of FIGS. **6** and **7**). In the example shown in FIGS. **44**, **45**, a matrix of fibers, e.g., 2×18 fibers, can be optically coupled to the optical connector part **882**. The matrix of fibers can have other configurations, such as a 3×12, 1×12, 3×12, 6×12, 12×12, 16×16, or 32×32 array of fibers. For example, the optical connector part **882** can have a configuration similar to the fiber coupling region **430** of FIG. **15** that is configured to couple 2×18 fibers, or any other number of fibers. The upper connector part **884** can also include alignment structures **886** (e.g., holes, grooves, posts) to receive corresponding mating structures of the second optical connector part.

(403) The optical module **880** can have any of various configurations, including an optical module containing silicon photonics integrated optics, indium phosphide integrated optics, one or more vertical-cavity surface-emitting lasers (VCSEL)s, one or more direct-detection optical receivers, or one or more coherent optical receivers. The optical module **880** can include any of the optical modules, co-packaged optical modules, integrated optical communication devices (e.g., **448**, **462**, **466**, or **472** of FIG. **17**, or **210** of FIG. **20**), integrated communication devices (e.g., **612** of FIG. **23**), or optical/electrical communication interfaces (e.g., **684** of FIG. **26**, **724** of FIG. **28**, or **760** of FIG. **29**) described in this specification and the documents incorporated by reference.
(404) The optical connector part **882** is inserted through an opening **888** of a substrate **890** and optically coupled to a photonic integrated circuit **896** mounted on the underside of the substrate **890**. The substrate **890** can be similar to the substrate **514** of FIG. **32**, and the photonic integrated circuit **896** can be similar to the photonic integrated circuit **524**. A first serializers/deserializers chip **892** and a second serializers/deserializers chip **894** are mounted on the substrate **890**, in which the chip **892** is positioned on one side of the optical connector part **882**, and the chip **894** is positioned on the other side of the optical connector part **882**. The first serializers/deserializers chip **892** can include circuitry similar to, e.g., the third serializers/deserializers module **398** and the fourth serializers/deserializers module **400** of FIG. **32**. The second serializers/deserializers chip **894** can include circuitry similar to, e.g., the first serializers/deserializers module **394** and the second serializers/deserializers module **396**. A second slab **898** (which can be similar to the second slab **518** of FIG. **32**) can be provided on the underside of the substrate **890** to provide a removable connection to a package substrate (e.g., **230**).
(405) FIGS. **46** and **47** show an exploded view and an assembled view, respectively, of a mechanical connector structure **900** built around the functional optical module **880** of FIGS. **44**, **45**. In this example embodiment, the mechanical connector structure **900** includes a lower mechanical part **902** and an upper mechanical part **904** that together receive the optical module **880**. Both lower and upper mechanical connector parts **902**, **904** can be made of a heat-conducting and rigid material, e.g., a metal.
(406) In some implementations, the upper mechanical part **904**, at its underside, is brought in thermal contact with the first serializers/deserializers chip **892** and the second serializers/deserializers chip **894**. The upper mechanical part **904** is also brought in thermal contact with the lower mechanical part **902**. The lower mechanical part **902** includes a removable latch mechanism, e.g., two wings **906** that can be elastically bent inwards (the movement of the wings **906** are represented by a double-arrow **908** in FIG. **47**), and each wing **906** includes a tongue **910** on an outer side.
(407) FIG. **48** is a diagram of a portion of the first grid structure **870** and the circuit board **862**. In some examples, a substrate (e.g., a ceramic substrate) can be used in place of the circuit board **862**. Grooves **920** are provided on the walls of the first grid structure **870**. As shown in the figure, the printed circuit board **862** (or substrate) has electrical contacts **876** that can be electrically coupled to electrical contacts on the second slab **898** of the optical module **880**. For example, the electrical contacts **876** can include an array of electrical contacts that has at least four rows and four columns of electrical contacts. For example, the array of electrical contacts can have ten or more rows or columns of electrical contacts. The electrical contacts **876** can be arranged in any two-dimensional pattern and do not necessarily have to be arranged in rows and columns. The circuit board **862** (or substrate) can also have three-dimensional features, such as on protruding elements or recessed elements, and the electrical contacts can be provided on the three-dimensional features. The optical module with connectors **868** can have three-dimensional features with electrical contacts that mate with the corresponding three-dimensional features with electrical contacts on the circuit board **862** (or substrate).
(408) Referring to FIG. **49**, when the lower mechanical part **902** is inserted into the first grid structure **870**, the tongues **910** (on the wings **906** of the lower mechanical part **902**) can snap into corresponding grooves **920** within the first grid structure **870** to mechanically hold the optical module **880** in place. The position of the tongues **910** on the wings **906** is selected such that when the mechanical connector structure **900** and the optical module **880** are inserted into the first grid structure **870**, the electrical connectors at the bottom of the second slab **898** are electrically coupled to the electrical contacts **876** on the printed circuit board **862** (or substrate). For example, the second slab **898** can include spring-loaded contacts that are mated with the contacts **876**.
(409) FIG. **50** shows the front-view of an assembled front module **860**. Three optical module with connectors (e.g., **868***a*, **868***b*, **868***c*) are inserted into the first grid structure **870**. In some embodiments, the optical modules **880** are arranged in a checkerboard pattern, whereby adjacent optical modules **880** and the corresponding mechanical connector structures **900** are rotated by 90 degrees such as to not allow any two wings to touch. This facilitates the removal of individual modules. In this example, the optical module with connector **868***a* is rotated 90 degrees relative to the optical module with connectors **868***b*, **868***c*.
(410) FIG. **51**A shows a first side view of the mechanical connector structure **900**. FIG. **51**B shows a cross-sectional view of the mechanical connector structure **900** along a plane **930** shown in FIG. **51**A. In some examples, the compression interposer (e.g., spring-loaded contacts) can be part of the receiving structure (e.g., mounted on the circuit board or substrate) as opposed to the removable module.
(411) FIG. **52**A shows a first side view of the mechanical connector structure **900** mounted within the first grid structure **870**. FIG. **52**B shows a cross-sectional view of the mechanical connector structure **900** mounted within the first grid structure **870** along a plane **940** shown in FIG. **52**A.
(412) FIG. **53** is a diagram of an assembly **958** that includes a fiber cable **956** that includes a plurality of optical fibers, an optical fiber connector **950**, the mechanical connector module **900**, and the first grid structure **870**. The optical fiber

connector **950** can be inserted into the mechanical connector module **900**, which can be further inserted into the first grid structure **870**. The printed circuit board **862** (or substrate) is attached to the first grid structure **870**, in which the electrical contacts **876** face electrical contacts **954** on the bottom side of the second slab **898** of the optical module **880**.

(413) FIG. **53** shows the individual components before they are connected. FIG. **54** is a diagram that shows the components after they are connected. The optical fiber connector **950** includes a lock mechanism **952** that disables the snap-in mechanism of the mechanical connector structure **900** so as to lock in place the mechanical connector structure **900** and the optical module **880**. In this example embodiment, the lock mechanism **952** includes studs on the optical fiber connector **950** that insert between the wings **906** and the upper mechanical part **904** of the mechanical connector module **900**, hence disabling the wings **906** from elastically bending inwards and consequentially locking the mechanical connector structure **900** and the optical module **880** in place. Further, the mechanical connector structure **900** includes a mechanism to hold the optical fiber connector **950** in place, such as a ball-detent mechanism as shown in the figure. When the optical fiber connector **950** is inserted into the mechanical connector structure **900**, spring-loaded balls **962** on the optical fiber connector **950** engage detents **964** in the wings **906** of the mechanical connector structure **900**. The springs push the balls **962** against the detents **964** and secure the optical fiber connector **950** in place.

(414) To remove the optical module **880** from the first grid structure **870**, the user can pull the optical fiber connector **950** and cause the balls **962** to disengage from the detents **964**. The user can then bend the wings **906** inwards so that the tongues **910** disengage from the grooves **920** on the walls of the first grid structure **870**.

(415) FIGS. **55**A and **55**B show perspective views of the mechanisms shown in FIGS. **53** and **54** before the optical fiber connector **950** is inserted into the mechanical connector structure **900**. As shown in FIG. **55**B, the lower side of the optical connector **950** includes alignment structures **960** that mate with the alignment structures **886** (FIG. **44**) on the upper connector part **884** of the optical module **880**. FIG. **55**B also shows the photonic integrated circuit **896** and the second slab **898** that includes electrical contacts (e.g., spring-loaded electrical contacts).

(416) FIG. **56** is a perspective view showing that the optical module **880** and the mechanical connector structure **900** are inserted into the first grid structure **870**, and the optical fiber connector **950** is separated from the mechanical connector structure **900**.

(417) FIG. **57** is a perspective view showing that the optical fiber connector **950** is mated with the mechanical connector structure **900**, locking the optical module **880** within the mechanical connector structure **900**.

(418) FIGS. **58**A to **58**D show an alternate embodiment in which an optical module with connector **970** includes a latch mechanism **972** that acts as a mechanical fastener that joins the optical module **880** to the printed circuit board **862** (or substrate) using the first grid structure **870** as a support. FIGS. **58**A and **58**B show various views of the optical module with connector **970** that includes the latch mechanism **972**. FIGS. **58**C and **58**D show various views of the optical module with connector **970** coupled to the printed circuit board **862** (or substrate) and the first grid structure **870**. For example, the user can easily attach or remove the optical module with connector **970** by pressing a lever **974** activating the latch mechanism **972**. The lever **974** is built in a way that it does not block the optical fibers (not shown in the figure) coming out of the optical module with connector **970**. Alternatively, an external tool can be used as a removable lever.

(419) FIG. **59** is a view of an optical module **1030** that includes an optical engine with a latch mechanism used to realize the compression and attachment of the optical engine to the printed circuit board. The module **1030** is similar to the example shown in FIG. **58**B but without the compression interposer. FIGS. **60**A and **60**B show an example latch mechanism that can be used for securing (with enough compression force) and removing the optical engine.

(420) FIGS. **60**A and **60**B show an example implementation of the lever **974** and the latch mechanism **972** in the optical module **1030**. FIG. **60**A shows an example in which the lever **974** is pushed down, causing the latch mechanism **972** to latch on to a support structure **976**, which can be part of the first grid structure **870**. FIG. **60**B shows an example in which the lever **974** is pulled up, causing the latch mechanism **972** to be released from the support structure **976**.

(421) FIG. **61** is a diagram of an example of a fiber cable connection design **980** that includes nested fiber optic cable and co-packaged optical module connections. In this design, a co-packaged optical module **982** is removably coupled to a co-packaged optical port **1000** formed in a support structure, such as the first grid structure **870**, and a fiber connector **983** is removably coupled to the co-packaged optical module **982**. The fiber connector **983** is coupled to a fiber cable **996** that includes a plurality of optical fibers. The fiber cable connection can be designed to be, e.g., MTP/MPO (Multi-fiber Termination Push-on/Multi-fiber Push On) compatible, or compatible to new standards as they emerge. Multi-fiber push on (MPO) connectors are commonly used to terminate multi-fiber ribbon connections in indoor environments and conforms to IEC-61754-7; EIA/TIA-604-5 (FOCIS 5) standards.

(422) In some implementations, the co-packaged optical module **982** includes a mechanical connector structure **984** and a smart optical assembly **986**. The smart optical assembly **986** includes, e.g., a photonic integrated circuit (e.g., **896** of FIG. **44**), and components for guiding light, power splitting, polarization management, optical filtering, and other light beam management before the photonic integrated circuit. The components can include, e.g., optical couplers, waveguides, polarization optics, filters, and/or lenses. Additional examples of the components that can be included in the co-packaged optical module **982** are described in U.S. published application US 2021/0286140. The mechanical connector structure **984** includes one or more fiber connector latches **988** and one or more co-packaged optical module latches **990**. The mechanical connector structure **984** can be inserted into the co-packaged optical port **1000** (e.g., formed in the first grid structure **870**), in which the co-packaged optical module latches **990** engage grooves **992** in the walls of the first grid structure **870**, thus securing the co-packaged optical module **982** to the co-packaged optical port **1000**, and causing the

electrical contacts of the smart optical assembly **986** to be electrically coupled to the electrical contacts **876** on the printed circuit board **862** (or substrate). When the fiber connector **983** is inserted into the mechanical connector structure **984**, the fiber connector latches **988** engage grooves **994** in the fiber connector **983**, thus securing the fiber connector **983** to the co-packaged optical module **982**, and causing the fiber cable **996** to be optically coupled to the smart optical assembly **986**, e.g., through optical paths in the fiber connector **983**.

(423) In some examples, the fiber connector **983** includes guide pins **998** that are inserted into holes in the smart optical assembly **986** to improve alignment of optical components (e.g., waveguides and/or lenses) in the fiber connector **983** to optical components (e.g., optical couplers and/or waveguides) in the smart optical assembly **986**. In some examples, the guide pins **998** can be chamfered shaped, or elliptical shaped that reduces wear.

(424) In some implementations, after the fiber connector **983** is installed in the co-packaged optical module **982**, the fiber connector **983** prevents the co-packaged optical module latches **990** from bending inwards, thus preventing the co-packaged optical module **982** from being inserted into, or released from, the co-packaged optical port **1000**. To couple the fiber cable **996** to the data processing system, the co-packaged optical module **982** is first inserted into the co-packaged optical port **1000** without the fiber connector **983**, then the fiber connector **983** is inserted into the mechanical connector structure **984**. To remove the fiber cable **996** from the data processing system, the fiber connector **983** can be removed from the mechanical connector structure **984** while the co-packaged optical module **982** is still coupled to the co-packaged optical port **1000**.

(425) In some implementations, the nested connection latches can be designed to allow the co-packaged optical module **982** to be inserted in, or removed from, the co-packaged optical port **1000** when a fiber cable is connected to the co-packaged optical module **982**.

(426) FIGS. **62** and **63** are diagrams showing cross-sectional views of an example of a fiber cable connection design **1010** that includes nested fiber optic cable and co-packaged optical module connections. FIG. **62** shows an example in which a fiber connector **1012** is removably coupled to a co-packaged optical module **1014**. FIG. **63** shows an example in which the fiber connector **1012** is separated from the co-packaged optical module **1014**.

(427) FIGS. **64** and **65** are diagrams showing additional cross-sectional views of the fiber cable connection design **1010**. The cross-sections are made along planes that vertically cut through the middle of the components shown in FIGS. **62** and **63**. FIG. **64** shows an example in which the fiber connector **1012** is removably coupled to the co-packaged optical module **1014**. FIG. **65** shows an example in which the fiber connector **1012** is separated from the co-packaged optical module **1014**.

(428) The following describes rack unit thermal architectures for rackmount systems (e.g., **560** of FIG. **22**, **600** of FIG. **23**, **630** of FIG. **24**, **680** of FIG. **26**, **720** of FIG. **28**, **750** of FIG. **29**, **860** of FIG. **43**) that include data processing chips (e.g., **572** of FIGS. **22**, **23**, **640** of FIG. **24**, **682** of FIG. **26**A, **722** of FIG. **28**, **758** of FIG. **29**, **864** of FIG. **43**) that are mounted on vertically oriented circuit boards that are substantially vertical to the bottom surfaces of the system housings or enclosures. In some implementations, the rack unit thermal architectures use air cooling to remove heat generated by the data processing chips. In these systems, the heat-generating data processing chips are positioned near the input/output interfaces, which can include, e.g., one or more of the integrated optical communication device **448**, **462**, **466**, or **472** of FIG. **17**, the integrated communication device **574** of FIG. **22** or **612** of FIG. **23**, the optical/electrical communication interface **644** of FIG. **24**, **684** of FIG. **26**, **724** of FIG. **28**, or **760** of FIG. **29**, or the optical module with connector **868** of FIG. **43**, that are positioned at or near the front panel to enable users to conveniently connect/disconnect optical transceivers to/from the rackmount systems. The rack unit thermal architectures described in this specification include mechanisms for increasing airflow across the surfaces of the data processing chips, or heat sinks thermally coupled to the data processing chips, taking into consideration that a substantial portion of the surface area on the front panel of the housing needs to be allocated to the input/output interfaces.

(429) The rackmount systems and rackmount devices described in this document can include, and are not limited to, e.g., rackmount computer servers, rackmount network switches, rackmount controllers, and rackmount signal processors.

(430) Referring to FIG. **67**, a data server **1140** suitable for installation in a standard server rack can include a housing **1042** that has a front panel **1034**, a rear panel **1036**, a bottom panel **1038**, a top panel, and side panels **1040**. For example, the housing **1042** can have a 2 rack unit (RU) form factor, having a width of about 482.6 mm (19 inches) and a height of 2 rack units. One rack unit is about 44.45 mm (approximately 1.75 inches). A printed circuit board **1042** is mounted on the bottom panel **1038**, and at least one data processing chip **1044** is electrically coupled to the printed circuit board **1042**. A microcontroller unit **1046** is provided to control various modules, such as power supplies **1048** and exhaust fans **1050**. In this example, the exhaust fans **1050** are mounted at the rear panel **1036**. For example, single mode optical connectors **1052** are provided at the front panel **1034** for connection to external optical cables. Optical interconnect cables **1036** transmit signals between the single mode optical connectors **1052** and the at least one data processing chip **1044**. The exhaust fans **1050** mounted at the rear panel **1036** cause the air to flow from the front side to the rear side of the housing **1042**. The directions of air flow are represented by arrows **1058**. Warm air inside the housing **1042** is vented out of the housing **1042** through the exhaust fans **1050** at the rear panel **1036**. In this example, the front panel **1034** does not include any fan in order to maximize the area used for the single mode optical connectors **1052**.

(431) For example, the data server **1300** can be a network switch server, and the at least one data processing chip **1044** can include at least one switch chip configured to process data having a total bandwidth of, e.g., about 51.2 Tbps. The at least one switch chip **1044** can be mounted on a substrate **1054** having dimensions of, e.g., about 100 mm×100 mm, and co-

packaged optical modules **1056** can be mounted near the edges of the substrate **1054**. The co-packaged optical modules **1056** convert input optical signals received from the optical interconnect cables **1036** to input electrical signals that are provided to the at least one switch chip **1044**, and converts output electrical signals from the at least one switch chip **1044** to output optical signals that are provided to the optical interconnect cables **1036**. When any of the co-packaged optical modules **1056** fails, the user needs to remove the network switch server **1030** from the server rack and open the housing **1042** in order to repair or replace the faulty co-packaged optical module **1056**.

(432) Referring to FIGS. **68**A and **68**B, in some implementations, a rackmount server **1060** includes a housing or case **1062** having a front panel **1064** (or face plate), a rear panel **1036**, a bottom panel **1038**, a top panel, and side panels **1040**. For example, the housing **1062** can have a form factor of 1 RU, 2 RU, 3 RU, or 4 RU, having a width of about 482.6 mm (19 inches) and a height of 1, 2, 3, or 4 rack units. A first printed circuit board **1066** is mounted on the bottom panel **1038**, and a microcontroller unit **1046** is electrically coupled to the first printed circuit board **1066** and configured to control various modules, such as power supplies **1048** and exhaust fans **1050**.

(433) In some implementations, the front panel **1064** includes a second printed circuit board **1068** that is oriented in a vertical direction, e.g., substantially perpendicular to the first circuit board **1066** and the bottom panel **1038**. In the following, the second printed circuit board **1068** is referred to as the vertical printed circuit board **1068**. The figures show that the second printed circuit board **1066** forms part of the front panel **1064**, but in some examples the second printed circuit board **1066** can also be attached to the front panel **1064**, in which the front panel **1064** includes openings to allow input/output connectors to pass through. The second printed circuit board **1066** includes a first side facing the front direction relative to the housing **1062** and a second side facing the rear direction relative to the housing **1062**. At least one data processing chip **1070** is electrically coupled to the second side of the vertical printed circuit board **1068**, and a heat dissipating device or heat sink **1072** is thermally coupled to the at least one data processing chip **1070**. In some examples, the at least one data processing chip **1070** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the printed circuit board **1068**. FIG. **68**C is a perspective view of an example of the heat dissipating device or heat sink **1072**. For example, the heat dissipating device **1072** can include a vapor chamber thermally coupled to heat sink fins. The exhaust fans **1050** mounted at the rear panel **1036** cause the air to flow from the front side to the rear side of the housing **1042**. The directions of air flow are represented by arrows **1078**. Warm air inside the housing **1042** is vented out of the housing **1042** through the exhaust fans **1050** at the rear panel **1036**.

(434) Co-packaged optical modules **1074** (also referred to as the optical/electrical communication interfaces) are attached to the first side (i.e., the side facing the front exterior of the housing **1062**) of the vertical printed circuit board **1068** for connection to external fiber cables **1076**. Each fiber cable **1076** can include an array of optical fibers. By placing the co-packaged optical modules **1074** on the exterior side of the front panel **1064**, the user can conveniently service (e.g., repair or replace) the co-packaged optical modules **1074** when needed. Each co-packaged optical module **1074** is configured to convert input optical signals received from the external fiber cable **1076** into input electrical signals that are transmitted to the at least one data processing chip **1070** through signal lines in or on the vertical printed circuit board **1068**. The co-packaged optical module **1074** also converts output electrical signals from the at least one data processing chip **1070** into output optical signals that are provided to the external fiber cables **1076**. Warm air inside the housing **1062** is vented out of the housing **1062** through the exhaust fans **1050** mounted at the rear panel **1036**.

(435) For example, the at least one data processing chip **1070** can include a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC). The rackmount server can be, and not limited to, e.g., a rackmount computer server, a rackmount switch, a rackmount controller, a rackmount signal processor, a rackmount storage server, a rackmount multi-purpose processing unit, a rackmount graphics processor, a rackmount tensor processor, a rackmount neural network processor, or a rackmount artificial intelligence accelerator. For example, each co-packaged optical module **1074** can include a module similar to the integrated optical communication device **448**, **462**, **466**, or **472** of FIG. **17**, the integrated optical communication device **210** of FIG. **20**, the integrated communication device **612** of FIG. **23**, the optical/electrical communication interface **684** of FIG. **26**, **724** of FIG. **28**, or **760** of FIG. **29**, the integrated optical communication device **512** of FIG. **32**, or the optical module with connector **868** of FIG. **43**. For example, each fiber cable **1076** can include the optical fibers **226** (FIGS. **2**, **4**), **272** (FIGS. **6**, **7**), **582** (FIG. **22**, **23**), or **734** (FIG. **28**), or the optical fiber cable **762** (FIG. **762**), **956** (FIG. **53**), or **996** (FIG. **61**).

(436) For example, the co-packaged optical module **1074** can include a first optical connector part (e.g., **456** of FIG. **17**, **578** of FIG. **22** or **23**, **746** of FIG. **28**) that is configured to be removably coupled to a second optical connector part (e.g., **458** of FIG. **17**, **580** of FIG. **22** or **23**, **748** of FIG. **28**) that is attached to the external fiber cable **1076**. For example, the co-packaged optical module **1074** includes a photonic integrated circuit (e.g., **450**, **464**, **468**, or **474** of FIG. **17**, **586** of FIG. **22**, **618** of FIG. **23**, or **726** of FIG. **28**) that is optically coupled to the first optical connector part. The photonic integrated circuit receives input optical signals from the first optical connector part and generates input electrical signals based on the input optical signals. At least a portion of the input electrical signals generated by the photonic integrated circuit are transmitted to the at least one data processing chip **1070** through electrical signal lines in or on the vertical printed circuit board **1068**. For example, the photonic integrated circuit can be configured to receive output electrical signals from the at least one data processing chip **1070** and generate output optical signals based on the output electrical signals. The output optical signals are transmitted through the first and second optical connector parts to the external fiber cable **1076**.

(437) In some examples, the fiber cable **1076** can include, e.g., 10 or more cores of optical fibers, and the first optical connector part is configured to couple 10 or more channels of optical signals to the photonic integrated circuit. In some examples, the fiber cable **1076** can include 100 or more cores of optical fibers, and the first optical connector part is configured to couple 100 or more channels of optical signals to the photonic integrated circuit. In some examples, the fiber cable **1076** can include 500 or more cores of optical fibers, and the first optical connector part is configured to couple 500 or more channels of optical signals to the photonic integrated circuit. In some examples, the fiber cable **1076** can include 1000 or more cores of optical fibers, and the first optical connector part is configured to couple 1000 or more channels of optical signals to the photonic integrated circuit.
(438) In some implementations, the photonic integrated circuit can be configured to generate first serial electrical signals based on the received optical signals, in which each first serial electrical signal is generated based on one of the channels of first optical signals. Each co-packaged optical module **1074** can include a first serializers/deserializers module that includes serializer units and deserializer units, in which the first serializers/deserializers module is configured to generate sets of first parallel electrical signals based on the first serial electrical signals and condition the electrical signals, and each set of first parallel electrical signals is generated based on a corresponding first serial electrical signal. Each co-packaged optical module **1074** can include a second serializers/deserializers module that includes serializer units and deserializer units, in which the second serializers/deserializers module is configured to generate second serial electrical signals based on the sets of first parallel electrical signals, and each second serial electrical signal is generated based on a corresponding set of first parallel electrical signals.
(439) In some examples, the rackmount server **1060** can include 4 or more co-packaged optical modules **1074** that are configured to be removably coupled to corresponding second optical connector parts that are attached to corresponding fiber cables **1076**. For example, the rackmount server **1060** can include 16 or more co-packaged optical modules **1074** that are configured to be removably coupled to corresponding second optical connector parts that are attached to corresponding fiber cables **1076**. In some examples, each fiber cable **1076** can include 10 or more cores of optical fibers. In some examples, each fiber cable **1076** can include 100 or more cores of optical fibers. In some examples, each fiber cable **1076** can include 500 or more cores of optical fibers. In some examples, each fiber cable **1076** can include 1000 or more cores of optical fibers. Each optical fiber can transmit one or more channels of optical signals. For example, the at least one data processing chip **1070** can include a network switch that is configured to receive data from an input port associated with a first one of the channels of optical signals, and forward the data to an output port associated with a second one of the channels of optical signals.
(440) In some implementations, the co-packaged optical modules **1074** are removably coupled to the vertical printed circuit board **1068**. For example, the co-packaged optical modules **1074** can be electrically coupled to the vertical printed circuit board **1068** using electrical contacts that include, e.g., spring-loaded elements, compression interposers, or land-grid arrays.
(441) Referring to FIGS. **69**A and **69**B, in some implementations, a rackmount server **1080** includes a housing **1082** having a front panel **1084**. The rackmount server **1080** is similar to the rackmount server **1060** of FIG. **68**A, except that one or more fans are mounted on the front panel **1084**, and one or more air louvers installed in the housing **1082** to direct air flow towards the heat dissipating device. For example, the rackmount server **1080** can include a first inlet fan **1086***a* mounted on the front panel **1084** to the left of the vertical printed circuit board **1068**, and a second inlet fan **1086***b* mounted on the front panel **1084** to the right of the vertical printed circuit board **1068**. The terms "right" and "left" refer to relative positions of components shown in the figure. It is understood that, depending on the orientation of a device having a first and second modules, a first module that is positioned to the "left" or "right" of a second module can in fact be to the "right" or "left" (or any other relative position) of the second module. For example, depending on the orientation of the rackmount server **1080**, the inlet fans can be positioned below and/or above the vertical printed circuit board **1068**. Depending on the shape of the rackmount server **1080**, it is possible to have inlet fans positioned left, right, below and/or above the vertical printed circuit board **1068**, or in any combination of those positions. One or more fans can be positioned in front of the plane that extends along the printed circuit board and designed to blow air towards components coupled to the front side of the printed circuit board, and one or more fans can be positioned to the rear of the plane that extends along the printed circuit board and designed to blow air towards components coupled to the back side of the printed circuit board. The inlet and exhaust fans operate in a push-pull manner, in which the inlet fans **1086***a* and **1086***b* (collectively referenced as **1086**) pull cool air into the housing **1082**, and the exhaust fans **1050** push warm air out of the housing **1082**. The inlet fans **1086** in the front panel or face plate **1064** and the exhaust fans **1050** on the backside of the rack generate a pressure gradient through the housing or case to improve air cooling compared to standard 1 RU implementations that include only backside exhaust fans.
(442) The inlet fans do not necessarily have to be attached to the front panel, and can also be positioned at a distance front the front panel. The vertical printed circuit board **1068** can be positioned at a distance from the front panel, and the position of the inlet fans can be adjusted accordingly to maximize the efficiency for transferring heat away from the heat sink **1072**.
(443) In some implementations, a left air louver **1088***a* and a right air louver **1088***b* are installed in the housing **1082** to direct airflow toward the heat dissipating device **1072**. The air louvers **1088***a*, **1088***b* (collectively referenced as **1088**) partition the space in the housing **1082** and force air to flow from the inlet fans **1086***a* and **1086***b*, pass over surfaces of fins of the heat dissipating device **1072**, and towards an opening **1090** between distal ends of the air louvers **1088**. The

directions of air flow near the inlet fans **1086***a* and **1086***b* are represented by arrows **1092***a* and **1092***b*. The air louvers **1088** increase the amount of air flows across the surfaces of the heat sink fins and enhance the efficiency of heat removal. The heat sink fins are oriented to extend along planes that are substantially parallel to the bottom surface **1038** of the housing **1082**. For example, the air louvers **1088** can have a curved shape, e.g., an S-shape as shown in the figure. The curved shape of the air louvers **1088** can be configured to maximize the efficiency of the heat sink. In some examples, the air louvers **1088** can also have a linear shape.
(444) For example, the heat sink can be a plate-fin heat sink, a pin-fin heat sink, or a plate-pin-fin heat sink. The pins can have a square or circular cross section. The heat sink configuration (e.g., pin pitch, length of pins or fins) and the louver configuration can be designed to optimize heat sink efficiency.
(445) For example, the co-packaged optical modules **1074** can be electrically coupled to the vertical printed circuit board **1068** using electrical contacts that include, e.g., spring-loaded elements, compression interposers, or land-grid arrays. For example, when compression interposers are used, the vertical circuit board **1068** can be positioned such that the face of compression interposers of the co-packaged optical module **1074** is coplanar with the face plate **1064** and the inlet fans **1086**.
(446) Referring to FIG. **70**, in some implementations, a rackmount server **1090** is similar to the rackmount server **1080** of FIG. **69**, which includes inlet fans mounted on the front panel. The inlet fans of the rackmount server **1090** are slightly rotated, as compared to the inlet fans of the rackmount server **1080** to improve efficiency of the heat sink. The rotational axes of the inlet fans, instead of being parallel to the front-to-rear direction relative to the housing **1082**, can be rotated slightly inwards. For example, the rotational axis of a left inlet fan **1092***a* can be rotated slightly clockwise and the rotational axis of a right inlet fan **1092***b* can be rotated slightly counter-clockwise, to enhance the air flow across the surfaces of the heat sink fins, further improving the efficiency of heat removal.
(447) In some implementations heat removal efficiency can be improved by positioning the vertical circuit board **1068** and the heat dissipating device **1072** further toward the rear of the housing so that a larger amount of air flows across the surface of the fins of the heat dissipating device **1072**.
(448) Referring to FIGS. **71**A to **71**B, a rackmount server **1100** includes a housing **1102** having a front panel or face plate **1104**, in which the portion of the face plate **1104** where the compression interposers for the co-packaged optical module **1074** are located are inset by a distance d with respect to the original face plate **1104**. The face plate **1104** has a recessed portion or an inset portion **1106** that is offset at a distance d (referred to as the “front panel inset distance”) toward the rear of the housing **1102** relative to the other portions (e.g., the portions on which the inlet fans **1086***a* and **1086***b* are mounted) of the front panel **1104**. The inset portion **1106** is referred to as the “recessed front panel,” “recessed face plate,” “front panel inset,” or “face plate inset.” The vertical printed circuit board **1068** is attached to the inset portion **1106**, which includes openings to allow the co-packaged optical modules **1074** to pass through. The inset portion **1106** is configured to have sufficient area to accommodate the co-packaged optical modules **1074**.
(449) By providing the inset portion **1106** in the front panel **1104**, the fins of the heat dissipating device **1072** can be more optimally positioned to be closer to the main air flow generated by the inlet fans **1086**, while maintaining serviceability of the co-packaged optical modules **1074**, e.g., allowing the user to repair or replace damaged co-packaged optical modules **1074** without opening the housing **1102**. The heat sink configuration (e.g., pin pitch, length of pins or fins) and the louver configuration can be designed to optimize heat sink efficiency. In addition, the front panel inset distance d can be optimized to improve heat sink efficiency.
(450) Referring to FIG. **72**, in some implementations, a rackmount server **1110** is similar to the rackmount server **1100** of FIG. **71**, except that the server **1110** includes a heat dissipating device **1112** that has fins **1114***a* and **1114***b* that extend beyond the edge of the vertical printed circuit board **1068** and closer to the inlet fans **1086***a*, **1086***b*, as compared to the fins in the example of FIG. **71**. The configuration of the fins (e.g., the shapes, sizes, and number of fins) can be selected to maximize the efficiency of heat removal.
(451) Referring to FIGS. **73**A and **73**B, in some implementations, a rackmount server **1120** includes a housing **1122** having a front panel **1124**, a rear panel **1036**, a bottom panel **1038**, a top panel, and side panels **1040**. The width and height of the housing **1122** can be similar to those of the housing **1062** of FIG. **68**A. The server **1120** includes a first printed circuit board **1066** that extends parallel to the bottom panel **1038**, and one or more vertical printed circuit boards, e.g., **1126***a* and **1126***b* (collectively referenced as **1126**), that are mounted perpendicular to the first printed circuit board **1066**. The server **1120** includes one or more inlet fans **1086** mounted on the front panel **1124** and one or more exhaust fans **1050** mounted on the rear panel **1036**. The air flow in the housing **1122** is generally in the front-to-rear direction. The directions of the air flows are represented by the arrows **1134**.
(452) Each vertical printed circuit board **1126** has a first surface and a second surface. The first surface defines the length and width of the vertical printed circuit board **1126**. The distance between the first and second surfaces defines the thickness of the vertical printed circuit board **1126**. The vertical printed circuit board **1126***a* or **1126***b* is oriented such that the first surface extends along a plane that is substantially parallel to the front-to-rear direction relative to the housing **1122**. At least one data processing chip **1128***a* or **1128***b* is electrically coupled to the first surface of the vertical printed circuit board **1126***a* or **1126***b*, respectively. In some examples, the at least one data processing chip **1128***a* or **1128***b* is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the printed circuit board **1126***a* or **1126***b*. A heat dissipating device **1130***a* or **1130***b* is thermally coupled to the at least one data processing chip **1128***a* or **1128***b*, respectively. The heat dissipating device **1130** includes fins that extend along planes that are substantially parallel to the

bottom panel **1038** of the housing **1122**. The heat sinks **1130***a* and **1130***b* are positioned directly behind to the inlet fans **1086***a* and **1086***b*, respectively, to maximize air flow across the fins and/or pins of the heat sinks **1130**.
(453) At least one co-packaged optical module **1132***a* or **1132***b* is mounted on the second side of the vertical printed circuit board **1126***a* or **1126***b*, respectively. The co-packaged optical modules **1132** are optically coupled, through optical interconnection links, to optical interfaces (not shown in the figure) mounted on the front panel **1124**. The optical interfaces are optically coupled to external fiber cables. The orientations of the vertical printed circuit boards **1126** and the fins of the heat dissipating devices **1130** are selected to maximize heat removal.
(454) Referring to FIGS. **74**A to **74**B, in some implementations, a rackmount server **1150** includes vertical printed circuit boards **1152***a* and **1152***b* (collectively referenced as **1152**) that have surfaces that extend along planes substantially parallel to the front-to-rear direction relative to the housing or case, similar to the vertical printed circuit boards **1126***a* and **1126***b* of FIG. **73**. The rackmount server **1150** includes a housing **1154** that has a modified front panel or face plate **1156** that has an inset portion **1158** configured to improve access and field serviceability of co-packaged optical modules **1160***a* and **1160***b* (collectively referenced as **1160**) that are mounted on the vertical printed circuit boards **1152***a* and **1152***b*, respectively. The inset portion **1158** is referred to as the “front panel inset” or “face plate inset.” The inset portion **1158** has a width w that is selected to enable hot-swap, in-field serviceability of the co-packaged optical modules **1160** to avoid the need to take the rackmount server **1150** out of service for maintenance.
(455) For example, the inset portion **1158** includes a first wall **1162**, a second wall **1164**, and a third wall **1166**. The first wall **1162** is substantially parallel to the second wall **1164**, and the third wall **1166** is positioned between the first wall **1162** and the second wall **1164**. For example, the first wall **1162** extends along a direction that is substantially parallel to the front-to-rear direction relative to the housing **1122**. The vertical printed circuit board **1152***a* is attached to the first wall **1162** of the inset portion **1158**, and the vertical printed circuit board **1152***b* is attached to the first wall **1162** of the inset portion **1158**. The first wall **1162** includes openings to allow the co-packaged optical modules **1160***a* to pass through, and the second wall **1164** includes openings to allow the co-packaged optical modules **1160***b* to pass through. For example, an inlet fan **1086***c* can be mounted on the third wall **1166**.
(456) Each vertical printed circuit board **1152** has a first surface and a second surface. The first surface defines the length and width of the vertical printed circuit board **1152**. The distance between the first and second surfaces defines the thickness of the vertical printed circuit board **1152**. The vertical printed circuit board **1152***a* or **1152***b* is oriented such that the first surface extends along a plane that is substantially parallel to the front-to-rear direction relative to the housing **1154**. At least one data processing chip **1170***a* or **1170***b* is electrically coupled to the first surface of the vertical printed circuit board **1152***a* or **1152***b*, respectively. In some examples, the at least one data processing chip **1170***a* or **1170***b* is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the printed circuit board **1152***a* or **1152***b*. A heat dissipating device **1168***a* or **1168***b* is thermally coupled to the at least one data processing chip **1170***a* or **1170***b*, respectively. The heat dissipating device **1168** includes fins that extend along planes that are substantially parallel to the bottom panel **1038** of the housing **1154**. The heat sinks **1168***a* and **1168***b* are positioned directly behind to the inlet fans **1086***a* and **1086***b*, respectively, to maximize air flow across the fins and/or pins of the heat sinks **1168***a* and **1168***b*.
(457) Referring to FIGS. **75**A to **75**B, in some implementations, a rackmount server **1180** includes a housing **1182** having a front panel **1184** that has an inset portion **1186** (referred to as the “front panel inset” or “face plate inset”). For example, the inset portion **1186** includes a first wall **1188** and a second wall **1190** that are oriented to make it easier for the user to connect or disconnect the fiber cables (e.g., **1076**) to the server **1180**, or to service the co-packaged optical modules **1074**. For example, the first wall **1188** can be at an angle θ.sub.1 relative to a nominal plane **1192** of the front panel **1184**, in which 0<θ.sub.1<90°. The second wall **1190** can be at an angle θ.sub.2 relative to the nominal plane **1192** of the front panel, in which 0<θ.sub.2<90°. The angles θ.sub.1 and θ.sub.2 can be the same or different. The nominal plane **1192** of the front panel **1184** is perpendicular to the side panels **1040** and the bottom panel.
(458) For example, a first vertical printed circuit board **1152***a* is attached to the first wall **1188**, and a second vertical printed circuit board **1152***b* is attached to the second wall **1190**. Comparing the rackmount server **1180** with the rackmount servers **1060** of FIG. **68**A, **1080** of FIG. **69**A, and **1100** of FIG. **71**, the server **1180** has a larger front panel area due to the angled front panel inset and can be connected to more fiber cables.
(459) Positioning the first and second walls **1188**, **1190** at an angle between 0 and 90° relative to the nominal plane of the front panel improves access and field serviceability of the co-packaged optical modules. Comparing the rackmount server **1180** with the rackmount server **1150** of FIG. **74**A, the server **1180** allows the user to more easily access the co-packaged optical modules that are positioned farther away from the nominal plane of the front panel. The angles θ.sub.1 and θ.sub.2 are selected to strike a balance between increasing the number of fiber cables that can be connected to the server and providing easy access to all of the co-packaged optical modules of the server. The front panel inset width and angle are configured to enable hot-swap, in-field serviceability to avoid taking the switch and rack out of service for maintenance.
(460) For examples, intake fans **1086***a* and **1086***b* can be mounted on the front panel **1184**. Outside air is drawn in by the intake fans **1086***a*, **1086***b*, passes through the surfaces of the fins and/or pins of the heatsinks **1168***a*, **1168***b*, and flows towards the rear of the housing **1182**. Examples of the flow directions for the air entering through the intake fans **1186***a* and **1186***b* are represented by arrows **1198***a*, **1198***b*, **1198***c*, and **1198***d*.
(461) Referring to FIGS. **75**B and **75**C, in some implementations, the front panel **1184** includes an upper air vent **1194***a* and baffles to direct outside air to enter through the upper air vent **1194***a*, flows downward and rearward such that the air passes over the surfaces of some of the fins and/or pins of the heat sinks **1186** (e.g., including the fins and/or pins closer to

the top of the heat sinks **1186**) and then flows toward an intake fan **1086***c* mounted at or near the distal or rear end of the front panel inset portion **1186**. The front panel **1184** includes a lower air vent **1194***b* and baffles to direct outside air to enter through the lower air vent **1194***b*, flows upward and rearward such that the air passes over the surfaces of some of the fins and/or pins of the heat sinks **1186** (e.g., including the fins and/or pins closer to the bottom of the heat sinks **1186**) and then flows toward the intake fan **1086***c*. Examples of the air flows through the upper and lower air vents **1194***a*, **1194***b* to the intake fan **1086***c* are represented by arrows **1196***a*, **1196***b*, **1196***c*, and **1196***d* in FIG. **75**C.
(462) For example, fiber cables connected to the co-packaged optical modules **1074** can block air flow for the intake fan **1086***c* if the intake fan **1086***c* is configured to receive air through openings directly in front of the intake fan **1086***c*. By using the upper air vent **1194***a*, the lower air vent **1194***b*, and the baffles to direct air flow as described above, the heat dissipating efficiency of the system can be improved (as compared to not having the air vents **1194** and the baffles).
(463) Referring to FIG. **76**, in some implementations, a network switch system **1210** includes a plurality of rackmount switch servers **1212** installed in a server rack **1214**. The network switch rack includes a top of the rack switch **1216** that routes data among the switch servers **1212** within the network switch system **1210**, and serves as a gateway between the network switch system **1210** and other network switch systems. The rackmount switch servers **1212** in the network switch system **1210** can be configured in a manner similar to any of the rackmount servers described above or below.
(464) In some implementations, the examples of rackmount servers shown in FIGS. **68**A, **69**A, and **70** can be modified by positioning the vertical printed circuit board behind the front panel. The co-packaged optical modules can be optically connected to fiber connector parts mounted on the front panel through short optical connection paths, e.g., fiber jumpers.
(465) Referring to FIGS. **77**A and **77**B, in some implementations, a rackmount server **1220** includes a housing **1222** having a front panel **1224**, a rear panel **1036**, a top panel **1226**, a bottom panel **1038**, and side panels **1040**. The front panel **1224** can be opened to allow the user to access components without removing the rackmount server **1220** from the rack. A vertically mounted printed circuit board **1230** is positioned substantially parallel to the front panel **1224** and recessed from the front panel **1224**, i.e., spaced apart at a small distance (e.g., less than 12 inches, or less than 6 inches, or less than 3 inches, or less than 2 inches) to the rear of the front panel **1224**. The printed circuit board **1230** includes a first side facing the front direction relative to the housing **1222** and a second side facing the rear direction relative to the housing **1222**. At least one data processing chip **1070** is electrically coupled to the second side of the vertical printed circuit board **1226**, and a heat dissipating device or heat sink **1072** is thermally coupled to the at least one data processing chip **1070**. In some examples, the at least one data processing chip **1070** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the printed circuit board **1226**.
(466) Co-packaged optical modules **1074** (also referred to as the optical/electrical communication interfaces) are attached to the first side (i.e., the side facing the front exterior of the housing **1222**) of the vertical printed circuit board **1230**. In some examples, the co-packaged optical modules **1074** are mounted on a substrate that is attached to the vertical printed circuit board **1230**, in which electrical contacts on the substrate are electrically coupled to corresponding electrical contacts on the vertical printed circuit board **1230**. In some examples, the at least one data processing chip **1070** is mounted on the rear side of the substrate, and the co-packaged optical modules **1074** are removably attached to the front side of the substrate, in which the substrate provides high speed connections between the at least one data processing chip **1070** and the co-packaged optical modules **1074**. For example, the substrate can be attached to a front side of the printed circuit board **1068**, in which the printed circuit board **1068** includes one or more openings that allow the at least one data processing chip **1070** to be mounted on the rear side of the substrate. The printed circuit board **1068** can provide from a motherboard electrical power to the substrate (and hence to the at least one data processing chip **1070** and the co-packaged optical modules **1074**, and allow the at least one data processing chip **1070** and the co-packaged optical modules **1074** to connect to the motherboard using low-speed electrical links. An array of co-packaged optical modules **1074** can be mounted on the vertical printed circuit board **1230** (or the substrate), similar to the examples shown in FIGS. **69**B and **71**B. The electrical connections between the co-packaged optical modules **1074** and the vertical printed circuit board **1070** (or the substrate) can be removable, e.g., by using land-grid arrays and/or compression interposers. The co-packaged optical modules **1074** are optically connected to first fiber connector parts **1232** mounted on the front panel **1224** through short fiber jumpers **1234***a*, **1234***b* (collectively referenced as **1234**). When the front panel **1224** is closed, the user can plug a second fiber connector part **1236** into the first fiber connector part **1232** on the front panel **1224**, in which the second fiber connector part **1236** is connected to an optical fiber cable **1238** that includes an array of optical fibers.
(467) In some implementations, the rackmount server **1220** is pre-populated with co-packaged optical modules **1074**, and the user does not need to access the co-packaged optical modules **1074** unless the modules need maintenance. During normal operation of the rackmount server **1220**, the user mostly accesses the first fiber connector parts **1232** on the front panel **1224** to connect to fiber cables **1238**.
(468) One or more intake fans, e.g., **1086***a*, **1086***b*, can be mounted on the front panel **1224**, similar to the examples shown in FIGS. **69**A and **70**. The positions and configurations of the intake fans **1086**, the heat sink **1072**, and the air louvers **1088***a*, **1088***b* are selected to maximize the heat transfer efficiency of the heat sink **1072**.
(469) The rackmount server **1220** can have a number of advantages. By placing the vertical printed circuit board **1230** at a recessed position inside the housing **1222**, the vertical printed circuit board **1230** is better protected by the housing **1222**, e.g., preventing users from accidentally bumping into the circuit board **1230**. By orienting the vertical printed circuit board **1230** substantially parallel to the front panel **1224** and mounting the co-packaged optical modules **1074** on the side of the circuit board **1230** facing the front direction, the co-packaged optical modules **1074** can be accessible to users for

maintenance without the need to remove the rackmount server **1220** from the rack.
(470) In some implementations, the front panel **1224** is coupled to the bottom panel **1038** using a hinge **1228** and configured such that the front panel **1224** can be securely closed during normal operation of the rackmount server **1220** and easily opened for maintenance. For example, if a co-packaged optical module **1074** fails, a technician can open and rotate the front panel **1224** down to a horizontal position to gain access to the co-packaged optical module **1074** to repair or replace it. For example, the movements of the front panel **1224** is represented by the bi-directional arrow **1250**. In some implementations, different fiber jumpers **1234** can have different lengths, depending on the distance between the parts that are connected by the fiber jumpers **1234**. For example, the distance between the co-packaged optical module **1074** and the first fiber connector part **1232** connected by the fiber jumper **1234***a* is less than the distance between the co-packaged optical module **1074** and the first fiber connector part **1232** connected by the fiber jumper **1234***b*, so the fiber jumper **1234***a* can be shorter than the fiber jumper **1234***b*. This way, by using fiber jumpers with appropriate lengths, it is possible to reduce the clutter caused by the fiber jumpers **1234** inside the housing **1222** when the front panel **1224** is closed and in its vertical position.
(471) In some implementations, the front panel **1224** can be configured to be opened and lifted upwards using lift-up hinges. This can be useful when the rackmount server is positioned near the top of the rack. In some examples, the front panel **1224** can be coupled to the side panel **1040** by using a hinge so that the front panel **1224** can be opened and rotated sideways. In some examples, the front panel can include a left front subpanel and a right front subpanel, in which the left front subpanel is coupled to the left side panel **1040** by using a first hinge, and the right front subpanel is coupled to the right panel **1040** by using a second hinge. The left front subpanel can be opened and rotated towards the left side, and the right front subpanel can be opened and rotated towards the right side. These various configurations for the front panel enable protection of the vertical printed circuit board **1230** and convenient access to the co-packaged optical modules **1074**.
(472) In some examples, the front panel can have an inset portion, similar to the example shown in FIG. **71**A, in which the vertical printed circuit board is in a recessed position relative to the inset portion of the front panel, i.e., at a small distance to the rear of the inset portion of the front panel. The front panel inset distance, the distance between the vertical printed circuit board and the front panel inset portion, and the air louver configuration can be selected to maximize the heat sink efficiency.
(473) Referring to FIG. **78**, in some implementations, a rackmount server **1240** can be similar to the rackmount server **1150** of FIG. **74**A, except that the vertical printed circuit boards are at recessed positions relative to the walls of the inset portion of the front panel. For example, a vertical printed circuit board **1152***a* is in a recessed position relative to a first wall **1242***a* of an inset portion **1244**, i.e., the vertical printed circuit board **1152***a* is spaced apart a small distance to the left from the first wall **1242***a*. A vertical printed circuit board **1152***b* is in a recessed position relative to a second wall **1242***b* of the inset portion **1244**, i.e., the vertical printed circuit board **1152***b* is spaced apart a small distance to the right from the second wall **1242***b*.
(474) For example, the first wall **1242***a* can be coupled to the bottom or top panel through hinges so that the first wall **1242***a* can be closed during normal operation of the rackmount server **1240** and opened for maintenance of the server **1240**. The distance w**2** between the first wall **1242***a* and the second wall **1242***b* is selected to be sufficiently large to enable the first wall **1242***a* and the second wall **1242***b* to be opened properly. This design has advantages similar to those of the rackmount server **1220** in FIGS. **77**A, **77**B.
(475) In some implementations, a rackmount server can be similar to the rackmount server **1180** shown in FIGS. **75**A to **75**C, except that the vertical printed circuit boards are at recessed positions relative to the walls of the inset portion of the front panel. For example, a first vertical printed circuit board is in a recessed position relative to the first wall **1188** of the inset portion **1186**, and a second vertical printed circuit board is in a recessed position relative to the second wall **1190** of the inset portion **1186**. For example, the first wall **1188** can be coupled to the bottom or top panel through hinges so that the first wall **1188** can be closed during normal operation of the rackmount server and opened for maintenance of the server. The angles θ.sub.1 and θ.sub.2 are selected to enable the first wall **1188** and the second wall **1190** to be opened properly. This design has advantages similar to those of the rackmount server **1220** in FIGS. **77**A, **77**B.
(476) A feature of the thermal architecture for the rackmount units (e.g., the rackmount servers **1060** of FIG. **68**A, **1090** of FIGS. **69**A, **70**, **1100** of FIGS. **71**A, **72**, **1120** of FIG. **73**A, **1150** of FIG. **74**A, **1180** of FIG. **75**A, **1220** of FIG. **77**B, and **1240** of FIG. **78**) described above is the use of co-packaged optical modules or optical/electrical communication interfaces that have higher bandwidth per module or interface, as compared to conventional designs. For example, each co-packaged optical module or optical/electrical communication interface can be coupled to a fiber cable that carries a large number of densely packed optical fiber cores. FIG. **9** shows an example of the integrated optical communication device **282** in which the optical signals provided to the photonic integrated circuit can have a total bandwidth of about 12.8 Tbps. By using co-packaged optical modules or optical/electrical communication interfaces that have higher bandwidth per module or interface, the number of co-packaged optical modules or optical/electrical communication interfaces required for a given total bandwidth for the rackmount unit is reduced, so the amount of area on the front panel of the housing reserved for connecting to optical fibers can be reduced. Therefore, it is possible to add one or more inlet fans on the front panel to improve thermal management while still maintaining or even increasing the total bandwidth of the rackmount unit, as compared to conventional designs.
(477) In some implementations, for the examples shown in FIGS. **72**, **74**A, **75**A, and **78**, and the variations in which the

vertical printed circuit boards are at recessed positions relative to the front panel, the shape of each of the top and bottom panels of the housing can have an inset portion at the front that corresponds to the inset portion of the front panel. This makes it more convenient to access the co-packaged optical modules or the optical connector parts mounted on the front panel without being hindered by the top and bottom panels. In some implementations, the server rack (e.g., **1214** of FIG. **76**) is designed such that front support structures of the server rack also have inset portions that correspond to the insert portions of the front panels of the rackmount servers installed in the server rack. For example, a custom server rack can be designed to install rackmount servers that all have the inset portions similar to the inset portion **1158** of FIG. **74**A. For example, a custom server rack can be designed to install rackmount servers that all have the inset portions similar to the inset portion **1186** of FIG. **75**A. In such examples, the inset portions extend vertically from the bottom-most server to the top-most server without any obstruction, making it easier for the user to access the co-packaged optical modules or optical connector parts.

(478) In some implementations, for the examples shown in FIGS. **72**, **74**A, **75**A, and **78**, and the variations in which the vertical printed circuit boards are at recessed positions relative to the front panel, the shape of the top and bottom panels of the housing can be similar to standard rackmount units, e.g., the top and bottom panels can have a generally rectangular shape.

(479) In the examples shown in FIGS. **68**A, **68**B, **69**A to **75**C, and **77**A to **78**, a grid structure similar to the grid structure **870** shown in FIG. **43** can be attached to the vertical printed circuit board. The grid structure can function as both (i) a heat spreader/heat sink and (ii) a mechanical holding fixture for the co-packaged optical modules (e.g., **1074**) or optical/electrical communication interfaces.

(480) FIGS. **96** to **97**B are diagrams of an example of a rackmount server **1820** that includes a vertically oriented circuit board **1822** positioned at a front portion of the rackmount server **1820**. FIG. **96** shows a top view of the rackmount server **1820**, FIG. **97**A shows a perspective view of the rackmount server **1820**, and FIG. **97**B shows a perspective view of the rackmount server **1820** with the top panel removed. The rackmount server **1820** has an active airflow management system that is configured to remove heat from a data processor during operation of the rackmount server **1820**.

(481) Referring to FIGS. **96**, **97**A, and **97**B, in some implementations, the rackmount server **1820** includes a housing **1824** that has a front panel **1826**, a left side panel **1828**, a right side panel **1840**, a bottom panel **1841**, a top panel **1843**, and a rear panel **1842**. The front panel **1826** can be similar to the front panels in the examples shown in FIGS. **68**A, **68**B, **69**A to **72**, **77**A, and **77**B. For example, the vertically oriented circuit board **1822** can be part of the front panel **1826**, or attached to the front panel **1826**, or positioned in a vicinity of the front panel **1826**, in which a distance between the circuit board **1822** and the front panel **1826** is not more than, e.g., 6 inches. A data processor **1844** (which can be, e.g., a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit) (see FIG. **99**) is mounted on the circuit board **1822**.

(482) A heat dissipating module **1846**, e.g., a heat sink, is thermally coupled to the data processor **1844** and configured to dissipate heat generated by the data processor **1828** during operation. The heat dissipating module **1846** can be similar to the heat dissipating device **1072** of FIGS. **68**A, **68**C, **69**A, **70**, and **71**A. In some examples, the heat dissipating module **1846** includes heat sink fins or pins having heat dissipating surfaces configured to optimize heat dissipation. In some examples, the heating dissipating module **1846** includes a vapor chamber thermally coupled to heat sink fins or pins. The rackmount server **1820** can include other components, such as power supply units, rear outlet fans, one or more additional horizontally oriented circuit boards, one or more additional data processors mounted on the horizontally oriented circuit boards, and one or more additional air louvers, that have been previously described in other embodiments of rackmount servers and are not repeated here.

(483) In some implementations, the active airflow management system includes an inlet fan **1848** that is positioned at a left side of the heat dissipating module **1846** and oriented to blow incoming air to the right toward the heat dissipating module **1846**. A front opening **1850** provides incoming air for the inlet fan **1848**. The front opening **1850** can be positioned to the left of the inlet fan **1848**. In the example of FIG. **96**, the circuit board **1822** is substantially parallel to the front panel **1826**, and the rotational axis of the inlet fan **1848** is substantially parallel to the plane of the circuit board **1822**. The inlet fan **1848** can also be oriented slightly differently. For example, the rotational axis of the inlet fan **1848** can be at an angle θ relative to the plane of the front panel **1826**, the angle θ being measured along a plane parallel to the bottom panel **1841**, in which $\theta \leq 45°$, or in some examples $\theta \leq 25°$, or in some examples $\theta \leq 5°$, or in some examples $\theta = 0°$.

(484) In some implementations, a baffle or an air louver **1852** (or internal panel or internal wall) is provided to guide the air entering the opening **1850** towards the inlet fan **1848**. An arrow **1854** shows the general direction of airflow from the opening **1850** to the inlet fan **1848**. In some examples, the air louver **1852** extends from the left side panel **1828** of the housing **1840** to a rear edge of the inlet fan **1848**. The air louver **1852** can be straight or curved. In some examples, the air louver **1852** can be configured to guide the inlet air blown from the inlet fan **1848** towards the heat dissipating module **1846**. For example, the air louver **1852** can extend from the left side panel **1828** to the left edge of the heat dissipating module **1846**. For example, the air louver **1852** can extend from the left side panel **1828** to a position at or near the rear of the heat dissipating module **1846**, in which the position can be anywhere from the left rear portion of the heat dissipating module **1846** to the right rear portion of the heat dissipating module **1846**. The air louver **1852** can extend from the bottom panel **1841** to the top panel **1843** in the vertical direction. An arrow **1856** shows the general direction of air flow through and out of the heating dissipating module **1846**.

(485) For example, the air louver **1852**, a front portion of the left side panel **1828**, the front panel **1826**, the circuit board **1822**, a front portion of the bottom panel **1841**, and a front portion of the top panel **1843** can form an air duct that guides the incoming cool air to flow across the heat dissipating surface of the heat dissipating module **1846**. Depending on the design, the air duct can extend to the left edge of the heat dissipating module **1846**, to a middle portion of the heat dissipating module **1846**, or extend approximately the entire length (from left to right) of the heat dissipating module **1846**.
(486) The inlet fan **1848** and the air louver **1852** are designed to improve airflow across the heat dissipating surface of the heat dissipating module **1846** to optimize or maximize heat dissipation from the data processor **1844** through the heat dissipating module **1846** to the ambient air. Different rackmount servers can have vertically mounted circuit boards with different lengths, can have data processors with different heat dissipation requirements, and can have heat dissipating modules with different designs. For example, the heat sink fins and/or pins can have different configurations. The inlet fan **1848** and the air louver **1852** can also have any of various configurations in order to optimize or maximize the heat dissipation from the data processor **1844**. In the example of FIG. **96**, the inlet fan **1848** directs air to flow generally in a direction (in this example, from left to right) that is parallel to the front panel across the heat dissipating surface of the heat dissipating module **1846**. In some implementations, the front opening can be positioned to the right side of the front panel, and the inlet fan can be positioned to the right side of the heat dissipating module and direct air to flow from right to left across the heat dissipating surface of the heat dissipating module. The air louver can be modified accordingly to optimize airflow and heat dissipation from the data processor.
(487) FIG. **98** is a diagram showing the front portion of the rackmount server **1820**. The baffle or air louver **1852**, a portion of the bottom panel **1841**, a portion of the top panel **1843**, and a portion of the left side panel **1828** form a duct that directs external air toward the inlet fan **1848**. A safety mechanism (not shown in the figure), such as a protective mesh, that allows air to pass through substantially freely while blocking larger objects from contacting the fan blades can be placed across the opening **1850**. In some implementations, an air filter can be provided in front of the inlet fan to reduce dust buildup inside the rackmount server.
(488) In some examples, orienting the inlet fan to face towards the side direction instead of the front direction (as in the examples shown in FIGS. **69**A and **71**A) can improve the safety and comfort of users operating the rackmount server **1820**. In some examples, orienting the inlet fan towards the side direction instead of the front direction can avoid the presence of a region in the heat dissipating module having little to no air flow. In the example of FIG. **71**A, the left and right inlet fans blow air toward the left and right side regions, respectively, of the heat dissipating device **1072**. The incoming air is drawn toward the rear of the heat dissipating module due to the air pressure gradient generated by the front and rear inlet fans. In some cases, the incoming air entering the left side of the heat dissipating device **1072** is drawn toward the rear of the heat dissipating device **1072** before reaching the middle part of the heat dissipating device **1072**. Similarly, the incoming air entering the right side of the heat dissipating device **1072** is drawn toward the rear of the heat dissipating device **1072** before reaching the middle part of the heat dissipating device **1072**. As a result, near the middle or middle-front region of the heat dissipating device **1072** there may be a region having little to no air flow, reducing the efficiency of heat dissipation. The design shown in FIGS. **96** to **98** avoids or reduces this problem.
(489) The front panel **1826** includes openings or interface ports **1860** that allow the rackmount server **1820** to be coupled to optical fiber cables and/or electrical cables. In some implementations, co-packaged optical modules **1870** can be inserted into the interface ports **1860**, in which the co-packaged optical modules **1870** function as optical/electrical communication interfaces for the data processor **1844**. The co-packaged optical modules have been described earlier in this document.
(490) FIG. **99** includes an upper diagram **1880** that shows a perspective front view of an example of the front panel **1826**, and a lower diagram **1882** that shows a perspective rear view of the front panel **1826**. The lower diagram **1882** shows the data processor **1844** mounted to the back side of the vertically oriented circuit board **1822**. The front panel **1826** includes openings or interface ports **1860** that allow insertion of communication interface modules, such as co-packaged optical modules, that provide interfaces between the data processor **1844** and external optical or electrical cables. The optical and electrical signal paths between the data processor **1844** and the co-packaged optical modules have been previously described in this document.
(491) FIG. **100** is a diagram of a top view of an example of a rackmount server **1890** that includes a vertically oriented circuit board **1822** positioned at a front portion of the rackmount server **1890**. A data processor **1844** is mounted on the circuit board **1822**, and a heat dissipating module **1846** is thermally coupled to the data processor **1844**. The rackmount server **1890** has an active airflow management system that is configured to remove heat from the data processor **1844** during operation. The rackmount server **1890** includes components that are similar to those of the rackmount server **1820** (FIG. **96**) and are not otherwise described here.
(492) In some implementations, the active airflow management system includes an inlet fan **1894** that is positioned at a left side of the heat dissipating module **1846** and oriented to blow inlet air to the right toward the heat dissipating module **1846**. A front opening **1850** allows incoming air to pass to the inlet fan **1894**. The front opening **1850** can be positioned to the left of the inlet fan **1894**. For example, the inlet fan **1894** can have a rotational axis that is at an angle θ relative to the front panel **1826**, in which $\theta \leq 45°$. In some examples, $\theta \leq 25°$. In some examples, $\theta \leq 5°$. In some examples, the circuit board **1822** is substantially parallel to the front panel **1826**, and the rotational axis of the inlet fan **1894** is substantially parallel to the circuit board **1822**. An inlet fan **1894**.

(493) In some implementations, a first baffle or air louver **1892** is provided to guide air from the opening **1850** towards the inlet fan **1894**, and from the inlet fan **1894** towards the heat dissipating module **1846**. A second baffle or air louver **1908** is provided to guide air from the right portion of the heat dissipating module **1846** toward the rear of the rackmount server **1890**. The first and second air louvers **1892**, **1894** can extend from the bottom panel to the top panel in the vertical direction.
(494) An arrow **1902** shows a general direction of airflow from the opening **1850** to the inlet fan **1894**. An arrow **1904** shows a general direction of airflow from the inlet fan **1894** to, and through, a center portion the heat dissipating module **1846**. An arrow **1906** shows a general direction of airflow through, and exiting, the right portion of the heat dissipating module **1846**. The first air louver **1892**, a front portion of the left panel, a front portion of the top panel, a front portion of the bottom panel, the front panel **1826**, the circuit board **1822**, and the second air louver **1908** in combination form a duct that channels the air to flow through the entire heat dissipating module **1846**, or a substantial portion of the heat dissipating module **1846**, thereby increasing the efficiency of heat dissipation from the data processor **1844**.
(495) In this example, the first air louver **1892** includes a left curved section **1896**, a middle straight section **1898**, and a right curved section **1900**. The left curved section **1896** extends from the left side panel to the inlet fan **1894**. The left curved section **1896** directs incoming air to turn from flowing in the front to rear direction to flowing in the left-to-right direction. The middle straight section **1898** is positioned to the rear of the heat dissipating module **1846** and extends from the inlet fan **1894** to beyond the center portion of the heat dissipating module **1846**. The middle straight section **1898** directs the air to flow generally in a left-to-right direction through a substantial portion (e.g., more than half) of the heat dissipating module **1846**. The right curved section **1900** and the second air louver **1908** in combination guide the air to turn from flowing in the left-to-right direction to flowing in a front to rear direction. The designs of the first and second air louvers **1892**, **1908** are selected to optimize the heat dissipation efficiency. The heat dissipating module **1846** can have a design that is different from what is shown in the figure, and the first and second air louvers **1892**, **1908** can also be modified accordingly.
(496) In the example of FIG. **100**, the inlet fan **1894** directs air to flow generally in a direction (in this example, from left to right) that is parallel to the front panel **1826** across the heat dissipating surface of the heat dissipating module **1846**. In some implementations, the front opening can be positioned to the right side of the front panel, and the inlet fan can be positioned to the right side of the heat dissipating module and direct air to flow from right to left across the heat dissipating surface of the heat dissipating module. The first and second air louvers can be modified accordingly to optimize airflow and heat dissipation from the data processor.
(497) Rackmount devices are typically installed in a rack such that the bottom panel is parallel to the horizontal direction, and the front panel has a width and a height in which the width is much larger than the height. For example, the housing of a rackmount device that has a 2 rack unit form factor can have a width of about 482.6 mm (19 inches) and a height of about 88.9 mm (3.5 inches). In some implementations, the rackmount device can be oriented differently, e.g., the housing can be rotated 90° about an axis that is parallel to the front-to-rear direction such that the nominal top and bottom panels become parallel to the vertical direction, and the nominal side panels become parallel to the horizontal direction. In some implementations, the housing can be turned an arbitrary angle θ about an axis that is parallel to the front-to-rear direction such that the nominal bottom panel is at the angle θ relative to the horizontal direction. For rackmount devices that are oriented such that the nominal bottom panel is not parallel to the horizontal direction, the inlet fan(s), the air louvers, and the heat sinks are designed to take into account that hot air rises in the upward direction. The inlet fan(s) is/are positioned at a lower position or lower positions than the heat sink and blow(s) incoming cool air upwards towards the heat sink.
(498) FIGS. **35**A to **37** show examples of optical communications systems **1250**, **1260**, **1270** in which in each system an optical power supply or photon supply provides optical power supply light to photonic integrated circuits hosted in multiple communication devices (e.g., optical transponders), and the optical power supply is external to the communication devices. The optical power supply can have its own housing, electrical power supply, and control circuitry, independent of the housings, electrical power supplies, and control circuitry of the communication devices. This allows the optical power supply to be serviced, repaired, or replaced independent of the communication devices. Redundant optical power supplies can be provided so that a defective external optical power supply can be repaired or replaced without taking the communication devices off-line. The external optical power supply can be placed at a convenient centralized location with a dedicated temperature environment (as opposed to being crammed inside the communication devices, which may have a high temperature). The external optical power supply can be built more efficiently than individual power supply units, as certain common parts such as monitoring circuitry and thermal control units can be amortized over many more communication devices. The following describes implementations of the fiber cabling for remote optical power supplies.
(499) FIG. **79** is a system functional block diagram of an example of an optical communication system **1280** that includes a first communication transponder **1282** and a second communication transponder **1284**. Each of the first and second communication transponders **1282**, **1284** can include one or more co-packaged optical modules described above. Each communication transponder can include, e.g., one or more data processors, such as network switches, central processing units, graphics processor units, tensor processing units, digital signal processors, and/or other application specific integrated circuits (ASICs). In this example, the first communication transponder **1282** sends optical signals to, and receives optical signals from, the second communication transponder **1284** through a first optical communication link **1290**. The one or more data processors in each communication transponder **1282**, **1284** process the data received from the

first optical communication link **1290** and outputs processed data to the first optical communication link **1290**. The optical communication system **1280** can be expanded to include additional communication transponders. The optical communication system **1280** can also be expanded to include additional communication between two or more external photon supplies, which can coordinate aspects of the supplied light, such as the respectively emitted wavelengths or the relative timing of the respectively emitted optical pulses.

(500) A first external photon supply **1286** provides optical power supply light to the first communication transponder **1282** through a first optical power supply link **1292**, and a second external photon supply **1288** provides optical power supply light to the second communication transponder **1284** through a second optical power supply link **1294**. In one example embodiment, the first external photon supply **1286** and the second external photon supply **1288** provide continuous wave laser light at the same optical wavelength. In another example embodiment, the first external photon supply **1286** and the second external photon supply **1288** provide continuous wave laser light at different optical wavelengths. In yet another example embodiment, the first external photon supply **1286** provides a first sequence of optical frame templates to the first communication transponder **1282**, and the second external photon supply **1288** provides a second sequence of optical frame templates to the second communication transponder **1284**. For example, as described in U.S. Pat. No. 11,153,670, each of the optical frame templates can include a respective frame header and a respective frame body, and the frame body includes a respective optical pulse train. The first communication transponder **1282** receives the first sequence of optical frame templates from the first external photon supply **1286**, loads data into the respective frame bodies to convert the first sequence of optical frame templates into a first sequence of loaded optical frames that are transmitted through the first optical communication link **1290** to the second communication transponder **1284**. Similarly, the second communication transponder **1284** receives the second sequence of optical frame templates from the second external photon supply **1288**, loads data into the respective frame bodies to convert the second sequence of optical frame templates into a second sequence of loaded optical frames that are transmitted through the first optical communication link **1290** to the first communication transponder **1282**.

(501) FIG. **80**A is a diagram of an example of an optical communication system **1300** that includes a first switch box **1302** and a second switch box **1304**. Each of the switch boxes **1302**, **1304** can include one or more data processors, such as network switches. The first and second switch boxes **1302**, **1304** can be separated by a distance greater than, e.g., 1 foot, 3 feet, 10 feet, 100 feet, or 1000 feet. The figure shows a diagram of a front panel **1306** of the first switch box **1302** and a front panel **1308** of the second switch box **1304**. In this example, the first switch box **1302** includes a vertical ASIC mount grid structure **1310**, similar to the grid structure **870** of FIG. **43**. A co-packaged optical module **1312** is attached to a receptor of the grid structure **1310**. The second switch box **1304** includes a vertical ASIC mount grid structure **1314**, similar to the grid structure **870** of FIG. **43**. A co-packaged optical module **1316** is attached to a receptor of the grid structure **1314**. The first co-packaged optical module **1312** communicates with the second co-packaged optical module **1316** through an optical fiber bundle **1318** that includes multiple optical fibers. Optional fiber connectors **1320** can be used along the optical fiber bundle **1318**, in which shorter sections of optical fiber bundles are connected by the fiber connectors **1320**.

(502) In some implementations, each co-packaged optical module (e.g., **1312**, **1316**) includes a photonic integrated circuit configured to convert input optical signals to input electrical signals that are provided to a data processor, and convert output electrical signals from the data processor to output optical signals. The co-packaged optical module can include an electronic integrated circuit configured to process the input electrical signals from the photonic integrated circuit before the input electrical signals are transmitted to the data processor, and to process the output electrical signals from the data processor before the output electrical signals are transmitted to the photonic integrated circuit. In some implementations, the electronic integrated circuit can include a plurality of serializers/deserializers configured to process the input electrical signals from the photonic integrated circuit, and to process the output electrical signals transmitted to the photonic integrated circuit. The electronic integrated circuit can include a first serializers/deserializers module having multiple serializer units and deserializer units, in which the first serializers/deserializers module is configured to generate a plurality of sets of first parallel electrical signals based on a plurality of first serial electrical signals provided by the photonic integrated circuit, and condition the electrical signals, in which each set of first parallel electrical signals is generated based on a corresponding first serial electrical signal. The electronic integrated circuit can include a second serializers/deserializers module having multiple serializer units and deserializer units, in which the second serializers/deserializers module is configured to generate a plurality of second serial electrical signals based on the plurality of sets of first parallel electrical signals, and each second serial electrical signal is generated based on a corresponding set of first parallel electrical signals. The plurality of second serial electrical signals can be transmitted toward the data processor.

(503) The first switch box **1302** includes an external optical power supply **1322** (i.e., external to the co-packaged optical module) that provides optical power supply light through an optical connector array **1324**. In this example, the optical power supply **1322** is located internal of the housing of the switch box **1302**. Optical fibers **1326** are optically coupled to an optical connector **1328** (of the optical connector array **1324**) and the co-packaged optical module **1312**. The optical power supply **1322** sends optical power supply light through the optical connector **1328** and the optical fibers **1326** to the co-packaged optical module **1312**. For example, the co-packaged optical module **1312** includes a photonic integrated circuit that modulates the power supply light based on data provided by a data processor to generate a modulated optical signal, and transmits the modulated optical signal to the co-packaged optical module **1316** through one of the optical

fibers in the fiber bundle **1318**.
(504) In some examples, the optical power supply **1322** is configured to provide optical power supply light to the co-packaged optical module **1312** through multiple links that have built-in redundancy in case of malfunction in some of the optical power supply modules. For example, the co-packaged optical module **1312** can be designed to receive N channels of optical power supply light (e.g., N1 continuous wave light signals at the same or at different optical wavelengths, or N1 sequences of optical frame templates), N1 being a positive integer, from the optical power supply **1322**. The optical power supply **1322** provides N1+M1 channels of optical power supply light to the co-packaged optical module **1312**, in which M1 channels of optical power supply light are used for backup in case of failure of one or more of the N1 channels of optical power supply light, M1 being a positive integer.
(505) The second switch box **1304** receives optical power supply light from a co-located optical power supply **1330**, which is, e.g., external to the second switch box **1304** and located near the second switch box **1304**, e.g., in the same rack as the second switch box **1304** in a data center. The optical power supply **1330** includes an array of optical connectors **1332**. Optical fibers **1334** are optically coupled to an optical connector **1336** (of the optical connectors **1332**) and the co-packaged optical module **1316**. The optical power supply **1330** sends optical power supply light through the optical connector **1336** and the optical fibers **1334** to the co-packaged optical module **1316**. For example, the co-packaged optical module **1316** includes a photonic integrated circuit that modulates the power supply light based on data provided by a data processor to generate a modulated optical signal, and transmits the modulated optical signal to the co-packaged optical module **1312** through one of the optical fibers in the fiber bundle **1318**.
(506) In some examples, the optical power supply **1330** is configured to provide optical power supply light to the co-packaged optical module **1316** through multiple links that have built-in redundancy in case of malfunction in some of the optical power supply modules. For example, the co-packaged optical module **1316** can be designed to receive N2 channels of optical power supply light (e.g., N2 continuous wave light signals at the same or at different optical wavelengths, or N2 sequences of optical frame templates), N2 being a positive integer, from the optical power supply **1322**. The optical power supply **1322** provides N2+M2 channels of optical power supply light to the co-packaged optical module **1312**, in which M2 channels of optical power supply light are used for backup in case of failure of one or more of the N2 channels of optical power supply light, M2 being a positive integer.
(507) FIG. **80**B is a diagram of an example of an optical cable assembly **1340** that can be used to enable the first co-packaged optical module **1312** to receive optical power supply light from the first optical power supply **1322**, enable the second co-packaged optical module **1316** to receive optical power supply light from the second optical power supply **1330**, and enable the first co-packaged optical module **1312** to communicate with the second co-packaged optical module **1316**. FIG. **80**C is an enlarged diagram of the optical cable assembly **1340** without some of the reference numbers to enhance clarity of illustration.
(508) The optical cable assembly **1340** includes a first optical fiber connector **1342**, a second optical fiber connector **1344**, a third optical fiber connector **1346**, and a fourth optical fiber connector **1348**. The first optical fiber connector **1342** is designed and configured to be optically coupled to the first co-packaged optical module **1312**. For example, the first optical fiber connector **1342** can be configured to mate with a connector part of the first co-packaged optical module **1312**, or a connector part that is optically coupled to the first co-packaged optical module **1312**. The first, second, third, and fourth optical fiber connectors **1342**, **1344**, **1346**, **1348** can comply with an industry standard that defines the specifications for optical fiber interconnection cables that transmit data and control signals, and optical power supply light.
(509) The first optical fiber connector **1342** includes optical power supply (PS) fiber ports, transmitter (TX) fiber ports, and receiver (RX) fiber ports. The optical power supply fiber ports provide optical power supply light to the co-packaged optical module **1312**. The transmitter fiber ports allow the co-packaged optical module **1312** to transmit output optical signals (e.g., data and/or control signals), and the receiver fiber ports allow the co-packaged optical module **1312** to receive input optical signals (e.g., data and/or control signals). Examples of the arrangement of the optical power supply fiber ports, the transmitter ports, and the receiver ports in the first optical fiber connector **1342** are shown in FIGS. **80**D, **89**, and **90**.
(510) FIG. **80**D shows an enlarged upper portion of the diagram of FIG. **80**B, with the addition of an example of a mapping of fiber ports **1750** of the first optical fiber connector **1342** and a mapping of fiber ports **1752** of the third optical fiber connector **1346**. FIG. **80**F shows an enlarged view of the diagram of FIG. **80**D. The power supply power ports are labeled ‘P’, the transmitter fiber ports are labeled ‘T’, and the receiver fiber ports are labeled ‘R’. Only some of the fiber ports are labeled in the figure. The mapping of fiber ports **1750** shows the positions of the transmitter fiber ports (e.g., **1753**), receiver fiber ports (e.g., **1755**), and power supply fiber ports (e.g., **1751**) of the first optical fiber connector **1342** when viewed in the direction **1754** into the first optical fiber connector **1342**. The mapping of fiber ports **1752** shows the positions of the power supply fiber ports (e.g., **1757**) of the third optical fiber connector **1346** when viewed in the direction **1756** into the third optical fiber connector **1346**.
(511) The second optical fiber connector **1344** is designed and configured to be optically coupled to the second co-packaged optical module **1316**. The second optical fiber connector **1344** includes optical power supply fiber ports, transmitter fiber ports, and receiver fiber ports. The optical power supply fiber ports provide optical power supply light to the co-packaged optical module **1316**. The transmitter fiber ports allow the co-packaged optical module **1316** to transmit output optical signals, and the receiver fiber ports allow the co-packaged optical module **1316** to receive input optical

signals. Examples of the arrangement of the optical power supply fiber ports, the transmitter ports, and the receiver ports in the second optical fiber connector **1344** are shown in FIGS. **80**E, **89**, and **90**.

(512) FIG. **80**E shows an enlarged lower portion of the diagram of FIG. **80**B, with the addition of an example of a mapping of fiber ports **1760** of the second optical fiber connector **1344** and a mapping of fiber ports **1762** of the fourth optical fiber connector **1348**. FIG. **80**G shows an enlarged view of the diagram of FIG. **80**E. The power supply power ports are labeled ‘P’, the transmitter fiber ports are labeled ‘T’, and the receiver fiber ports are labeled ‘R’. Only some of the fiber ports are labeled in the figure. The mapping of fiber ports **1760** shows the positions of the transmitter fiber ports (e.g., **1763**), receiver fiber ports (e.g., **1765**), and power supply fiber ports (e.g., **1761**) of the second optical fiber connector **1344** when viewed in the direction **1764** into the second optical fiber connector **1344**. The mapping of fiber ports **1762** shows the positions of the power supply fiber ports (e.g., **1767**) of the fourth optical fiber connector **1348** when viewed in the direction **1766** into the fourth fiber connector **1348**.

(513) The third optical connector **1346** is designed and configured to be optically coupled to the power supply **1322**. The third optical connector **1346** includes optical power supply fiber ports (e.g., **1757**) through which the power supply **1322** can output the optical power supply light. The fourth optical connector **1348** is designed and configured to be optically coupled to the power supply **1330**. The fourth optical connector **1348** includes optical power supply fiber ports (e.g., **1762**) through which the power supply **1322** can output the optical power supply light.

(514) In some implementations, the optical power supply fiber ports, the transmitter fiber ports, and the receiver fiber ports in the first and second optical fiber connectors **1342**, **1344** are designed to be independent of the communication devices, i.e., the first optical fiber connector **1342** can be optically coupled to the second switch box **1304**, and the second optical fiber connector **1344** can be optically coupled to the first switch box **1302** without any re-mapping of the fiber ports. Similarly, the optical power supply fiber ports in the third and fourth optical fiber connectors **1346**, **1348** are designed to be independent of the optical power supplies, i.e., if the first optical fiber connector **1342** is optically coupled to the second switch box **1304**, the third optical fiber connector **1346** can be optically coupled to the second optical power supply **1330**. If the second optical fiber connector **1344** is optically coupled to the first switch box **1302**, the fourth optical fiber connector **1348** can be optically coupled to the first optical power supply **1322**.

(515) The optical cable assembly **1340** includes a first optical fiber guide module **1350** and a second optical fiber guide module **1352**. The optical fiber guide module depending on context is also referred to as an optical fiber coupler or splitter because the optical fiber guide module combines multiple bundles of fibers into one bundle of fibers, or separates one bundle of fibers into multiple bundles of fibers. The first optical fiber guide module **1350** includes a first port **1354**, a second port **1356**, and a third port **1358**. The second optical fiber guide module **1352** includes a first port **1360**, a second port **1362**, and a third port **1364**. The fiber bundle **1318** extends from the first optical fiber connector **1342** to the second optical fiber connector **1344** through the first port **1354** and the second port **1356** of the first optical fiber guide module **1350** and the second port **1362** and the first port **1360** of the second optical fiber guide module **1352**. The optical fibers **1326** extend from the third optical fiber connector **1346** to the first optical fiber connector **1342** through the third port **1358** and the first port **1354** of the first optical fiber guide module **1350**. The optical fibers **1334** extend from the fourth optical fiber connector **1348** to the second optical fiber connector **1344** through the third port **1364** and the first port **1360** of the second optical fiber guide module **1352**.

(516) A portion (or section) of the optical fibers **1318** and a portion of the optical fibers **1326** extend from the first port **1354** of the first optical fiber guide module **1350** to the first optical fiber connector **1342**. A portion of the optical fibers **1318** extend from the second port **1356** of the first optical fiber guide module **1350** to the second port **1362** of the second optical fiber guide module **1352**, with optional optical connectors (e.g., **1320**) along the paths of the optical fibers **1318**. A portion of the optical fibers **1326** extend from the third port **1358** of the first optical fiber connector **1350** to the third optical fiber connector **1346**. A portion of the optical fibers **1334** extend from the third port **1364** of the second optical fiber connector **1352** to the fourth optical fiber connector **1348**.

(517) The first optical fiber guide module **1350** is designed to restrict bending of the optical fibers such that the bending radius of any optical fiber in the first optical fiber guide module **1350** is greater than the minimum bending radius specified by the optical fiber manufacturer to avoid excess optical light loss or damage to the optical fiber. For example, the minimum bend radii can be 2 cm, 1 cm, 5 mm, or 2.5 mm. Other bend radii are also possible. For example, the fibers **1318** and the fibers **1326** extend outward from the first port **1354** along a first direction, the fibers **1318** extend outward from the second port **1356** along a second direction, and the fibers **1326** extend outward from the third port **1358** along a third direction. A first angle is between the first and second directions, a second angle is between the first and third directions, and a third angle is between the second and third directions. The first optical fiber guide module **1350** can be designed to limit the bending of optical fibers so that each of the first, second, and third angles is in a range from, e.g., 30° to 180°.

(518) For example, the portion of the optical fibers **1318** and the portion of the optical fibers **1326** between the first optical fiber connector **1342** and the first port **1354** of the first optical fiber guide module **1350** can be surrounded and protected by a first common sheath **1366**. The optical fibers **1318** between the second port **1356** of the first optical fiber guide module **1350** and the second port **1362** of the second optical fiber guide module **1352** can be surrounded and protected by a second common sheath **1368**. The portion of the optical fibers **1318** and the portion of the optical fibers **1334** between the second optical fiber connector **1344** and the first port **1360** of the second optical fiber guide module **1352** can be surrounded and protected by a third common sheath **1369**. The optical fibers **1326** between the third optical fiber

connector **1346** and the third port **1358** of the first optical fiber guide module **1350** can be surrounded and protected by a fourth common sheath **1367**. The optical fibers **1334** between the fourth optical fiber connector **1348** and the third port **1364** of the second optical fiber guide module **1352** can be surrounded and protected by a fifth common sheath **1370**. Each of the common sheaths can be laterally flexible and/or laterally stretchable, as described in, e.g., U.S. patent application Ser. No. 16/822,103.

(519) One or more optical cable assemblies **1340** (FIGS. **80**B, **80**C) and other optical cable assemblies (e.g., **1400** of FIG. **82**B, **82**C, **1490** of FIG. **84**B, **84**C) described in this document can be used to optically connect switch boxes that are configured differently compared to the switch boxes **1302**, **1304** shown in FIG. **80**A, in which the switch boxes receive optical power supply light from one or more external optical power supplies. For example, in some implementations, the optical cable assembly **1340** can be attached to a fiber-optic array connector mounted on the outside of the front panel of an optical switch, and another fiber-optic cable then connects the inside of the fiber connector to a co-packaged optical module that is mounted on a circuit board positioned inside the housing of the switch box. The co-packaged optical module (which includes, e.g., a photonic integrated circuit, optical-to-electrical converters, such as photodetectors, and electrical-to-optical converters, such as laser diodes) can be co-packaged with a switch ASIC and mounted on a circuit board that can be vertically or horizontally oriented. For example, in some implementations, the front panel is mounted on hinges and a vertical ASIC mount is recessed behind it. See the examples in FIGS. **77**A, **77**B, and **78**. The optical cable assembly **1340** provides optical paths for communication between the switch boxes, and optical paths for transmitting power supply light from one or more external optical power supplies to the switch boxes. The switch boxes can have any of a variety of configurations regarding how the power supply light and the data and/or control signals from the optical fiber connectors are transmitted to or received from the photonic integrated circuits, and how the signals are transmitted between the photonic integrated circuits and the data processors.

(520) One or more optical cable assemblies **1340** and other optical cable assemblies (e.g., **1400** of FIG. **82**B, **82**C, **1490** of FIG. **84**B, **84**C) described in this document can be used to optically connect computing devices other than switch boxes. For example, the computing devices can be server computers that provide a variety of services, such as cloud computing, database processing, audio/video hosting and streaming, electronic mail, data storage, web hosting, social networking, supercomputing, scientific research computing, healthcare data processing, financial transaction processing, logistics management, weather forecasting, or simulation, to list a few examples. The optical power light required by the optoelectronic modules of the computing devices can be provided using one or more external optical power supplies. For example, in some implementations, one or more external optical power supplies that are centrally managed can be configured to provide the optical power supply light for hundreds or thousands of server computers in a data center, and the one or more optical power supplies and the server computers can be optically connected using the optical cable assemblies (e.g., **1340**, **1400**, **1490**) described in this document and variations of the optical cable assemblies using the principles described in this document.

(521) FIG. **81** is a system functional block diagram of an example of an optical communication system **1380** that includes a first communication transponder **1282** and a second communication transponder **1284**, similar to those in FIG. **79**. The first communication transponder **1282** sends optical signals to, and receives optical signals from, the second communication transponder **1284** through a first optical communication link **1290**. The optical communication system **1380** can be expanded to include additional communication transponders.

(522) An external photon supply **1382** provides optical power supply light to the first communication transponder **1282** through a first optical power supply link **1384**, and provides optical power supply light to the second communication transponder **1284** through a second optical power supply link **1386**. In one example, the external photon supply **1282** provides continuous wave light to the first communication transponder **1282** and to the second communication transponder **1284**. In one example, the continuous wave light can be at the same optical wavelength. In another example, the continuous wave light can be at different optical wavelengths. In yet another example, the external photon supply **1282** provides a first sequence of optical frame templates to the first communication transponder **1282**, and provides a second sequence of optical frame templates to the second communication transponder **1284**. Each of the optical frame templates can include a respective frame header and a respective frame body, and the frame body includes a respective optical pulse train. The first communication transponder **1282** receives the first sequence of optical frame templates from the external photon supply **1382**, loads data into the respective frame bodies to convert the first sequence of optical frame templates into a first sequence of loaded optical frames that are transmitted through the first optical communication link **1290** to the second communication transponder **1284**. Similarly, the second communication transponder **1284** receives the second sequence of optical frame templates from the external photon supply **1382**, loads data into the respective frame bodies to convert the second sequence of optical frame templates into a second sequence of loaded optical frames that are transmitted through the first optical communication link **1290** to the first communication transponder **1282**.

(523) FIG. **82**A is a diagram of an example of an optical communication system **1390** that includes a first switch box **1302** and a second switch box **1304**, similar to those in FIG. **80**A. FIG. **82**F shows an enlarged view of a portion of the diagram of FIG. **82**A, including the switch box **1302** and a portion of the fiber bundle **1318**. The first switch box **1302** includes a vertical ASIC mount grid structure **1310**, and a co-packaged optical module **1312** is attached to a receptor of the grid structure **1310**. The second switch box **1304** includes a vertical ASIC mount grid structure **1314**, and a co-packaged optical module **1316** is attached to a receptor of the grid structure **1314**. The first co-packaged optical module **1312** communicates with the second co-packaged optical module **1316** through an optical fiber bundle **1318** that includes

multiple optical fibers.
(524) As discussed above in connection with FIGS. **80**A to **80**E, the first and second switch boxes **1302**, **1304** can have other configurations. For example, horizontally mounted ASICs can be used. A fiber-optic array connector attached to a front panel can be used to optically connect the optical cable assembly **1340** to another fiber-optic cable that connects to a co-packaged optical module mounted on a circuit board inside the switch box. The front panel can be mounted on hinges and a vertical ASIC mount can be recessed behind it. The switch boxes can be replaced by other types of server computers.
(525) In an example embodiment, the first switch box **1302** includes an external optical power supply **1322** that provides optical power supply light to both the co-packaged optical module **1312** in the first switch box **1302** and the co-packaged optical module **1316** in the second switch box **1304**. In another example embodiment, the optical power supply can be located outside the switch box **1302** (cf. **1330**, FIG. **80**A). The optical power supply **1322** provides the optical power supply light through an optical connector array **1324**. Optical fibers **1392** are optically coupled to an optical connector **1396** and the co-packaged optical module **1312**. The optical power supply **1322** sends optical power supply light through the optical connector **1396** and the optical fibers **1392** to the co-packaged optical module **1312** in the first switch box **1302**. Optical fibers **1394** are optically coupled to the optical connector **1396** and the co-packaged optical module **1316**. The optical power supply **1322** sends optical power supply light through the optical connector **1396** and the optical fibers **1394** to the co-packaged optical module **1316** in the second switch box **1304**.
(526) FIG. **82**B shows an example of an optical cable assembly **1400** that can be used to enable the first co-packaged optical module **1312** to receive optical power supply light from the optical power supply **1322**, enable the second co-packaged optical module **1316** to receive optical power supply light from the optical power supply **1322**, and enable the first co-packaged optical module **1312** to communicate with the second co-packaged optical module **1316**. FIG. **82**C is an enlarged diagram of the optical cable assembly **1400** without some of the reference numbers to enhance clarity of illustration.
(527) The optical cable assembly **1400** includes a first optical fiber connector **1402**, a second optical fiber connector **1404**, and a third optical fiber connector **1406**. The first optical fiber connector **1402** is similar to the first optical fiber connector **1342** of FIGS. **80**B, **80**C, **80**D, and is designed and configured to be optically coupled to the first co-packaged optical module **1312**. The second optical fiber connector **1404** is similar to the second optical fiber connector **1344** of FIGS. **80**B, **80**C, **80**E, and is designed and configured to be optically coupled to the second co-packaged optical module **1316**. The third optical connector **1406** is designed and configured to be optically coupled to the power supply **1322**. The third optical connector **1406** includes first optical power supply fiber ports (e.g., **1770**, FIG. **82**D) and second optical power supply fiber ports (e.g., **1772**). The power supply **1322** outputs optical power supply light through the first optical power supply fiber ports to the optical fibers **1392**, and outputs optical power supply light through the second optical power supply fiber ports to the optical fibers **1394**. The first, second, and third optical fiber connectors **1402**, **1404**, **1406** can comply with an industry standard that defines the specifications for optical fiber interconnection cables that transmit data and control signals, and optical power supply light.
(528) FIG. **82**D shows an enlarged upper portion of the diagram of FIG. **82**B, with the addition of an example of a mapping of fiber ports **1774** of the first optical fiber connector **1402** and a mapping of fiber ports **1776** of the third optical fiber connector **1406**. FIG. **82**G shows an enlarged view of the diagram of FIG. **82**D. The power supply power ports are labeled ‘P’, the transmitter fiber ports are labeled ‘T’, and the receiver fiber ports are labeled ‘R’. Only some of the fiber ports are labeled in the figure. The mapping of fiber ports **1774** shows the positions of the transmitter fiber ports (e.g., **1778**), receiver fiber ports (e.g., **1780**), and power supply fiber ports (e.g., **1782**) of the first optical fiber connector **1402** when viewed in the direction **1784** into the first optical fiber connector **1402**. The mapping of fiber ports **1776** shows the positions of the power supply fiber ports (e.g., **1770**, **1772**) of the third optical fiber connector **1406** when viewed in the direction **1786** into the third optical fiber connector **1406**. In this example, the third optical fiber connector **1406** includes 8 optical power supply fiber ports.
(529) In some examples, optical connector array **1324** of the optical power supply **1322** can include a first type of optical connectors that accept optical fiber connectors having 4 optical power supply fiber ports, as in the example of FIG. **80**D, and a second type of optical connectors that accept optical fiber connectors having 8 optical power supply fiber ports, as in the example of FIG. **82**D. In some examples, if the optical connector array **1324** of the optical power supply **1322** only accepts optical fiber connectors having 4 optical power supply fiber ports, then a converter cable can be used to convert the third optical fiber connector **1406** of FIG. **82**D to two optical fiber connectors, each having 4 optical power supply fiber ports, that is compatible with the optical connector array **1324**.
(530) FIG. **82**E shows an enlarged lower portion of the diagram of FIG. **82**B, with the addition of an example of a mapping of fiber ports **1790** of the second optical fiber connector **1404**. FIG. **82**H shows an enlarged view of the diagram of FIG. **82**E. The power supply power ports are labeled ‘P’, the transmitter fiber ports are labeled ‘T’, and the receiver fiber ports are labeled ‘R’. Only some of the fiber ports are labeled in the figure. The mapping of fiber ports **1790** shows the positions of the transmitter fiber ports (e.g., **1792**), receiver fiber ports (e.g., **1794**), and power supply fiber ports (e.g., **1796**) of the second optical fiber connector **1404** when viewed in the direction **1798** into the second optical fiber connector **1404**.
(531) The port mappings of the optical fiber connectors shown in FIGS. **80**D, **80**E, **82**D, and **82**E are merely examples. Each optical fiber connector can include a greater number or a smaller number of transmitter fiber ports, a greater number

or a smaller number of receiver fiber ports, and a greater number or a smaller number of optical power supply fiber ports, as compared to those shown in FIGS. **80**D, **80**E, **82**D, and **82**E. The arrangement of the relative positions of the transmitter, receiver, and optical power supply fiber ports can also be different from those shown in FIGS. **80**D, **80**E, **82**D, and **82**E.

(532) The optical cable assembly **1400** includes an optical fiber guide module **1408**, which includes a first port **1410**, a second port **1412**, and a third port **1414**. The optical fiber guide module **1408** depending on context is also referred as an optical fiber coupler (for combining multiple bundles of optical fibers into one bundle of optical fiber) or an optical fiber splitter (for separating a bundle of optical fibers into multiple bundles of optical fibers). The fiber bundle **1318** extends from the first optical fiber connector **1402** to the second optical fiber connector **1404** through the first port **1410** and the second port **1412** of the optical fiber guide module **1408**. The optical fibers **1392** extend from the third optical fiber connector **1406** to the first optical fiber connector **1402** through the third port **1414** and the first port **1410** of the optical fiber guide module **1408**. The optical fibers **1394** extend from the third optical fiber connector **1406** to the second optical fiber connector **1404** through the third port **1414** and the second port **1412** of the optical fiber guide module **1408**.

(533) A portion of the optical fibers **1318** and a portion of the optical fibers **1392** extend from the first port **1410** of the optical fiber guide module **1408** to the first optical fiber connector **1402**. A portion of the optical fibers **1318** and a portion of the optical fibers **1394** extend from the second port **1412** of the optical fiber guide module **1408** to the second optical fiber connector **1404**. A portion of the optical fibers **1394** extend from the third port **1414** of the optical fiber connector **1408** to the third optical fiber connector **1406**.

(534) The optical fiber guide module **1408** is designed to restrict bending of the optical fibers such that the radius of curvature of any optical fiber in the optical fiber guide module **1408** is greater than the minimum radius of curvature specified by the optical fiber manufacturer to avoid excess optical light loss or damage to the optical fiber. For example, the optical fibers **1318** and the optical fibers **1392** extend outward from the first port **1410** along a first direction, the optical fibers **1318** and the optical fibers **1394** extend outward from the second port **1412** along a second direction, and the optical fibers **1392** and the optical fibers **1394** extend outward from the third port **1414** along a third direction. A first angle is between the first and second directions, a second angle is between the first and third directions, and a third angle is between the second and third directions. The optical fiber guide module **1408** is designed to limit the bending of optical fibers so that each of the first, second, and third angles is in a range from, e.g., 30° to 180°.

(535) For example, the portion of the optical fibers **1318** and the portion of the optical fibers **1392** between the first optical fiber connector **1402** and the first port **1410** of the optical fiber guide module **1408** can be surrounded and protected by a first common sheath **1416**. The optical fibers **1318** and the optical fibers **1394** between the second optical fiber connector **1404** and the second port **1412** of the optical fiber guide module **1408** can be surrounded and protected by a second common sheath **1418**. The optical fibers **1392** and the optical fibers **1394** between the third optical fiber connector **1406** and the third port **1414** of the optical fiber guide module **1408** can be surrounded and protected by a third common sheath **1420**. Each of the common sheaths can be laterally flexible and/or laterally stretchable.

(536) FIG. **83** is a system functional block diagram of an example of an optical communication system **1430** that includes a first communication transponder **1432**, a second communication transponder **1434**, a third communication transponder **1436**, and a fourth communication transponder **1438**. Each of the communication transponders **1432**, **1434**, **1436**, **1438** can be similar to the communication transponders **1282**, **1284** of FIG. **79**. The first communication transponder **1432** communicates with the second communication transponder **1434** through a first optical link **1440**. The first communication transponder **1432** communicates with the third communication transponder **1436** through a second optical link **1442**. The first communication transponder **1432** communicates with the fourth communication transponder **1438** through a third optical link **1444**.

(537) An external photon supply **1446** provides optical power supply light to the first communication transponder **1432** through a first optical power supply link **1448**, provides optical power supply light to the second communication transponder **1434** through a second optical power supply link **1450**, provides optical power supply light to the third communication transponder **1436** through a third optical power supply link **1452**, and provides optical power supply light to the fourth communication transponder **1438** through a fourth optical power supply link **1454**.

(538) FIG. **84**A is a diagram of an example of an optical communication system **1460** that includes a first switch box **1462** and a remote server array **1470** that includes a second switch box **1464**, a third switch box **1466**, and a fourth switch box **1468**. The first switch box **1462** includes a vertical ASIC mount grid structure **1310**, and a co-packaged optical module **1312** is attached to a receptor of the grid structure **1310**. The second switch box **1464** includes a co-packaged optical module **1472**, the third switch box **1466** includes a co-packaged optical module **1474**, and the third switch box **1468** includes a co-packaged optical module **1476**. The first co-packaged optical module **1312** communicates with the co-packaged optical modules **1472**, **1474**, **1476** through an optical fiber bundle **1478** that later branches out to the co-packaged optical modules **1472**, **1474**, **1476**.

(539) In one example embodiment, the first switch box **1462** includes an external optical power supply **1322** that provides optical power supply light through an optical connector array **1324**. In another example embodiment, the optical power supply can be located external to switch box **1462** (cf. **1330**, FIG. **80**A). Optical fibers **1480** are optically coupled to an optical connector **1482**, and the optical power supply **1322** sends optical power supply light through the optical connector **1482** and the optical fibers **1480** to the co-packaged optical modules **1312**, **1472**, **1474**, **1476**.

(540) FIG. **84**B shows an example of an optical cable assembly **1490** that can be used to enable the optical power supply

1322 to provide optical power supply light to the co-packaged optical modules **1312**, **1472**, **1474**, **1476**, and enable the co-packaged optical module **1312** to communicate with the co-packaged optical modules **1472**, **1474**, **1476**. The optical cable assembly **1490** includes a first optical fiber connector **1492**, a second optical fiber connector **1494**, a third optical fiber connector **1496**, a fourth optical fiber connector **1498**, and a fifth optical fiber connector **1500**. The first optical fiber connector **1492** is configured to be optically coupled to the co-packaged optical module **1312**. The second optical fiber connector **1494** is configured to be optically coupled to the co-packaged optical module **1472**. The third optical fiber connector **1496** is configured to be optically coupled to the co-packaged optical module **1474**. The fourth optical fiber connector **1498** is configured to be optically coupled to the co-packaged optical module **1476**. The fifth optical fiber connector **1500** is configured to be optically coupled to the optical power supply **1322**. FIG. **84**C is an enlarged diagram of the optical cable assembly **1490**.

(541) Optical fibers that are optically coupled to the optical fiber connectors **1500** and **1492** enable the optical power supply **1322** to provide the optical power supply light to the co-packaged optical module **1312**. Optical fibers that are optically coupled to the optical fiber connectors **1500** and **1494** enable the optical power supply **1322** to provide the optical power supply light to the co-packaged optical module **1472**. Optical fibers that are optically coupled to the optical fiber connectors **1500** and **1496** enable the optical power supply **1322** to provide the optical power supply light to the co-packaged optical module **1474**. Optical fibers that are optically coupled to the optical fiber connectors **1500** and **1498** enable the optical power supply **1322** to provide the optical power supply light to the co-packaged optical module **1476**.

(542) Optical fiber guide modules **1502**, **1504**, **1506**, and common sheaths are provided to organize the optical fibers so that they can be easily deployed and managed. The optical fiber guide module **1502** is similar to the optical fiber guide module **1408** of FIG. **82**B. The optical fiber guide modules **1504**, **1506** are similar to the optical fiber guide module **1350** of FIG. **80**B. The common sheaths gather the optical fibers in a bundle so that they can be more easily handled, and the optical fiber guide modules guide the optical fibers so that they extend in various directions toward the devices that need to be optically coupled by the optical cable assembly **1490**. The optical fiber guide modules restrict bending of the optical fibers such that the bending radiuses are greater than minimum values specified by the optical fiber manufacturers to prevent excess optical light loss or damage to the optical fibers.

(543) The optical fibers **1480** that extend from the include optical fibers that extend from the optical **1482** are surrounded and protected by a common sheath **1508**. At the optical fiber guide module **1502**, the optical fibers **1480** separate into a first group of optical fibers **1510** and a second group of optical fibers **1512**. The first group of optical fibers **1510** extend to the first optical fiber connector **1492**. The second group of optical fibers **1512** extend toward the optical fiber guide modules **1504**, **1506**, which together function as a 1:3 splitter that separates the optical fibers **1512** into a third group of optical fibers **1514**, a fourth group of optical fibers **1516**, and a fifth group of optical fibers **1518**. The group of optical fibers **1514** extend to the optical fiber connector **1494**, the group of optical fibers **1516** extend to the optical fiber connector **1496**, and the group of optical fibers **1518** extend to the optical fiber connector **1498**. In some examples, instead of using two 1:2 split optical fiber guide modules **1504**, **1506**, it is also possible to use a 1:3 split optical fiber guide module that has four ports, e.g., one input port and three output ports. In general, separating the optical fibers in a 1:N split (N being an integer greater than 2) can occur in one step or multiple steps.

(544) FIG. **85** is a diagram of an example of a data processing system (e.g., data center) **1520** that includes N servers **1522** spread across K racks **1524**. In this example, there are 6 racks **1524**, and each rack **1524** includes 15 servers **1522**. Each server **1522** directly communicates with a tier 1 switch **1526**. The left portion of the figure shows an enlarged view of a portion **1528** of the system **1520**. A server **1522***a* directly communicates with a tier 1 switch **1526***a* through a communication link **1530***a*. Similarly, servers **1522***b*, **1522***c* directly communicate with the tier 1 switch **1526***a* through communication links **1530***b*, **1530***c*, respectively. The server **1522***a* directly communicates with a tier 1 switch **1526***b* through a communication link **1532***a*. Similarly, servers **1522***b*, **1522***c* directly communicate with the tier 1 switch **1526***b* through communication links **1532***b*, **1532***c*, respectively. Each communication link can include a pair of optical fibers to allow bi-directional communication. The system **1520** bypasses the conventional top-of-rack switch and can have the advantage of higher data throughput. The system **1520** includes a point-to-point connection between every server **1522** and every tier 1 switch **1526**. In this example, there are 4 tier 1 switches **1526**, and 4 fiber pairs are used per server **1522** for communicating with the tier 1 switches **1526**. Each tier-1 switch **1526** is connected to N servers, so there are N fiber pairs connected to each tier-1 switch **1526**.

(545) Referring to FIG. **86**, in some implementations, a data processing system (e.g., data center) **1540** includes tier-1 switches **1526** that are co-located in a rack **1540** separate from the N servers **1522** that are spread across K racks **1524**. Each server **1522** has a direct link to each of the tier-1 switches **1526**. In some implementations, there is one fiber cable **1542** (or a small number<<N/K of fiber cables) from the tier-1 switch rack **1540** to each of the K server racks **1524**.

(546) FIG. **87**A is a diagram of an example of a data processing system **1550** that includes N=1024 servers **1552** spread across K=32 racks **1554**, in which each rack **1554** includes N/K=1024/32=32 servers **1552**. There are 4 tier-1 switches **1556** and an optical power supply **1558** that is co-located in a rack **1560**.

(547) Optical fibers connect the servers **1552** to the tier-1 switches **1556** and the optical power supply **1558**. In this example, a bundle **1562** of 9 optical fibers is optically coupled to a co-packaged optical module **1564** of a server **1552**, in which 1 optical fiber provides the optical power supply light, and 4 pairs of (a total of 8) optical fibers provide 4 bi-directional communication channels, each channel having a 100 Gbps bandwidth, for a total of 4×100 Gbps bandwidth in each direction. Because there are 32 servers **1552** in each rack **1554**, there are a total of 256+32=288 optical fibers that

extend from each rack **1554** of servers **1552**, in which 32 optical fibers provide the optical power supply light, and 256 optical fibers provide 128 bi-directional communication channels, each channel having a 100 Gbps bandwidth.

(548) For example, at the server rack side, optical fibers **1566** (that are connected to the servers **1552** of a rack **1554**) terminate at a server rack connector **1568**. At the switch rack side, optical fibers **1578** (that are connected to the switch boxes **1556** and the optical power supply **1558**) terminate at a switch rack connector **1576**. An optical fiber extension cable **1572** is optically coupled to the server rack side and the switch rack side. The optical fiber extension cable **1572** includes 256+32=288 optical fibers. The optical fiber extension cable **1572** includes a first optical fiber connector **1570** and a second optical fiber connector **1574**. The first optical fiber connector **1570** is connected to the server rack connector **1568**, and the second optical fiber connector **1574** is connected to the switch rack connector **1576**. At the switch rack side, the optical fibers **1578** include 288 optical fibers, of which 32 optical fibers **1580** are optically coupled to the optical power supply **1558**. The 256 optical fibers that carry 128 bi-directional communication channels (each channel having a 100 Gbps bandwidth in each direction) are separated into four groups of 64 optical fibers, in which each group of 64 optical fibers is optically coupled to a co-packaged optical module **1582** in one of the switch boxes **1556**. The co-packaged optical module **1582** is configured to have a bandwidth of 32×100 Gbps=3.2 Tbps in each direction (input and output). Each switch box **1556** is connected to each server **1552** of the rack **1554** through a pair of optical fibers that carry a bandwidth of 100 Gbps in each direction.

(549) The optical power supply **1558** provides optical power supply light to co-packaged optical modules **1582** at the switch boxes **1556**. In this example, the optical power supply **1558** provides optical power supply light through 4 optical fibers to each co-packaged optical module **1582**, so that a bundle **1581** having a total of 16 optical fibers is used to provide the optical power supply light to the 4 switch boxes **1556**. A bundle of optical fibers **1584** is optically coupled to the co-packaged optical module **1582** of the switch box **1556**. The bundle of optical fibers **1584** includes 64+16=80 fibers. In some examples, the optical power supply **1558** can provide additional optical power supply light to the co-packaged optical module **1582** using additional optical fibers. For example, the optical power supply **1558** can provide optical power supply light to the co-packaged optical module **1582** using 32 optical fibers with built-in redundancy.

(550) In some implementations, the server rack on which the servers **1552** are mounted is provided with a server rack connector **1568** attached to the server rack chassis, and an optical fiber cable system that includes the optical fibers **1566** optically connected to the server rack connector **1568**, in which the optical fibers **1566** divides into separate bundles **1562** of optical fibers that are optically connected to the servers **1552**.

(551) Similarly, the server rack on which the switch boxes **1556** are mounted is provided with switch rack connectors **1576** attached to the switch rack chassis, and corresponding optical fiber cable systems that each includes the optical fibers **1578** optically connected to the corresponding switch rack connector **1576**, in which the optical fibers **1578** divides into separate bundles of optical fibers that are optically connected to the switch boxes **1556** and the optical power supply **1558**. For example, a switch rack that is configured to connect up to 32 racks of servers **1552** can include 32 built-in switch rack connectors **1576**, and 32 corresponding optical fiber cable systems that are optically connected to 32 co-packaged optical modules in each of the switch boxes **1556**, and 32 laser sources in the optical power supply **1556**.

(552) When an operator sets up a first rack of servers **1552**, the operator connects the bundles **1562** of optical fibers (that is provided with the first server rack) to the servers **1552** in the first rack, connects the optical fiber connector **1570** of a first optical fiber extension cable **1572** to the server rack connector **1568** at the first server rack, and connects the optical fiber connector **1574** of the first optical fiber extension cable **1572** to a first one of the switch rack connectors **1578** at the switch rack. When the operator sets up a second rack of servers **1552**, the operator connects the bundles **1562** of optical fibers (that is provided with the second server rack) to the servers **1552** in the second rack, connects the optical fiber connector **1570** of a second optical fiber extension cable **1572** to the server rack connector **1568** at the second server rack, and connects the optical fiber connector **1574** of the second optical fiber extension cable **1572** to a second one of the switch rack connectors **1578**, and so forth.

(553) In some implementations, the optical power supply **1558** can be any optical power supply described above, and the power supply light can include any control signals and/or optical frame templates described above.

(554) Referring to FIG. **87**B, the data processing system **1550** includes an optical fiber guide module **1590** that helps organize the optical fibers so that they are directed to the appropriate directions. The optical fiber guide module **1590** also restricts bending of the optical fibers to be within the specified limits to prevent excess optical light loss or damage to the optical fibers. The optical fiber guide module **1590** includes a first port **1592**, a second port **1594**, and a third port **1596**. The optical fibers that extend outward from the first port **1592** are optically coupled to the switch rack connector **1576**. The optical fibers that extend outward from the second port **1594** are optically coupled to the switch boxes. The optical fibers that extend outward from the third port **1596** are optically coupled to the optical power supply **1558**.

(555) The following figures show enlarged portions of FIG. **87**A to more clearly illustrate how the optical power supply is distributed from the optical power supply **1558** to the co-packaged optical modules **1564**, and how the data from the servers **1552** are switched by the switch boxes **1556**. FIG. **136**A shows the same modules as FIG. **87**A. FIGS. **136**B, **136**D, and **136**F show enlarged portions **13600**, **13602**, and **13604**, respectively, of the data processing system **1550** shown in FIG. **136**A. FIG. **136**C shows an enlarged portion **13606** of the portion **13600** in FIG. **136**B.

(556) Referring to FIGS. **136**B and **136**C, the bundle **1562** of 9 optical fibers is optically coupled to the co-packaged optical module **1564** of the server **1552**. The bundle **1562** of 9 optical fibers includes a bundle **13162** of 8 data optical fibers and 1 power supply optical fiber **13610** that provides the power supply light. In this example, the bundle **13162** of 8

data fibers includes 4 pairs **13612** of optical fibers that provide 4 bi-directional communication channels, each channel having a 100 Gbps bandwidth, for a total of 4×100 Gbps bandwidth in each direction. In FIGS. **87**A, **87**B, **136**A, **136**B, and **136**C, the optical fiber connectors are not shown. The optical fiber connectors are shown in FIG. **137**.

(557) Referring to FIG. **136**D, 32 bundles **1562** of optical fibers extend from the switch rack connector **1576** toward the servers **1552**, in which each bundle **1562** includes 9 optical fibers as shown in FIG. **136**C. Only 4 bundles **1562** of optical fibers are shown in the figure. The bundle **1562** of 9 optical fibers includes a bundle **13162** of 8 data optical fibers and 1 power supply optical fiber **13610**. The bundle **13612** of 8 data fibers extend from the switch rack connector **1576** toward the switch boxes **1556**. The power supply optical fiber **13610** extend towards the optical power supply **1558**. Power supply optical fibers **13616** extend from the optical power supply **1558** toward the switch boxes **1556** and are used to carry power supply light to the switch boxes **1556**. In this example, a bundle **13618** of 48 power supply optical fibers are used to carry power supply light from the optical power supply **1558** to the servers **1552** and the switch boxes **1556**. The bundle **13618** of power supply optical fibers includes a bundle **13620** of 32 power supply optical fibers **13612** that provide power supply light to the 32 servers **1552**, and a bundle **13622** of 16 power supply optical fibers **13616** that provide power supply light to the 4 switch boxes **1556**, in which each switch box **1556** receives power supply light from 4 power supply optical fibers **13616**.

(558) FIG. **136**E shows the portion **13602** with the optical fiber guide module **1590**. The optical fiber guide module **1590** includes the first port **1592**, the second port **1594**, and the third port **1596**. The optical fibers that extend outward from the first port **1592** are optically coupled to the switch rack connector **1576**. The optical fibers that extend outward from the second port **1594** are optically coupled to the switch boxes **1556**. The optical fibers that extend outward from the third port **1596** are optically coupled to the optical power supply **1558**.

(559) FIG. **136**F shows an enlarged view of the portion **13604** of the data processing system **1550** in FIG. **136**A. FIG. **136**G shows an enlarged portion **13626** of the portion **13604** in FIG. **136**F. FIG. **136**H shows an enlarged portion **13628** of the portion **13626** in FIG. **136**G. Referring to FIGS. **136**F, **136**G, and **136**H, in this example, a bundle **13630** of optical fibers includes the 32 bundles **13612** (see FIG. **136**D) of data optical fibers optically connected to the 32 servers **1552**, respectively, and the bundle **13622** (see FIG. **136**D) of 16 power supply optical fibers optically connected to the optical power supply **1558**. Each bundle **13612** of data optical fibers includes 8 data optical fibers. The 8 data optical fibers of the first bundle **13612** (connected to the first server **1552**) are optically connected to the 4 switch boxes **1556**, in which a first pair **13632** of data optical fibers are optically connected to a first co-packaged optical module **13624** of the first switch box **1556**, a second pair **13634** of data optical fibers are optically connected to a first co-packaged optical module **13624** of the second switch box **1556**, a third pair **13636** of data optical fibers are optically connected to a first co-packaged optical module **13624** of the third switch box **1556**, and a fourth pair **13638** of data optical fibers are optically connected to a first co-packaged optical module **13624** of the fourth switch box **1556**. Each co-packaged optical module **13624** is also optically connected to 4 power supply optical fibers **13616** (see FIG. **136**D). Each co-packaged optical module **13624** is optically connected to a bundle **13632** of optical fibers that include 64 data optical fibers (optically connected to the 32 servers **1552**) and 4 power supply optical fibers (connected to the optical power supply **1558**).

(560) The 8 data optical fibers of the second bundle **13612** (optically connected to the second server **1552**) are optically connected to the 4 switch boxes **1556** in a similar manner, in which a first pair of data optical fibers are optically connected to a second co-packaged optical module of the first switch box **1556**, a second pair of data optical fibers are optically connected to a second co-packaged optical module of the second switch box **1556**, a third pair of data optical fibers are optically connected to a second co-packaged optical module of the third switch box **1556**, and a fourth pair of data optical fibers are optically connected to a second co-packaged optical module of the fourth switch box **1556**, and so forth.

(561) For example, each co-packaged optical module **13624** in the switch box **1556** is optically connected to a total of 64 data optical fibers from the 32 servers **1552**. Each co-packaged optical module **13624** is optically connected to a pair of data optical fibers from each server **1552**, allowing the co-packaged optical module **13624** to be in optical communication with every one of the 32 servers **1552** in a server rack. For example, each switch box **1556** can include 32 co-packaged optical modules **13624**, in which each co-packaged optical module **13624** is in optical communication with 32 servers in a server rack, and different co-packaged optical modules **13624** are in optical communication with the servers in different server racks. This way, each server **1552** is in optical communication with each of the 4 switch boxes **1556**, and each switch box **1556** is in optical communication with every server **1552** in every server rack.

(562) Each co-packaged optical module **13624** in the switch box **1556** is also optically connected to 4 power supply optical fibers **13616** (see FIG. **136**D). Each co-packaged optical module **13624** can be optically connected to any number of power supply optical fibers, depending on the amount of power supply light needed for the operation of optical modulators in the co-packaged optical module **13624**. For example, each co-packaged optical module can be optically connected through multiple power supply optical fibers to multiple optical power supplies to provide redundancy and increase reliability. The co-packaged optical modules **13624** of the switch boxes **1556** receive power supply light from a remote optical power supply **1558** that is located outside of the housings of the switch boxes **1556** and optically connected to the co-packaged optical modules **13624** through power supply optical fibers **13616**. In some implementations, this allows management and service of the optical power supply **1558** to be independent of the switch boxes **1556**. The optical power supply **1558** can have a thermal environment that is different from that of the switch boxes **1556**. For example, the optical power supply **1558** can be placed in an enclosure that is equipped with an active thermal control system to ensure

that the laser sources operate in an environment with a stable temperature. This way, the laser sources are not affected by the thermal fluctuations caused by the operations of the switch boxes **1556**.

(563) FIGS. **136**A to **136**H show the optical fiber connections between the switch boxes **1556** and one rack of 32 servers **1552**. The other racks of servers can be optically connected to the switch boxes **1556** and the optical power supply **1558** in a similar manner. This way, each switch box **1556** is capable of switching or transmitting data between any two servers **1552** among the multiple racks of servers.

(564) FIGS. **87**A, **87**B, and **136**A to **136**H show an example of optical fiber cable configuration for optically connecting the co-packaged optical modules or optical interfaces of multiple servers to co-packaged optical modules or optical interfaces of switch boxes, and providing power supply light from a remote optical power supply to the co-packaged optical modules of the servers and the switch boxes. Referring to FIG. **137**, in some implementations, an optical fiber cable **13700** configured to optically connect the servers **1552**, the switch boxes **1556**, and the optical power supply **1558** includes three main segments: (i) a first segment **13702** that includes optical fiber connectors **13708** that are optically coupled to the co-packaged optical modules of the servers **1552**, (ii) a second segment **13704** includes optical fiber connectors **13710** and **13722** that are optically coupled to the co-packaged optical modules of the switch boxes **1556** and the optical power supply **1558**, and (iii) a third segment **13706** that is optically connected between the first segment **13702** and the second segment **13704**. The third segment **13706** functions as an optical fiber extension cable.

(565) In some implementations, the first segment **13702** includes an optical fiber connector **13712** that is optically coupled to an optical fiber connector **13714** of the third segment **13706**. The first segment **13702** includes 32 optical fiber connectors **13708** that are optically coupled to 32 servers **1552**. The optical fiber connector **13712** includes 32 power supply fiber ports, 128 transmitter fiber ports, and 128 receiver fiber ports, and each optical fiber connector **13708** includes 1 power supply fiber port, 4 transmitter fiber ports, and 4 receiver fiber ports. The second segment **13704** includes an optical fiber connector **13718** that is optically coupled to an optical fiber connector **13720** of the third segment **13706**.

(566) In some implementations, the second segment **13704** includes 4 optical fiber connectors **13710** that are optically coupled to 4 switch boxes **1556** and 1 optical fiber connector **13722** that is optically coupled to the optical power supply **1558**. The optical fiber connector **13720** includes 32 power supply fiber ports, 128 transmitter fiber ports, and 128 receiver fiber ports. The optical fiber connector **13722** includes 48 power supply fiber ports. Each optical fiber connector **13710** includes 4 power supply fiber ports, 32 transmitter fiber ports, and 32 receiver fiber ports.

(567) The number of power supply fiber ports, transmitter fiber ports, and receiver fiber ports described above are used as examples only, it is possible to have different numbers of power supply fiber ports, transmitter fiber ports, and receiver fiber ports depending on application. It is also possible to have different numbers of optical fiber connectors **13708**, **13710**, and **13722** depending on application.

(568) For example, when a data center is set up to include a first rack of servers **1552** and a rack of switch boxes **1556** and optical power supply **1558**, the optical fiber cable **13700** can be used to optically connect the servers **1552** in the first rack to the switch boxes **1556** and the optical power supply **1558**. When a second rack of servers **1552** is set up in the data center, another optical fiber cable **13700** can be used to optically connect the servers **1552** in the second rack to the switch boxes **1556** and the optical power supply **1558**, and so forth.

(569) Referring to FIG. **138**, in some implementations, a data processing system **13800** uses wavelength division multiplexing (WMD) to transmit signals having multiple wavelengths (e.g., w**1**, w**2**, w**3**, w**4**) in the optical fibers, thereby reducing the number of optical fibers needed between the servers **1552** and the switch boxes **1556** for a given bandwidth, or increasing the bandwidth for a given number of optical fibers. In this example, "w**1**" represents the first wavelength, "w**2**" represents the second wavelength, "w**3**" represents the third wavelength, and "w**4**" represents the fourth wavelength, and so forth.

(570) In this example, the data processing system **13800** includes N=1024 servers **13802** spread across K=32 racks **13804**, in which each rack **13804** includes N/K=1024/32=32 servers **13802**. There are 4 tier-1 switches **13806** and an optical power supply **13808** that is co-located in a rack **13810**.

(571) Optical fibers connect the servers **13802** to the tier-1 switches **13806** and the optical power supply **13808**. In this example, a bundle **13812** of 3 optical fibers is optically coupled to a co-packaged optical module **113814** of a server **13802**, in which 1 optical fiber provides the optical power supply light, and 1 pair of optical fibers provide 4 bi-directional communication channels by using 4 different wavelengths per fiber, each channel having a 100 Gbps bandwidth, for a total of 4×100 Gbps bandwidth in each direction. Because there are 32 servers **13802** in each rack **13804**, there are a total of 64+32=96 optical fibers that extend from each rack **13804** of servers **13802**, in which 32 optical fibers provide the optical power supply light, and 64 optical fibers provide 128 bi-directional communication channels using 4 different wavelengths, each channel having a 100 Gbps bandwidth.

(572) For example, at the server rack side, optical fibers **13816** (that are connected to the servers **153802** of a rack **13804**) terminate at a server rack connector **13818**. At the switch rack side, optical fibers **13820** (that are connected to the switch boxes **13806** and the optical power supply **13808**) terminate at a switch rack WDM translator **13822**. The switch rack WDM translator **13822** includes 4×4 wavelength/space shuffle matrices. A 4×4 wavelength/space shuffle matrix shuffles the WDM signals between 4 servers and 4 switch boxes **13806** so that (i) 4 signals having 4 different wavelengths from a server **13802** are sent to 4 switch boxes **13806**, (ii) 4 single-wavelength signals from 4 different servers **13802** are sent to a single switch box **13806**, (iii) 4 signals having 4 different wavelengths from a switch box **13806** are sent to 4 different

servers **13802**, and (iv) 4 single-wavelength signals from 4 different switch boxes **13806** are sent to a single server **13802**. The switch rack WDM translator **13822** is described in more detail below.

(573) An optical fiber extension cable **13824** is optically coupled to the server rack side and the switch rack side. The optical fiber extension cable **13824** includes 64+32=96 optical fibers. The optical fiber extension cable **13824** includes a first optical fiber connector **13826** and a second optical fiber connector **13828**. The first optical fiber connector **13826** is connected to the server rack connector **13818**, and the second optical fiber connector **13828** is connected to the switch rack WDM translator **13822**. At the switch rack side, the optical fibers **13820** include 72 optical fibers, of which 8 optical fibers **13832** are optically coupled to the optical power supply **13808**. The 64 optical fibers that carry 128 bi-directional communication channels (each channel having a 100 Gbps bandwidth in each direction) are separated into four groups of 16 optical fibers, in which each group of 16 optical fibers is optically coupled to a co-packaged optical module **13834** in one of the switch boxes **13806**. The co-packaged optical module **13834** is configured to have a bandwidth of 32×100 Gbps=3.2 Tbps in each direction (input and output). Each switch box **13806** is connected to each server **13802** of the rack **13804** through a pair of optical fibers that carry a bandwidth of 100 Gbps in each direction. In this example, each switch box **13806** is capable of switching data from the 32 servers **13802**, and each switch box **13806** has a 32×32×100 Gbps=102 Tbps bandwidth.

(574) The optical power supply **13810** provides optical power supply light to co-packaged optical modules **13834** at the switch boxes **13806**. In this example, the optical power supply **13808** provides optical power supply light through 2 optical fibers to each co-packaged optical module **13834**, so that a total of 8 optical fibers are used to provide the optical power supply light to the 4 switch boxes **13834**. A bundle of optical fibers **13836** is optically coupled to the co-packaged optical module **13834** of the switch box **13806**. The bundle of optical fibers **13836** includes 16+2=18 fibers. In some examples, the optical power supply **13808** can provide additional optical power supply light to the co-packaged optical module **13834** using additional optical fibers. For example, the optical power supply **13808** can provide optical power supply light to the co-packaged optical module **13834** using 4 optical fibers with built-in redundancy.

(575) An optical fiber guide module, similar to the module **1590** in FIG. **87**B, can be provided to help organize the optical fibers so that they are directed to the appropriate directions.

(576) In some implementations, the server rack on which the servers **13802** are mounted is provided with a server rack connector **13818** attached to the server rack chassis, and an optical fiber cable system that includes the optical fibers **13816** optically connected to the server rack connector **13818**, in which the optical fibers **13816** divide into separate bundles **13812** of optical fibers that are optically connected to the servers **13802**.

(577) In some implementations, the server rack on which the switch boxes **13806** are mounted is provided with switch rack WDM translators **13822** attached to the switch rack chassis, and corresponding optical fiber cable systems that each includes the optical fibers **13820** optically connected to the corresponding switch rack WDM translator **13822**, in which the optical fibers **13820** divide into separate bundles of optical fibers that are optically connected to the switch boxes **13806** and the optical power supply **13808**. For example, a switch rack that is configured to connect up to 32 racks of servers **13802** can include 32 built-in switch rack WDM translators **13822**, and 32 corresponding optical fiber cable systems that are optically connected to 32 co-packaged optical modules in each of the switch boxes **13806**, and 32 laser sources in the optical power supply **13808**.

(578) When an operator sets up a first rack of servers **13802**, the operator connects the bundles **13812** of optical fibers (that is provided with the first server rack) to the servers **13802** in the first rack, connects the optical fiber connector **13826** of a first optical fiber extension cable **13824** to the server rack connector **13826** at the first server rack, and connects the optical fiber connector **13828** of the first optical fiber extension cable **13824** to a first one of the switch rack WDM translators **13822** at the switch rack. When the operator sets up a second rack of servers **13802**, the operator connects the bundles **13812** of optical fibers (that is provided with the second server rack) to the servers **13802** in the second rack, connects the optical fiber connector **13826** of a second optical fiber extension cable **13824** to the server rack connector **13818** at the second server rack, and connects the optical fiber connector **13828** of the second optical fiber extension cable **13824** to a second one of the switch rack WDM translators **13822**, and so forth.

(579) In some implementations, the optical power supply **13808** can be any optical power supply described above, and the power supply light can include any control signals and/or optical frame templates described above.

(580) FIG. **139**A is a diagram of the switch rack WDM translator **13822**, which includes wavelength/space shuffle matrices **13970** that shuffle the WDM signals so that (i) a WDM signal from a server **13802** is demultiplexed into 4 single-wavelength signals that are sent to 4 different switch boxes **13806**, (ii) 4 single-wavelength signals from different servers **13802** are multiplexed into a WDM signal that is sent to a single switch box **13806**, (iii) a WDM signal from a switch box **13806** is demultiplexed into 4 single-wavelength signals that are sent to 4 different servers **13802**, and (iv) 4 single-wavelength signals from different switch boxes **13806** are multiplexed into a WDM signal that is sent to a single server **13802**.

(581) FIG. **139**B is a diagram of the wavelength/space shuffle matrix **13970**. In the example shown in FIGS. **139**A and **139**B, the WDM signals use four different wavelengths (e.g., w**1**, w**2**, w**3**, w**4**), and the switch rack WDM translator **13822** uses 4×4 wavelength/space shuffle matrices **13970**. It is also possible to use a different number of wavelengths, such as 2, 3, 5, 6, 7, 8, . . . , 16, 40, 88, 96, or 120, etc., different wavelengths. If the WDM signals are configured to have N different wavelengths, N×N wavelength/space shuffle matrices can be used to shuffle the N signals carried by the N different wavelengths.

(582) In this example, the switch rack WDM translator **13822** includes eight 4×4 wavelength/space shuffle matrices **13970** to process the WDM signals from and to the 32 servers **13802**. A first 4×4 wavelength/space shuffle matrix **13970** includes 4 multiplexer/demultiplexers **13972***a*, **13972***b*, **13972***c*, **13972***d* (collectively referenced as **13972**) that process the WDM signals from and to servers **1** to **4**. A second 4×4 wavelength/space shuffle matrix **13970** includes 4 multiplexer/demultiplexers that process the WDM signals from and to servers **5** to **8**. A third 4×4 wavelength/space shuffle matrix **13970** includes 4 multiplexer/demultiplexers that process the WDM signals from and to servers **9** to **12**, and so forth. The first 4×4 wavelength/space shuffle matrix **13970** includes 4 multiplexer/demultiplexers **13974***a*, **13974***b*, **13974***c*, **13974***d* (collectively referenced as **13974**) that process the WDM signals from and to switches **1** to **4**. The second 4×4 wavelength/space shuffle matrix **13970** includes 4 multiplexer/demultiplexers that process the WDM signals from and to switches **5** to **8**. The third 4×4 wavelength/space shuffle matrix **13970** includes 4 multiplexer/demultiplexers that process the WDM signals from and to switches **9** to **12**, and so forth.
(583) In the first 4×4 wavelength/space shuffle matrix **13970**, the multiplexer/demultiplexer **13972***a* receives WDM signals from server **1** through optical fiber **13976***a***1**, and sends WDM signals to server **1** through optical fiber **13976***a***2**. The multiplexer/demultiplexer **13972***b* receives WDM signals from server **2** through optical fiber **13976***b***1**, and sends WDM signals to server **2** through optical fiber **13976***b***2**. The multiplexer/demultiplexer **13972***c* receives WDM signals from server **3** through optical fiber **13976***c***1**, and sends WDM signals to server **3** through optical fiber **13976***c***2**. The multiplexer/demultiplexer **13972***d* receives WDM signals from server **4** through optical fiber **13976***d***1**, and sends WDM signals to server **4** through optical fiber **13976***d***2**.
(584) The multiplexer/demultiplexer **13974***a* receives WDM signals from switch **1** through optical fiber **13978***a***1**, and sends WDM signals to switch **1** through optical fiber **13978***a***2**. The multiplexer/demultiplexer **13974***b* receives WDM signals from switch **2** through optical fiber **13978***b***1**, and sends WDM signals to switch **2** through optical fiber **13978***b***2**. The multiplexer/demultiplexer **13974***c* receives WDM signals from switch **3** through optical fiber **13978***c***1**, and sends WDM signals to switch **3** through optical fiber **13978***c***2**. The multiplexer/demultiplexer **13974***d* receives WDM signals from switch **4** through optical fiber **13978***d***1**, and sends WDM signals to switch **4** through optical fiber **13978***d***2**.
(585) The following describes the signal paths from the servers **13802** to the switches **13806**. The multiplexer/demultiplexer **13972***a* demultiplexes the WDM signal received from server **1** and provides a signal having the wavelength w**1** to the multiplexer/demultiplexer **13974***a*, provides a signal having the wavelength w**2** to the multiplexer/demultiplexer **13974***b*, provides a signal having the wavelength w**3** to the multiplexer/demultiplexer **13974***c*, and provides a signal having the wavelength w**4** to the multiplexer/demultiplexer **13974***d.*
(586) The multiplexer/demultiplexer **13972***b* demultiplexes the WDM signal received from server **2** and provides a signal having the wavelength w**1** to the multiplexer/demultiplexer **13974***b*, provides a signal having the wavelength w**2** to the multiplexer/demultiplexer **13974***c*, provides a signal having the wavelength w**3** to the multiplexer/demultiplexer **13974***d*, and provides a signal having the wavelength w**4** to the multiplexer/demultiplexer **13974***a.*
(587) The multiplexer/demultiplexer **13972***c* demultiplexes the WDM signal received from server **3** and provides a signal having the wavelength w**1** to the multiplexer/demultiplexer **13974***c*, provides a signal having the wavelength w**2** to the multiplexer/demultiplexer **13974***d*, provides a signal having the wavelength w**3** to the multiplexer/demultiplexer **13974***a*, and provides a signal having the wavelength w**4** to the multiplexer/demultiplexer **13974***b.*
(588) The multiplexer/demultiplexer **13972***d* demultiplexes the WDM signals received from server **4** and provides a signal having the wavelength w**1** to the multiplexer/demultiplexer **13974***d*, provides a signal having the wavelength w**2** to the multiplexer/demultiplexer **13974***a*, provides a signal having the wavelength w**3** to the multiplexer/demultiplexer **13974***b*, and provides a signal having the wavelength w**4** to the multiplexer/demultiplexer **13974***c.*
(589) The multiplexer/demultiplexer **13974***a* receives a signal having the wavelength w**1** from the multiplexer/demultiplexer **13972***a*, receives a signal having the wavelength w**2** from the multiplexer/demultiplexer **13972***d*, receives a signal having the wavelength w**3** from the multiplexer/demultiplexer **13972***c*, receives a signal having the wavelength w**4** from the multiplexer/demultiplexer **13972***b*, combines the signals having the wavelengths w**1**, w**2**, w**3**, w**4** into a WDM signal having wavelengths w**1**, w**2**, w**3**, w**4**, and sends the WDM signal to switch **1** through the optical fiber **13978***a***1**.
(590) The multiplexer/demultiplexer **13974***b* receives a signal having the wavelength w**1** from the multiplexer/demultiplexer **13972***b*, receives a signal having the wavelength w**2** from the multiplexer/demultiplexer **13972***a*, receives a signal having the wavelength w**3** from the multiplexer/demultiplexer **13972***d*, receives a signal having the wavelength w**4** from the multiplexer/demultiplexer **13972***c*, combines the signals having the wavelengths w**1**, w**2**, w**3**, w**4** into a WDM signal having wavelengths w**1**, w**2**, w**3**, w**4**, and sends the WDM signal to switch **2** through the optical fiber **13978***b***1**.
(591) The multiplexer/demultiplexer **13974***c* receives a signal having the wavelength w**1** from the multiplexer/demultiplexer **13972***c*, receives a signal having the wavelength w**2** from the multiplexer/demultiplexer **13972***b*, receives a signal having the wavelength w**3** from the multiplexer/demultiplexer **13972***a*, receives a signal having the wavelength w**4** from the multiplexer/demultiplexer **13972***d*, combines the signals having the wavelengths w**1**, w**2**, w**3**, w**4** into a WDM signal having wavelengths w**1**, w**2**, w**3**, w**4**, and sends the WDM signal to switch **3** through the optical fiber **13978***c***1**.
(592) The multiplexer/demultiplexer **13974***d* receives a signal having the wavelength w**1** from the multiplexer/demultiplexer **13972***d*, receives a signal having the wavelength w**2** from the multiplexer/demultiplexer

13972*c*, receives a signal having the wavelength w**3** from the multiplexer/demultiplexer **13972***b*, receives a signal having the wavelength w**4** from the multiplexer/demultiplexer **13972***a*, combines the signals having the wavelengths w**1**, w**2**, w**3**, w**4** into a WDM signal having wavelengths w**1**, w**2**, w**3**, w**4**, and sends the WDM signal to switch **4** through the optical fiber **13978***d***1**.

(593) The following describes the signal paths from the switches **13806** to the servers **13802**. The multiplexer/demultiplexer **13974***a* receives a WDM signal from switch **1**, demultiplexes the WDM signal, and provides a signal having the wavelength w**1** to the multiplexer/demultiplexer **13972***a*, provides a signal having the wavelength w**2** to the multiplexer/demultiplexer **13972***d*, provides a signal having the wavelength w**3** to the multiplexer/demultiplexer **13972***c*, and provides a signal having the wavelength w**4** to the multiplexer/demultiplexer **13972***b*.

(594) The multiplexer/demultiplexer **13974***b* receives a WDM signal from switch **2**, demultiplexes the WDM signal, and provides a signal having the wavelength w**1** to the multiplexer/demultiplexer **13972***b*, provides a signal having the wavelength w**2** to the multiplexer/demultiplexer **13972***a*, provides a signal having the wavelength w**3** to the multiplexer/demultiplexer **13974***d*, and provides a signal having the wavelength w**4** to the multiplexer/demultiplexer **13974***c*.

(595) The multiplexer/demultiplexer **13974***c* receives a WDM signal from switch **3**, demultiplexes the WDM signal, and provides a signal having the wavelength w**1** to the multiplexer/demultiplexer **13972***c*, provides a signal having the wavelength w**2** to the multiplexer/demultiplexer **13972***b*, provides a signal having the wavelength w**3** to the multiplexer/demultiplexer **13972***a*, and provides a signal having the wavelength w**4** to the multiplexer/demultiplexer **13972***d*.

(596) The multiplexer/demultiplexer **13974***d* receives a WDM signal from switch **4**, demultiplexes the WDM signal, and provides a signal having the wavelength w**1** to the multiplexer/demultiplexer **13972***d*, provides a signal having the wavelength w**2** to the multiplexer/demultiplexer **13972***c*, provides a signal having the wavelength w**3** to the multiplexer/demultiplexer **13972***b*, and provides a signal having the wavelength w**4** to the multiplexer/demultiplexer **13972***a*.

(597) The multiplexer/demultiplexer **13972***a* receives a signal having the wavelength w**1** from the multiplexer/demultiplexer **13974***a*, receives a signal having the wavelength w**2** from the multiplexer/demultiplexer **13974***b*, receives a signal having the wavelength w**3** from the multiplexer/demultiplexer **13974***c*, receives a signal having the wavelength w**4** from the multiplexer/demultiplexer **13974***d*, combines the signals having the wavelengths w**1**, w**2**, w**3**, w**4** into a WDM signal having wavelengths w**1**, w**2**, w**3**, w**4**, and sends the WDM signal to sever **1** through the optical fiber **13976***a***2**.

(598) The multiplexer/demultiplexer **13972***b* receives a signal having the wavelength w**1** from the multiplexer/demultiplexer **13974***b*, receives a signal having the wavelength w**2** from the multiplexer/demultiplexer **13974***c*, receives a signal having the wavelength w**3** from the multiplexer/demultiplexer **13974***d*, receives a signal having the wavelength w**4** from the multiplexer/demultiplexer **13974***a*, combines the signals having the wavelengths w**1**, w**2**, w**3**, w**4** into a WDM signal having wavelengths w**1**, w**2**, w**3**, w**4**, and sends the WDM signal to sever **2** through the optical fiber **13976***b***2**.

(599) The multiplexer/demultiplexer **13972***c* receives a signal having the wavelength w**1** from the multiplexer/demultiplexer **13974***c*, receives a signal having the wavelength w**2** from the multiplexer/demultiplexer **13974***d*, receives a signal having the wavelength w**3** from the multiplexer/demultiplexer **13974***a*, receives a signal having the wavelength w**4** from the multiplexer/demultiplexer **13974***b*, combines the signals having the wavelengths w**1**, w**2**, w**3**, w**4** into a WDM signal having wavelengths w**1**, w**2**, w**3**, w**4**, and sends the WDM signal to sever **3** through the optical fiber **13976***c***2**.

(600) The multiplexer/demultiplexer **13972***d* receives a signal having the wavelength w**1** from the multiplexer/demultiplexer **13974***d*, receives a signal having the wavelength w**2** from the multiplexer/demultiplexer **13974***a*, receives a signal having the wavelength w**3** from the multiplexer/demultiplexer **13974***b*, receives a signal having the wavelength w**4** from the multiplexer/demultiplexer **13974***c*, combines the signals having the wavelengths w**1**, w**2**, w**3**, w**4** into a WDM signal having wavelengths w**1**, w**2**, w**3**, w**4**, and sends the WDM signal to sever **4** through the optical fiber **13976***d***2**.

(601) 16 data optical fibers are used to connect the switch rack WDM translator **13822** to a co-packaged optical module of a switch **13806**. Each of 8 data optical fiber transmits a WDM signal have 4 wavelengths carrying signals from 4 servers **13802** to the switch **13806**. Each of 8 data optical fiber transmits a WDM signal have 4 wavelengths carrying signals from the switch **13806** to 4 servers **13802**.

(602) In some implementations, the power supply optical fibers pass through the switch rack WDM translator **13822** without being affected by the wavelength/space shuffle matrices **13970**. In some implementations, the power supply optical signals do not pass through the switch rack WDM translator **13822**, in which the power supply optical fibers are combined with the data fibers at a location external to the WDM translator **13822**.

(603) The WDM translator **13822** includes a first interface that is optically coupled to the plurality of optical fibers that are optically to the servers **13802**. The WDM translator **13822** includes a second interface that is optically coupled to the plurality of optical fibers that are optically to the switches **13806** and the optical power supply **13808**. In FIG. **139**, the first interface is shown at the left side of the WDM translator **13822**, and the second interface is shown at the right side of the WDM translator **13822**. The first interface includes a first set of optical fiber ports, a second set of optical fiber ports,

and a first set of power supply fiber ports. The first set of optical fiber ports are optically coupled to optical fibers that transmit WDM signals to the servers **13802**. The second set of optical fiber ports are optically coupled to optical fibers that transmit WDM signals from the servers **13802**. The first set of power supply fiber ports are optically coupled to optical fibers that provide power supply light to the photonic integrated circuits of the servers **13802**.
(604) The second interface of the WDM translator **13822** includes a third set of optical fiber ports, a fourth set of optical fiber ports, and a second set of power supply fiber ports. The third set of optical fiber ports are optically coupled to optical fibers that transmit WDM signals to the switches **13806**. The fourth set of optical fiber ports are optically coupled to optical fibers that transmit WDM signals from the switches **13806**. The second set of power supply fiber ports are optically coupled to optical fibers that are optically coupled to the optical power supply **13808**.
(605) The first set of optical fiber ports and the second set of optical fiber ports are optically coupled to the multiplexer/demultiplexers **13972** of the wavelength/space shuffle matrix **13970**. The third set of optical fiber ports and the fourth set of optical fiber ports are optically coupled to the multiplexer/demultiplexers **13974** of the wavelength/space shuffle matrix **13970**. The first set of power supply fiber ports are optically coupled to the second set of power supply fiber ports, in which the power supply light is transmitted from the optical power supply **13808** to the servers **13802** through the second set of power supply fiber ports and the first set of power supply fiber ports.
(606) In the signal paths from the servers **13802** to the switches **13806**, each multiplexer/demultiplexer **13972** functions as a demultiplexer that demultiplexes a WDM signal (from a corresponding server **13802**) having multiple wavelengths into the component signals, in which each component signal has a single wavelength, and the different component signals are sent to different switches **13806**. Each multiplexer/demultiplexer **13974** functions as re-multiplexer that multiplexes the component signals from different servers **13802** into a WDM signal having multiple wavelengths that is sent to a corresponding switch **13806**.
(607) In the signal paths from the switches **13806** to the servers **13802**, each multiplexer/demultiplexer **13974** functions as a demultiplexer that demultiplexes a WDM signal (from a corresponding switch **13806**) having multiple wavelengths into the component signals, in which each component signal has a single wavelength, and the different component signals are sent to different servers **13802**. Each multiplexer/demultiplexer **13972** functions as re-multiplexer that multiplexes the component signals from different switches **13806** into a WDM signal having multiple wavelengths that is sent to a corresponding server **13802**.
(608) In some implementations, the data processing system includes N switches **13806** and uses WDM signals that include N different wavelengths w**1**, w**2**, . . . , wn that are transmitted between the servers **13802** and the switches **13806**. In this example, the WDM translator includes N×N wavelength/space shuffle matrices. The first interface of the WDM translator includes a first set of optical fiber ports that output WDM signals having N wavelengths to the servers **13802**, a second set of optical fiber ports that receive WDM signals having N wavelengths from the servers **13802**, and a first set of power supply fiber ports that provide power supply light to the photonic integrated circuits of the servers **13802**. The second interface of the WDM translator includes a third set of optical fiber ports that output WDM signals having N wavelengths to the switches **13806**, a fourth set of optical fiber ports that receive WDM signals having N wavelengths from the switches **13806**, and a second set of power supply fiber ports that are optically coupled to the optical power supply module **13808**.
(609) In some implementations, the optical power supply **13808** provides power supply light having multiple wavelengths that correspond to the wavelengths in the WDM signals transmitted by the servers **13802** and the switches **13806**. Any technique for providing power supply light for supporting photonic integrated circuits that process WDM signals can be used.
(610) The following describes the components of the data processing system **13800** in greater detail. FIG. **140**A shows the same data processing system **13800** of FIG. **138**. FIGS. **140**B, **140**D, and **140**F show enlarged portions **13900**, **13902**, and **13904**, respectively, of the data processing system **13800**. FIG. **140**C shows an enlarged portion **13906** of the portion **13900** in FIG. **140**B.
(611) Referring to FIGS. **140**B and **140**C, the bundle **13812** of 3 optical fibers is optically coupled to the co-packaged optical module **13814** of a server **13802**. The bundle **13812** of 3 optical fibers includes a power supply optical fiber **13840** for transmitting power supply light from the optical power supply **13810** to the co-packaged optical module **13814**. The bundle **13812** further includes a pair of data optical fibers **13842** that each carry WDM signals having 4 different wavelengths w**1**, w**2**, w**3**, and w**4**. For example, the pair of data optical fibers **13842** provide 4 bi-directional communication channels, each channel having a 100 Gbps bandwidth, for a total of 4×100 Gbps bandwidth in each direction. In FIGS. **138**, **140**A, **140**B, and **140**C, the optical fiber connectors that are used to connect the optical fibers to the co-packaged optical module are not shown.
(612) Referring to FIG. **140**D, 32 bundles **13812** of optical fibers extend from the switch rack connector **13828** toward the 32 servers **13802**, in which each bundle **13812** includes 3 optical fibers as shown in FIG. **140**C. Only 4 bundles **13812** of optical fibers are shown in the figure. Each bundle **13812** of 3 optical fibers includes a pair **13842** of data optical fibers and 1 power supply optical fiber **13840**. The WDM signals transmitted from the 32 servers **13802** in the 32 pairs **13842** of data optical fibers are shuffled by the switch rack WDM translator **13822**, which sends the shuffled WDM signals through 32 pairs **13852** of data optical fibers toward the switch boxes **13806**.
(613) The power supply optical fiber **13840** extends towards the optical power supply **13810**. Power supply optical fibers **13844** extend from the optical power supply **13810** toward the switch boxes **13806** and are used to carry power supply

light to the switch boxes **13806**. In this example, a bundle **13846** of 40 power supply optical fibers are used to carry power supply light from the optical power supply **13810** to the servers **13802** and the switch boxes **13806**. The bundle **13846** of power supply optical fibers includes a bundle **13848** of 32 power supply optical fibers **13840** that provide power supply light to the 32 servers **13802**, and a bundle **13850** of 8 power supply optical fibers **13844** that provide power supply light to the 4 switch boxes **13806**, in which each switch box **13806** receives power supply light from 2 power supply optical fibers **13844**.

(614) FIG. **140**E shows the portion **13902** with an optical fiber guide module **13854**. The optical fiber guide module **13854** includes a first port **13856**, a second port **13858**, and a third port **13860**. The optical fibers that extend outward from the first port **13856** are optically coupled to the switch rack WDM translator **13822**. The optical fibers that extend outward from the second port **13858** are optically coupled to the switch boxes **13806**. The optical fibers that extend outward from the third port **13860** are optically coupled to the optical power supply **13810**.

(615) FIG. **140**F shows an enlarged view of the portion **13904** of the data processing system **13800** in FIG. **140**A. FIG. **140**G shows an enlarged portion **13908** of the portion **13904** in FIG. **140**F. FIG. **140**H shows an enlarged portion **13910** of the portion **13908** in FIG. **140**G. Referring to FIGS. **140**F, **140**G, and **140**H, in this example, a bundle **13912** of optical fibers includes the 32 pairs **13852** of data optical fibers optically connected to the switch rack WDM translator **13822**, and the bundle **13850** of 8 power supply optical fibers optically connected to the optical power supply **13810**.

(616) The bundle **13912** of optical fibers includes eight pairs of data optical fibers and a pair of power supply optical fibers that are optically coupled to a co-packaged optical module **13914** of the first switch box **13806**, eight pairs of data optical fibers and a pair of power supply optical fibers that are optically coupled to a co-packaged optical module **13914** of the second switch box **13806**, eight pairs of data optical fibers and a pair of power supply optical fibers that are optically coupled to a co-packaged optical module **13914** of the third switch box **13806**, and eight pairs of data optical fibers and a pair of power supply optical fibers that are optically coupled to a co-packaged optical module **13914** of the fourth switch box **13806**.

(617) Among the eight pairs of data optical fibers that are optically coupled to each switch box **13806**, the first pair of data optical fibers carry WDM signals from and to servers **1** to **4**, the second pair of data optical fibers carry WDM signals from and to servers **5** to **8**, the third pair of data optical fibers carry WDM signals from and to servers **9** to **12**, and so forth. This allows the co-packaged optical module **13914** to communicate with every one of the 32 servers **13802** in a server rack. For example, each switch box **13806** can include 32 co-packaged optical modules **13914**, in which each co-packaged optical module **13914** is capable of communicating with 32 servers in a server rack, and different co-packaged optical modules **13914** are capable of communicating with the servers in different server racks. This way, each server **13802** is in optical communication with each of the 4 switch boxes **13806**, and each switch box **13806** is in optical communication with every one of the 32 servers **13802** in every one of the 32 server racks.

(618) In this example, each co-packaged optical module **1391** in the switch box **13806** is optically connected to 2 power supply optical fibers **13844** (see FIG. **140**D). Each co-packaged optical module **1391** can be optically connected to any number of power supply optical fibers, depending on the amount of power supply light needed for the operation of optical modulators in the co-packaged optical module **1391**. For example, each co-packaged optical module can be optically connected through multiple power supply optical fibers to multiple optical power supplies to provide redundancy and increase reliability. The co-packaged optical modules **13914** of the switch boxes **13806** receive power supply light from a remote optical power supply **13808** that is located outside of the housings of the switch boxes **13806** and optically connected to the co-packaged optical modules **13914** through power supply optical fibers **13844**. In some implementations, this allows management and service of the optical power supply **13808** to be independent of the switch boxes **13806**. The optical power supply **13808** can have a thermal environment that is different from that of the switch boxes **13806**. For example, the optical power supply **13808** can be placed in an enclosure that is equipped with an active thermal control system to ensure that the laser sources operate in an environment with a stable temperature. This way, the laser sources are not affected by the thermal fluctuations caused by the operations of the switch boxes **13806**.

(619) FIGS. **140**A to **140**H show the optical fiber connections between the switch boxes **13806** and one rack of 32 servers **13802**. The other racks of servers can be optically connected to the switch boxes **13806** and the optical power supply **13808** in a similar manner. This way, each switch box **13806** is capable of switching or transmitting data between any two server **13802** among the multiple racks of servers.

(620) FIGS. **138** and **140**A to **140**H show an example of optical fiber cable configuration in a WDM data processing system for optically connecting the co-packaged optical modules or optical interfaces of multiple servers to co-packaged optical modules or optical interfaces of switch boxes, and providing power supply light from a remote optical power supply to the co-packaged optical modules of the servers and the switch boxes. Referring to FIG. **141**, in some implementations, an optical fiber cable **14100** configured to optically connect the servers **13802**, the switch boxes **13806**, and the optical power supply **13808** includes three main segments: (i) a first segment **14102** that includes optical fiber connectors **14108** that are optically coupled to the co-packaged optical modules of the servers **13802**, (ii) a second segment includes optical fiber connectors **14110** that are optically coupled to the co-packaged optical modules of the switch boxes **13806**, and an optical fiber connector **14112** that is optically coupled to the optical power supply **13808**, and (iii) an optical fiber extension cable **14106** that is optically connected between the first segment **14102** and the second segment **14104**.

(621) In some implementations, the first segment **14102** includes an optical fiber connector **14114** that is optically coupled

to an optical fiber connector **14116** of the optical fiber extension cable **14106**. The first segment **14102** includes 32 optical fiber connectors **14108** that are optically coupled to the 32 servers **13802**. The optical fiber connector **14114** includes 32 power supply fiber ports, 32 transmitter fiber ports, and 32 receiver fiber ports. Each optical fiber connector **14108** includes 1 power supply fiber port, 1 transmitter fiber port, and 1 receiver fiber port. The second segment **14104** includes a switch rack WDM translator **14118** that is optically coupled to an optical fiber connector **14120** of the optical fiber extension cable **14106**.

(622) In some implementations, the second segment **14104** includes 4 optical fiber connectors **14110** that are optically coupled to 4 switch boxes **13806** and 1 optical fiber connector **14112** that is optically coupled to the optical power supply **13808**. The switch rack WDM translator **14118** includes 32 power supply fiber ports, 32 transmitter fiber ports, and 32 receiver fiber ports. The optical fiber connector **14112** includes 40 power supply fiber ports. Each optical fiber connector **14110** includes 2 power supply fiber ports, 8 transmitter fiber ports, and 8 receiver fiber ports.

(623) The number of power supply fiber ports, transmitter fiber ports, and receiver fiber ports described above are used as examples only, it is possible to have different numbers of power supply fiber ports, transmitter fiber ports, and receiver fiber ports depending on application. It is also possible to have different numbers of optical fiber connectors **14108**, **14110**, and **14112** depending on application.

(624) The data processing system **13800** of FIG. **138** uses 4 wavelengths over a fiber pair as opposed to 4 parallel spatial paths over 8 fibers used in the data processing system **1550** of FIG. **87**A. The data processing system **13800** of FIG. **138** includes a switch-to-rack WDM translator that has combinations of demultiplexers and multiplexers that function as wavelength/space shuffle matrices. In some implementations, it is possible to replace the server-to-rack connector **13818** with a server-to-rack WDM translator that has combinations of demultiplexers and multiplexers that function as wavelength/space shuffle matrices. In this example, the switch-to-rack WDM translator **13822** can be replaced with an optical fiber connector. Thus, the WDM translator that includes combinations of demultiplexers and multiplexers that function as wavelength/space shuffle matrices can be placed either near the servers **13802** or near the switches **13806**.

(625) FIG. **88** is a diagram of an example of the connector port mapping for an optical fiber interconnection cable **1600**, which includes a first optical fiber connector **1602**, a second optical fiber connector **1604**, optical fibers **1606** that transmit data and/or control signals between the first and second optical fiber connectors **1602**, **1604**, and optical fibers **1608** that transmit optical power supply light. Each optical fiber terminates at an optical fiber port **1610**, which can include, e.g., lenses for focusing light entering or exiting the optical fiber port **1610**. The first and second optical fiber connectors **1602**, **1604** can be, e.g., the optical fiber connectors **1342** and **1344** of FIGS. **80**B, **80**C, the optical fiber connectors **1402** and **1404** of FIGS. **82**B, **82**C, or the optical fiber connectors **1570** and **1574** of FIG. **87**A. The principles for designing the optical fiber interconnection cable **1600** can be used to design the optical cable assembly **1340** of FIGS. **80**B, **80**C, the optical cable assembly **1400** of FIGS. **82**B, **82**C, and the optical cable assembly **1490** of FIGS. **84**B, **84**C.

(626) In the example of FIG. **88**, each optical fiber connector **1602** or **1604** includes 3 rows of optical fiber ports, each row including 12 optical fiber ports. Each optical fiber connector **1602** or **1604** includes 4 power supply fiber ports that are connected to optical fibers **1608** that are optically coupled to one or more optical power supplies. Each optical fiber connector **1602** or **1604** includes 32 fiber ports (some of which are transmitter fiber ports, and some of which are receiver fiber ports) that are connected to the optical fibers **1606** for data transmission and reception.

(627) In some implementations, the mapping of the fiber ports of the optical fiber connectors **1602**, **1604** are designed such that the interconnection cable **1600** can have the most universal use, in which each fiber port of the optical fiber connector **1602** is mapped to a corresponding fiber port of the optical fiber connector **1604** with a 1-to-1 mapping and without transponder-specific port mapping that would require fibers **1606** to cross over. This means that for an optical transponder that has an optical fiber connector compatible with the interconnection cable **1600**, the optical transponder can be connected to either the optical fiber connector **1602** or the optical fiber connector **1604**. The mapping of the fiber ports is designed such that each transmitter port of the optical fiber connector **1602** is mapped to a corresponding receiver port of the optical fiber connector **1604**, and each receiver port of the optical fiber connector **1602** is mapped to a corresponding transmitter port of the optical fiber connector **1604**.

(628) FIG. **89** is a diagram showing an example of the fiber port mapping for an optical fiber interconnection cable **1660** that includes a pair of optical fiber connectors, i.e., a first optical fiber connector **1662** and a second optical fiber connector **1664**. FIG. **142** is an enlarged view of the diagram of FIG. **89**. The power supply fiber ports are labeled ‘P’, the transmitter fiber ports are labeled ‘T’, and the receiver fiber ports are labeled ‘R’. Only some of the fiber ports are labeled in the figure. The optical fiber connectors **1662** and **1664** are designed such that either the first optical fiber connector **1662** or the second optical fiber connector **1664** can be connected to a given communication transponder that is compatible with the optical fiber interconnection cable **1660**. The diagram shows the fiber port mapping when viewed from the outer edge of the optical fiber connector into the optical fiber connector (i.e., toward the optical fibers in the interconnection cable **1660**).

(629) The first optical fiber connector **1662** includes transmitter fiber ports (e.g., **1614***a*, **1616***a*), receiver fiber ports (e.g., **1618***a*, **1620***a*), and optical power supply fiber ports (e.g., **1622***a*, **1624***a*). The second optical fiber connector **1664** includes transmitter fiber ports (e.g., **1614***b*, **1616***b*), receiver fiber ports (e.g., **1618***b*, **1620***b*), and optical power supply fiber ports (e.g., **1622***b*, **1624***b*). For example, assume that the first optical fiber connector **1662** is connected to a first optical transponder, and the second optical fiber connector **1664** is connected to a second optical transponder. The first optical transponder transmits first data and/or control signals through the transmitter ports (e.g., **1614***a*, **1616***a*) of the first

optical fiber connector **1662**, and the second optical transponder receives the first data and/or control signals from the corresponding receiver fiber ports (e.g., **1618***b*, **1620***b*) of the second optical fiber connector **1664**. The transmitter ports **1614***a*, **1616***a* are optically coupled to the corresponding receiver fiber ports **1618***b*, **1620***b* through optical fibers **1628**, **1630**, respectively. The second optical transponder transmits second data and/or control signals through the transmitter ports (e.g., **1614***b*, **1616***b*) of the second optical fiber connector **1664**, and the first optical transponder receives the second data and/or control signals from the corresponding receiver fiber ports (**1618***a*, **1620***a*) of the first optical fiber connector **1662**. The transmitter port **1616***b* is optically coupled to the corresponding receiver fiber port **1620***a* through an optical fiber **1632**.
(630) A first optical power supply transmits optical power supply light to the first optical transponder through the power supply fiber ports of the first optical fiber connector **1662**. A second optical power supply transmits optical power supply light to the second optical transponder through the power supply fiber ports of the second optical fiber connector **1664**. The first and second power supplies can be different (such as the example of FIG. **80**B) or the same (such as the example of FIG. **82**B).
(631) In the following description, when referring to the rows and columns of fiber ports of the optical fiber connector, the uppermost row is referred to as the 1.sup.st row, the second uppermost row is referred to as the 2.sup.nd row, and so forth. The leftmost column is referred to as the 1.sup.st column, the second leftmost column is referred to as the 2.sup.nd column, and so forth.
(632) For an optical fiber interconnection cable having a pair of optical fiber connectors (i.e., a first optical fiber connector and a second optical fiber connector) to be universal, i.e., either one of the pair of optical fiber connectors can be connected to a given optical transponder, the arrangement of the transmitter fiber ports, the receiver fiber ports, and the power supply fiber ports in the optical fiber connectors have a number of properties. These properties are referred to as the “universal optical fiber interconnection cable port mapping properties.” The term “mapping” here refers to the arrangement of the transmitter fiber ports, the receiver fiber ports, and the power supply fiber ports at particular locations within the optical fiber connector. The first property is that the mapping of the transmitter, receiver, and power supply fiber ports in the first optical fiber connector is the same as the mapping of the transmitter, receiver, and power supply fiber ports in the second optical fiber connector (as in the example of FIG. **89**).
(633) In the example of FIG. **89**, the individual optical fibers connecting the transmitter, receiver, and power supply fiber ports in the first optical fiber connector to the transmitter, receiver, and power supply fiber ports in the second optical fiber connector are parallel to one another.
(634) In some implementations, each of the optical fiber connectors includes a unique marker or mechanical structure, e.g., a pin, that is configured to be at the same spot on the co-packaged optical module, similar to the use of a “dot” to denote “pin **1**” on electronic modules. In some examples, such as those shown in FIGS. **89** and **90**, the larger distance from the bottom row (the third row in the examples of FIGS. **89** and **90**) to the connector edge can be used as a “marker” to guide the user to attach the optical fiber connector to the co-packaged optical module connector in a consistent manner.
(635) The mapping of the fiber ports of the optical fiber connectors of a “universal optical fiber interconnection cable” has a second property: When mirroring the port map of an optical fiber connector and replacing each transmitter port with a receiver port as well as replacing each receiver port with a transmitter port in the mirror image, the original port mapping is recovered. The mirror image can be generated with respect to a reflection axis at either connector edge, and the reflection axis can be parallel to the row direction or the column direction. The power supply fiber ports of the first optical fiber connector are mirror images of the power supply fiber ports of the second optical fiber connector.
(636) The transmitter fiber ports of the first optical fiber connector and the receiver fiber ports of the second optical fiber connector are pairwise mirror images of each other, i.e., each transmitter fiber port of the first optical fiber connector is mirrored to a receiver fiber port of the second optical fiber connector. The receiver fiber ports of the first optical fiber connector and the transmitter fiber ports of the second optical fiber connector are pairwise mirror images of each other, i.e., each receiver fiber port of the first optical fiber connector is mirrored to a transmitter fiber port of the second optical fiber connector.
(637) Another way of looking at the second property is as follows: Each optical fiber connector is transmitter port-receiver port (TX-RX) pairwise symmetric and power supply port (PS) symmetric with respect to one of the main or center axes, which can be parallel to the row direction or the column direction. For example, if an optical fiber connector has an even number of columns, the optical fiber connector can be divided along a center axis parallel to the column direction into a left half portion and a right half portion. The power supply fiber ports are symmetric with respect to the main axis, i.e., if there is a power supply fiber port in the left half portion of the optical fiber connector, there will also be a power supply fiber port at the mirror location in the right half portion of the optical fiber connector. The transmitter fiber ports and the receiver fiber ports are pairwise symmetric with respect to the main axis, i.e., if there is a transmitter fiber port in the left half portion of the optical fiber connector, there will be a receiver fiber port at a mirror location in the right half portion of the optical fiber connector. Likewise, if there is a receiver fiber port in the left half portion of the optical fiber connector, there will be a transmitter fiber port at a mirror location in the right half portion of the optical fiber connector.
(638) For example, if an optical fiber connector has an even number of rows, the optical fiber connector can be divided along a center axis parallel to the row direction into an upper half portion and a lower half portion. The power supply fiber ports are symmetric with respect to the main axis, i.e., if there is a power supply fiber port in the upper half portion of the optical fiber connector, there will also be a power supply fiber port at the mirror location in the lower half portion of the

optical fiber connector. The transmitter fiber ports and the receiver fiber ports are pairwise symmetric with respect to the main axis, i.e., if there is a transmitter fiber port in the upper half portion of the optical fiber connector, there will be a receiver fiber port at a mirror location in the lower half portion of the optical fiber connector. Likewise, if there is a receiver fiber port in the upper half portion of the optical fiber connector, there will be a transmitter fiber port at a mirror location in the lower half portion of the optical fiber connector.

(639) The mapping of the transmitter fiber ports, receiver fiber ports, and power supply fiber ports follow a symmetry requirement that can be summarized as follows: (i) Mirror all ports on either one of the two connector edges. (ii) Swap TX (transmitter) and RX (receiver) functionality on the mirror image. (iii) Leave mirrored PS (power supply) ports as PS ports. (iv) The resulting port map is the same as the original one.

Essentially, a viable port map is TX-RX pairwise symmetric and PS symmetric with respect to one of the main axes.

(640) The properties of the mapping of the fiber ports of the optical fiber connectors can be mathematically expressed as follows: Port matrix M with entries PS=0, TX=+1, RX=−1; Column-mirror operation custom character; Row-mirror operation custom characterM; A viable port map either satisfies −custom character=M or −custom characterM=M.

(641) In some implementations, if a universal optical fiber interconnection cable has a first optical fiber connector and a second optical fiber connector that are mirror images of each other after swapping the transmitter fiber ports to receiver fiber ports and swapping the receiver fiber ports to transmitter fiber ports in the mirror image, and the mirror image is generated with respect to a reflection axis parallel to the column direction, as in the example of FIG. **89**, then each optical fiber connector should be TX-RX pairwise symmetric and PS symmetric with respect to a center axis parallel to the column direction. If a universal optical fiber interconnection cable has a first optical fiber connector and a second optical fiber connector that are mirror images of each other after swapping the transmitter and receiver fiber ports in the mirror image, and the mirror image is generated with respect to a reflection axis parallel to the row direction, as in the example of FIG. **90**, then each optical fiber connector should be TX-RX pairwise symmetric and PS symmetric with respect to a center axis parallel to the row direction.

(642) In some implementations, a universal optical fiber interconnection cable: a. Comprises n_trx strands of TX/RX fibers and n_p strands of power supply fibers, in which 0≤n_p≤n_trx. b. The n_trx strands of TX/RX fibers are mapped 1:1 from a first optical fiber connector to the same port positions on a second optical fiber connector through the optical fiber cable, i.e. the optical fiber cable can be laid out in a straight manner without leading to any cross-over fiber strands. c. Those connector ports that are not 1:1 connected by TX/RX fibers may be connected to power supply fibers via a break-out cable.

(643) In some implementations, a universal optical module connector has the following properties: a. Starting from a connector port map PM**0**. b. First mirror port map PM**0** either across the row dimension or across the column dimension. c. Mirroring can be done either across a column axis or across a row axis. d. Replace TX ports by RX ports and vice versa. e. If at least one mirrored and replaced version of the port map again results in the starting port map PM**0**, the connector is called a universal optical module connector.

(644) In FIG. **89**, the arrangement of the transmitter, receiver, and power supply fiber ports in the first optical fiber connector **1662**, and the arrangement of the transmitter, receiver, and power supply fiber ports in the second optical fiber connector **1664** have the two properties described above. First property: When looking into the optical fiber connector (from the outer edge of the connector inward toward the optical fibers), the mapping of the transmitter, receiver, and power supply fiber ports in the first optical fiber connector **1662** is the same as the mapping of the transmitter, receiver, and power supply fiber ports in the optical fiber connector **1664**. Row **1**, column **1** of the optical fiber connector **1662** is a power supply fiber port (**1622***a*), and row **1**, column **1** of the optical fiber connector **1664** is also a power supply fiber port (**1622***b*). Row **1**, column **3** of the optical fiber connector **1662** is a transmitter fiber port (**1614***a*), and row **1**, column **3** of the optical fiber connector **1664** is also a transmitter fiber port (**1614***b*). Row **1**, column **10** of the optical fiber connector **1662** is a receiver fiber port (**1618***a*), and row **1**, column **10** of the optical fiber connector **1664** is also a receiver fiber port (**1618***b*), and so forth.

(645) The optical fiber connectors **1662** and **1664** have the second universal optical fiber interconnection cable port mapping property described above. The port mapping of the optical fiber connector **1662** is a mirror image of the port mapping of the optical fiber connector **1664** after swapping each transmitter port to a receiver port and swapping each receiver port to a transmitter port in the mirror image. The mirror image is generated with respect to a reflection axis **1626** at the connector edge that is parallel to the column direction. The power supply fiber ports (e.g., **1662***a*, **1624***a*) of the optical fiber connector **1662** are mirror images of the power supply fiber ports (e.g., **1622***b*, **1624***b*) of the optical fiber connector **1664**. The transmitter fiber ports (e.g., **1614***a*, **1616***a*) of the optical fiber connector **1662** and the receiver fiber ports (e.g., **1618***b*, **1620***b*) of the optical fiber connector **1664** are pairwise mirror images of each other, i.e., each transmitter fiber port (e.g., **1614***a*, **1616***a*) of the optical fiber connector **1662** is mirrored to a receiver fiber port (e.g., **1618***b*, **1620***b*) of the optical fiber connector **1664**. The receiver fiber ports (e.g., **1618***a*, **1620***a*) of the optical fiber connector **1662** and the transmitter fiber ports (e.g., **1618***b*, **1620***b*) of the optical fiber connector **1664** are pairwise mirror images of each other, i.e., each receiver fiber port (e.g., **1618***a*, **1620***a*) of the optical fiber connector **1662** is mirrored to a transmitter fiber port (e.g., **1618***b*, **1620***b*) of the optical fiber connector **1664**.

(646) For example, the power supply fiber port **1622***a* at row **1**, column **1** of the optical fiber connector **1662** is a mirror image of the power supply fiber port **1624***b* at row **1**, column **12** of the optical fiber connector **1664** with respect to the reflection axis **1626**. The power supply fiber port **1624***a* at row **1**, column **12** of the optical fiber connector **1662** is a

mirror image of the power supply fiber port **1622***b* at row **1**, column **1** of the optical fiber connector **1664**. The transmitter fiber port **1614***a* at row **1**, column **3** of the optical fiber connector **1662** and the receiver fiber port **1618***b* at row **1**, column **10** of the optical fiber connector **1604** are pairwise mirror images of each other. The receiver fiber port **1618***a* at row **1**, column **10** of the optical fiber connector **1662** and the transmitter fiber port **1614***b* at row **1**, column **3** of the optical fiber connector **1664** are pairwise mirror images of each other. The transmitter fiber port **1616***a* at row **3**, column **3** of the optical fiber connector **1662** and the receiver fiber port **1620***b* at row **3**, column **10** of the optical fiber connector **1664** are pairwise mirror images of each other. The receiver fiber port **1620***a* at row **3**, column **10** of the optical fiber connector **1662** and the transmitter fiber port **1616***b* at row **3**, column **3** of the optical fiber connector **1664** are pairwise mirror images of each other.
(647) In addition, and as an alternate view of the second property, each optical fiber connector **1662**, **1664** is TX-RX pairwise symmetric and PS symmetric with respect to the center axis that is parallel to the column direction. Using the first optical fiber connector **1662** as an example, the power supply fiber ports (e.g., **1622***a*, **1624***a*) are symmetric with respect to the center axis, i.e., if there is a power supply fiber port in the left half portion of the first optical fiber connector **1662**, there will also be a power supply fiber port at the mirror location in the right half portion of the first optical fiber connector **1662**. The transmitter fiber ports and the receiver fiber ports are pairwise symmetric with respect to the main axis, i.e., if there is a transmitter fiber port in the left half portion of the first optical fiber connector **1662**, there will be a receiver fiber port at a mirror location in the right half portion of the first optical fiber connector **1662**. Likewise, if there is a receiver fiber port in the left half portion of the optical fiber connector **1662**, there will be a transmitter fiber port at a mirror location in the right half portion of the optical fiber connector **1662**.
(648) If the port mapping of the first optical fiber connector **1662** is represented by port matrix M with entries PS=0, TX=+1, RX=−1, then −custom character=M, in which custom character represents the column-mirror operation, e.g., generating a mirror image with respect to the reflection axis **1626**.
(649) FIG. **90** is a diagram showing another example of the fiber port mapping for an optical fiber interconnection cable **1670** that includes a pair of optical fiber connectors, i.e., a first optical fiber connector **1672** and a second optical fiber connector **1674**. FIG. **143** is an enlarged view of the diagram of FIG. **90**. The power supply power ports are labeled ‘P’, the transmitter fiber ports are labeled ‘T’, and the receiver fiber ports are labeled ‘R’. Only some of the fiber ports are labeled in the figure. In the diagram, the port mapping for the second optical fiber connector **1674** is the same as that of optical fiber connector **1672**. The optical fiber interconnection cable **1670** has the two universal optical fiber interconnection cable port mapping properties described above.
(650) First property: The mapping of the transmitter, receiver, and power supply fiber ports in the first optical fiber connector **1672** is the same as the mapping of the transmitter, receiver, and power supply fiber ports in the second optical fiber connector **1674**.
(651) Second property: The port mapping of the first optical fiber connector **1672** is a mirror image of the port mapping of the second optical fiber connector **1674** after swapping each transmitter port to a receiver port and swapping each receiver port to a transmitter port in the mirror image. The mirror image is generated with respect to a reflection axis **1640** at the connector edge parallel to the row direction.
(652) Alternative view of the second property: Each of the first and second optical fiber connectors **1672**, **1674** is TX-RX pairwise symmetric and PS symmetric with respect to the central axis that is parallel to the row direction. For example, the optical fiber connector **1672** can be divided in two halves along a central axis parallel to the row direction. The power supply fiber ports (e.g., **1678**, **1680**) are symmetric with respect to the center axis. The transmitter fiber ports (e.g., **1682**, **1684**) and the receiver fiber ports (e.g., **1686**, **1688**) are pairwise symmetric with respect to the center axis, i.e., if there is a transmitter fiber port (e.g., **1682** or **1684**) in the upper half portion of the first optical fiber connector **1672**, then there will be a receiver fiber port (e.g., **1686**, **1688**) at a mirror location in the lower half of the optical fiber connector **1672**. Likewise, if there is a receiver fiber port in the upper half portion of the optical fiber connector **1672**, then there is a transmitter fiber port at a mirror location in the lower half portion of the optical fiber connector **1672**. In the example of FIG. **90**, the middle row **1690** should all be power supply fiber ports.
(653) In general, if the port mapping of the first optical fiber connector is a mirror image of the port mapping of the second optical fiber connector after swapping the transmitter and receiver ports in the mirror image, the mirror image is generated with respect to a reflection axis at the connector edge parallel to the row direction (as in the example of FIG. **90**), and there is an odd number of rows in the port matrix, then the center row should all be power supply fiber ports. If the port mapping of the first optical fiber connector is a mirror image of the port mapping of the second optical fiber connector after swapping the transmitter and receiver ports in the mirror image, the mirror image is generated with respect to a reflection axis at the connector edge parallel to the column direction, and there is an odd number of columns in the port matrix, then the center column should all be power supply fiber ports.
(654) FIG. **91** is a diagram of an example of a viable port mapping for an optical fiber connector **1700** of a universal optical fiber interconnection cable. FIG. **144** shows the diagram of FIG. **91** in which the power supply power ports **1702** are labeled ‘P’, the transmitter fiber ports **1704** are labeled ‘T’, and the receiver fiber ports **1706** are labeled ‘R’. The optical fiber connector **1700** includes power supply fiber ports (e.g., **1702**), transmitter fiber ports (e.g., **1704**), and receiver fiber ports (e.g., **1706**). The optical fiber connector **1700** is TX-RX pairwise symmetric and PS symmetric with respect to the center axis that is parallel to the column direction.
(655) FIG. **92** is a diagram of an example of a viable port mapping for an optical fiber connector **1710** of a universal

optical fiber interconnection cable. FIG. **145** shows the diagram of FIG. **92** in which the power supply power ports are labeled ‘P’, the transmitter fiber ports are labeled ‘T’, and the receiver fiber ports are labeled ‘R’. The optical fiber connector **1710** includes power supply fiber ports (e.g., **1712**), transmitter fiber ports (e.g., **1714**), and receiver fiber ports (e.g., **1716**). The optical fiber connector **1710** is TX-RX pairwise symmetric and PS symmetric with respect to the center axis that is parallel to the column direction.

(656) FIG. **93** is a diagram of an example of a port mapping for an optical fiber connector **1720** that is not appropriate for a universal optical fiber interconnection cable. FIG. **146** shows the diagram of FIG. **93** in which the power supply power ports are labeled ‘P’, the transmitter fiber ports are labeled ‘T’, and the receiver fiber ports are labeled ‘R’. The optical fiber connector **1720** includes power supply fiber ports (e.g., **1722**), transmitter fiber ports (e.g., **1724**), and receiver fiber ports (e.g., **1726**). The optical fiber connector **1720** is not TX-RX pairwise symmetric with respect to the center axis that is parallel to the column direction, or the center axis that is parallel to the row direction.

(657) FIG. **94** is a diagram of an example of a viable port mapping for a universal optical fiber interconnection cable that includes a pair of optical fiber connectors, i.e., a first optical fiber connector **1800** and a second optical fiber connector **1802**. The power supply power ports are labeled ‘P’, the transmitter fiber ports are labeled ‘T’, and the receiver fiber ports are labeled ‘R’. The mapping of the transmitter, receiver, and power supply fiber ports in the first optical fiber connector **1800** is the same as the mapping of the transmitter, receiver, and power supply fiber ports in the second optical fiber connector **1802**. The port mapping of the first optical fiber connector **1800** is a mirror image of the port mapping of the second optical fiber connector **1802** after swapping the transmitter and receiver ports in the mirror image. The mirror image is generated with respect to a reflection axis **1804** at the connector edge parallel to the column direction. The optical fiber connector **1800** is TX-RX pairwise symmetric and PS symmetric with respect to the center axis **1806** that is parallel to the column direction.

(658) FIG. **95** is a diagram of an example of a viable port mapping for a universal optical fiber interconnection cable that includes a pair of optical fiber connectors, i.e., a first optical fiber connector **1810** and a second optical fiber connector **1812**. The power supply power ports are labeled ‘P’, the transmitter fiber ports are labeled ‘T’, and the receiver fiber ports are labeled ‘R’. The mapping of the transmitter, receiver, and power supply fiber ports in the first optical fiber connector **1810** is the same as the mapping of the transmitter, receiver, and power supply fiber ports in the second optical fiber connector **1812**. The port mapping of the first optical fiber connector **1810** is a mirror image of the port mapping of the second optical fiber connector **1812** after swapping the transmitter and receiver ports in the mirror image. The mirror image is generated with respect to a reflection axis **1814** at the connector edge parallel to the column direction. The optical fiber connector **1810** is TX-RX pairwise symmetric and PS symmetric with respect to the center axis **1816** that is parallel to the column direction.

(659) In the example of FIG. **95**, the first, third, and fifth rows each has an even number of fiber ports, and the second and fourth rows each has an odd number of fiber ports. In general, a viable port mapping for a universal optical fiber interconnection cable can be designed such that an optical fiber connector includes (i) rows that all have even numbers of fiber ports, (ii) rows that all have odd numbers of fiber ports, or (iii) rows that have mixed even and odd numbers of fiber ports. A viable port mapping for a universal optical fiber interconnection cable can be designed such that an optical fiber connector includes (i) columns that all have even numbers of fiber ports, (ii) columns that all have odd numbers of fiber ports, or (iii) columns that have mixed even and odd numbers of fiber ports.

(660) The optical fiber connector of a universal optical fiber interconnection cable does not have be a rectangular shape as shown in the examples of FIGS. **89**, **90**, **92** to **95**. The optical fiber connectors can also have an overall triangular, square, pentagonal, hexagonal, trapezoidal, circular, oval, or n-sided polygon shape, in which n is an integer larger than 6, as long as the arrangement of the transmitter, receiver, and power supply fiber ports in the optical fiber connectors have the three universal optical fiber interconnection cable port mapping properties described above.

(661) In the examples of FIGS. **80**A, **82**A, **84**A, and **87**A, the switch boxes (e.g., **1302**, **1304**) includes co-packaged optical modules (e.g., **1312**, **1316**) that is optically coupled to the optical fiber interconnection cables or optical cable assemblies (e.g., **1340**, **1400**, **1490**) through fiber array connectors. For example, the fiber array connector can correspond to the first optical connector part **213** in FIG. **20**. The optical fiber connector (e.g., **1342**, **1344**, **1402**, **1404**, **1492**, **1498**) of the optical cable assembly can correspond to the second optical connector part **223** in FIG. **20**. The port map (i.e., mapping of power supply fiber ports, transmitter fiber ports, and receiver fiber ports) of the fiber array connector (which is optically coupled to the photonic integrated circuit) is a mirror image of the port map of the optical fiber connector (which is optically coupled to the optical fiber interconnection cable). The port map of the fiber array connector refers to the arrangement of the power supply, transmitter, and receiver fiber ports when viewed from an external edge of the fiber array connector into the fiber array connector.

(662) As described above, universal optical fiber connectors have symmetrical properties, e.g., each optical fiber connector is TX-RX pairwise symmetric and PS symmetric with respect to one of the main or center axes, which can be parallel to the row direction or the column direction. The fiber array connector also has the same symmetrical properties, e.g., each fiber array connector is TX-RX pairwise symmetric and PS symmetric with respect to one of the main or center axes, which can be parallel to the row direction or the column direction.

(663) In some implementations, a restriction can be imposed on the port mapping of the optical fiber connectors of the optical cable assembly such that the optical fiber connector can be pluggable when rotated by 180 degrees, or by 90 degrees in the case of a square connector. This results in further port mapping constraints.

(664) FIG. **101** is a diagram of an example of an optical fiber connector **1910** having a port map **1912** that is invariant against a 180-degree rotation. Rotating the optical fiber connector **1910** 180 degrees results in a port map **1914** that is the same as the port map **1912**. The port map **1912** also satisfies the second universal optical fiber interconnection cable port mapping property, e.g., the optical fiber connector is TX-RX pairwise symmetric and PS symmetric with respect to the center axis parallel to the column direction.
(665) FIG. **102** is a diagram of an example of an optical fiber connector **1920** having a port map **1922** that is invariant against a 90-degree rotation. Rotating the optical fiber connector **1920** 180 degrees results in a port map **1924** that is the same as the port map **1922**. The port map **1922** also satisfies the second universal optical fiber interconnection cable port mapping property, e.g., the optical fiber connector is TX-RX pairwise symmetric and PS symmetric with respect to the center axis parallel to the column direction.
(666) FIG. **103**A is a diagram of an example of an optical fiber connector **1930** having a port map **1932** that is TX-RX pairwise symmetric and PS symmetric with respect to the center axis parallel to the column direction. When mirroring the port map **1932** to generate a mirror image **1934** and replacing each transmitter port with a receiver port as well as replacing each receiver port with a transmitter port in the mirror image **1934**, the original port map **1932** is recovered. The mirror image **1934** is generated with respect to a reflection axis at the connector edge parallel to the column direction.
(667) Referring to FIG. **103**B, the port map **1932** of the optical fiber connector **1930** is also TX-RX pairwise symmetric and PS symmetric with respect to the center axis parallel to the row direction. When mirroring the port map **1932** to generate a mirror image **1936** and replacing each transmitter port with a receiver port as well as replacing each receiver port with a transmitter port in the mirror image **1936**, the original port map **1932** is recovered. The mirror image **1936** is generated with respect to a reflection axis at the connector edge parallel to the row direction.
(668) In the examples of FIGS. **69**A to **78**, **96** to **98**, and **100**, one or more fans (e.g., **1086**, **1092**, **1848**, **1894**) blow air across the heatsink (e.g., **1072**, **1114**, **1130**, **1168**, **1846**) thermally coupled to the data processor (e.g., **1844**). The co-packaged optical modules can generate heat, in which some of the heat can be directed toward the heatsink and dissipated through the heatsink. To further improve heat dissipation from the co-packaged optical modules, in some implementations, the rackmount system includes two fans placed side-by-side, in which a first fan blows air toward the co-packaged optical modules that are mounted on a front side of the printed circuit board (e.g., **1068**), and a second fan blows air toward the heatsink that is thermally coupled to the data processor mounted on a rear side of the printed circuit board.
(669) In some implementations, the one or more fans can have a height that is smaller than the height of the housing (e.g., **1824**) of the rackmount server (e.g., **1820**). The co-packaged optical modules (e.g., **1074**) can occupy a region on the printed circuit board (e.g., **1068**) that extends in the height direction greater than the height of the one or more fans. One or more baffles can be provided to guide the cool air from the one or more fans or intake air duct to the heatsink and the co-packaged optical modules. One or more baffles can be provided to guide the warm air from the heatsink and the co-packaged optical modules to an air duct that directs the air toward the rear of the housing.
(670) When the one or more fans have a height that is smaller than the height of the housing (e.g., **1824**), the space above and/or below the one or more fans can be used to place one or more remote laser sources. The remote laser sources can be positioned near the front panel and also near the co-packaged optical modules. This allows the remote laser sources to be serviced conveniently.
(671) FIG. **104** shows a top view of an example of a rackmount device **1940**. The rackmount device **1940** includes a vertically oriented printed circuit board or substrate **1230** positioned at a distance behind a front panel **1224** that can be closed during normal operation of the device, and opened for maintenance of the device, similar to the configuration of the rackmount server **1220** of FIG. **77**A. A data processing chip **1070** is electrically coupled to the rear side of the vertical printed circuit board or substrate **1230**, and a heat dissipating device or heat sink **1072** is thermally coupled to the data processing chip **1070**. Co-packaged optical modules **1074** are attached to the front side (i.e., the side facing the front exterior of the housing **1222**) of the vertical printed circuit board or substrate **1230**. A first fan **1942** is provided to blow air across the co-packaged optical modules **1074** at the front side of the printed circuit board or substrate **1230**. A second fan **1944** is provided to blow air across the heatsink **1072** to the rear of the printed circuit board or substrate **1230**. The first and second fans **1942**, **1944** are positioned at the left of the printed circuit board or substrate **1230**. Cooler air (represented by arrows **1946**) is directed from the first and second fans **1942**, **1944** toward the heatsink **1072** and the co-packaged optical modules **1074**. Warmer air (represented by arrows **1948**) is directed from the heatsink **1072** and the co-packaged optical modules **1074** through an air duct **1950** positioned at the right of the printed circuit board or substrate **1230** toward the rear of the housing.
(672) FIG. **105** shows a front view of an example of the rackmount device **1940** when the front panel **1224** is opened to allow access to the co-packaged optical modules **1074**. The first and second fans **1942**, **1944** have a height that is smaller than the height of the region occupied by the co-packaged optical modules **1074**. A first baffle **1952** directs the air from the fan **1942** to the region where the co-packaged optical modules **1074** are mounted, and a second baffle **1954** directs the air from the region where the co-packaged optical modules **1074** are mounted to the air duct **1950**.
(673) In this example, the first and second fans **1942**, **1944** have a height that is smaller than the height of the housing of the rackmount device **1940**. Remote laser sources **1956** can be positioned above and below the fans. Remote laser sources **1956** can also be positioned above and below the air duct **1950**.
(674) For example, a switch device having a 51.2 Tbps bandwidth can use thirty-two 1.6 Tbps co-packaged optical modules. Two to four power supply fibers (e.g., **1326** in FIG. **80**A) can be provided for each co-packaged optical module,

and a total of 64 to 128 power supply fibers can be used to provide optical power to the 32 co-packaged optical modules. One or two laser modules at 500 mW each can be used to provide the optical power to each co-packaged optical module, and 32 to 64 laser modules can be used to provide the optical power to the 32 co-packaged optical modules. The 32 to 64 laser modules can be fitted in the space above and below the fans **1942**, **1944** and the air duct **1950**.

(675) For example, the area **1958***a* above the fans **1942**, **1944** can have an area (measured along a plane parallel to the front panel) of about 16 cm×5 cm and can fit about 28 QSFP cages, and the area **1958***b* below the fans can have an area of about 16 cm×5 cm and can fit about 28 QSFP cages. The area **1958***c* above the air duct **1950** can have an area of about 8 cm×5 cm and can fit about 12 QSFP cages, and the area **1958***d* below the air duct **1950** can have an area of about 8 cm×5 cm and can fit about 12 QSFP cages. Each QSFP cage can include a laser module. In this example, a total of 80 QSFP cages can be fit above and below the fans and the air duct, allowing 80 laser modules to be positioned near the front panel and near the co-packaged optical modules, making it convenient to service the laser modules in the event of malfunction or failure.

(676) Referring to FIGS. **106** and **107**, an optical cable assembly **1960** includes a first fiber connector **1962**, a second fiber connector **1964**, and a third fiber connector **1966**. The first fiber connector **1962** can be optically connected to the co-packaged optical module **1074**, the second fiber connector **1964** can be optically connected to the laser module, and the third fiber connector **1966** can be optically connected to the fiber connector part (e.g., **1232** of FIG. **77**A) at the front panel **1224**. The first fiber connector **1962** can have a configuration similar to that of the fiber connector **1342** of FIGS. **80**C, **80**D. The second fiber connector **1964** can have a configuration similar to that of the fiber connector **1346**. The third fiber connector **1966** can have a configuration similar to that of the first fiber connector **1962** but without the power supply fiber ports. The optical fibers **1968** between the first fiber connector **1962** and the third fiber connector **1966** perform the function of the fiber jumper **1234** of FIG. **77**A.

(677) FIG. **108** is a diagram of an example of a rackmount device **1970** that is similar to the rackmount device **1940** of FIGS. **104**, **105**, **107**, except that the optical axes of the laser modules **1956** are oriented at an angle θ relative to the front-to-rear direction, 0<θ<90°. This can reduce the bending of the optical fibers that are optically connected to the laser modules **1956**.

(678) FIG. **109** is a diagram showing the front view of the rackmount device **1970**, with the optical cable assembly **1960** optically connected to modules of the rackmount device **1970**. When the laser modules **1956** are oriented at an angle θ relative to the front-to-rear direction, 0<θ<90°, fewer laser modules **1956** can be placed in the spaces above and below the fans **1942**, **1944** and the air duct **1950**, as compared to the example of FIGS. **104**, **105**, **107**, in which the optical axes of the laser modules **1956** are oriented parallel to the front-to-rear direction. In the example of FIG. **109**, a total of 64 laser modules are placed in the spaces above and below the fans **1942**, **1944** and the air duct **1950**.

(679) FIG. **110** is a top view diagram of an example of a rackmount device **1980** that is similar to the rackmount device **1940** of FIGS. **104**, **105**, **107**, except that the optical axes of the laser modules **1956** are oriented parallel to the front panel **1224**. This can reduce the bending of the optical fibers that are optically connected to the laser modules **1956**.

(680) FIG. **111** is a front view diagram of the rackmount device **1980**, with the optical cable assembly **1960** optically connected to modules of the rackmount device **1980**. The laser modules **1956***a* are positioned to the left side of the space above and below the fans **1942**, **1944**. Sufficient space (e.g., **1982**) is provided at the right of the laser modules **1956***a* to allow the user to conveniently connect or disconnect the fiber connectors **1964** to the laser modules **1956***a*. The laser modules **1956***b* are positioned above and below the air duct **1950**. Sufficient space (e.g., **1984**) is provided at the left of the laser modules **1956***b* to allow the user to conveniently connect or disconnect the fiber connectors **1964** to the laser modules **1956***b*.

(681) Referring to FIG. **112**, a table **1990** shows example parameter values of the rackmount device **1940**.

(682) FIGS. **113** and **114** show another example of a rackmount device **2000** and example parameter values.

(683) FIGS. **115** and **116** are a top view and a front view, respectively, of the rackmount device **2000**. An upper baffle **2002** and a lower baffle **2004** are provided to guide the air flowing from the fans **1942**, **1944** to the heatsink **1072** and the co-packaged optical modules **1074**, and from the heatsink **1072** and the co-packaged optical modules **1074** to the air duct **1950**. In this example, portions of the upper and lower baffles **2002**, **2004** form portions of the upper and lower walls of the air duct **1950**.

(684) The upper baffle **2002** includes a cutout or opening **2006** that allows optical fibers **2008** to pass through. As shown in FIG. **116**, the optical fibers **2008** extend from the co-packaged optical modules **1074** upward, through the cutout or opening **2006** in the upper baffle **2002**, and extend toward the laser modules **1956** along the space above the upper baffle **2002**. The upper baffle **2002** allows the optical fibers **2008** to be better organized to reduce the obstruction to the air flow caused by the optical fibers **2008**. The lower baffle **2004** has a similar cutout or opening to help organize the optical fibers that are optically connected to the laser modules located in the space below the fans **1942**, **1944**.

(685) FIG. **117** is a top view diagram of a system **11700** that includes a front panel **11702**, which can be rotatably coupled to the lower panel by a hinge. FIG. **117** shows the front panel **11702** in the open position. The front panel **11702** includes an air inlet grid **11704** and an array of fiber connector parts **11706**. Each fiber connector part **11706** can be optically coupled to the third fiber connector **1966** of the cable assembly **1960** of FIG. **106**. In some implementations, the hinged front panel includes a mechanism that shuts off the remote laser source modules **1956**, or reduces the power to the remote laser source modules **1956**, once the front panel **11702** is opened. This prevents the technicians from being exposed to harmful radiation. In this example, the laser source modules **1956** and the optical fibers for providing the power supply

light are disposed behind the front panel **11702**.

(686) FIG. **118** is a diagram of an example of a system **2120** that includes a recirculating reservoir **2122** that circulates a coolant **2124** to carry heat away from the data processor **2126**, which for example can be a switch integrated circuit. In this example, the data processor **2126** is mounted at the back side of the substrate and obscured from view. In this example, the data processor **2126** is immersed in the coolant **2124**, and the inlet fan **2128** is used to blow air across the surface of the co-packaged optical modules **2130** to a heat dissipating device thermally coupled to the co-packaged optical modules.

(687) FIGS. **119** to **122** are examples that provide heat dissipating solutions for co-packaged optical modules, taking into consideration the locations of "hot aisles" in data centers. FIG. **119** shows a top view of an environment **11900**, e.g., in a data center, in which multiple rackmount servers **11902** are installed. The servers **11902** include inlet fans **11904** positioned at the front **11906** and outlet fans **11908** at the rear **11910**. Cold air enters the housing **11912** from the front **11906**, the air is warmed by the heat generating electronic devices in the server **11902**, and hot air is blown out of the housing **11912** from the rear **11910**. The aisle in the data center in front of the servers **11902** is referred to as the "cold aisle" **11914**, and the aisle to the rear of the servers **11902** is referred to as the "hot aisle" **11916**.

(688) FIG. **120** is a simulation of the thermal properties of the rackmount server **11902** in which the heat sink **1846** is thermally coupled a second heat sink **17202** through heat pipes **17204**. In this simulation, the temperature distribution of the server **11902** ranged from about 27° C. to about 110.5° C. The region **11920** where the inlet fans **11904** are located has a temperature of about 27° C., which is the room temperature used in the simulation. The junction **11922** between the data processor and the heat sink has a temperatures below 105° C., which shows that the thermal design used in this example can provide adequate cooling to the data processor electrically coupled to the vertical circuit board positioned near the front panel.

(689) Referring to FIG. **121**, in some implementations, in case it is desirable that fiber cabling be implemented on the back side of a rack (where the hot aisle is located), a rackmount server **12000** can include a duct **12002** inside the housing **11912** to transfer cold air to the co-packaged optical modules **12004** that are now mounted on the back side. In this example, one or more inlet fans **12006** are provided at the front of the duct **12002**, and one or more fans **12008** are provided at the rear of the duct **12002** to blow the air toward the heat sink **12010** thermally coupled to the data processor, and to the co-packaged optical modules **12004**.

(690) Referring to FIG. **122**, in some implementations, a rackmount server **12100** includes fiber jumper cables **12102** that connect the co-packaged optical modules **12004** that are still facing the front aisle (towards the cold aisle **11914**) to a back-panel **12104** facing the hot aisle **11916**.

(691) Referring to FIG. **123**, in some implementations, a vertically mounted processor blade **12300** can include a substrate **12302** having a first side **12304** and a second side **12306**. The substrate **12302** can be made of, e.g., one or more ceramic materials, or organic "high density build-up" (HDBU). For example, the substrate **12302** can be a printed circuit board. An electronic processor **12308** is mounted on the first side **12304** of the substrate **12302**, in which the electronic processor **12308** is configured to process or store data. For example, the electronic processor **12308** can be a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC). For example, the electronic processor **12308** can be a memory device or a storage device. In this context, processing of data includes writing data to, or reading data from, the memory or storage device, and optionally performing error correction. The memory device can be, e.g., random access memory (RAM), which can include, e.g., dynamic RAM (DRAM) or static RAM (SRAM). The storage device can include, e.g., solid state memory or drive, which can include, e.g., one or more non-volatile memory (NVM) Express® (NVMe) SSD (solid state drive) modules, or Intel® Optane™ persistent memory. The example of FIG. **123** shows one electronic processor **12308**, through there can also be multiple electronic processors **12308** mounted on the substrate **12302**.

(692) The vertically mounted processor blade **12300** includes one or more optical interconnect modules or co-packaged optical modules **12310** mounted on the second side **12306** of the substrate **12302**. For example, the optical interconnect module **12310** includes an optical port configured to receive optical signals from an external optical fiber cable, and a photonic integrated circuit configured to generate electrical signals based on the received optical signals, and transmit the electrical signals to the electronic processor **12308**. The photonic integrated circuit can also be configured to generate optical signals based on electrical signals received from the electronic processor **12308**, and transmit the optical signals to the external optical fiber cable. The optical interconnect module or co-packaged optical module **12310** can be similar to, e.g., the integrated optical communication device **262** of FIG. **6**; **282** of FIGS. **7-9**; **462**, **466**, **448**, **472** of FIG. **17**; **612** of FIG. **23**; **684** of FIG. **26**; **704** of FIG. **27**; **724** of FIG. **28**; the co-packaged optical module **1074** of FIGS. **68**A, **69**A, **70**, **71**A; **1132** of FIG. **73**A; **1160** of FIG. **74**A; **1074** of FIGS. **75**A, **75**B, **77**A, **77**B, **104**, **107**, **109**, **116**; **1312** of FIGS. **80**A, **82**A, **84**A; or **1564**, **1582** of FIG. **87**A. In the example of FIG. **123**, the optical interconnect module or co-packaged optical module **12310** does not necessarily have to include serializers/deserializers (SerDes), e.g., **216**, **217** of FIGS. **2** to **8** and **10** to **12**. The optical interconnect module or co-packaged optical module **12310** can include the photonic integrated circuit **12314** without any serializers/deserializers. For example, the serializers/deserializers can be mounted on the substrate separate from the optical interconnect module or co-packaged optical module **12310**.

(693) For example, the substrate **12302** can include electrical connectors that extend from the first side **12304** to the second side **12306** of the substrate **12302**, in which the electrical connectors pass through the substrate **12302** in a

thickness direction. For example, the electrical connectors can include vias of the substrate **12302**. The optical interconnect module **12310** is electrically coupled to the electronic processor **12308** by the electrical connectors.
(694) For example, the vertically mounted processor blade **12300** can include an optional optical fiber connector **12312** for connection to an optical fiber cable bundle. The optical fiber connector **12312** can be optically coupled to the optical interconnector modules **12310** through optical fiber cables **12314**. The optical fiber cables **12314** can be connected to the optical interconnect modules **12310** through a fixed connector (in which the optical fiber cable **12314** is securely fixed to the optical interconnect module **12310**) or a removable connector in which the optical fiber cable **12314** can be easily detached from the optical interconnect module **12310**, such as with the use of an optical connector part **266** as shown in FIG. **6**. The removable connector can include a structure similar to the mechanical connector structure **900** of FIGS. **46**, **47** and **51**A to **57**.
(695) For example, the substrate **12302** can be positioned near the front panel of the housing of the server that includes the vertically mounted processor blade **12300**, or away from the front panel and located anywhere inside the housing. For example, the substrate **12302** can be parallel to the front panel of the housing, perpendicular to the front panel, or oriented in any angle relative to the front panel. For example, the substrate **12302** can be oriented vertically to facilitate the flow of hot air and improve dissipation of heat generated by the electronic processor **12308** and/or the optical interconnect modules **12310**.
(696) For example, the optical interconnect module or co-packaged optical module **12310** can receive optical signals through vertical or edge coupling. FIG. **123** shows an example in which the optical fiber cables are vertically coupled to the optical interconnect modules or co-packaged optical modules **12310**. It is also possible to connect the optical fiber cables to the edges of the optical interconnect modules or co-packaged optical modules **12310**. For example, optical fibers in the optical fiber cable can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc.
(697) For example, the optical interconnect modules **12310** can receive optical power from an optical power supply, such as **1322** of FIG. **80**A, **1558** of FIG. **87**A. For example, the optical interconnect modules **12310** can include one or more of optical coupling interfaces **414**, demultiplexers **419**, splitters **415**, multiplexers **418**, receivers **421**, or modulators **417** of FIG. **20**.
(698) FIG. **124** is a top view of an example of a rack system **12400** that includes several vertically mounted processor blades **12300**. The vertically mounted processor blades **12300** can be positioned such that the optical fiber connectors **12312** are near the front of the rack system **12400** (which allows external optical fiber cables to be optically coupled to the front of the rack system **12400**), or near the back of the rack system **12400** (which allows external optical fiber cables to be optically coupled to the back of the rack system **12400**). Several rack systems **12400** can be stacked vertically similar to the example shown in FIG. **76**, in which the server rack **1214** includes several servers **1212** stacked vertically, or the example shown in FIG. **87**A, in which several servers **1552** are stacked vertically in a rack **1554**. For example, the optical interconnect modules **12310** can receive optical power from an optical power supply, such as **1558** of FIG. **87**A.
(699) In some implementations, the vertically mounted processor blades **12300** can include blade pairs, in which each blade pair includes a switch blade and a processor blade. The electronic processor of the switch blade includes a switch, and the electronic processor of the processor blade is configured to process data provided by the switch. For example, the electronic processor of the processor blade is configured to send processed data to the switch, which switches the processed data with other data, e.g., data from other processor blades.
(700) In the examples shown in FIGS. **123** and **124**, the optical interconnect module or co-packaged optical module **12310** is mounted on the second side of the substrate **12302**. In some implementations, the optical interconnect module **12310** or the optical fiber cable **12314** extends through or partially through an opening in the substrate **12302**, similar to the example shown in FIGS. **35**A to **35**C. The photonic integrated circuit in the optical interconnect module **12310** is electrically coupled to the electronic processor **12308** or to another electronic circuit, such as a serializers/deserializers module positioned at or near the first side of the substrate **12302**. The optical interconnect module **12310** and the optical fiber cable **12314** define a signal path that allows a signal from the optical fiber cable **12314** to be transmitted from the second side of the substrate **12302** through the opening to the electronic processor **12308**. The signal is converted from an optical signal to an electric signal by the photonic integrated circuit, which defines part of the signal path. This allows the optical fiber cables to be positioned on the second side of the substrate **12302**.
(701) In the example of FIG. **104**, the printed circuit board or substrate **1230** is positioned a short distance from the front panel **1224** to improve air flow between the printed circuit board or substrate **1230** and the front panel **1224** to help dissipate heat generated by the co-packaged optical modules **1074**. The following describes a mechanism that allows the user to conveniently connect the co-packaged optical module to an optical fiber cable using a pluggable module that has a rigid structure that spans the distance between the co-packaged optical modules and the front panel.
(702) Referring to FIG. **125**A, in some implementations, a rackmount server **12370** can have a hinge-mounted front panel, similar to the example shown in FIG. **77**A. The rackmount server **12370** includes a housing **12372** having a top panel **12374**, a bottom panel **12376**, and a front panel **12378** that is coupled to the bottom panel **12376** using a hinge **12324**. A vertically mounted substrate **12880** is positioned substantially perpendicular to the bottom panel **12376** and recessed from the front panel **12378**. The substrate **12380** includes a first side facing the front direction relative to the housing **12372** and

a second side facing the rear direction relative to the housing **12372**. At least one electronic processor or data processing chip **12382** is electrically coupled to the second side of the vertical substrate **12380**, and a heat dissipating device or heat sink **12384** is thermally coupled to the at least one data processing chip **12382**. Co-packaged optical modules **12386** (or optical interconnect modules) are attached to the first side of the vertical substrate **12380**. The substrate **12380** provides high-speed connections between the co-packaged optical modules **12386** and the data processing chip **12382**. The co-packaged optical module **12386** is optically connected to a first fiber connector part **12388**, which is optically connected through a fiber pigtail **12320** to one or more second fiber connector parts **12322** mounted on the front panel **12378**.
(703) In the example of FIG. **125**A, the front panel **12378** is rotatably connected to the bottom panel by the hinge **12324**. In other examples, the front panel **12378** can be rotatably connected to the top panel **12374** or the side panel **12376** so as to flap upwards or to flap sideways when opened.
(704) For example, the electronic processor or data processing chip **12382** can be a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC). For example, the electronic processor **12382** can be a memory device or a storage device. In this context, processing of data includes writing data to, or reading data from, the memory or storage device, and optionally performing error correction. The memory device can be, e.g., random access memory (RAM), which can include, e.g., dynamic RAM (DRAM) or static RAM (SRAM). The storage device can include, e.g., solid state memory or drive, which can include, e.g., one or more non-volatile memory (NVM) Express® (NVMe) SSD (solid state drive) modules, or Intel® Optane™ persistent memory. The example of FIG. **125**A shows one electronic processor **12382**, through there can also be multiple electronic processors **12382** mounted on the substrate **12380**. In some examples, the substrate **12380** can also be replaced by a circuit board.
(705) The co-packaged optical module (or optical interconnect module) **12386** can be similar to, e.g., the integrated optical communication device **262** of FIG. **6**; **282** of FIGS. **7-9**; **462**, **466**, **448**, **472** of FIG. **17**; **612** of FIG. **23**; **684** of FIG. **26**; **704** of FIG. **27**; **724** of FIG. **28**; the co-packaged optical module **1074** of FIGS. **68**A, **69**A, **70**, **71**A; **1132** of FIG. **73**A; **1160** of FIG. **74**A; **1074** of FIGS. **75**A, **75**B, **77**A, **77**B, **104**, **107**, **109**, **116**; **1312** of FIGS. **80**A, **82**A, **84**A; or **1564**, **1582** of FIG. **87**A. In the example of FIG. **125**A, the optical interconnect module or co-packaged optical module **12386** does not necessarily have to include serializers/deserializers (SerDes), e.g., **216**, **217** of FIGS. **2** to **8** and **10** to **12**. The optical interconnect module or co-packaged optical module **12386** can include the photonic integrated circuit without any serializers/deserializers. For example, the serializers/deserializers can be mounted on the circuit board separate from the optical interconnect module or co-packaged optical module **12386**.
(706) FIG. **130** is a side view of an example of a rackmount server **13000** that has a hinge-mounted front panel. The rackmount server **13000** includes a housing **13002** having a top panel **13004**, a bottom panel **13006**, and an upper swivel front panel **13008** that is coupled to a lower fixed front panel **13030** using a hinge **13010**. In some examples, the hinge can be attached to the side panel so that the front panel is opened horizontally. A horizontally mounted host printed circuit board **13012** is attached to the bottom panel **13006**. A vertically mounted printed circuit board **13014**, which can be, e.g., a daughter-card, is positioned substantially vertically and perpendicular to the bottom panel **13006** and recessed from the front panel **13008**. A package substrate **13016** is attached to the front side of the vertical printed circuit board **13014**. At least one electronic processor or data processing chip **13018** is electrically coupled to the rear side of the package substrate **13016**, and a heat dissipating device or heat sink **13020** is thermally coupled to the at least one data processing chip **13018**. Co-packaged optical modules **13022** (or optical interconnect modules) are removably attached to the front side of the package substrate **13016**. The package substrate **13016** provides high-speed connections between the co-packaged optical modules **13022** and the data processing chip **13018**. The co-packaged optical module **13022** is optically connected to a first fiber connector part **13024**, which is optically connected through a fiber pigtail **13026** to one or more second fiber connector parts **13028** attached to the back side of the front panel **13008**. The second fiber connector parts **13028** can be optically connected to optical fiber cables that pass through openings in the hinged front panel **13008**.
(707) For example, the fiber connector **13028** can be connected to the backside of the front panel **13008** during replacement of the CPO module **13022**. The CPO module **13022** can be unplugged from the connector (e.g., an LGA socket) on the package substrate **13016**, and be disconnected from the first fiber connector part **13024**.
(708) For example, one or more rows of pluggable external laser sources (ELS) **13032** can be in standard pluggable form factor accessible from the lower fixed part **13030** of the front panel with rear blind-mate connectors. Optical fibers **13034** transmit the power supply light from the laser sources **13032** to the CPO modules **13022**. The external laser sources **13032** are electrically connected to a conventionally (horizontal) oriented system printed circuit board or the vertically oriented daughterboard. In this example, the row(s) of pluggable external laser sources **13032** is/are positioned below the datapath optical connection. The pluggable external laser sources **13032** do not need to connect to the CPO substrate because there are no high-speed signals that require proximity.
(709) In some implementations, as shown in FIG. **131**, external laser sources can be located behind the hinged front panel (not user accessible without opening the door) and can then be front-mating similar to typical optical pluggables. FIG. **131** is a top view of an example of a rackmount server **13100** that is similar to the rackmount server **13000** of FIG. **130** except that one or more rows of external laser sources **13102** are placed inside the housing **13002**. Optical fibers **13034** transmit the power supply light from the laser sources **13102** to the CPO modules **13022**.
(710) FIG. **132** is a diagram of an example of the optical cable **13026** that optically couples the CPO modules **13022** to the optical fiber cables at the front panel **13008**. The optical cable **13026** includes a first multi-fiber push on (MPO) connector

13200, a laser supply MPO connector **13202**, four datapath MPO connectors **13204**, and a jumper cable **13206** that includes optical fibers that optically connect the MPO connectors. In this example, the optical cable **13026** supports a total bandwidth of 1.6 Tb/s, including 16 full-duplex 400 G DR4+ signals (100 G per fiber) plus 4 external laser source (ELS) connections.

(711) The first MPO connector **13200** is optically coupled to the CPO module **13022** and includes, e.g., 36 fiber ports (e.g., 3 rows of fiber ports, each row having 12 fiber ports, similar to the fiber ports shown in FIGS. **80**D, **80**E, **82**D, **82**E, **89** to **93**), which includes 4 power supply fiber ports and 32 data fiber ports. The laser supply MPO connector **13202** is optically coupled to the external laser source, such as **13032** (FIG. **130**) or **13102** (FIG. **131**). The datapath MPO connectors **13204** are optically coupled to external optical fiber cables. For example, each external optical fiber cable can support a 400 GBASE-DR4 link, so the four datapath MPO connectors **13204** can support 16 full-duplex 400 G DR4+ signals (100 G per fiber). The jumper cable **13206** fans the MPO connector **13200** out to datapath MPOs **13204** on the front panel **13008** (e.g., 4×400 G DR4+ using 4×1×12 MPOs or 2×800 G DR8+ using 2×2×12 MPOs) and the laser supply MPO **13202**. For example, the optical cable **13026** can be DR-16+ (e.g., 1.6 Tb/s at 100 G per fiber, gray optics, ˜2 km reach). This architecture also supports FR-n (WDM).

(712) In this example, the CPO module **13022** is configured to support 4×400 Gb/s=1.6 Tb/s data rate. The jumper cable **13206** includes four (4) power supply optical fibers **13034** that optically connect four (4) power supply fiber ports of the laser supply MPO connector **13202** to the corresponding power supply fiber ports of the first MPO connector **13200**. The jumper cable **13206** includes four (4) sets of eight (8) data optical fibers **13208**. The eight (8) data optical fibers **13208** optically connect eight (8) transmit or receive fiber ports of each datapath MPO connector **13204** to the corresponding transmit or receive fiber ports of the first MPO connector **13200**. For example, the power supply optical fibers **13034** can be polarization maintaining optical fibers. The fan-out cable **13206** can handle multiple functions including merging the external laser source and data paths, splitting of external light source between multiple CPO modules **13022**, and handling polarization. Regarding the force requirement on the CPO module's connector, the optical connector leverages an MPO type connection and can have a similar or smaller force as compared to a standard MPO connector.

(713) Referring to FIG. **125**B, in some implementations, a rackmount server **12420** has a front panel **12402** (which can be, e.g., fixed) and a vertically mounted substrate **12380** recessed from the front panel **12402**. The front panel **12402** has openings that allow pluggable modules **12404** to be inserted. Each pluggable module **12404** includes a co-packaged optical module **12386**, one or more multi-fiber push on (MPO) connectors **12406**, a fiber guide **12408** that mechanically connects the co-packaged optical module **12386** to the one or more multi-fiber push on connectors **12406**, and a fiber pigtail **12410** that optically connects the co-packaged optical module **12386** to the one or more multi-fiber push on connectors **12406**. For example, the length of the fiber guide **12408** is designed such that when the pluggable module **12404** is inserted into the opening of the front panel **12402** and the co-packaged optical module **12386** is electrically coupled to the vertically mounted substrate **12380**, the one or more multi-fiber push on connectors **12406** are near the front panel, e.g., flush with, or slightly protrude from, the front panel **12402** so that the user can conveniently attach external fiber optic cables. For example, the front face of the connectors **12406** can be within an inch, or half an inch, or one-fourth of an inch, of the front surface of the front panel **12402**.

(714) For example, the housing **12372** can include guide rails or guide cage **12412** that help guide the pluggable modules **12404** so that the electrical connectors of the co-packaged optical modules **12386** are aligned with the electrical connectors on the printed circuit board.

(715) In some implementations, the rackmount server **12420** has inlet fans mounted near the front panel **12402** and blow air in a direction substantially parallel to the front panel **12402**, similar to the examples shown in FIGS. **96** to **98**, **100**, **104**, **105**, **107** to **116**. The height h**1** of the fiber guide **12408** (measured along a direction perpendicular to the bottom panel) can be designed to be smaller than the height h**2** of the multi-fiber push on connectors **12406** so that there is space **12412** between adjacent fiber guides **12408** (in the vertical direction) to allow air to flow between the fiber guides **12408**. The fiber guide **12408** can be a hollow tube with inner dimensions sufficiently large to accommodate the fiber pigtail **12410**. The fiber guide **12408** can be made of metal or other thermally conductive material to help dissipate heat generated by the co-packaged optical module **12386**. The fiber guide **12408** can have arbitrary shapes, e.g., to optimize thermal properties. For example, the fiber guide **12408** can have side openings, or a web structure, to allow air to flow pass the fiber guide **12408**. The fiber guide **12408** is designed to be sufficiently rigid to enable the pluggable module **12404** to be inserted and removed from the rackmount server **12420** multiple times (e.g., several hundred times, several thousand times) under typical usage without deformation.

(716) FIG. **126**A includes various views of an example of a rackmount server **12500** that includes CPO front-panel pluggable modules **12502**. Each pluggable module **12502** includes a co-packaged optical module **12504** that is optically coupled to one or more array connectors, such as multi-fiber push on connectors **12506**, through a fiber pigtail **12508**. In this example, each co-packaged optical module **12504** is optically coupled to 2 array connectors **12506**. The pluggable module **12502** includes a rigid fiber guide **12510** that approximately spans the distance between the front panel and the vertically mounted printed circuit board or substrate.

(717) A front view **12512** (at the upper right of FIG. **126**A) shows an example of a front panel **12514** with an upper group of array connectors **12516**, a lower group of array connectors **12518**, a left group of array connectors **12520**, and a right group of array connectors **12522**. Each rectangle in the groups of array connectors **12516**, **12518**, **12520**, **12522** in the front view **12512** represents an array connector **12506**. In this example, each group of array connectors **12516**, **12518**,

12520, **12522** includes 16 array connectors **12506**.

(718) A front view **12524** (at the middle right of FIG. **126**A) shows an example of a recessed vertically mounted printed circuit board or substrate **12526** on which an application specific integrated circuit (ASIC) or data processing chip **12382** is mounted on the rear side and not shown in the front view **12524**. The printed circuit board or substrate **12526** has an upper group of electrical contacts **12528**, a lower group of electrical contacts **12530**, a left group of electrical contacts **12532**, and a right group of electrical contacts **12534**. Each rectangle in the groups of electrical contacts **12528**, **12530**, **12532**, **12534** in the front view **12524** represents an array of electrical contacts associated with one co-packaged optical module **12504**. In this example, each group of electrical contacts **12528**, **12530**, **12532**, **12534** includes 8 arrays of electrical contacts that are configured to be electrically coupled to the electrical contacts of 8 co-packaged optical modules **12504**. In this example, each co-packaged optical module **12504** is optically coupled to two array connectors **12506**, so the number of rectangles shown in the front view **12512** is twice the number of squares shown in the front view **12524**. The front panel **12514** includes openings that allow insertion of the pluggable modules **12502**. In this example, each opening has a size that can accommodate two array connectors **12506**.

(719) A top view **12536** (at the lower right of FIG. **126**A) of the front portion of the rackmount server **12500** shows a top view of the pluggable modules **12506**. In the top view **12536**, the two leftmost pluggable modules **12538** include co-packaged optical modules **12504** that are electrically coupled to the electrical contacts in the left group of electrical contacts **12532** shown in the front view **12524**, and include array connectors **12506** in the left group of array connectors **12520** shown in the front view **12512**. In the top view **12536**, the two right-most pluggable modules **12540** include co-packaged optical modules **12504** that are electrically coupled to the electrical contacts in the right group of electrical contacts **12534** shown in the front view **12524**, and include array connectors **12506** in the right group of array connectors **12522** shown in the front view **12512**. In the top view **12536**, the four middle pluggable modules **12542** include co-packaged optical modules **12504** that are electrically coupled to the electrical contacts in the upper group of electrical contacts **12528** shown in the front view **12524**, and include array connectors **12506** in the upper group of array connectors **12516** shown in the front view **12512**.

(720) The front view **12524** (at the middle right of FIG. **126**A) shows a first inlet fan **12544** that blows air from left to right across the space between the front panel **12514** and the printed circuit board or substrate **12526**. The top view **12536** (at the lower right of FIG. **126**A) shows the first inlet fan **12544** and a second inlet fan **12546**. The first inlet fan **12544** is mounted at the front side of the printed circuit board or substrate **12526** and blows air across the pluggable modules **12502** to help dissipate the heat generated by the co-packaged optical modules **12504**. The second inlet fan **12546** is mounted at the rear side of the printed circuit board or substrate **12526** and blows air across the data processing chip **12382** and the heat dissipating device **12314**.

(721) As shown in the front view **12512** (at the upper right of the FIG. **126**A), the front panel **12514** includes an opening **12548** that provides incoming air for the front inlet fans **12544**, **12546**. A protective mesh or grid can be provided at the opening **12548**.

(722) A left side view **12550** (at the middle left of FIG. **126**A) of the front portion of the rackmount server **12500** shows pluggable modules **12552** that correspond to the upper group of array connectors **12516** in the front view **12512** and the upper group of electrical contacts **12528** in the front view **12524**. The left side view **12550** also shows pluggable modules **12554** that correspond to the lower group of array connectors **12518** in the front view **12512** and the lower group of electrical contacts **12530** in the front view **12524**. As shown in the left side view **12550**, guide rails or guide cage **12556** can be provided to help guide the pluggable modules **12502** so that the electrical connectors of the co-packaged optical modules **12504** are aligned with the electrical contacts on the printed circuit board or substrate **12526**. The pluggable modules **12502** can be fastened at the front panel **12514**, e.g., using clip mechanisms.

(723) A left side view **12558** of the front portion of the rackmount server **12500** shows pluggable modules **12560** that correspond to the left group of array connectors **12520** in the front view **12512** and the left group of electrical contacts **12532** in the front view **12524**.

(724) In this example, the fiber guides **12510** for the pluggable modules **12502** that correspond to the left and right groups of array connectors **12520**, **12522**, and the left and right groups of electrical contacts **12532**, **12534** are designed to have smaller heights so that there are gaps between adjacent fiber guides **12510** in the vertical direction to allow air to flow through.

(725) In some implementations, each co-packaged optical module can receive optical signals from a large number of fiber cores, and each co-packaged optical module can be optically coupled to external fiber optic cables through three or more array connectors that occupy an overall area at the front panel that is larger than the overall area occupied by the co-packaged optical module on the printed circuit board.

(726) Referring to FIG. **126**B, in some implementations, a rackmount server **12600** is designed to use pluggable modules **12602** having a spatial fan-out design. Each pluggable module **12602** includes a co-packaged optical module **12604** that is optically coupled, through a fiber pigtail **12606**, to one or more array connectors **12608** that have an overall area larger than the area of the co-packaged optical module **12604**. The area is measured along the plane parallel to the front panel. In this example, each co-packaged optical module **12604** is optically coupled to 4 array connectors **12608**. The pluggable module **12602** includes a tapered fiber guide **12610** that is narrower near the co-packaged optical module **12604** and wider near the array connectors **12608**.

(727) A front view **12612** (at the upper right of FIG. **126**B) shows an example of a front panel **12614** that can

accommodate an array of 128 array connectors **12608** arranged in 16 rows and 8 columns. The front view **12524** (at the middle right of FIG. **126**B) of the recessed printed circuit board or substrate **12526** and the top view (at the lower right of FIG. **126**B) of the front portion of the rackmount server **12600** are similar to corresponding views in FIG. **126**A.

(728) A left side view **12616** (at the middle left of FIG. **126**B) shows an example of pluggable modules **12602** that have co-packaged optical modules that are connected to the upper and lower groups of electrical contacts on the printed circuit board or substrate **12526**. A left side view **12618** (at the lower left of FIG. **126**B) shows an example of pluggable modules **12602** that have co-packaged optical modules that are connected to the left group of electrical contacts on the printed circuit board or substrate **12526**. As shown in the left side view **12618**, guide rails or guide cage **12620** can be provided to help guide the pluggable modules **12602** so that the electrical contacts of the co-packaged optical modules **12604** are aligned with corresponding electrical contacts on the printed circuit board or substrate **12526**.

(729) For example, the rackmount server **12420**, **12500**, **12600** can be provided to customers with or without the pluggable modules. The customer can insert as many pluggable modules as needed.

(730) Referring to FIG. **127**, in some implementations, a CPO front panel pluggable module **12700** can include a blind mate connector **12702** that is designed receive optical power supply light. A portion of the fiber pigtail **12714** is optically coupled to the blind mate connector **12702**. FIG. **127** includes a side view **12704** of a rackmount server **12706** that includes laser sources **12708** that provide optical power supply light to the co-packaged optical modules **12710** in the pluggable modules **12700**. The laser sources **12708** are optically coupled, through optical fibers **12712**, to optical connectors **12714** that are configured to mate with the blind-mate connectors **12702** on the pluggable modules **12700**. When the pluggable module **12700** is inserted into the rackmount server **12706**, the electrical contacts of the co-packaged optical module **12710** contacts the corresponding electrical contacts on the printed circuit board or substrate **12526**, and the blind-mate connector **12702** mates with the optical connector **12714**. This allows the co-packaged optical module **12710** to receive optical signals from external fiber optic cables and the optical power supply light through the fiber pigtail **12714**.

(731) In some implementations, to prevent the light from the laser source **12708** from harming operators of the rackmount server **12706**, a safety shut-off mechanism is provided. For example, a mechanical shutter can be provided on disconnection of the blind-mate connector **12702** from the optical connector **12712**. As another example, electrical contact sensing can be used, and the laser can be shut off upon detecting disconnection of the blind-mate connector **12702** from the optical connector **12712**.

(732) Referring to FIG. **128**, in some implementations, one or more photon supplies **12800** can be provided in the fiber guide **12408** to provide power supply light to the co-packaged optical module **12386** through one or more power supply optical fibers **12802**. The one or more photon supplies **12800** can be selected to have a wavelength (or wavelengths) and power level (or power levels) suitable for the co-packaged optical module **12386**. Each photon supply **12800** can include, e.g., one or more diode lasers having the same or different wavelengths.

(733) Electrical connections (not shown in the figure) can be used to provide electrical power to the one or more photon supplies **12800**. In some implementations, the electrical connections are configured such that when the co-packaged optical module **12386** is removed from the substrate **12380**, the electrical power to the one or more photon supplies **12800** is turned off. This prevents light from the one or more photon supplies **12800** from harming operators. Additional signals lines (not shown in the figure) can provide control signals to the photon supply **12800**. In some embodiments, electrical connections to the photon supplies **12800** are made to the system through the CPO module **12386**. In some embodiments, electrical connections to the photon supplies **12800** use parts of the fiber guide **12408**, which in some embodiments is made from electrically conductive materials. In some embodiments, the fiber guide **12408** is made of multiple parts, some of which are made from electrically conductive materials and some of which are made from electrically insulating materials. In some embodiments, two electrically conductive parts are mechanically connected but electrically separated by an electrical insulating part. For example, a first electrically conductive part of the fiber guide **12408** can be connected to the positive power input terminal of the photon supply (e.g., laser), and a second electrically conductive part of the fiber guide **12408** can be connected to the ground terminal of the photon supply.

(734) For example, the photon supply **12800** is thermally coupled to the fiber guide **12408**, and the fiber guide **12408** can help dissipate heat from the photon supply **12800**. As described in more detail below, in some examples, the photon supply can be thermally coupled to a heat dissipating device that is isolated from the other heat dissipating device(s) thermally coupled to other heat-generating electronic circuitry in the pluggable module so as to maintain the photon supply at a more stable temperature or lower temperature.

(735) Referring to FIG. **167**, in some implementations, a pluggable module **16700** includes a co-packaged optical module **16702**, one or more multi-fiber push on (MPO) connectors **16704**, a fiber guide **16706** that mechanically connects the co-packaged optical module **16702** to the one or more multi-fiber push on connectors **16704**, a fiber array pigtail **16708** that optically connects the co-packaged optical module **16702** to the one or more multi-fiber push on connectors **16704**, a photon supply (e.g., one or more lasers) **16710**, and one or more power supply optical fibers **16712** that transmit the optical power supply light from the photon supply **16710** to the co-packaged optical module **16702**. The pluggable module **16700** includes a CPO module heat sink **16714** and a laser heat sink **16716** that are thermally isolated from each other. The fiber guide **16706** can be made of thermally conductive material, such as metal, and functions as the CPO module heat sink **16714**. The CPO module heat sink **16714** is thermally coupled to the CPO module **16702**, and the laser heat sink **16716** is thermally coupled to the laser(s) of the photon supply **16710**. For example, an air gap can be provided between

the CPO module heat sink **16714** and the laser heat sink **16716** to thermally isolate the laser heat sink **16716** from the CPO module heat sink **16714**. In some implementations, a thermally insulating material **16718** can be positioned between the CPO module heat sink **16714** and the laser heat sink **16716**.

(736) For example, the CPO module heat sink **16714** and the laser heat sink **16716** can be made of a material having a thermal conductivity greater than 50 W/mK (Watts per meter-Kelvin), preferably greater than 100 W/mK, and more preferably greater than 200 W/mK. For example, the CPO module heat sink **16714** and the laser heat sink **16716** can be made of a metal or a metal alloy, such as aluminum, aluminum alloy, brass, copper, zinc, or a combination of the above. The CPO module heat sink **16714** and the laser heat sink **16716** can have fins to increase the heat dissipation surface area. The CPO module heat sink **16714** and the laser heat sink **16716** can be made of the same material or different materials. For example, the thermally insulating material **16718** can have a thermal conductivity less than 10 W/mK, preferably less than 1 W/mK. For example, the thermally insulating material **16718** can be made of quartz, silicone rubber, or plastic. The thermal conductivity values, the heat sink materials, and thermally insulating materials described above are merely examples, other values and materials can also be used.

(737) For example, the CPO module **16702** has a first side that is optically coupled to a two-dimensional arrangement (e.g., 2D array) of optical fibers of the fiber array pigtail **16708**. The CPO module **16702** has a second side that has (or is coupled to) a two-dimensional array electrical interface **16720** that includes a two-dimensional arrangement (e.g., 2D array) of electrical contacts **16722**. By using the two-dimensional arrangement of optical fibers and two-dimensional arrangement of electrical contacts, the CPO module **16702** enables a high throughput data path between external optical fiber cable(s) and the data processor (e.g., **12382** of FIGS. **125**B, **129**). In order to process the large amount of data passing through the CPO module **16702**, the electronic circuitry that processes the electrical signals can generate a significant amount of heat. During operation of the pluggable module **16700**, the heat generated by the electronic circuity for processing the electrical signals associated with the CPO module **16712** can vary significantly depending on the amount of data passing through the CPO module **16712** at a given time. Thus, the temperature of the CPO module heat sink **16714** can vary greatly. Thermally isolating the laser heat sink **16716** from the CPO module heat sink **16712** can have one or more of the following advantages. First, the laser heat sink **16716** can be at a lower temperature compared to the CPO module heat sink **16712** during operation of the pluggable module **16700**, allowing the laser(s) of the photon supply **16710** to operate at a lower temperature (as compared to thermally coupling the laser(s) to the CPO module heat sink **16712**). Second, the laser(s) of the photon supply **16710** can operate at a more consistent temperature and be less influenced by the variations in the temperature of the other electronic circuitry in the pluggable module.

(738) The technique of thermally isolating the heat sink for the laser(s) of the photon supply from the heat sink(s) for the other heat-generating electronic circuitry in the pluggable module can be applied to other types of pluggable modules, such as pluggable modules that have form factors that comply with common industry standards, such as SFP (small form-factor pluggable), SFP+ (or 10 Gb SFP), SFP28, OSFP (octal SFP), OSFP-XD (OSFP extra dense), QSFP (quad small form-factor pluggable), QSFP+, QSFP28, QSFP56, or QSFP-DD (quad small form-factor pluggable double density).

(739) Referring to FIGS. **168**A and **169**, in some implementations, a pluggable module **16800** has a form factor that complies with, e.g., SFP, SFP+ (or 10 Gb SFP), SFP28, OSFP, OSFP-XD, QSFP, QSFP+, QSFP28, QSFP56, or QSFP-DD standard. The pluggable module **16800** includes a photonic integrated circuit (which is obscured from view in the figure) and two laser modules **16802** that provide optical power supply light to the photonic integrated circuit. The photonic integrated circuit is thermally coupled to the housing **16804**, which is thermally conductive. The housing **16804** is thermally coupled to a CPO module heat sink **16806** mounted on a top side of the housing **16804**. The laser modules **16802** are thermally coupled to a laser heat sink **16808** using thermal gap pads (TIM) **16810**. The laser heat sink **16808** is thermally isolated from the CPO module heat sink **16806**, either by an air gap, or by a thermally insulating material **16812**.

(740) FIG. **168**B is a diagram of an example of the top panel **16814** of the housing **16804** of the pluggable module **16800**. The top panel **16814** defines an opening **16816** that enables a thermal path to be provided between the laser modules **16802** and the laser heat sink **16808**. In some examples, the lower portion of the laser heat sink **16808** and/or the thermal gap pads **16810** extend through the opening **16816** without touching the top panel **16814**. In some examples, the thermally insulating material **16812** is provided between the laser heat sink **16808** and the top panel **16814**.

(741) FIG. **170**A is a diagram showing the results of a simulation demonstrating the thermal isolation of the laser modules **16802**. FIG. **170**B is a diagram of a cross section of the laser heat sink **16808** used in the simulation. In this simulation, the air flow is about 2.5 CFM, and each of the laser modules is rated at 1.25 W. Table 1 below shows, for various ambient air temperatures, the temperature of the pluggable module case or housing, and the temperature of the laser heat sink.

(742) TABLE-US-00001 TABLE 1 Air temperature Air flow Case temperature Laser heat sink (° C.) (CFM) (° C.) temperature (° C.) 27.0 2.5 45.1 43.8 35.0 2.5 53.1 51.7 40.0 2.5 57.9 56.5

(743) Table 1 shows that in this example, the laser heat sink **16808** can have a temperature that is about 1.3° C. to 1.4° C. lower than the case temperature during operation of the pluggable module **16800**.

(744) The parameter values described above, such as air flow CFM, laser power wattage, and ambient air temperature, are used as examples only. It is understood that the pluggable modules can operate in different environmental conditions, such as having different ambient temperatures and air flows. The amount of heat generated by the electronic circuitry can vary. The design of the CPO module heat sink and the laser heat sink can vary, such as having different configurations (e.g., geometry and number) of fins to increase the heat dissipation surface area. The material of the housing and heat sinks can

vary, such as using aluminum, aluminum alloy, copper, copper alloy, or a combination of the above. In some examples, the CPO module heat sink can be integrated with the housing and made of a same material. For example, the material for the housing with the integrated CPO module heat sink can be the same or different from the material for the laser heat sink.

(745) In some examples, by adjusting the design of the heat sinks, the temperature of the laser heat sink can be much lower than the temperature of the CPO module heat sink. For example, compared with the simulation results shown in Table 1, in another thermal simulation in which fins are added to the heat sinks to increase the heat dissipation surface area, and aluminum was used for the heat sinks, using an air flow of 2.5 CFM and an ambient temperature of 27° C., under certain power conditions for the electronic circuitry (including, e.g., driver, transimpedance amplifier, digital signal processor, microcontroller, and DC/DC converters) and the laser modules, the CPO module heat sink can have a temperature of about 67.6° C., while the laser heat sink can have a lower temperature of about 39.0° C. In another thermal simulation, by adjusting the material of the housing and the integrated CPO module heat sink to use an aluminum alloy, the CPO module heat sink can have a temperature of about 69.5° C., while the laser heat sink can have a lower temperature of about 38.5° C.

(746) When we say that the laser heat sink is thermally isolated from the CPO module heat sink, we mean that an air gap or a thermally insulating material is provided between the laser heat sink and the CPO module heat sink to significantly reduce the amount of heat transferred between the CPO module heat sink and the laser heat sink.

(747) In some examples, the CPO module **12386** is coupled to spring-loaded elements or compression interposers mounted on the substrate **12380**. The force required to press the CPO module **12386** into the spring-loaded elements or the compression interposers can be large. The following describes mechanisms to facilitate pressing the CPO module **12386** into the spring-loaded elements or the compression interposers.

(748) Referring to FIG. **129**, in some implementations, a rackmount server **12920** includes a substrate **12380** that is attached to a printed circuit board **12906**, which has an opening to allow the data processing chip **12382** to protrude or partially protrude through the opening and be attached to the substrate **12380**. The printed circuit board **12906** can have many functions, such as providing support for a large number of electrical power connections for the data processing chip **12382**. The CPO module **12386** can be mounted on the substrate **12380** through a CPO mount or a front lattice **12902**. A bolster plate **12914** is attached to the rear side of the printed circuit board **12906**. Both the substrate **12380** and the printed circuit board **12906** are sandwiched between the CPO mount or front lattice **12902** and the bolster plate **12914** to provide mechanical strength so that CPO modules **12386** can exert the required pressure onto the substrate **12380**. Guide rails/cage **12900** extend from the front panel **12904** or the front portion of the fiber guide **12408** to the bolster plate **12914** and provide rigid connections between the CPO mount **12902** and the front panel **12904** or the front portion of the fiber guide **12408**.

(749) Clamp mechanisms **12908**, such as screws, are used to fasten the guide rails/cage **12900** to the front portion of the fiber guide **12408**. After the CPO module **12386** is initially pressed into the spring-loaded elements or the compression interposers, the screws **12908** are tightened, which pulls the guide rails/cage **12900** forward, thereby pulling the bolster plate **12914** forward and provide a counteracting force that pushes the spring-loaded elements or the compression interposers in the direction of the CPO module **12386**. Springs **12910** can be provided between the guide rails **12900** and the front portion of the fiber guide **12408** to provide some tolerance in the positioning of the front portion of the fiber guide **12408** relative to the guide rails **12900**.

(750) The right side of FIG. **129** shows front views of the guide rails/cage **12900**. For example, the guide rails **12900** can include multiple rods (e.g., four rods) that are arranged in a configuration based on the shape of the front portion of the fiber guide **12408**. If the front portion of the fiber guide **12408** has a square shape, the four rods of the guide rails **12900** can be positioned near the four corners of the front portion of the squared-shaped fiber guide **12408**. In some examples, a guide cage **12912** can be provided to enclose the guide rails **12900**. The guide rails **12900** can also be used without the guide cage **12912**.

(751) In the examples of FIGS. **125**B and **129**, the data processing chip **12382** includes electrical contacts that are electrically coupled to electrical contacts on a first side (the side facing toward the rear panel of the housing) of the substrate **12380**, and the co-packaged optical modules **12386** include electrical contacts that are coupled to electrical contacts on a second side (the side facing toward the front panel of the housing) of the substrate **12380**. At least some of the electrical contacts on the first side of the substrate **12380** are electrically coupled to at least some of the electrical contacts on the second side of the substrate **12380**, e.g., through contact vias that pass through the substrate **12380**. This allows the electrical signals to be transmitted between the co-packaged optical module(s) **12386** and the data processor **12382**. In this document, when we say that the electrical signals generated by the co-packaged optical modules are transmitted to the data processor **12382**, it is understood that the electrical signals from the co-packaged optical module(s) **12386** are not necessarily transmitted directly to the data processor **12382**. Rather, the electrical signals from the co-packaged optical module(s) **12386** can be further processed, such as by amplifiers, serializers/deserializers, retimers, and/or digital signal processors. The electrical signals from the co-packaged optical modules(s) can be modulated or demodulated, or the modulation method can be changed, prior to being sent to the data processor. Similarly, when we say that the electrical signals generated by the data processor are transmitted to the co-packaged optical module(s), it is understood that the electrical signals from the data processor are not necessarily transmitted directly to the co-packaged optical module(s). Rather, the electrical signals from the data processor can be further processed, such as by amplifiers, serializers/deserializers, retimers, and/or digital signal processors. The electrical signals from the data processor can be

modulated or demodulated, or the modulation method can be changed, prior to being sent to the co-package optical module(s). The substrate **12380** can be, e.g., a ceramic substrate, an organic high density build-up substrate, or a silicon substrate. In some examples, the substrate **12380** can be replaced by a printed circuit board.

(752) In some implementations, the data processing chip **12380** has electrical contacts that are electrically coupled to electrical contacts on a first side of a first substrate, and the co-packaged optical modules **12386** has electrical contacts that are coupled to electrical contacts on a first side of a second substrate. Electrical contacts on a second side of the first substrate are electrically coupled to electrical contacts on a second side of the second substrate. This allows electrical signals to be transmitted between the data processing chip and the co-packaged optical modules. In some examples, the first substrate can be replaced by a first printed circuit board. In some examples, the second substrate can be replaced by a second printed circuit board.

(753) In some implementations, the data processor is electrically coupled to a first substrate, the co-packaged optical module(s) are electrically coupled to a second substrate, the first substrate is mounted to a first side of a third substrate, and the second substrate is mounted to a second side of the third substrate. The third substrate can function as an interposer in which the arrangement of the electrical contacts on a first side of the third substrate can be different from the arrangement of electrical contacts on a second side of the third substrate. In some examples, one or more of the first, second, and third substrates can be replaced by one or more printed circuit boards.

(754) In some implementations, there can be any number of substrate(s) and/or printed circuit board(s) between the data processor and the co-packaged optical module(s). Each substrate can be, e.g., a ceramic substrate, an organic high density build-up substrate, or a silicon substrate. Each substrate or printed circuit board can function as an interposer in which the arrangement of the electrical contacts on a first side of the substrate or printed circuit board can be different from the arrangement of electrical contacts on a second side of the substrate or printed circuit board. On each substrate or printed circuit board, there can be mounted additional modules, such as amplifiers, filters, retimers, serializers, deserializers, modulators, demodulators, power supply modules, or digital signal processors.

(755) Each co-packaged optical module **12386** can include a substrate, a photonic integrated circuit mounted on the substrate, and one or more electronic integrated circuits. One or more electronic integrated circuits can be mounted on the photonic integrated circuit, and one or more electronic integrated circuits can be mounted on the substrate beside the photonic integrated circuit. Additional examples of the arrangements of the photonic integrated circuit and the electronic integrated circuits are shown in FIGS. **162** to **166**D.

(756) The following describes examples of rackmount servers having various thermal solutions to assist in dissipating heat generated from the data processors and the co-packaged optical modules coupled to the vertically oriented circuit boards or substrates positioned near the front panel.

(757) FIG. **133** shows a top view diagram **17600** and a side view diagram **17601** of a rackmount server **17602** that has a hinged front panel **17604** having front panel fiber connectors **13028** (see FIG. **130**). Co-packaged optical modules **13022** are optically coupled to the fiber connectors **13028** through fiber pigtails **13026**. Pluggable external laser sources (ELS) **13032** provide power supply light that are transmitted through optical fibers **13034** to the CPO modules **13022**. The server **17602** is similar to the server **11700** of FIG. **117**, except that the air inlet grid **17608** is larger, and the external laser sources **13032** (or **1956** of FIG. **105**) have front-to-back airflow cooling through the use of two extra fans behind the laser sources **13032**. In this example, an inlet fan **17604** blows air in the left-to-right direction toward the CPO modules **13022**. A second fan **17606***a* and a third fan **17606***b* are positioned behind the laser sources **13032** to direct air flow to assist in cooling the laser sources **13032**.

(758) FIG. **134** shows a top view **17700** of an example of a rackmount server **17702**, a VASIC-plane front view **17704** of the rackmount server **17702**, and a front-panel front view **17706** of the rackmount server **17702**. In this example, front-connected recessed remote laser sources **17708** are placed behind the left-to-right fans **17710**. The configuration of the inlet fans **17710** results in unidirectional airflow, as represented by the arrows **17712**. The VASIC-plane front view **17704** shows the front view when the hinged front panel is opened and lowered. The front-panel front view **17706** shows the air inlet grid **17714** and front panel fiber connectors **17716**. For example, the connectors **17716** can include 64 LC connectors (providing a bandwidth of 1.6 Tbps FR16) or 128 MPO connectors (providing a bandwidth of 400 Gbps DR4).

(759) FIG. **135** shows a top view **17800** of an example of a rackmount server **17802** in which the external laser sources **17804** are mounted below the VASIC-plane and directly accessible from the front panel for easy front-panel access/serviceability. A VASIC-plane front view **17806** shows the front view when the hinged front panel is opened and lowered. A front-panel front view **17808** shows the air intake grid, the front panel fiber connectors, and the external laser sources.

(760) In the examples shown in FIGS. **69**A to **76**, **77**B, **78**, **96** to **98**, **100**, **104**, **108**, **110**, **112**, **113**, **117**, **119**, **121**, **122**, **126**A, **126**B, and **133** to **135**, one of the inlet fans is mounted, attached, or coupled to the front panel, or positioned very close to the front panel. In some implementations, depending on the position of the circuit board or substrate on which the data processor and the co-packaged optical modules are coupled, the inlet fan nearest the front panel can be positioned at a distance from the front panel, e.g., a few inches from the front panel, or within one-fourth of the distance between the front panel and the rear panel (which can correspond to the depth of the housing). Here, the distance between the fan and the front panel refers to the distance between the tip of the fan blade and the front panel. The fan blade rotates during operation, so when we say that the distance between the fan and the front panel is within one-fourth the distance between the front panel and the rear panel, we mean that the fan is positioned near the front panel in which at least a portion of a

fan blade of the fan is within one-fourth of the distance between the front and rear panels for at least some time period during operation of the fan.

(761) The following describes an example in which the communication interface(s) support memory modules mounted in smaller circuit boards that are electrically coupled to a larger circuit board positioned near the front panel. FIG. **147** shows a top view of an example of a system **16200** that includes a vertically oriented circuit board **16202** (also referred to as a carrier card) that is substantially parallel to the front panel **16204**. Several memory modules **16206** are electrically coupled to the circuit board **16202**, e.g., using sockets, such as DIMM (dual in line memory module) sockets. Each memory module **16202** includes a circuit board **16208** and one or more memory integrated circuits **16210**, which can be mounted on one side or both sides of the circuit board **16208**. One or more optical interface modules **16212** (e.g., co-packaged optical modules) are electrically coupled to the circuit board **16202** and function as the interface between the memory modules **16206** and one or more communication optical fiber cables **16214**. For example, each optical interface module **16214** can support up to 1.6 Tbps bandwidth. When N optical interface modules **16214** are used (N being a positive integer), the total bandwidth can be up to N×1.6 Tbps. One or more fans **16216** can be mounted near the front panel **16204** to assist in removing heat generated by the various components (e.g., the optical interface modules **16212** and the memory modules **16206**) coupled to the circuit board **16202**. The technologies for implementing the optical interface modules **16212** and configuring the fans **16216** and airflows for optimizing heat removal have been described above and not repeated here.

(762) FIG. **148** is an enlarged diagram of the carrier card **16202**, the optical interface module(s) **16212**, and the memory modules **16206**. In this example, the memory modules **16206** are mounted on both the front side and the rear side of the carrier card **16202**. It is also possible mount the memory modules **16206** to just the front side, or just the rear side, of the carrier card **16202**. In some examples, heat sinks are thermally attached to the memory chips **16210**.

(763) In some implementations, the memory modules **16206** on the carrier card **16202** can be used as, e.g., computer memory, disaggregated memory, or a memory pool. For example, the system **16200** can provide a large memory bank or memory pool that is accessible by more than one central processing unit. A data processing system can be implemented as a spatially co-located solution, e.g., 4 sets of the memory modules **16206** supporting 4 processors sitting in a common box or housing. A data processing system can also be implemented as a spatially separated solution, e.g., a rack full of processors, connected by optical fiber cables to another rack full of DIMMs (or other memory). In this example, the rack full of memory modules can include multiple systems **16200**. For example, the system **16200** is useful for implementing memory disaggregation to decouple physical memory allocated to virtual servers (e.g., virtual machines or containers or executors) at their initialization time from the runtime management of the memory. The decoupling allows a server under high memory usage to use the idle memory either from other servers hosted on the same physical node (node level memory disaggregation) or from remote nodes in the same cluster (cluster level memory disaggregation).

(764) FIG. **149** is a front view of an example of the carrier card **16202**, the optical interface module(s) **16212**, and the memory modules **16206**. In this example, three rows of memory modules **16206** are attached to the circuit board **16202**. The number of memory modules **16206** can vary depending on application. The orientation of the memory modules **16206** can also be modified depending on how the system is configured. For example, instead of orienting the memory modules **16206** to extend in the vertical direction as shown in FIG. **164**, the memory modules **16206** can also be oriented to extend in the horizontal direction, or at an angle between 0° to 90° relative to the horizontal direction, in order to optimize air flow and heat dissipation.

(765) FIG. **150** is a front view of an example of the carrier card **16202** with two optical interface modules **16212**, and memory modules **16206**. FIGS. **149** and **150**, as well as many other figures, are not drawn to scale. The optical interface modules **16212** can be much smaller than what is shown in the figure, and many more optical interface modules **16212** can be attached to the circuit board **16202**. For example, the optical interface module **16212** can be positioned in the space **16218** (shown in dashed lines) between the four memory modules **16206**. In some examples, the memory modules **16206** can interface directly with the optical interface module **16212**.

(766) Referring to FIG. **151**, in some implementations, one or more memory controllers or switches **16600** (e.g. Compute Express Link (CXL) controller(s)) is/are electrically coupled to the carrier card **16202** and configured to aggregate the traffic from the memory modules **16206**. For example, the memory controller(s) or switch(es) **16600** can be implemented as an integrated circuit mounted on the rear side of the carrier card **16202**, opposite to the optical interface module(s) **16212**. Electrical traces are provided on or in the circuit board **16202** to connect the memory modules **16206** to the CXL controller/switch(es) **16600**, and the CXL controller/switch(es) **16600** then aggregate the traffic from the memory modules **16206** and interface them to the CPO module **16212**.

(767) The carrier card **16202** and the memory modules **16206** can be any of a variety of sizes depending on the available space in the housing. The capacity of the memory modules **16206** can vary depending on application. As memory technology improves in the future, it is expected that the capacity of the memory modules **16206** will increase in the future. For example, the carrier card **16202** can have dimensions of 20 cm×20 cm, each memory module **16206** can have dimensions of 10 cm×2 cm, and each memory module can have a capacity of 64 GB. A spacing of 6 mm can be provided between memory modules **16206**. The memory modules **16206** can occupy both sides of the carrier card **16202**. In this example, the carrier card **16202** has a height of 20 cm and can support 2 rows of memory modules **16206**, with each memory module **16206** extending 10 cm in the vertical direction. With a carrier card width of 20 cm and a 6 mm spacing between memory modules **16206**, there can be about 32 memory modules per row, and about 64 memory modules per side

of the carrier card **16202**. When the memory modules are mounted on both sides of the carrier card **16202**, there can be up to a total of about 128 memory modules **16206** per carrier card. With up to 64 GB capacity for each memory module **16206**, the carrier card **16202** can support up to about 8 TB memory in a space approximately the size of 1,600 cm.sup.3.
(768) While this disclosure includes references to illustrative embodiments, this specification is not intended to be construed in a limiting sense. Various modifications of the described embodiments, as well as other embodiments within the scope of the disclosure, which are apparent to persons skilled in the art to which the disclosure pertains are deemed to lie within the principle and scope of the disclosure, e.g., as expressed in the following claims.
(769) For example, the techniques described above for improving the operations of systems that include rackmount servers (see FIGS. **76**, **85** to **87**B) can also be applied to systems that include blade servers. In the examples shown in FIGS. **80**A and **82**A, each of the switch boxes **1302** and **1304** can be replaced with any type of data processing device, such as a data processing device that includes one or more of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC). For example, in FIG. **84**A, each of the switch boxes **1462**, **1464**, **1466**, and **1468** can be replaced with any type of data processing device, such as a data processing device that includes one or more of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC).
(770) In some implementations, the devices **1464**, **1466**, and **1468** can be rackmount servers mounted on a same rack, the switch box **1462** can be a top-of-rack switch **1462**, and the servers (e.g., **1464**, **1466**, **1468**) in the rack communicate with each other through the top-of-rack switch **1462**. In this example, the co-packaged optical modules or optical communication interfaces are configured to receive power supply light provided by the optical power supply **1322** and/or **1332**.
(771) For example, in FIGS. **85** and **86**, each of the servers **1522** (e.g., **1522***a*, **1522***b*, **1522***c*) can be any type of server that includes one or more of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC). For example, one or more of the servers **1522** can be data storage servers, and one or more of the servers **1522** can be data processing servers that execute application programs that access (e.g., read and write) data stored in the data storage servers.
(772) For example, in FIGS. **87**A and **87**B, each of the servers **1552** can be any type of server that includes one or more of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC). For example, each of the switch boxes **1556** can be replaced with any type of high-bandwidth data processing system, such as a data processing system that includes one or more of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC).
(773) For example, in FIG. **138**, each of the servers **13802** can be any type of server that includes one or more of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC). For example, each of the switch boxes **13806** can be replaced with any type of high-bandwidth data processing system, such as a data processing system that includes one or more of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC).
(774) For example, the data processing system **1550** of FIG. **87**A and the data processing system **13800** of FIG. **138** can implement a high-speed, high-bandwidth data processing system that includes one or more high-speed, high-bandwidth data processors that access large memory banks or memory pools through optical communication links. For example, one or more of the switch boxes **1556** of FIG. **87**A can be replaced with one or more of the rack systems **12400** of FIG. **124** that include several vertically mounted processor blades **12300**. One or more of the servers **1552** can be one or more of the storage systems **16200** that include vertically oriented circuit boards **16202** on which several memory modules **16206** are mounted. One or more of the optical fiber cables **13700** (FIG. **137**) can be used to optically connect the one or more rack systems **12400** to one or more storage systems **16200**. The co-packaged optical modules or optical communication interfaces at the one or more rack systems **12400** and the one or more storage systems **16200** can receive power supply light provided by an external laser source, such as the optical power supply **1558**.
(775) Similarly, one or more of the switch boxes **13806** of FIG. **138** can be replaced with one or more of the rack systems **12400** of FIG. **124** that include several vertically mounted processor blades **12300**. One or more of the servers **13802** can be one or more of the storage systems **16200** that include vertically oriented circuit boards **16202** on which several memory modules **16206** are mounted. One or more of the optical fiber cables **14100** (FIG. **141**) can be used to optically connect the one or more rack systems **12400** to one or more storage systems **16200**. The co-packaged optical modules or optical communication interfaces at the one or more rack systems **12400** and the one or more storage systems **16200** can receive power supply light provided by an external laser source, such as the optical power supply **13808**.

(776) For example, the processor blades **12300** of the rack systems **12400** can include data processors that implement a variety of services, such as cloud computing, database processing, audio/video hosting and streaming, electronic mail, data storage, web hosting, social networking, supercomputing, scientific research computing, healthcare data processing, financial transaction processing, logistics management, weather forecasting, simulation, hosting virtual worlds, or hosting one or more metaverses, to list a few examples. Such services may require fast access to large amounts of data. For example, implementing a metaverse platform may require access to vast amounts of stored data that are used to simulate virtual worlds and interactions among users and objects in the virtual worlds. Such data can be stored across multiple storage systems **16200** across multiple racks. The optical fiber cables **13700** allow the processor blades **12300** to access the data stored in the storage systems **16200** through high-bandwidth optical links.
(777) In some implementations, optical transceiver modules can have form factors that comply with common industry standards, such as SFP (small form-factor pluggable), SFP+ (or 10 Gb SFP), SFP28, OSFP (octal SFP), OSFP-XD (OSFP extra dense), QSFP (quad small form-factor pluggable), QSFP+, QSFP28, QSFP56, or QSFP-DD (quad small form-factor pluggable double density).
(778) Referring to FIG. **152**A, in some implementations, a pluggable optical module **15200** is configured to be plugged into a cage mounted on a circuit board of a host device, such as a network switch box or a server computer. The pluggable optical module **15200** includes a housing (or case) **15202**, a user fiber connector **15204**, a fiber harness **15206**, a circuit board or substrate **15208**, a receiver photonic integrated circuit **15210**, a transmitter photonic integrated circuit **15211**, a receiver fiber connector **15212**, a transmitter fiber connector **15213**, a receiver application specific integrated circuit (ASIC) **15214**, a transmitter ASIC **15215**, an electrical connector (or connector tongue) **15216**, and a handle **15230**.
(779) For example, the handle **15230** enables the user to conveniently push the pluggable optical module **15200** into the corresponding cage, or pull the pluggable optical module **15200** from the cage. The user fiber connector **15204** is configured to be optically connected to a fiber-optic cable provided by the user. The fiber harness **15206** includes several optical fibers that optically connect the user fiber connector **15204** to the receiver fiber connector **15212** and the transmitter fiber connector **15213**, which can be, e.g., turning mirrors. The receiver fiber connector **15212** couples input light beams from the optical fibers of the fiber harness **15206** to optical couplers (e.g., v-groove couplers, grating couplers, etc.) on the receiver photonic integrated circuit **15210**. The transmitter fiber connector **15213** couples output light beams from the optical couplers (e.g., v-groove couplers, grating couplers, etc.) on the transmitter photonic integrated circuit **15211** to optical fibers of the fiber harness **15206**.
(780) For example, the receiver photonic integrated circuit **15210** is configured to convert the input optical signals to electrical signals, which are processed by the receiver ASIC **15214**. The transmitter photonic integrated circuit **15211** is configured to convert the output electrical signals from the transmitter ASIC **15215** to output optical signals. The output optical signals are sent through the transmitter fiber connector **15213**, the fiber harness **15206**, and the user fiber connector **15204** to the optical fiber cable.
(781) The receiver ASIC **15214** and the transmitter ASIC **15215** can perform a number of functions, such as digital signal processing for preparing the electrical signals in a format suitable for conversion to optical signals (e.g., PAM4 DSP equalization), signal quality monitoring, electrical interface (sometimes the electrical signals can have the same data rate as the optical signals, sometimes the data rate of the electrical signals are increased using a 2:1 gearbox to twice their rate before converting to optical signals). In some examples, the output electrical signals from the receiver photonic integrated circuit **15210** are amplified by a transimpedance amplifier (TIA) before being sent to the receiver ASIC **15214**, and the electrical signals output from the transmitter ASIC **15215** are amplified by driver amplifiers before being sent to the transmitter photonic integrated circuit **15211**.
(782) In some examples, the transimpedance amplifiers are integrated into the receiver ASIC **15214**, and the driver amplifiers are integrated into the transmitter ASIC **15215**. In some examples, the receiver ASIC **15214** and the transmitter ASIC **15215** are integrated into a single electrical integrated circuit. In some examples, the receiver photonic integrated circuit **15210** and the transmitter photonic integrated circuit **15211** are integrated into a single photonic integrated circuit. In some examples, the receiver photonic integrated circuit **15210**, the transmitter photonic integrated circuit **15211**, the receiver ASIC **15214**, and the transmitter **15215** are formed on a single semiconductor substrate. In some examples, some or all of the electronic functionality (e.g., electronic amplification, signal conditioning, control loop functionality, monitoring functionality, etc.) is monolithically integrated into one or more of the photonic integrated circuits.
(783) The processed electrical signals are sent to the host device through the electrical connector **15216** as input electrical signals for the host device. The electrical connector **15216** can include, e.g., an Ethernet interface, a CMIS (common management interface specification) interface, an SPI (serial peripheral interface) interface an I.sup.2C (inter-integrated circuit) interface, etc. For example, the electrical connector **15216** is formed by a portion of a printed circuit board with contact pads. The electrical connector **15216** can be, e.g., an edge connector, connector tongue, or connector card. An example pinout specification for the contact pads is provided in FIG. **158**.
(784) In some implementations, the pluggable optical module **15200** complies with a small form factor pluggable module specification, which can be, e.g., SFP, SFP+, 10 Gb SFP, SFP28, OSFP, OSFP-XD, QSFP, QSFP+, QSFP28, QSFP56, or QSFP-DD. For example, the small form factor pluggable module specification can be “Specification for OSFP OCTAL SMALL FORM FACTOR PLUGGABLE MODULE,” Rev 4.0, May 28, 2021, available from OSFP MSA. For example, the small form factor pluggable module specification can be “QSFP-DD/QSFP-DD800/QSFP112 Hardware Specification for QSFP Double Density 8× and QSFP 4× Pluggable Transceivers,” Revision 6.01, May 28, 2021, available from QSFP-

DD MSA.

(785) In the description of the pluggable optical module, the optical connector **15204** is said to be closer to the front side relative to the electrical connector **15216**, and the electrical connector **15216** is said to be closer to the rear side relative to the optical connector **15204**. The longitudinal or axial direction **15218** extends parallel to the front-to-rear direction. To facilitate discussion of the orientation of various components of the pluggable module **15200**, a coordinate system is used in which the z-direction is parallel to the longitudinal direction **15218** (or lengthwise direction), the x-direction is parallel to the width direction, and the y-direction is parallel to the height direction of the housing of the pluggable optical module **15200**. For example, the bottom surface of the housing of the pluggable optical module **15200** extends substantially parallel to the x-z plane, and side walls of the housing extend substantially parallel to the y-z plane. In some examples, the length of the pluggable optical module is at least 50% greater than the width and at least 50% greater than the height. In some examples, the length of the pluggable optical module is at least twice the width and at least twice the height.

(786) FIG. **152**B is a cross-sectional diagram of the pluggable optical module **15200** along a plane parallel to the x-y plane and perpendicular to the longitudinal direction **15218**. In some implementations, the housing **15202** includes a bottom wall **15220**, an upper wall **15222**, and side walls **15224***a*, **15224***b*. A heat sink **15226** is provided on the exterior of the upper wall **15222**. For example, the circuit board (or substrate) **15208** is substantially parallel to the bottom wall **15220**, the receiver and transmitter photonic integrated circuits **15210**, **15211** are mounted on the circuit board (or substrate) **15208**. The receiver and transmitter fiber connectors **15212**, **15213** are mounted on the receiver and transmitter photonic integrated circuits **15210**, **15211**, respectively. In this example, the circuit board (or substrate) **15208** has a width that is slightly smaller than the width of the inner bottom wall, and a length that is at least 50% larger than the width. The dimensions of the circuit board (or substrate) **15208** naturally lends to its orientation being substantially parallel to the bottom wall and extending lengthwise in the longitudinal direction **15218**.

(787) FIG. **153**A is a side view (along a plane parallel to the y-z plane) of an example pluggable optical module **15200**. FIG. **153**B is a perspective view of a rear portion **15240** of the pluggable optical module **15200**. The pluggable optical module **15200** includes an edge connector (also referred to as a connector tongue or paddle card) **15404** that includes several connector pads **15408** on the top and bottom sides of the paddle card **15404**. For example, the paddle card **15404** can be part of the circuit board **15208**. The connector pads **15408** of the pluggable optical module **15400** are configured to be electrically coupled to the connector pads of a receptacle mounted on the circuit board of the host device.

(788) FIG. **153**C is a side view cross-sectional diagram (along a plane parallel to the y-z plane) of the pluggable optical module **15200** plugged into a cage **15402**, in which the edge connector or paddle card **15404** is mated with a receptacle **15406**. In this example, the edge connector **15404** is part of the substrate or circuit board **15208** on which the photonic integrated circuits **15210** and **15211** are mounted.

(789) In the example of FIGS. **152**A to **153**C, the pluggable optical module **15200** uses a flat arrangement of photonic integrated circuits, application specific integrated circuits, and fiber coupling. In some examples, such a configuration limits the pluggable optical module **15200** to using one-dimensional fiber arrays (e.g., in the form of ribbons) that are coupled into the photonic integrated circuits either horizontally (e.g., using v-grooves) or vertically (e.g., using one-dimensional arrays of grating couplers). In order to significantly scale capacity, it may be useful to adopt two-dimensional arrays of fiber interfaces to connect two-dimensional arrays of optical fibers to the photonic integrated circuits. However, it is difficult to implement two-dimensional interfaces in the pluggable optical module **15200** because the vertical spacing between the substrate or circuit board **15208** and the upper wall **15222** of the housing **15202** is too small for implementing two-dimensional fiber array interfacing.

(790) The inventors realized that by orienting the substrate or circuit board vertically, i.e., perpendicular to the bottom surface of the housing **15202**, it is possible to implement two-dimensional fiber array interfacing to the photonic integrated circuits, thereby significantly increasing the bandwidth supported by the pluggable optical module. Furthermore, as the technology for manufacturing the photonic integrated circuits and the electrical integrated circuits improve, the photonic integrated circuits and the electrical integrated circuits can be made smaller, and some electronic integrated circuits can be stacked on the photonic integrated circuit, resulting in a co-packaged optical module that can be mounted on the vertically oriented substrate or circuit board.

(791) In some implementations, the photonic integrated circuit can be mounted on a vertical substrate or circuit board substantially parallel to the front face of the pluggable optical module. This configuration allows a vertical two-dimensional fiber array to be coupled to the photonic integrate circuit without any fiber bends or turning mirrors.

(792) Other configurations are also possible. In some implementations, in a second configuration, the photonic integrated circuit is mounted on a horizontal substrate or circuit board (parallel to the bottom surface of the housing) that is positioned lower than the edge connector (or paddle card or connector tongue) **15404**. This configuration provides more space between the substrate or circuit board and the upper wall **15222**, allowing for fiber bends from vertical to horizontal, or a turning mirror solution from vertical to horizontal, for two-dimensional fiber arrays.

(793) In some implementations, in a third configuration, the photonic integrated circuit is mounted on a vertical substrate or circuit board that is oriented parallel to a side wall of the housing. This allows a vertical two dimensional fiber array to be coupled to the photonic integrated circuit with a large fiber bend radius taking up the entire width of the module.

(794) FIG. **154**A is a rear view of an example OSFP optical transceiver module (or "OSFP module") **15300**, showing the electrical connector **15216** positioned in the space defined by the inner walls of the housing **15304** of the OSFP module **15300**. The space defined by the inner walls of the housing **15304** has dimensions of about 7.2 mm×20.58 mm along a

plane parallel to the x-y plane.
(795) FIG. **154**B is a rear view of an example OSFP-XD optical transceiver module (or "OSFP-XD module") **15302**, showing the electrical connector **15216** positioned in the space defined by the inner walls of the housing **15306** of the OSFP-XD module **15302**. The space defined by the inner walls of the housing **15306** has dimensions of about 6.2 mm×20.58 mm along a plane parallel to the x-y plane.
(796) FIG. **154**C is a diagram of an example co-packaged optical module **15310** that can fit inside the housing **15304** of the OSFP module **15300** with the substrate or circuit board oriented vertically (relative to the horizontal bottom wall of the housing). The co-packaged optical module **15310** includes a photonic integrated circuit **15312** mounted on a substrate (or circuit board) **15314**, in which a first set of electrical integrated circuits **15316** are mounted on the photonic integrated circuit **15312**, and a second set of electrical integrated circuits **15318** are mounted on the substrate **15314** adjacent to the photonic integrated circuit **15312**. A micro optics connector **15322** optically couples the photonic integrated circuit **15312** to an optical fiber cable.
(797) FIG. **154**D is a diagram of an example co-packaged optical module **15320** that can fit inside the housing **15306** of the OSFP-XD module **15302** with the substrate or circuit board oriented vertically (relative to the horizontal bottom wall of the housing). The co-packaged optical module **15320** can be similar to the co-packaged optical module **15310** (FIG. **153**C) except that the dimensions of the components are slightly different in order to fit inside the housing **15306** of the OSFP-XD module **15302**. Additional details of the co-packaged optical modules **15310** and **15320** are provided later in this document.
(798) FIG. **154**E is a cross-sectional diagram of an example OSFP pluggable optical module **15330** in which the co-packaged optical module **15310** is fitted in the space defined by the housing **15304**, with the substrate **15314** of the co-packaged optical module **15310** substantially perpendicular to the bottom surface of the housing **15304** of the OSFP pluggable optical module **15330**. The top surface of the photonic integrated circuit **15312** is substantially perpendicular to the bottom surface of the housing **15304**, allowing a two-dimensional array of optical fibers to be coupled to the photonic integrated circuit **15312**.
(799) FIG. **154**F is a cross-sectional diagram of an example OSFP-XD pluggable optical module **15340** in which the co-packaged optical module **15320** is fitted in the space defined by the housing **15306**, with the substrate **15314** of the co-packaged optical module **15320** substantially perpendicular to the bottom surface of the housing **15304** of the OSFP pluggable optical module **15340**. The top surface of the photonic integrated circuit **15312** is substantially perpendicular to the bottom surface of the housing **15304**, allowing a two-dimensional array of optical fibers to be coupled to the photonic integrated circuit **15312**.
(800) FIG. **155**A is a side view cross-sectional diagram (along a plane parallel to the y-z plane) of an example pluggable optical module **15500** plugged into a cage **15402**, in which an edge connector **15404** (e.g., a paddle card) is mated with a receptacle **15406**. The pluggable optical module **15500** includes an optical connector **15204**, a fiber harness **15508**, a co-packaged optical module **15502**, a connector module **15504** (or tongue module or connector card), and flexible radio frequency (RF) cables **15506**. The co-packaged optical module **15502** includes a photonic integrated circuit **15312** having a surface that faces the front side of the pluggable optical module **15500**. The fiber harness **15508** optically couples the optical connector **15204** to the photonic integrated circuit **15312**. The fiber harness **15508** includes a bundle of optical fibers that are coupled to an optical fiber connector **15510** that is coupled to the photonic integrated circuit **15312**. The optical fiber connector **15510** includes a two-dimensional array of fiber ports, which can be variations of the two-dimensional array of fiber ports shown in FIGS. **89** to **95**, either with or without the power supply fiber ports.
(801) FIG. **155**E is a diagram of an example of the fiber port mapping for the optical fiber connector **15510**. The transmitter fiber ports **1704** are labeled 'T', and the receiver fiber ports **1706** are labeled 'R'. The fiber port mapping shown in FIG. **155**E is merely an example, other fiber port mappings can also be used.
(802) The flexible RF cables **15506** (e.g., available from Molex, Lisle, IL, or similar cables) electrically couple the co-packaged optical module **15502** to the connector module **15504**. For example, the co-packaged optical module **15502** can be similar to the co-packaged optical module **15310** (FIG. **153**C) or **15320** (FIG. **153**D). For example, the connector module **15504** includes a paddle card **15404** that includes several connector pads **15408** on the top and bottom sides of the paddle card **15404**. First terminations of the flexible RF cables **15506** are connected to first connector parts **15512**, and second terminations of the flexible RF cables **15506** are connected to second connector parts **15514**. The first connector parts **15512** are electrically coupled to the photonic integrated circuit **15312** through conductive vias in the substrate **15314**. The second connector parts **15514** are electrically coupled to the conductive pads **15408** through conductive traces in or on the paddle card **15404**. Use of the flexible RF cables **15506** allows the co-packaged optical module **15502** to be mechanically decoupled from the connector module **15504**. This allows the co-packaged optical module **15502** and the connector module **15504** to be used in different pluggable optical modules having different lengths.
(803) FIG. **155**B is a front view (along a plane parallel to the x-y plane) of the co-packaged optical module **15502**. FIG. **155**C is a rear view of the connector module **15504**. FIG. **155**D is a side view of the co-packaged optical module **15502**, the flexible RF cables **15506**, and the connector module **15504**.
(804) FIG. **156**A is a side view cross-sectional diagram (along a plane parallel to the y-z plane) of an example pluggable optical module **15600** plugged into a cage **15402**, in which an edge connector **15404** (e.g., a paddle card) is mated with a receptacle **15406**. The pluggable optical module **15600** includes an optical connector **15204**, a fiber harness **15508**, a co-packaged optical module **15502**, and a connector module **15602**. In this example, the connector module **15602** is

mechanically and electrically coupled to the co-packaged optical module **15502**. The pluggable optical module **15600** can be used in situations where it is not necessary to mechanically decouple the pluggable optical module **15600** from the connector module **15602**.

(805) FIG. **156**B is a front view (along a plane parallel to the x-y plane) of the co-packaged optical module **15502**. FIG. **156**C is a rear view of the connector module **15602**. FIG. **156**D is a side view of the co-packaged optical module **15502** and the connector module **15602**. The connector module **15602** includes the paddle card **15404**, which has several connector pads **15408** on the top and bottom sides of the paddle card **15404**.

(806) FIG. **157**A is a side view cross-sectional diagram (along a plane parallel to the y-z plane) of an example pluggable optical module **15700** plugged into a cage **15402**, in which an edge connector **15404** (e.g., a paddle card) is mated with a receptacle **15406**. FIG. **157**B is a top view cross-sectional diagram (along a plane parallel to the x-z plane) of the pluggable optical module **15700**. As shown in FIG. **157**B, in some examples, the connector pads **15408** extend in a direction parallel to the longitudinal direction or the z-axis. FIG. **157**C is a side view (along a plane parallel to the y-z plane) of the pluggable optical module **15700**.

(807) The pluggable optical module **15700** includes one or more laser sources **15702** that provide power supply light through one or more optical fibers **15704** to the photonic circuit(s) of the co-packaged optical module **15502**. In the example of FIGS. **157**A and **157**B, the pluggable optical module **15700** includes two laser sources **15702**. It is also possible to have a different number of laser source(s).

(808) The pluggable optical module **15700** includes a fiber harness **15706** that optically couples the optical connector **15204** to the photonic integrated circuit **15312**. The fiber harness **15706** includes a bundle of fibers **15708** that are coupled to an optical fiber connector **15710** that is coupled to the photonic integrated circuit **15312**. The fiber harness **15706** includes the optical fibers **15704**, which are also optically connected to the optical fiber connector **15710**. Power supply light is transmitted from the laser sources **15702** through the optical fibers **15704** and the power supply fiber ports of the optical fiber connector **15710** to the photonic integrated circuit **15312**.

(809) FIG. **157**D is a diagram of an example of the fiber port mapping for the optical fiber connector **15710**. The power supply power ports **1702** are labeled ‘P’, the transmitter fiber ports **1704** are labeled ‘T’, and the receiver fiber ports **1706** are labeled ‘R’. The fiber port mapping shown in FIG. **157**D is merely an example, other fiber port mappings can also be used.

(810) FIG. **158** shows an example of the OSFP module pinout configuration. A diagram **15800** shows the pinout configuration for the top side of the paddle card, and a diagram **15802** shows the pinout configuration for the bottom side of the paddle card.

(811) FIG. **159**A is a perspective view of an example 1×1 cage **15900** mounted on a circuit board **15902**. FIG. **159**B is a perspective view of an example 1×4 cage **15904** mounted on a circuit board **15906**. FIG. **159**C is a perspective view of an example OSFP module **15908** inserted into the 1×1 cage **15900**.

(812) FIG. **160** is a top view cross-sectional diagram of an example pluggable optical module **16000** that includes a co-packaged optical module **16004** mounted on a side wall **16006**. A tongue-to-board connector **16008** is provided to mechanically and electrically couple the substrate (or circuit board) of the co-packaged optical module **16004** to the paddle card (or tongue) **15404**. There is sufficient space inside the cage **16002** for the optical fibers of the fiber harness **15706** to bend and be vertically coupled to the photonic integrated circuit **15312**.

(813) FIG. **161** is a side view cross-sectional diagram of an example pluggable optical module **16100** plugged into a cage **15402**, in which an edge connector **15404** (e.g., a paddle card) is mated with a receptacle **15406**. The pluggable optical module **16100** includes a housing **15306** and a co-packaged optical module **16102**. The co-packaged optical module **16102** includes a photonic integrated circuit **15312** mounted on a substrate (or circuit board) **16104** that is oriented parallel to the bottom surface of the housing and is positioned lower than the edge connector (or paddle card or connector tongue) **15404**. A tongue-to-board connector **16106** is provided to mechanically and electrically couple the substrate (or circuit board) **16104** to the edge connector (or paddle card or connector tongue) **15404**. In some examples, the substrate **16104** and the edge connector **15404** can be mechanically coupled by solder joints. A fiber harness **16104** optically connects the optical connector **15204** to a fiber bend or turning mirror **16108** that interfaces with the photonic integrated circuit **15312**.

(814) The following describes examples of co-packaged optical modules that can be used in the pluggable optical modules, e.g., **15330** (FIG. **154**E), **15340** (FIG. **153**F), **15500** (FIG. **155**A), **15600** (FIG. **156**A), **15700** (FIG. **157**A), **16000** (FIG. **160**), **16100** (FIG. **161**). The co-packaged optical modules described below can also be used as the co-packaged optical modules **12386** (FIGS. **125**A, **125**B, **128**, **129**), **12504** (FIG. **126**A), **12604** (FIG. **126**B), **12710** (FIG. **127**), **13022** (FIGS. **130**, **131**, **133** to **135**).

(815) Referring to FIG. **162**, a co-packaged optical module **16700** includes a substrate **16702** and a photonic integrated circuit **16704** mounted on the substrate **16702**. A lens array **16706** and a micro optics connector **16708** optically couples the photonic integrated circuit **16704** to an optical fiber cable. The lens array **16706** and the micro optics connector **16708** will be referred to as the optical connector. A first set of one or more integrated circuits **16710** are mounted on the top side of the photonic integrated circuit **16704** using, e.g., copper pillars, or solder bumps. The first set of one or more integrated circuits **16710** is positioned adjacent to or near the optical connector. For example, two or more integrated circuits **16710** can be positioned on two or more sides of the optical connector, surrounding or partially surrounding the optical connector. A second set of integrated circuits **16712** is mounted on the substrate **16702** and electrically coupled to the photonic integrated circuit **16704**.

(816) For example, each integrated circuit **16710** (mounted on the photonic integrated circuit **16704**) can include an electrical drive amplifier or a transimpedance amplifier. Each integrated circuits **16712** (mounted on the substrate) can include a SerDes or a DSP chip or a combination of SerDes/DSP chips.
(817) FIGS. **163**A and **163**B show perspective views of an example of the co-packaged optical module **16700**. FIG. **163**A shows the substrate **16702**, the photonic integrated circuit **16704**, the first set of electrical integrated circuits **16710** mounted on the photonic integrate circuit **16704**, and a second set of electrical integrated circuits **16712** mounted on the substrate **16702**. FIG. **163**B shows the same components as those shown in the left diagram, with the addition of a smart connector **16800** that connects to an optical fiber cable, and a socket **16802** that electrically couples to the electrical contacts on the bottom side of the substrate **16702**. The socket **16802** can be on another substrate or circuit board **16804**.
(818) FIGS. **164**A and **164**B shows additional examples of perspective views of the co-packaged optical module **16700**. FIG. **165** shows a top view of an example of the placement of the electrical integrated circuits **16710** on the photonic integrated circuit **16704**. In this example, the lens array **16706** is positioned near the center of the photonic integrated circuit **16704**, and the electrical integrated circuits **16710** are placed at the north, south, east, and west positions relative to the lens array **16706**. By placing the electrical integrated circuits **16710** on top of the photonic integrated circuit **16704** and surrounding the lens array **16706** (or any other type of optical connector), the co-packaged optical module **16700** can be made more compact. Furthermore, the conductive traces between the electrical integrated circuits **16710** and active components in the photonic integrated circuit **16704** can be made shorter, resulting in better performance, e.g., higher data rate, higher signal-to-noise ratio, and lower power required to transmit the signals, as compared to a configuration in which the electrical signals have to travel longer distances.
(819) There are several ways to package the electrical integrated circuits and the photonic integrated circuit in order to achieve a compact, small-size, and energy efficient co-packaged optical module. FIG. **166**A shows an example in which a photonic integrated circuit **16704** has an active layer **17100** that is positioned near the top surface of the photonic integrated circuit **16704**. The fiber connection **17102** (which can include, e.g., a 2D array of focusing lenses) is coupled to the fiber connection **17102** from the top side. The top side in FIG. **166**A corresponds to the front side of the photonic integrated circuit facing the optical connector **15204** in FIG. **155**A. For example, grating couplers in the active PIC layer **17100** can be positioned under the fiber connection **17102** to couple the optical signals from the fiber connection **17102** into optical waveguides on the active PIC layer **17100**, and from the optical waveguides out to the fiber connection **17102**. The electrical integrated circuits **16710** are mounted on the top side of the photonic integrated circuit **16704** and are coupled to the active PIC layer **17100** through contact pads and optionally short conductive traces. For example, the active PIC layer **17100** can include photodetectors that convert the optical signals received from the fiber connection **17102** to electrical current signals that are transmitted to the drivers and transimpedance amplifiers in the electrical integrated circuits **16710**. Similarly, the electrical integrated circuits **16710** can send electrical signals to the electro-optic modulators in the active PIC layer **17100** that convert the electrical signals to optical signals that are output through the fiber connection **17102**.
(820) FIG. **166**B shows an example in which the electrical integrated circuits **16710** are coupled to the bottom surface of the photonic integrated circuit **16704** and electrically coupled to the active PIC layer **17100** using through silicon vias **17104**. The bottom surface of the photonic integrated circuit **16704** in FIG. **166**B corresponds to the rear surface of photonic integrated circuit **15312** in FIG. **155**A. The through silicon vias **17104** provide signal conduction paths in the thickness direction through the silicon die or substrate of the photonic integrated circuit **16704**. The drivers and transimpedance amplifiers in the electrical integrated circuits **16710** can be positioned directly under the photonic integrated circuit active components, such as the photodiodes and the electro-optic modulators, so that the shortest electrical signal paths can be used between the photonic integrated circuit **16704** and the electrical integrated circuits **16710**.
(821) FIG. **166**C shows an example in which the fiber connection **17102** is coupled to the photonic integrated circuit **16704** through the bottom side (in a configuration referred to as “backside illumination”), such that the optical signals from the fiber connection **17102** pass through the silicon die or substrate before being received by the photodetectors in the active PIC layer **17100**. The bottom side of the photonic integrated circuit **16704** in FIG. **166**C corresponds to the front side of the photonic integrated circuit facing the optical connector **15204** in FIG. **155**A. The modulators in the active PIC layer **17100** transmit modulated optical signals through the silicon die or substrate to the fiber connection **17102**. The portion of the active PIC layer **17100** directly above the fiber connection **17102** can include grating couplers. The photodetectors and modulators are positioned at a distance from the grating couplers. The electrical integrated circuits **16710** are positioned directly above or near the photodetectors and the modulators, so the locations of the electrical integrated circuits **16710** relative to the active PIC layer **17100** in the example of FIG. **166**C will be similar to those in the example of FIG. **166**A.
(822) FIG. **166**D shows an example in which backside illumination is used, and the electrical integrated circuits **16710** are coupled to the bottom side of the photonic integrated circuit **16704**. The bottom side of the photonic integrated circuit **16704** of FIG. **166**D corresponds to the front side of the photonic integrated circuit in FIG. **155**A. The electrical integrated circuits **16710** are electrically coupled to the active components (e.g., photodetectors and electro-optic modulators) in the active PIC layer **17100** using through silicon vias **17104**, similar to the example in FIG. **166**B.
(823) In some implementations, an integrated circuit is configured to surround or partially surround the vertical fiber connector. For example, the integrated circuit can have an L-shape that surrounds two sides of the vertical fiber connector

(e.g., two of north, east, south, and west sides). For example, the integrated circuit can have a U-shape that surrounds three sides of the vertical fiber connector (e.g., three of north, east, south, and west sides). For example, the integrated circuit can have an opening in the center region to allow the vertical fiber connector to pass through, in which the integrated circuit completely surrounds the vertical fiber connector. The dimensions of the opening in the integrated circuit are selected to allow the optical fiber connector to pass through to enable an optical fiber to be optically coupled to the photonic integrated circuit. For example, the integrated circuit with an opening in the center region can have a circular or polygonal shape at the outer perimeter. A feature of the integrated circuit mounted on the same surface as the vertical fiber connector is that it takes advantage of the space available on the surface of the photonic integrated circuit that is not occupied by the vertical fiber connector so that the electrical integrated circuit can be placed near or adjacent to the active components (e.g., photodetectors and/or modulators) of the photonic integrated circuit.
(824) In some implementations, an integrated circuit defining an opening can be manufactured by the following process: Step 1: Use semiconductor lithography to form an integrated circuit on a semiconductor die (or wafer or substrate), in which a first interior region of the semiconductor die does not have integrated circuit component intended to be used for the final integrated circuit (but can have components intended to be used for other products). Step 2: Use a laser (or any other suitable cutting tool) to cut an opening in the first interior region of the semiconductor die. Step 3: Place the semiconductor die on a lower mold resin that defines an opening in an interior region. A lead frame or electrical connectors are attached to the lower mold resin. Step 4: Wire bond electrical contacts on the semiconductor die to the lead frame or electrical connectors attached to the lower mold resin. Step 5: Attach an upper mold resin to the lower mold resin, and enclose the semiconductor die between the lower and upper mold resins. The upper mold resin defines an opening in an interior region that corresponds to the opening in the lower mold resin. In some examples, the footprint of the semiconductor die is within the footprint of the lower/upper mold resins so that the semiconductor die is completely enclosed inside the lower and upper mold resins. In some examples, the lower and/or upper mold resin can have additional openings, and the opening(s) in the lower and/or upper mold resins can be configured to expose one or more portions of the semiconductor die.
(825) An integrated circuit having an L-shape or a U-shape can be manufactured using a similar process. For example, in step 1, circuitry is formed in an L-shaped or U-shaped footprint. In step 2, the laser or cutting tool cuts the die according to the L-shape or U-shape footprint. In steps 3 and 5, a lower mold resin and an upper mold resin having the desired L-shape or U-shape are used.
(826) Additional details of the components used in the data processing systems described in this document, e.g., the co-packaged optical modules, the optical modules, the optical communication interfaces, the photonic integrated circuits, the electronic integrated circuits, etc., can be found in U.S. patent application Ser. No. 17/478,483, filed on Sep. 17, 2021, published as US20220159860; U.S. patent application Ser. No. 17/495,338, filed on Oct. 6, 2021; U.S. patent application Ser. No. 17/531,470, filed on Nov. 19, 2021; U.S. patent application Ser. No. 17/592,232, filed on Feb. 3, 2022; PCT application PCT/US2021/021953, filed on Mar. 11, 2021, published as WO 2021/183792; PCT application PCT/US2021/022730, filed on Mar. 17, 2021, published as WO 2021/188648; PCT application PCT/US2021/027306, filed on Apr. 14, 2021, published as WO 2021/211725; PCT application PCT/US2021/035179, filed on Jun. 1, 2021, published as WO 2021/247521, PCT application PCT/US2021/050945, filed on Sep. 17, 2021, published as WO 2022/061160, PCT application PCT/US2021/053745, filed on Oct. 6, 2021, published as WO 2022/076539, PCT application PCT/US2021/060215, filed on Nov. 19, 2021, published as WO 2022/109349, U.S. Pat. Nos. 11,287,585, 11,194,109, and 11,153,670. The entire contents of the above patent applications, patent application publications, and patents are incorporated by reference.
(827) Some embodiments can be implemented as circuit-based processes, including possible implementation on a single integrated circuit.
(828) Unless explicitly stated otherwise, each numerical value and range should be interpreted as being approximate as if the word “about” or “approximately” preceded the value or range.
(829) It will be further understood that various changes in the details, materials, and arrangements of the parts which have been described and illustrated in order to explain the nature of this disclosure can be made by those skilled in the art without departing from the scope of the disclosure, e.g., as expressed in the following claims.
(830) The use of figure numbers and/or figure reference labels in the claims is intended to identify one or more possible embodiments of the claimed subject matter in order to facilitate the interpretation of the claims. Such use is not to be construed as necessarily limiting the scope of those claims to the embodiments shown in the corresponding figures.
(831) Although the elements in the following method claims, if any, are recited in a particular sequence with corresponding labeling, unless the claim recitations otherwise imply a particular sequence for implementing some or all of those elements, those elements are not necessarily intended to be limited to being implemented in that particular sequence.
(832) Reference herein to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment can be included in at least one embodiment of the disclosure. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive of other embodiments. The same applies to the term “implementation.”
(833) Unless otherwise specified herein, the use of the ordinal adjectives “first,” “second,” “third,” etc., to refer to an object of a plurality of like objects merely indicates that different instances of such like objects are being referred to, and

is not intended to imply that the like objects so referred-to have to be in a corresponding order or sequence, either temporally, spatially, in ranking, or in any other manner.
(834) Also for purposes of this description, the terms "couple," "coupling," "coupled," "connect," "connecting," or "connected" refer to any manner known in the art or later developed in which energy is allowed to be transferred between two or more elements, and the interposition of one or more additional elements is contemplated, although not required. Conversely, the terms "directly coupled," "directly connected," etc., imply the absence of such additional elements.
(835) As used herein in reference to an element and a standard, the term compatible means that the element communicates with other elements in a manner wholly or partially specified by the standard, and would be recognized by other elements as sufficiently capable of communicating with the other elements in the manner specified by the standard. The compatible element does not need to operate internally in a manner specified by the standard.
(836) The described embodiments are to be considered in all respects as only illustrative and not restrictive. In particular, the scope of the disclosure is indicated by the appended claims rather than by the description and figures herein. All changes that come within the meaning and range of equivalency of the claims are to be embraced within their scope.
(837) The description and drawings merely illustrate the principles of the disclosure. It will thus be appreciated that those of ordinary skill in the art will be able to devise various arrangements that, although not explicitly described or shown herein, embody the principles of the disclosure and are included within its spirit and scope. Furthermore, all examples recited herein are principally intended expressly to be only for pedagogical purposes to aid the reader in understanding the principles of the disclosure and the concepts contributed by the inventor(s) to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions. Moreover, all statements herein reciting principles, aspects, and embodiments of the disclosure, as well as specific examples thereof, are intended to encompass equivalents thereof.
(838) The functions of the various elements shown in the figures, including any functional blocks labeled or referred to as "processors" and/or "controllers," can be provided through the use of dedicated hardware as well as hardware capable of executing software in association with appropriate software. When provided by a processor, the functions can be provided by a single dedicated processor, by a single shared processor, or by a plurality of individual processors, some of which can be shared. Moreover, explicit use of the term "processor" or "controller" should not be construed to refer exclusively to hardware capable of executing software, and can implicitly include, without limitation, digital signal processor (DSP) hardware, network processor, application specific integrated circuit (ASIC), field programmable gate array (FPGA), read only memory (ROM) for storing software, random access memory (RAM), and non-volatile storage. Other hardware, conventional and/or custom, can also be included. Similarly, any switches shown in the figures are conceptual only. Their function can be carried out through the operation of program logic, through dedicated logic, through the interaction of program control and dedicated logic, or even manually, the particular technique being selectable by the implementer as more specifically understood from the context.
(839) As used in this application, the term "circuitry" can refer to one or more or all of the following: (a) hardware-only circuit implementations (such as implementations in only analog and/or digital circuitry); (b) combinations of hardware circuits and software, such as (as applicable): (i) a combination of analog and/or digital hardware circuit(s) with software/firmware and (ii) any portions of hardware processor(s) with software (including digital signal processor(s)), software, and memory(ies) that work together to cause an apparatus, such as a mobile phone or server, to perform various functions); and (c) hardware circuit(s) and or processor(s), such as a microprocessor(s) or a portion of a microprocessor(s), that requires software (e.g., firmware) for operation, but the software does not need to be present when it is not needed for operation." This definition of circuitry applies to all uses of this term in this application, including in any claims. As a further example, as used in this application, the term circuitry also covers an implementation of merely a hardware circuit or processor (or multiple processors) or portion of a hardware circuit or processor and its (or their) accompanying software and/or firmware. The term circuitry also covers, for example and if applicable to the particular claim element, a baseband integrated circuit or processor integrated circuit for a mobile device or a similar integrated circuit in server, a cellular network device, or other computing or network device.
(840) It should be appreciated by those of ordinary skill in the art that any block diagrams herein represent conceptual views of illustrative circuitry embodying the principles of the disclosure.
(841) Although the present invention is defined in the attached claims, it should be understood that the present invention can also be defined in accordance with the following sets of embodiments:
First Set of Embodiments
(842) Embodiment 1: A system comprising: a housing having a front panel; a first substrate that is positioned at a distance from the front panel, in which a data processor is mounted on the first substrate; and a pluggable module comprising an optical module, at least one first optical connector, a first fiber optic cable that is optically coupled between the optical module and the first optical connector, and a fiber guide that is positioned between the optical module and the first optical connector and provides mechanical support for the optical module and the first optical connector; wherein the optical module is configured to receive optical signals from the first optical connector and generate electrical signals based on the received optical signals, and the electrical signals or processed versions of the electrical signals are transmitted to the data processor; and wherein the pluggable module has a shape that enables the pluggable module to pass through an opening in the front panel to enable the optical module to be coupled to the first substrate.
(843) Embodiment 2: The system of embodiment 1 in which the first optical connector is configured to mate with a

corresponding optical connector of an external fiber optic cable.
(844) Embodiment 3: The system of embodiment 1 or 2 in which the first optical connector comprises a multi-fiber push on (MPO) connector.
(845) Embodiment 4: The system of any of embodiments 1 to 3 in which the fiber guide has a length configured such that when the pluggable module is inserted through the opening in the front panel and the optical module is coupled to the first substrate or a module mounted on the first substrate, the at least one first optical connector is in a vicinity of the front panel to enable a user to attach at least one external fiber optic cable to the at least one first optical connector.
(846) Embodiment 5: The system of any of embodiments 1 to 3 in which the fiber guide has a length configured such that when the pluggable module is inserted through the opening in the front panel and the optical module is coupled to the first substrate or a module mounted on the first substrate, the at least one first optical connector has a front surface that is flush with, or protrudes from, a front surface of the front panel to enable a user to attach at least one external fiber optic cable to the at least one first optical connector.
(847) Embodiment 6: The system of any of embodiments 1 to 3 in which the fiber guide has a length configured such that when the pluggable module is inserted through the opening in the front panel and the optical module is coupled to the first substrate or a module mounted on the first substrate, the at least one first optical connector has a front face that is within an inch of a front surface of the front panel.
(848) Embodiment 7: The system of any of embodiments 1 to 6 in which the fiber guide has a length of at least one inch.
(849) Embodiment 8: The system of any of embodiments 1 to 6 in which the fiber guide has a length of at least two inches.
(850) Embodiment 9: The system of any of embodiments 1 to 6 in which the fiber guide has a length of at least four inches.
(851) Embodiment 10: The system of any of embodiments 1 to 9 in which the pluggable module comprises at least two first optical connectors, and each first optical connector is configured to be mated with a second optical connector of an external fiber optic cable.
(852) Embodiment 11: The system of any of embodiments 1 to 6 in which the pluggable module comprises at least four first optical connectors, and each first optical connector is configured to be mated with a second optical connector of an external fiber optic cable.
(853) Embodiment 12: The system of any of embodiments 1 to 11 in which the first fiber optic cable comprises a fiber pigtail.
(854) Embodiment 13: The system of any of embodiments 1 to 11 in which the first substrate has a main surface that is oriented at an angle in a range of 0 to 45 degrees relative to the front panel.
(855) Embodiment 14: The system of embodiment 13 in which the first substrate is oriented parallel to the front panel.
(856) Embodiment 15: The system of any of embodiments 1 to 14 in which the first substrate has a first side and a second side that is opposite the first side, the data processor comprises electrical contacts that are electrically coupled to electrical contacts on the first side of the first substrate, the pluggable module comprises electrical contacts that are electrically coupled to electrical contacts on the second side of the first substrate, and at least some of the electrical contacts on the first side of the first substrate are electrically coupled to at least some of the electrical contacts on the second side of the first substrate.
(857) Embodiment 16: The system of embodiment 15 in which the first substrate comprises at least one of a ceramic substrate, an organic high density build-up substrate, or a silicon substrate.
(858) Embodiment 17: The system of any of embodiments 1 to 14 in which the system comprises a second substrate, the data processor comprises electrical contacts that are electrically coupled to electrical contacts on the first substrate, the pluggable module comprises electrical contacts that are electrically coupled to electrical contacts on the second substrate, and at least some of the electrical contacts on the first substrate are electrically coupled to at least some of the electrical contacts on the second substrate.
(859) Embodiment 18: The system of embodiment 17 in which the first substrate is mounted on a first side of a third substrate or circuit board, and the second substrate is mounted on a second side of the third substrate or circuit board.
(860) Embodiment 19: The system of embodiment 18 in which each of the first and second substrate comprises at least one of a ceramic substrate, an organic high density build-up substrate, or a silicon substrate.
(861) Embodiment 20: The system of any of embodiments 1 to 19, comprising an inlet fan mounted near the front panel and configured to increase an air flow across a surface of at least one of (i) the optical module, or (ii) a heat dissipating device thermally coupled to the optical module.
(862) Embodiment 21: The system of embodiment 20, comprising two or more pluggable modules, in which each pluggable module comprises an optical module, at least one first optical connector, a first fiber optic cable that is optically coupled between the optical module and the first optical connector, and a fiber guide that is positioned between the optical module and the first optical connector; wherein the fiber guides are configured to allow air blown from the inlet fan to flow past the fiber guides and carry away heat generated by the optical module.
(863) Embodiment 22: The system of any of embodiments 1 to 21, comprising a laser module configured to provide optical power to the optical module.
(864) Embodiment 23: The system of embodiment 22, comprising a second optical connector optically coupled to the laser module, wherein the pluggable module comprises a third optical connector that is configured to mate with the second

optical connector when the pluggable module is coupled to the first substrate, and wherein the first optical connector is optically coupled to the optical module to enable the optical module to receive the optical power from the laser module.
(865) Embodiment 24: The system of embodiment 22 or 23, comprising a first heat dissipating device and a second heat dissipating device, the first heat dissipating device is thermally isolated from the second heat dissipating device, the first heat dissipating device is thermally coupled to the optical module, and the second heat dissipating device is thermally coupled to the laser module.
(866) Embodiment 25: The system of any of embodiments 1 to 23 in which the fiber guide comprise at least one of metal or a thermal conductive material.
(867) Embodiment 26: The system of any of embodiments 1 to 25 in which the fiber guide comprises a hollow tube.
(868) Embodiment 27: The system of any of embodiments 1 to 26 in which the fiber guide is rigid along a direction from the first optical connector to the optical module and has a strength sufficient to withstand a compression force exerted on the pluggable module when the pluggable module is inserted through the opening in the front panel and coupled to the first substrate.
(869) Embodiment 28: The system of any of embodiments 1 to 27 in which the fiber guide has a spatial fan-out design such that a first portion of the fiber guide near the optical module has a smaller dimension compared to the dimension of a second portion of the fiber guide near the at least one first optical connector.
(870) Embodiment 29: The system of embodiment 28 in which the at least one first optical connector has an overall footprint that is larger than a footprint of the optical module.
(871) Embodiment 30: The system of any of embodiments 1 to 29 in which the data processor comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, an application specific integrated circuit (ASIC), or a storage device.
(872) Embodiment 31: The system of any of embodiments 1 to 30 in which a photon supply is disposed in, on, or near the fiber guide, and the photon supply is configured to provide optical power supply light to the optical module.
(873) Embodiment 32: The system of embodiment 31 in which the photon supply is thermally coupled to an inner surface or an outer surface of the fiber guide, and the fiber guide is configured to assist in dissipating heat from the photon supply.
(874) Embodiment 33: The system of any of embodiments 1 to 32, comprising guide rails configured to guide the optical module as the optical module move from a first position near the front panel to a second position near the first substrate.
(875) Embodiment 34: The system of embodiment 33 in which the optical module comprises a co-packaged optical module comprising a photonic integrated circuit and one or more electrical integrated circuits that condition electrical signals transmitted to or from the photonic integrated circuit.
(876) Embodiment 35: The system of embodiment 34, comprising a co-packaged optical module (CPO) mount attached to the first substrate, and the guide rails are configured to provide rigid connections between the CPO mount and the front panel or a front portion of the fiber guide.
(877) Embodiment 36: The system of embodiment 34 or 35 in which the photonic integrated circuit comprises at least one of (i) a photodetector to convert optical signals to electrical signals, or (ii) a modulator to convert electrical signals to optical signals.
(878) Embodiment 37: The system of embodiment 33 in which the system comprises a co-packaged optical module (CPO) mount and a bolster plate, the co-packaged optical module is mounted on the first substrate through the CPO mount, and the bolster plate is positioned to the rear of the substrate and configured to exert a force in a front direction when the guide rails are fastened to a front portion of the fiber guide or to the front panel.
(879) Embodiment 38: The system of any of embodiments 1 to 37 in which the optical module has a first side and a second side, the first fiber optical cable has a first end that has a two-dimensional arrangement of optical fiber cores, the first side of the optical module is optically coupled to the two-dimensional arrangement of optical fiber cores, and the second side of the optical module has a two-dimensional arrangement of electrical contacts that are configured to mate with a two-dimensional arrangement of electrical contacts on the first substrate.
(880) Embodiment 39: The system of embodiment 38 in which the two-dimensional arrangement of electrical contacts of the optical module comprise at least two rows of electrical contacts, and each row includes at least two electrical contacts.
(881) Embodiment 40: The system of embodiment 39 in which the two-dimensional arrangement of electrical contacts of the optical module comprise at least four rows of electrical contacts, and each row includes at least four electrical contacts.
(882) Embodiment 41: The system of embodiment 40 in which the two-dimensional arrangement of electrical contacts of the optical module comprise at least ten rows of electrical contacts, and each row includes at least ten electrical contacts.
(883) Embodiment 42: An apparatus comprising: a pluggable module comprising a co-packaged optical module, at least one first optical connector, a first fiber optic cable that is optically coupled between the co-packaged optical module and the first optical connector, and a fiber guide that is positioned between the co-packaged optical module and the first optical connector and provides mechanical support for the co-packaged optical module and the first optical connector; wherein the co-packaged optical module is configured to receive optical signals from the at least one first optical connector, and generate electronic signals based on the optical signals.
(884) Embodiment 43: The apparatus of embodiment 42 in which the fiber guide comprise at least one of metal or a thermal conductive material.
(885) Embodiment 44: The apparatus of embodiment 42 or 43 in which the fiber guide comprises a hollow tube.

(886) Embodiment 45: The apparatus of any of embodiments 42 to 44 in which the fiber guide is rigid along a direction from the first optical connector to the co-packaged optical module and has a strength sufficient to withstand a compression force exerted on the pluggable module when the pluggable module is inserted through an opening in a front panel of a housing and coupled to the substrate.
(887) Embodiment 46: The apparatus of any of embodiments 42 to 45 in which the fiber guide has a spatial fan-out design such that a first portion of the fiber guide near the co-packaged optical module has a smaller dimension compared to the dimension of a second portion of the fiber guide near the at least one first optical connector.
(888) Embodiment 47: The apparatus of any of embodiments 42 to 45 in which the at least one first optical connector has an overall footprint that is larger than a footprint of the co-packaged optical module.
(889) Embodiment 48: The system of any of embodiments 42 to 47 in which the co-packaged optical module has a first side and a second side, the first fiber optical cable has a first end that has a two-dimensional arrangement of optical fiber cores, the first side of the optical module is optically coupled to the two-dimensional arrangement of optical fiber cores, and the second side of the optical module has a two-dimensional arrangement of electrical contacts.
(890) Embodiment 49: The apparatus of embodiment 48 in which the two-dimensional arrangement of electrical contacts of the optical module comprise at least two rows of electrical contacts, and each row includes at least two electrical contacts.
(891) Embodiment 50: The apparatus of embodiment 49 in which the two-dimensional arrangement of electrical contacts of the optical module comprise at least four rows of electrical contacts, and each row includes at least four electrical contacts.
(892) Embodiment 51: The apparatus of embodiment 50 in which the two-dimensional arrangement of electrical contacts of the optical module comprise at least ten rows of electrical contacts, and each row includes at least ten electrical contacts.
(893) Embodiment 52: A rackmount server comprising: a housing having a front panel and a rear panel, in which the front panel defines an opening, and the rear panel is at a first distance from the front panel; and a substrate that is positioned at a second distance from the front panel, in which the second distance is less than one-third of the first distance, a data processor is mounted on the substrate, the substrate has a main surface that is oriented at an angle in a range of 0 to 45 degrees relative to the front panel; wherein at least one of (i) the substrate has electrical contacts that are configured to the electrically coupled to electrical contacts of a co-packaged optical module, or (ii) a first module is mounted on the substrate, in which the first module has electrical contacts that are configured to the electrically coupled to electrical contacts of a co-packaged optical module.
(894) Embodiment 53: The rackmount server of embodiment 52 in which the substrate is oriented substantially parallel to the front panel.
(895) Embodiment 54: The rackmount server of embodiment 52 or 53 in which the opening in the front panel is configured to allow a pluggable module that includes the co-packaged optical module to be inserted through the opening to enable the co-packaged optical module to be electrically coupled to the electrical contacts on the substrate or the electrical contacts on the first module mounted on the substrate.
(896) Embodiment 55: The rackmount server of embodiment 53, comprising the pluggable module.
(897) Embodiment 56: The rackmount server of embodiment 55, in which the pluggable module comprises the co-packaged optical module, at least one first optical connector, a first fiber optic cable that is optically coupled between the co-packaged optical module and the first optical connector, and a fiber guide that is positioned between the co-packaged optical module and the first optical connector and provides mechanical support for the co-packaged optical module and the first optical connector.
(898) Embodiment 57: The rackmount server of embodiment 56 in which the co-packaged optical module is configured to receive optical signals from the first optical connector, generate electrical signals based on the received optical signals, and transmit the electrical signals to the data processor.
(899) Embodiment 58: The rackmount server of any of embodiments 52 to 57 in which the data processor comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, an application specific integrated circuit (ASIC), or a storage device.
(900) Embodiment 59: The rackmount server of any of embodiments 52 to 58 in which the substrate has a two-dimensional arrangement of electrical contacts that are configured to be electrically coupled to a two-dimensional arrangement of electrical contacts of the co-package optical module.
(901) Embodiment 60: The rackmount server of embodiment 59 in which the two-dimensional arrangement of electrical contacts of the substrate comprise at least two rows of electrical contacts, and each row includes at least two electrical contacts.
(902) Embodiment 61: The rackmount server of embodiment 60 in which the two-dimensional arrangement of electrical contacts of the substrate comprise at least four rows of electrical contacts, and each row includes at least four electrical contacts.
(903) Embodiment 62: The rackmount server of embodiment 61 in which the two-dimensional arrangement of electrical contacts of the substrate comprise at least ten rows of electrical contacts, and each row includes at least ten electrical contacts.

(904) Embodiment 63: The rackmount server of any of embodiments 52 to 58 in which the substrate has a plurality of groups of two-dimensional arrangement of electrical contacts that are configured to be electrically coupled to a corresponding plurality of groups of two-dimensional arrangement of electrical contacts of co-package optical modules.
(905) Embodiment 64: The rackmount server of embodiment 63 in which the plurality of groups of two-dimensional arrangement of electrical contacts comprises at least four groups of two-dimensional arrangement of electrical contacts, and each group of two-dimensional arrangement of electrical contacts comprise at least four rows of electrical contacts, and each row includes at least four electrical contacts.
(906) Embodiment 65: The rackmount server of embodiment 64 in which the plurality of groups of two-dimensional arrangement of electrical contacts comprises at least ten groups of two-dimensional arrangement of electrical contacts, and each group of two-dimensional arrangement of electrical contacts comprise at least ten rows of electrical contacts, and each row includes at least ten electrical contacts.
(907) Embodiment 66: A system comprising: a first substrate comprising least one of a ceramic substrate, an organic high density build-up substrate, or a silicon substrate; a data processor mounted on a rear side of the first substrate; a co-packaged optical module that is removably coupled to a front side of the first substrate and configured to receive optical signals from an optical connector, generate electrical signals based on the received optical signals, and transmit the electrical signals to the data processor; and a printed circuit board attached to the rear side of the first substrate, in which the printed circuit board includes an opening, and the data processor protrudes or partially protrudes through the opening, and the printed circuit board provides electrical power to the data processor through signal lines or traces in or on the first substrate.
(908) Embodiment 67: An apparatus comprising an optical transceiver module comprising: a photonic integrated circuit configured to perform at least one of (i) converting optical signals to electrical signals, or (ii) converting electrical signals to optical signals; at least one optical connector, in which the photonic integrated circuit is configured to receive optical signals from the at least one optical connector or transmit optical signals to the at least one optical connector; a plurality of electrical contacts, in which the photonic integrated circuit is configured to receive electrical signals from the plurality of electrical contacts or provide electrical signals to the plurality of electrical contacts; at least one electronic component positioned in an electrical signal path between the photonic integrated circuit and the plurality of electrical contacts and configured to process electrical signals sent to or from the photonic integrated circuit; at least one laser configured to provide optical power supply light to the photonic integrated circuit; and a first thermal path and a second thermal path, in which the second thermal path is thermally isolated from the first thermal path, the first thermal path enables heat from the at least one laser to be conducted outside of the optical module, and the second thermal path enables heat from the at least one electronic component to be conducted outside of the optical module.
(909) Embodiment 68: The apparatus of embodiment 67 in which the optical transceiver module comprises a pluggable optical transceiver module, the plurality of electrical contacts of the pluggable optical transceiver module are configured to be removably and electrically coupled to corresponding electrical contacts of a data processing apparatus.
(910) Embodiment 69: The apparatus of embodiment 67 in which the plurality of electrical contacts of the optical transceiver module are configured to be fixedly and electrically coupled to corresponding electrical contacts of a data processing apparatus.
(911) Embodiment 70: The apparatus of any of embodiments 67 to 69 in which the at least one electronic component comprises at least one of a serializer, a deserializer, a serializer/deserializer, a digital signal processor, a driver module, or an amplifier module.
(912) Embodiment 71: The apparatus of any of embodiments 67 to 70 in which the at least one laser is positioned closer to the at least one optical connector and farther away from the plurality of electrical contacts.
(913) Embodiment 72: The apparatus of any of embodiments 67, 68, 70, and 71 in which the optical transceiver module has a form factor that complies with at least one of SFP (small form-factor pluggable), SFP+ (or 10 Gb SFP), SFP28, OSFP (octal SFP), OSFP-XD (OSFP extra dense), QSFP (quad small form-factor pluggable), QSFP+, QSFP28, QSFP56, or QSFP-DD (quad small form-factor pluggable double density) standard.
(914) Embodiment 73: The apparatus of any of embodiments 67 to 72 in which the at least one optical connector has a first end that has a two-dimensional arrangement of optical fiber cores, and the photonic integrated circuit is optically coupled to the two-dimensional arrangement of optical fiber cores using a two-dimensional arrangement of optical couplers.
(915) Embodiment 74: The apparatus of any of embodiments 67 to 73 in which the optical transceiver module comprises a housing, the at least one electrical component and the at least one laser are positioned inside the housing, the housing defines an opening, wherein the optical transceiver module comprises a first heat dissipating device and a second heat dissipating device, the second heat dissipating device is thermally isolated from the first heat dissipating device, the second heat dissipating device is thermally coupled to the housing, wherein the first thermal path extends from the at least one laser through the opening defined by the housing to the first heat dissipating device, and the second thermal path extends from the at least one electrical component through the housing to the second heat dissipating device.
(916) Embodiment 74a: The apparatus of embodiment 74 in which the optical transceiver module provides an air gap between the first heat dissipating device and the second heat dissipating device.
(917) Embodiment 74b: The apparatus of embodiment 74 in which the optical transceiver module includes a thermally insulating material positioned between the first heat dissipating device and the second heat dissipating device.

(918) Embodiment 74c: The apparatus of embodiment 74b in which each of the heat dissipating device and the second heat dissipating device is made of a material having a thermal conductivity greater than 50 W/mK.
(919) Embodiment 74d: The apparatus of embodiment 74c in which each of the heat dissipating device and the second heat dissipating device is made of a material having a thermal conductivity greater than 100 W/mK.
(920) Embodiment 74e: The apparatus of embodiment 74d in which each of the heat dissipating device and the second heat dissipating device is made of a material having a thermal conductivity greater than 200 W/mK.
(921) Embodiment 74f: The apparatus of any of embodiments 74b to 74e in which the thermally insulating material has a thermal conductivity less than 10 W/mK.
(922) Embodiment 74g: The apparatus of embodiment 74f in which the thermally insulating material has a thermal conductivity less than 1 W/mK.
(923) Embodiment 75: The apparatus of any of embodiments 67 to 73 in which the optical transceiver module comprises a fiber guide that is positioned between the photonic integrated circuit and the at least one optical connector and provides mechanical support for the first optical connector and the photonic integrated circuit or a module that includes the photonic integrated circuit.
(924) Embodiment 76: The apparatus of embodiment 75 in which the fiber guide comprise at least one of metal or a thermal conductive material.
(925) Embodiment 77: The apparatus of embodiment 75 or 76 in which the fiber guide comprises a hollow tube.
(926) Embodiment 78: The apparatus of any of embodiments 75 to 77 in which the fiber guide is rigid along a direction from the at least one optical connector to the photonic integrated circuit or the module that includes the photonic integrated circuit and has a strength sufficient to withstand a compression force exerted on the optical transceiver module to cause the optical transceiver module to engage a receptor of another apparatus and cause the plurality of electrical contacts to be electrically coupled to corresponding electrical contacts of the other apparatus.
(927) Embodiment 79: The apparatus of any of embodiments 75 to 78 in which the fiber guide has a spatial fan-out design such that a first portion of the fiber guide near the photonic integrated circuit has a smaller dimension compared to the dimension of a second portion of the fiber guide near the at least one optical connector.
(928) Embodiment 80: The apparatus of any of embodiments 75 to 79 in which the plurality of electrical contacts comprise a two-dimensional arrangement of electrical contacts.
(929) Embodiment 81: The apparatus of embodiment 80 in which the two-dimensional arrangement of electrical contacts of the optical module comprise at least two rows of electrical contacts, and each row includes at least two electrical contacts.
(930) Embodiment 82: The apparatus of embodiment 81 in which the two-dimensional arrangement of electrical contacts of the optical module comprise at least four rows of electrical contacts, and each row includes at least four electrical contacts.
(931) Embodiment 83: The apparatus of embodiment 82 in which the two-dimensional arrangement of electrical contacts of the optical module comprise at least ten rows of electrical contacts, and each row includes at least ten electrical contacts.
(932) Embodiment 84: A rackmount server comprising a plurality of the systems of any of embodiments 1 to 41 and 66.
(933) Embodiment 85: A rackmount server comprising a plurality of the apparatuses of any of embodiments 42 to 51 and 67 to 83.
(934) Embodiment 86: A data center comprising a plurality of the rackmount servers of any of embodiments 52 to 64, 84, and 85.
(935) Embodiment 87: A method comprising: providing a data processing server comprising a housing having a front panel that defines an opening; providing a substrate positioned in the housing spaced apart from the front panel, in which a data processor is electrically coupled to a rear side of the substrate; providing a pluggable module comprising an optical module, at least one first optical connector, a first fiber optic cable that is optically coupled between the optical module and the first optical connector, and a fiber guide that is positioned between the optical module and the first optical connector and provides mechanical support for the optical module and the first optical connector; optically coupling an external fiber optic cable to the optical connector of the pluggable module; inserting the pluggable module through the opening in the front panel and electrically coupling a two-dimensional arrangement of electrical contacts of the optical module with a corresponding two-dimensional arrangement of electrical contacts on a front side of the substrate; and establishing a communication path between the data processor and the external fiber optic cable through the pluggable module.
(936) Embodiment 88: The method of embodiment 87, comprising transmitting data between the data processor and the external fiber optic cable through the pluggable module with a bandwidth of at least 500 Gbps.
(937) Embodiment 89: The method of embodiment 88, comprising transmitting data between the data processor and the external fiber optic cable through the pluggable module with a bandwidth of at least 1 Tbps.
(938) Embodiment 90: The method of any of embodiments 87 to 89 in which the front panel defines a plurality of openings, and the front side of the substrate comprises a plurality of groups of two-dimensional arrangements of electrical contacts; wherein the method comprises: providing a plurality of the pluggable modules; optically coupling a plurality of external fiber optic cables to the optical connectors of the pluggable modules; inserting the pluggable modules through the openings in the front panel and electrically coupling groups of two-dimensional arrangements of electrical contacts of the

optical modules with corresponding groups of two-dimensional arrangements of electrical contacts on the front side of the substrate; and establishing communication paths between the data processor and the external fiber optic cables through the pluggable modules.
(939) Embodiment 91: The method of embodiment 90 in which the plurality of pluggable modules comprise at least 10 pluggable modules, and the method comprises transmitting data between the data processor and the external fiber optic cables through the pluggable modules with an aggregate bandwidth of at least 5 Tbps.
(940) Embodiment 92: The method of embodiment 91 in which the plurality of pluggable modules comprise at least 30 pluggable modules, and the method comprises transmitting data between the data processor and the external fiber optic cables through the pluggable modules with an aggregate bandwidth of at least 15 Tbps.
(941) Embodiment 93: A method including: operating the system of any of embodiments 1 to 41 and 66.
(942) Embodiment 94: A method including: operating the apparatus of any of embodiments 42 to 51 and 67 to 83.
(943) Embodiment 95: A method including: operating the rackmount server of any of embodiments 52 to 64, 84, and 85.
(944) Embodiment 96: A method including: operating the data center of embodiment 86.
(945) Embodiment 97: A method including: assembling the system of any of embodiments 1 to 41 and 66.
(946) Embodiment 98: A method including: assembling the apparatus of any of embodiments 42 to 51 and 67 to 83.
(947) Embodiment 99: A method including: assembling the rackmount server of any of embodiments 52 to 64, 84, and 85.
(948) Embodiment 100: A method including: assembling the data center of embodiment 86.
(949) The following is a second set of embodiments. The embodiment numbers below refer to those in the second set of embodiments.
(950) Embodiment 1: An apparatus comprising: a pluggable optical module comprising: a fiber connector configured to be optically coupled to an optical fiber cable; an optical module comprising a photonic integrated circuit having a first surface, in which a plurality of optical couplers are provided at the first surface of the photonic integrated circuit; a fiber harness optically coupled between the fiber connector and the first surface of the photonic integrated circuit, in which the fiber harness comprises a plurality of optical fibers and an optical fiber connector, the optical fiber connector is configured to optically couple the plurality of optical fibers to the first surface of the photonic integrated circuit, the optical fiber connector comprises a two-dimensional arrangement of fiber ports, the two-dimensional arrangement of fiber ports and the optical couplers at the first surface of the photonic integrated circuit are configured to enable light signals to be transmitted between the photonic integrated circuit and the plurality of optical fibers; and an edge connector having conductive pads configured to be electrically coupled to conductive pads of a receptacle when the edge connector is mated with the receptacle, in which the conductive pads of the edge connector are electrically coupled to the optical module.
(951) Embodiment 2: The apparatus of embodiment 1 in which the two-dimensional arrangement of fiber ports comprise at least two rows of fiber ports, and each row includes at least eight fiber ports.
(952) Embodiment 3: The apparatus of embodiment 2 in which the two-dimensional arrangement of fiber ports comprise at least three rows of fiber ports, and each row includes at least eight fiber ports.
(953) Embodiment 4: The apparatus of embodiment 3 in which the two-dimensional arrangement of fiber ports comprise at least four rows of fiber ports, and each row includes at least eight fiber ports.
(954) Embodiment 5: The apparatus of any of embodiments 1 to 4 in which the pluggable optical module complies with a small form factor pluggable module specification comprising at least one of SFP (small form-factor pluggable), SFP+, 10 Gb SFP, SFP28, OSFP (octal SFP), OSFP-XD (OSFP extra dense), QSFP (quad small form-factor pluggable), QSFP+, QSFP28, QSFP56, or QSFP-DD (quad small form-factor pluggable double density).
(955) Embodiment 6: The apparatus of any of embodiments 1 to 5 in which the pluggable optical module has a length not more than 200 mm, a width not more than 50 mm, and a height not more than 26 mm.
(956) Embodiment 7: The apparatus of any of embodiments 1 to 6 in which the pluggable optical module comprises a housing having an inner upper wall and an inner lower wall, the edge connector has an upper surface extending along a first plane that is at a first distance d**1** relative to the inner upper wall, the edge connector has a lower surface extending along a second plane that is at a second distance d**2** relative to the inner lower wall, wherein the fiber harness is substantially vertically coupled to the first surface of the photonic integrated circuit such that light from the fiber harness is directed toward the first surface of the photonic integrated circuit at an angle θ**1** relative to a direction vertical to the first surface of the photonic integrated circuit, 0<θ**1**<10°, the fiber harness when extending from the first surface of the photonic integrated circuit and bending to a direction parallel to the first surface requires a clearance distance of at least d**3** so as to not damage the optical fibers in the fiber harness, and wherein d**1**<d**3**, and d**2**<d**3**.
(957) Embodiment 8: The apparatus of embodiment 7 in which the housing has a first inner side wall and a second inner side wall, the substrate or circuit board is attached to the first inner side wall, wherein a distance from the first surface of the photonic integrated circuit to the second inner side wall is d**4**, and d**3**<d**4**.
(958) Embodiment 9: The apparatus of embodiment 7 in which the first surface of the photonic integrated circuit is oriented at an angle θ**2** relative to the inner upper wall, and 45°<θ**2**<135°.
(959) Embodiment 10: The apparatus of embodiment 9 in which 70°<θ**2**<110°.
(960) Embodiment 11: The apparatus of embodiment 10 in which 80°<θ**2**<100°.
(961) Embodiment 12: The apparatus of embodiment 11 in which 85°<θ**2**<95°.
(962) Embodiment 13: The apparatus of embodiment 9 in which the photonic integrated circuit is mounted on a substrate or circuit board that is electrically coupled to the edge connector by one or more flexible cables.

(963) Embodiment 14: The apparatus of embodiment 7 in which the photonic integrated circuit is mounted on an upper surface of a substrate or circuit board, the edge connector has an upper surface and a lower surface, the lower surface of the edge connector is attached to the upper surface of the substrate or circuit board, the upper surface of the substrate or circuit board is at a distance d**4** relative to the inner upper wall of the housing, and d**3**<d**4**.
(964) Embodiment 15: The apparatus of any of embodiments 1 to 14 in which the photonic integrated circuit is configured to perform at least one of (i) convert optical signals received from the optical fiber cable to electrical signals that are transmitted to the edge connector, or (ii) convert electrical signals that are received from the edge connector to optical signals that are transmitted to the optical fiber cable.
(965) Embodiment 16: The apparatus of any of embodiments 1 to 15 in which the optical module comprises a first set of at least two electrical integrated circuits that are mounted on the first surface of the photonic integrated circuit.
(966) Embodiment 17: The apparatus of embodiment 16 in which the first set of at least two electrical integrated circuits comprise two electrical integrated circuits that are positioned on opposite sides of the optical fiber connector along a plane parallel to the first surface of the photonic integrated circuit.
(967) Embodiment 18: The apparatus of embodiment 17 in which the first set of at least one electrical integrated circuit comprises four electrical integrated circuits that surround four sides of the optical fiber connector along the plane parallel to the first surface of the photonic integrated circuit.
(968) Embodiment 19: The apparatus of any of embodiments 16 to 18 in which the optical module comprises: a substrate or circuit board, in which the photonic integrated circuit is mounted on the substrate or circuit board, and a second set of at least one electrical integrated circuit mounted on the substrate or circuit board and electrically coupled to the photonic integrated circuit through one or more signal conductors and/or traces.
(969) Embodiment 20: The apparatus of any of embodiments 16 to 19 in which the photonic integrated circuit comprises at least one of a photodetector or an optical modulator, and the first set of at least one integrated circuit comprises at least one of a transimpedance amplifier configured to amplify a current generated by the photodetector or a driver configured to drive the optical modulator.
(970) Embodiment 21: The apparatus of embodiment 19 or 20 in which the second set of at least one electrical integrated circuit comprises a serializers/deserializers module.
(971) Embodiment 22: The apparatus of any of embodiments 1 to 21 in which the pluggable optical module comprises at least one laser source that is configured to provide power supply light to the photonic integrated circuit.
(972) Embodiment 23: The apparatus of embodiment 22 in which the fiber harness comprises at least one optical fiber that optically couples the at least one laser source to the photonic integrated circuit.
(973) Embodiment 24: The apparatus of embodiment 23 in which the optical fiber connector comprises at least one power supply fiber port.
(974) Embodiment 25: The apparatus of any of embodiments 1 to 24, comprising a second circuit board and a cage mounted on the second circuit board, in which the pluggable optical module is plugged into the cage, and the receptacle is located inside the cage.
(975) Embodiment 26: The apparatus of embodiment 25, comprising a server computer comprising a first data processor, in which the second circuit board is part of the server computer, the pluggable optical module is configured to provide a communication interface that enables the first data processor to communicate with a second data processor through the optical fiber cable.
(976) Embodiment 27: The apparatus of embodiment 26 in which the first data processor comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC).
(977) Embodiment 28: The apparatus of embodiment 27, comprising at least one of a supercomputer, an autonomous vehicle, or a robot, wherein the supercomputer, the autonomous vehicle, or the robot comprises the server computer.
(978) Embodiment 29: The apparatus of any of embodiments 26 to 28 in which the server computer comprises a plurality of cages of embodiment 25 and a plurality of pluggable optical modules of any of embodiments 1 to 25, the plurality of pluggable optical modules are plugged into the plurality of cages, each pluggable optical module is plugged into a corresponding cage.
(979) Embodiment 30: A system comprising: a data center comprising: a plurality of server computers of embodiment 26 or 27; and a plurality of pluggable optical modules of any of embodiments 1 to 24; wherein each server computer communicates with one or more other server computers through one or more optical fiber cables and the plurality of pluggable optical modules.
(980) Embodiment 31: An apparatus comprising: a pluggable optical module comprising: a fiber connector configured to be optically coupled to an optical fiber cable; an optical module comprising: a photonic integrated circuit having a first surface; and a first set of at least two electrical integrated circuits that are mounted on the first surface of the photonic integrated circuit; a fiber harness optically coupled between the fiber connector and the first surface of the photonic integrated circuit; and an edge connector having conductive pads configured to be electrically coupled to conductive pads of a receptacle when the edge connector is mated with the receptacle, in which the conductive pads of the edge connector are electrically coupled to the optical module.
(981) Embodiment 32: The apparatus of embodiment 31 in which the pluggable optical module complies with a small

form factor pluggable module specification comprising at least one of SFP (small form-factor pluggable), SFP+, 10 Gb SFP, SFP28, OSFP (octal SFP), OSFP-XD (OSFP extra dense), QSFP (quad small form-factor pluggable), QSFP+, QSFP28, QSFP56, or QSFP-DD (quad small form-factor pluggable double density).
(982) Embodiment 33: The apparatus of embodiment 31 or 32 in which the fiber harness comprises an optical connector that is coupled to the photonic integrated circuit, the first set of at least two electrical integrated circuits comprise two electrical integrated circuits that are positioned on opposite sides of the optical connector along a plane parallel to the first surface of the photonic integrated circuit.
(983) Embodiment 34: The apparatus of embodiment 33 in which the first set of at least one electrical integrated circuit comprises four electrical integrated circuits that surround four sides of the optical connector along the plane parallel to the first surface of the photonic integrated circuit.
(984) Embodiment 35: The apparatus of any of embodiments 31 to 34 in which the optical module comprises: a substrate or circuit board, in which the photonic integrated circuit is mounted on the substrate or circuit board, and a second set of at least one electrical integrated circuit mounted on the substrate or circuit board and electrically coupled to the photonic integrated circuit through one or more signal conductors and/or traces.
(985) Embodiment 36: The apparatus of any of embodiments 31 to 35 in which the photonic integrated circuit comprises at least one of a photodetector or an optical modulator, and the first set of at least one integrated circuit comprises at least one of a transimpedance amplifier configured to amplify a current generated by the photodetector or a driver configured to drive the optical modulator.
(986) Embodiment 37: The apparatus of embodiment 35 or 36 in which the second set of at least one electrical integrated circuit comprises a serializers/deserializers module.
(987) Embodiment 38: The apparatus of any of embodiments 31 to 37 in which the pluggable optical module comprises a housing having an inner bottom wall, an inner upper wall, and inner side walls, wherein the inner bottom, upper, and side walls define a space to accommodate the optical module; wherein the optical module is oriented relative to the housing such that the first surface of the photonic integrated circuit is at an angle between 45° to 135° relative to the bottom surface of the housing.
(988) Embodiment 39: The apparatus of embodiment 38 in which the optical module is oriented relative to the housing such that the first surface of the photonic integrated circuit is at an angle between 70° to 110° relative to the bottom surface of the housing.
(989) Embodiment 40: The apparatus of embodiment 39 in which the optical module is oriented relative to the housing such that the first surface of the photonic integrated circuit is at an angle between 80° to 100° relative to the bottom surface of the housing.
(990) Embodiment 41: The apparatus of embodiment 40 in which the optical module is oriented relative to the housing such that the first surface of the photonic integrated circuit is at an angle between 85° to 95° relative to the bottom surface of the housing.
(991) Embodiment 42: The apparatus of any of embodiments 35 to 37 in which the pluggable optical module comprises a housing having an inner upper wall and an inner lower wall, the edge connector has an upper surface extending along a first plane that is at a first distance d**1** relative to the inner upper wall, the edge connector has a lower surface extending along a second plane that is at a second distance d**2** relative to the inner lower wall, wherein the fiber harness is substantially vertically coupled to the first surface of the photonic integrated circuit such that light from the fiber harness is directed toward the first surface of the photonic integrated circuit at an angle θ**1** relative to a direction vertical to the first surface of the photonic integrated circuit, 0<θ**1**<10°, the fiber harness when extending from the first surface of the photonic integrated circuit and bending to a direction parallel to the first surface requires a clearance distance of at least d**3** so as to not damage the optical fibers in the fiber harness, and wherein d**1**<d**3**, and d**2**<d**3**.
(992) Embodiment 43: The apparatus of embodiment 42 in which the housing has a first inner side wall and a second inner side wall, the substrate or circuit board is attached to the first inner side wall, wherein a distance from the first surface of the photonic integrated circuit to the second inner side wall is d**4**, and d**3**<d**4**.
(993) Embodiment 44: The apparatus of any of embodiments 31 to 43 in which the photonic integrated circuit is configured to perform at least one of (i) convert optical signals received from the optical fiber cable to electrical signals that are transmitted to the edge connector, or (ii) convert electrical signals that are received from the edge connector to optical signals that are transmitted to the optical fiber cable.
(994) Embodiment 45: A method comprising: transmitting signals between an optical fiber cable and a data processing apparatus through a pluggable optical module having a photonic integrated circuit, including: transmitting optical signals between the optical fiber cable and the photonic integrated circuit through a fiber harness and a plurality of optical couplers provided at a first surface of the photonic integrated circuit; and transmitting electrical signals between the photonic integrated circuit and the data processing apparatus through an edge connector of the pluggable optical module; wherein the fiber harness comprises a plurality of optical fibers and an optical fiber connector that optically couples the plurality of optical fibers to the plurality of optical couplers at the first surface of the photonic integrated circuit; and wherein the optical fiber connector comprises a two-dimensional arrangement of fiber ports that are optically coupled to the optical couplers at the first surface of the photonic integrated circuit.

Claims

1. A system comprising: a housing for a rackmount server having a front panel; a first substrate that is positioned at a distance from the front panel, in which a data processor is mounted on the first substrate, wherein the first substrate is spaced apart from the front panel at a distance less than 12 inches, and the first substrate has a main surface that is oriented at an angle in a range of 0 to 45 degrees relative to the front panel; and a pluggable module comprising an optical module, at least one first optical connector, a first fiber optic cable that is optically coupled between the optical module and the first optical connector, wherein the pluggable module has a bandwidth of at least 10 gigabits per second; wherein the optical module is configured to receive optical signals from the first optical connector and generate electrical signals based on the received optical signals, and the electrical signals or processed versions of the electrical signals are transmitted to the data processor; wherein the pluggable module has a shape that enables the pluggable module to pass through an opening in the front panel to enable the optical module to be coupled to the first substrate; and wherein the data processor is capable of processing data at a rate of at least 100 gigabits per second.
2. The system of claim 1 in which the first optical connector is configured to mate with a corresponding optical connector of an external fiber optic cable.
3. The system of claim 1 in which the first optical connector comprises a multi-fiber push on (MPO) connector.
4. The system of claim 1 in which the pluggable module comprises at least two first optical connectors, and each first optical connector is configured to be mated with a second optical connector of an external fiber optic cable.
5. The system of claim 1 in which the pluggable module comprises at least four first optical connectors, and each first optical connector is configured to be mated with a second optical connector of an external fiber optic cable.
6. The system of claim 1 in which the first fiber optic cable comprises a fiber pigtail.
7. The system of claim 1 in which the first substrate is oriented parallel to the front panel.
8. The system of claim 1 in which the first substrate has a first side and a second side that is opposite the first side, the data processor comprises electrical contacts that are electrically coupled to electrical contacts on the first side of the first substrate, the pluggable module comprises electrical contacts that are electrically coupled to electrical contacts on the second side of the first substrate, and at least some of the electrical contacts on the first side of the first substrate are electrically coupled to at least some of the electrical contacts on the second side of the first substrate.
9. The system of claim 8 in which the first substrate comprises at least one of a ceramic substrate, an organic high density build-up substrate, or a silicon substrate.
10. The system of claim 1 in which the system comprises a second substrate, the data processor comprises electrical contacts that are electrically coupled to electrical contacts on the first substrate, the pluggable module comprises electrical contacts that are electrically coupled to electrical contacts on the second substrate, and at least some of the electrical contacts on the first substrate are electrically coupled to at least some of the electrical contacts on the second substrate.
11. The system of claim 10 in which the first substrate is mounted on a first side of a third substrate or circuit board, and the second substrate is mounted on a second side of the third substrate or circuit board.
12. The system of claim 11 in which each of the first and second substrate comprises at least one of a ceramic substrate, an organic high density build-up substrate, or a silicon substrate.
13. The system of claim 1, comprising a laser module configured to provide optical power to the optical module.
14. The system of claim 13, comprising a second optical connector optically coupled to the laser module, wherein the pluggable module comprises a third optical connector that is configured to mate with the second optical connector when the pluggable module is coupled to the first substrate, and wherein the first optical connector is optically coupled to the optical module to enable the optical module to receive the optical power from the laser module.
15. The system of claim 13, comprising a first heat dissipating device and a second heat dissipating device, the first heat dissipating device is thermally isolated from the second heat dissipating device, the first heat dissipating device is thermally coupled to the optical module, and the second heat dissipating device is thermally coupled to the laser module.
16. The system of claim 1 wherein the pluggable module comprises a fiber guide that is positioned between the optical module and the first optical connector and provides mechanical support for the optical module and the first optical connector.
17. The system of claim 1 wherein the optical module has a first side and a second side, the first fiber optical cable has a first end that has a two-dimensional arrangement of optical fiber cores, the first side of the optical module is optically coupled to the two-dimensional arrangement of optical fiber cores, and the second side of the optical module has a two-dimensional arrangement of electrical contacts that are configured to mate with a two-dimensional arrangement of electrical contacts on the first substrate.
18. The system of claim 17 wherein the two-dimensional arrangement of electrical contacts of the optical module comprise at least two rows of electrical contacts, and each row includes at least two electrical contacts.
19. The system of claim 18 wherein the two-dimensional arrangement of electrical contacts of the optical module comprise at least four rows of electrical contacts, and each row includes at least four electrical contacts.
20. The system of claim 19 wherein the two-dimensional arrangement of electrical contacts of the optical module comprise at least ten rows of electrical contacts, and each row includes at least ten electrical contacts.
21. The system of claim 16 wherein the data processor comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, an application specific integrated circuit (ASIC), or a storage device.
22. The system of claim 16 in which the fiber guide has a length configured such that when the pluggable module is

inserted through the opening in the front panel and the optical module is coupled to the first substrate or a module mounted on the first substrate, the at least one first optical connector is in a vicinity of the front panel to enable a user to attach at least one external fiber optic cable to the at least one first optical connector.
23. The system of claim 16 in which the fiber guide has a length configured such that when the pluggable module is inserted through the opening in the front panel and the optical module is coupled to the first substrate or a module mounted on the first substrate, the at least one first optical connector has a front surface that is flush with, or protrudes from, a front surface of the front panel to enable a user to attach at least one external fiber optic cable to the at least one first optical connector.
24. The system of claim 16 in which the fiber guide has a length configured such that when the pluggable module is inserted through the opening in the front panel and the optical module is coupled to the first substrate or a module mounted on the first substrate, the at least one first optical connector has a front face that is within an inch of a front surface of the front panel.
25. The system of claim 16 in which the fiber guide has a length of at least one inch.
26. The system of claim 16 in which the fiber guide has a length of at least two inches.
27. The system of claim 16 in which the fiber guide has a length of at least four inches.
28. The system of claim 16 wherein the fiber guide has a length configured such that when the pluggable module is inserted through the opening in the front panel and the optical module is coupled to the first substrate or a module mounted on the first substrate, the at least one first optical connector has a front face that is within an inch of a front surface of the front panel.
29. The system of claim 28, comprising two or more pluggable modules, in which each pluggable module comprises an optical module, at least one first optical connector, a first fiber optic cable that is optically coupled between the optical module and the first optical connector, and a fiber guide that is positioned between the optical module and the first optical connector; wherein the fiber guides are configured to allow air blown from an inlet fan to flow past the fiber guides and carry away heat generated by the optical modules.
30. The system of claim 16 wherein the fiber guide comprises at least one of metal or a thermal conductive material.
31. The system of claim 16 wherein the fiber guide comprises a hollow tube.
32. The system of claim 16 wherein the fiber guide is rigid along a direction from the first optical connector to the optical module and has a strength sufficient to withstand a compression force exerted on the pluggable module when the pluggable module is inserted through the opening in the front panel and coupled to the first substrate.
33. The system of claim 16 wherein the fiber guide has a spatial fan-out design such that a first portion of the fiber guide near the optical module has a smaller dimension compared to the dimension of a second portion of the fiber guide near the at least one first optical connector.
34. The system of claim 33 wherein the at least one first optical connector has an overall footprint that is larger than a footprint of the optical module.
35. The system of claim 16 wherein a photon supply is disposed in, on, or near the fiber guide, and the photon supply is configured to provide optical power supply light to the optical module.
36. The system of claim 35 wherein the photon supply is thermally coupled to an inner surface or an outer surface of the fiber guide, and the fiber guide is configured to assist in dissipating heat from the photon supply.
37. The system of claim 16, comprising guide rails configured to guide the optical module as the optical module moves from a first position near the front panel to a second position near the first substrate.
38. The system of claim 37 wherein the optical module comprises a co-packaged optical module comprising a photonic integrated circuit and one or more electrical integrated circuits that condition electrical signals transmitted to or from the photonic integrated circuit.
39. The system of claim 38, comprising a co-packaged optical module (CPO) mount attached to the first substrate, and the guide rails are configured to provide rigid connections between the CPO mount and the front panel or a front portion of the fiber guide.
40. The system of claim 38 wherein the photonic integrated circuit comprises at least one of (i) a photodetector to convert optical signals to electrical signals, or (ii) a modulator to convert electrical signals to optical signals.
41. The system of claim 37 wherein the system comprises a co-packaged optical module (CPO) mount and a bolster plate, the co-packaged optical module is mounted on the first substrate through the CPO mount, and the bolster plate is positioned to the rear of the substrate and configured to exert a force in a front direction when the guide rails are fastened to a front portion of the fiber guide or to the front panel.
42. The system of claim 1, in which the first substrate has a main surface that is oriented at an angle in a range of 0 to 45 degrees relative to the front panel.
43. The system of claim 1, comprising a laser module configured to provide optical power to the optical module.
44. The system of claim 43, comprising a first heat dissipating device and a second heat dissipating device, the first heat dissipating device is thermally isolated from the second heat dissipating device, the first heat dissipating device is thermally coupled to the optical module, and the second heat dissipating device is thermally coupled to the laser module.
45. The system of claim 1 in which the fiber guide comprises at least one of metal or a thermal conductive material.
46. The system of claim 1 in which the fiber guide comprises a hollow tube.
47. The system of claim 1 in which the fiber guide is rigid along a direction from the first optical connector to the optical

module and has a strength sufficient to withstand a compression force exerted on the pluggable module when the pluggable module is inserted through the opening in the front panel and coupled to the first substrate.

48. The system of claim 1 in which the fiber guide has a spatial fan-out design such that a first portion of the fiber guide near the optical module has a smaller dimension compared to the dimension of a second portion of the fiber guide near the at least one first optical connector.

49. The system of claim 48 in which the at least one first optical connector has an overall footprint that is larger than a footprint of the optical module.

50. The system of claim 1 in which the data processor comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, an application specific integrated circuit (ASIC), or a storage device.

51. The system of claim 1 in which a photon supply is disposed in, on, or near the fiber guide, and the photon supply is configured to provide optical power supply light to the optical module.

52. The system of claim 51 in which the photon supply is thermally coupled to an inner surface or an outer surface of the fiber guide, and the fiber guide is configured to assist in dissipating heat from the photon supply.

53. The system of claim 1, comprising guide rails configured to guide the optical module as the optical module moves from a first position near the front panel to a second position near the first substrate.

54. A rackmount server comprising: a housing having a front panel and a rear panel, in which the front panel defines an opening, and the rear panel is at a first distance from the front panel; and a substrate that is positioned at a second distance from the front panel, in which the second distance is less than one-third of the first distance, a data processor is mounted on the substrate, the substrate has a main surface that is oriented at an angle in a range of 0 to 45 degrees relative to the front panel; wherein at least one of (i) the substrate has electrical contacts that are configured to be electrically coupled to electrical contacts of a co-packaged optical module, or (ii) a first module is mounted on the substrate, in which the first module has electrical contacts that are configured to be electrically coupled to electrical contacts of a co-packaged optical module.

55. The rackmount server of claim 54 in which the opening in the front panel is configured to allow a pluggable module that includes the co-packaged optical module to be inserted through the opening to enable the co-packaged optical module to be electrically coupled to the electrical contacts on the substrate or the electrical contacts on the first module mounted on the substrate.